18th INTERNATIONAL CONFERENCE ON SCIENCE AND INNOVATIVE ENGINEERING

—18 ICSIE 2026—

PROCEEDINGS OF THE HYBRID CONFERENCE

Hybrid

11th and 12th OCTOBER 2026



Organised By

PRINCE DR. K. VASUDEVAN
COLLEGE OF ENGINEERING AND TECHNOLOGY
Chennai, India

In Collaboration With

OSIET, INDIA

MANIPAL UNIVERSITY COLLEGE, MALAYSIA

18th INTERNATIONAL CONFERENCE ON "SCIENCE AND INNOVATIVE ENGINEERING – 2026" (ICSIE – 2026)

October 11th – 12th, 2026

ORGANIZED BY

ORGANIZATION OF SCIENCE AND INNOVATIVE ENGINEERING & TECHNOLOGY (OSIET)

Chennai, India.

Website: www.ijsiet.org.in

In Association with

PRINCE DR. K. VASUDEVAN COLLEGE OF ENGINEERING AND TECHNOLOGY

Medavakkam - Mambakkam Road, Ponmar, Chennai – 600 127

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In Collaboration

Manipal University College Malaysia (Melaka Campus)

Melaka, Malaysia

https://manipal.edu.my/

ICSIE 2026

Proceeding of 18th International Conference on Science and Innovative Engineering

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(OSIET), Chennai

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CHAIRMAN'S MESSAGE

I am delighted to welcome you all to the International Conference on Science and Innovative Engineering" (ICSIE 2026) which brings together experts and academics from around the world. The conference this year has brought together a tremendous and rich diversity of authors and speakers from universities, government and industry to share ideas and new perspectives on a wide range of Communications, Electronics, Networking, Computer Science, Mechanical and Electricals research and technologies. Addressing the new technical and business issues are essential to advance in today's engineering and technological environments. In order to provide an outstanding technical level for the presentations at the conference, we have invited more than 15 distinguished experts in the Engineering and Technology field to participate in the Advisory Committee. Academic excellence has always been the hall mark of our institute and we are committed to provide a comprehensive education which seeks to develop the students into academically proficient, morally upright and socially well integrated individuals. I would like to express my thanks to all authors for their outstanding contributions and in particular the members of the program board for their competent evaluation of the large number of submissions

We are very pleased with the quality, depth, and breadth of this year's technical program. I wish you a most enjoyable experience at the Conference.

Dr.K.Vasudevan

Chairman
Prince Group of Educational Institutions

VICE CHAIRMAN'S MESSAGE



The words "Engineering" and "Technology" are not just words but are vital disciplines that guide the world by bringing huge changes to the current way of living. Modern Engineering and Technological innovations are making people think that nothing is impossible in this world. Everything can be achieved with an extremely powerful vision and a path to attain that vision. If a person is doing a marvel, keep in mind that you too can outshine them with perseverance. Dr. A. P. J Abdul Kalam said "Dream is not something that comes when you sleep, it is something that doesn't let you sleep", so, dream a lot about how to bring changes and try to make the changes come true. Remember earning money is not life; it is just a part of life which can be easily earned when you do it with dedication. Science and technology have become so closely intertwined, and I hope the conference will help you to reinforce each other.

We are very much elated in welcoming you all to our college for the 17th International Conference on Science and Innovative Engineering and I wish the students all the very best for their presentations.

Dr. V.Vishnu Karthick,
Vice Chairman,
Prince Shri Venkateshwara Padmavathy Engineering College.

ADMINISTRATIVE OFFICER'S MESSAGE



The world is growing fast in fact it grows in such a way that a thing invented yesterday becomes obsolete today, which means there is a plethora of competition in each and every discipline of engineering and technology. This scenario goes well with a famous quote of a philosopher Heraclitus, who said, "Change is the only constant in this world." Nowadays people need new inventions for their daily works so that they can do it with ease. People not only need to do their activities with ease but they also need it to be done within a period of time. As an engineer we need to solve problems that are left unsolved. As a way of showcasing the talents of young minds to find solutions we have organised this 18th International Conference on Science and Innovative Engineering.

I welcome you all to this event, my warm wishes to the students who are going to present their papers and also hearty thanks for the staff who have made this event a great success.

Er. K. Parthasarathy

Administrative Officer

Prince Group of Educational Institutions

PRINCIPAL'S MESSAGE



I am indeed most delighted to chair this 18th International Conference on Science and Innovative Engineering (ICSIE 2026). The sharp, clear sighted vision and precise decision making powers of our management has benefited our college to stay competitive. The Pedagogy at the Institute is Modern where a variety of learning, behavioural tools are used in quality pursuance of knowledge, development of skills, attitudes, and values complemented by academia-industry interface imparting uniqueness to our programme. The Institute focuses on the holistic development of its students through variety of methodologies and extracurricular activities the whole year round. Today, we live in an era of incredibly rapid technological change. Technology has dominated our lives and we now have the ever evolving technology at our finger tips. Symposium, Seminars and International Conferences and Workshops are organized in the institute for their overall development. Teaching and Research are the two primary activities through which we fulfil our Mission and Objective. The Institute takes pride in welcoming the National and International participants for 18th International Conference.

With a firm foundation of the past and high hopes for a bright future, I wish everyone good luck and prosperity ahead. May we grow many technological wings!

Dr. T.Sunder Selwyn
Principal
Prince Dr. K. Vasudevan College of Engineering and Technology.

MANIPAL
UNIVERSITY COLLEGE
MALAYSIA



Greetings to all esteemed faculty, researchers, industry leaders, and students attending the 18th International Conference on Science and Innovative Engineering (ICSIE 2026) at Prince Dr. K. Vasudevan College of Engineering and Technology, Chennai, India. I am honored to be delivering the keynote address at this prestigious event on October 11th & 12th. This conference presents a remarkable platform to explore the latest advancements and trends shaping the landscape of engineering, technology, and management. This conference fosters a dynamic environment where all researchers can engage in thought-provoking discussions, share ground-breaking research findings, and forge meaningful collaborations. These interactions will undoubtedly spark innovation and propel us towards a future brimming with possibilities.

Whether the delegates are seasoned professional or a budding researcher, I believe that ICSIE 2026 will offer an unique opportunity to learn from each other, share everyone's perspectives, and build lasting connections. I am eager to connect with all of the delegates and witness the innovative ideas that emerge from this vibrant gathering. Together, let us push the boundaries of knowledge and create a future where engineering, technology, and management work in harmony for the betterment of our world.

I look forward to a stimulating and productive conference!!

DR.ANTONY V.SAMROT
DIRECTOR (RESEARCH, INNOVATION AND POSTGRADUATE STUDIES)
MANIPAL UNIVERSITY COLLEGE MALAYSIA
MELAKA, MALAYSIA

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
Autonomous Institute | Affiliated to Osmania University



Technical Education is the backbone of every nation and is the stepping stone for a country to move into the niche of a developed nation. India Entrepreneurs are emerging as a global Entrepreneur, Indian figuring increasingly in the list of the richest persons in the world. India's knowledge power making India as a preferred destination for outsourcing knowledge services from India. India is fast emerging as a destination for world class R&D centres and innovation hub. Young friends they are the signs of even a brighter tomorrow for India and its people.

You must therefore be highly excited to make your own contributions to the growth and development of India as a dream.

As a Keynote speaker of the 18th International Conference on Science and Innovative Engineering (ICSIE 2026), I congratulate all the participants who have contributed their technical articles in the proceedings.

DR. G. N. R. PRASAD
ASSOCIATE PROFESSOR
DEPARTMENT OF MCA
CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY
GANDIPET, TELANGANA, INDIA

ABOUT INSTITUTION

Prince Educational Society was established in 1978 by our Founder - Chairman Dr. K. Vasudevan, M.A., B.Ed., Ph.D. Going down memory lane in the seventies, our Chairman had realized the need for an Institution which will serve as a role model and stand apart from other Educational Institutions.

VISION

The main aim of Prince Dr. K. Vasudevan College of Engineering & Technology is to meet the challenges and demand of the highly advanced scientific and technological fields and to prepare to meet the man power needs of the world.

MISSION

To provide goal-oriented, quality-based and value-added education through state-of-the-art technology on a par with international standards. To promote nation - building activities in science, technology, humanities and management through research. To create and sustain a community of learning that sticks on to social, ethical, ecological, cultural and economic.

About PDKVCET

Prince Dr.K.Vasudevan College of Engineering and Technology has emerged as one of the fastest growing institutions of higher learning in India. We started out as a modest institute in 2009, to serve the needs of the local community. Today, as we enter the 17th year, as a flagship institute, we continue to be in service of country and society. Prince Dr.K.Vasudevan College of Engineering and Technolgy has a plethora of traditions that make the institute a unique and truly special place. Our institution that produces the next generation of leaders and advances tomorrow's thinking. Prince Educational Society was established in 1978 by our Founder - Chairman Dr. K. Vasudevan, M.A., B.Ed., Ph.D. Going down memory lane in the seventies, our Chairman had realized the need for an Institution which will serve as a role model and stand apart from other Educational Institutions. The setting up of Prince Matriculation Higher Secondary School, Nanganallur, Chennai, helped in the evolution of Prince Matriculation Higher Secondary School, Madipakkam, Chennai, which met this need. In order to serve the poor and the needy, he started a Tamil Medium School as well. Today this educationist, Industrialist and pioneer heads an Arts and Science College and an Engineering College in Chennai. These Institutions aim at imparting quality education in the fields of engineering, arts and science.

OUR GROUP OF INSTITUTIONS

Prince Shri Venkateshwara Padmavathy Engineering College

Prince Dr. K. Vasudevan College of Engineering & Technology

Prince Shri Venkateshwara Arts and Science College

Prince Matriculation Higher Secondary School,7,Kannagi Street, Madipakkam to Puzhuthivakkam, Chennai 091

Prince Sri Vari Vidyalaya CBSE School,93 ,College Road , Nanganallur, Chennai 114.

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Prince Matriculation Higher Secondary School, 67&68, College Road, Nanganallur, Chennai 114.

ICSIE 2026

PATRON-IN-CHIEF

Thiru. Dr. K. Vasudevan, M.A., B.Ed., Ph.D,

Chairman, Prince Group of Educational Institutions.

Thiru. Dr.V.VishnuKarthik, MD, Vice-Chairman, Prince Group of Educational Institutions.

SECRETARY

Er.K.Parthasarathy, B.E,

Administrative Officer, Prince Group of Educational Institutions.

CONFERENCE CHAIR

Dr.Sunder Selwyn, Principal,
Prince Dr.K.Vasudevan College of Engineering

KEYNOTE SPEAKERS

Dr.Antony V.Samrot, Director (Research, Innovation and Postgraduate Studies) Manipal University College Malaysia, Melaka, Malaysia

Dr. G. N. R. PRASAD, Associate Professor, Chaitanya Bharathi Institute of Technology, Gandipet, Telangana, India

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103	ICSIEOCT260617	DESIGN AND ANALYSIS OF A PROSTHETIC LIMB WITH SHOCK ABSORPTION AND LIGHT WEIGHT COMPOSITE MATERIALS
104	ICSIEOCT260615	DERMOSCOPIC IMAGE ANALYSIS FOR SKIN CANCER ASSESSMENT: VISUAL FEATURE INTERPRETATION AND AN ENHANCED PREPROCESSING FRAMEWORK
105	ICSIEOCT260610	EARLY DETECTION OF DIABETIC RETINOPATHY USING DEEP LEARNING
106	ICSIEOCT260562	INTELLISTUDY: AN AI-POWERED WEBCAM COMPANION FOR SMART LAPTOP STUDY MONITORING
107	ICSIEOCT260561	SMART AUTOMATION SYSTEM FOR A SMALL SCALE INDUSTRIAL OVEN
108	ICSIEOCT260677	DEVELOPMENT OF A PLC-BASED INTELLIGENT INTERLOCKING AND PROCESS CONTROL SYSTEM FOR INDUSTIAL HOIST CRANE AUTOMATION
109	ICSIEOCT260807	A HYBRID ANN-LSTM FRAMEWORK FOR ACCURATE CARDIOVASCULAR DISEASE DETECTION
110	ICSIEOCT260654B	MULTIMODAL ARTIFICIAL INTELLIGENCE FOR PLANT DISEASE DETECTION: INTEGRATING CNNS, VISION TRANSFORMERS, BERT, GENERATIVE MODELS, AND AGENTIC AI
111	ICSIEOCT260646	YOLOV8N-BASED DETECTION OF MARINE DEBRIS AND BIOLOGICAL OBJECTS IN UNDERWATER IMAGES
112	ICSIEOCT2600477	A LEAKAGE-FREE MATERIALS INFORMATICS FRAMEWORK FOR AUTOMATED ASTM E112 GRAIN SIZE CLASSIFICATION AND HARDNESS PREDICTION IN TI-6AL-4V ALLOYS
113	ICSIEOCT260604	MONITORING SPATIOTEMPORAL DROUGHT DYNAMICS USING AN ENTROPY-BASED SENTINEL-1 SAR INDEX AND GOOGLE EARTH ENGINE (GEE): A CASE STUDY OF PATNA DISTRICT, INDIA
114	ICSIEOCT260603	ROBUST MULTICLASS LEUKEMIA CLASSIFICATION USING HYBRID FEATURE REDUCTION AND ENSEMBLE LEARNING
115	ICSIEOCT260903	A DATA-DRIVEN DECISION SUPPORT SYSTEM FOR HEART DISEASE CLASSIFICATION USING MULTI-PARAMETRIC CLINICAL ATTRIBUTES
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117	ICSIEOCT2600347	PREDICTING E-COMMERCE PURCHASE INTENTIONS USING ENSEMBLE LEARNING AND EXPLAINABLE AI
118	ICSIEOCT260887	DESIGN AND DEVELOPMENT OF A SOLAR-POWERED AUTOMATIC IRRIGATION SYSTEM USING SOIL MOISTURE-BASED CONTROL
119	ICSIEOCT2600483	DEEP LEARNING TECHNIQUES FOR MULTI-CLASS CLASSIFICATIONS OF DIABETIC RETINOPATHY SEVERITY
120	ICSIEOCT260649	SMART COMMUNITY TOOL & EQUIPMENT SHARING LIBRARY USING MOBILE APP

121	ICSIEOCT260714	ANALYZING AND IDENTIFYING SOCIAL MEDIA ADDICTION THROUGH COMPUTATIONAL MODELS AND DATA-DRIVEN APPROACHES-A SYSTEMATIC LITERATURE REVIEW
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125	ICSIEOCT260819	DESIGN AND DEVELOPMENT OF IOT-INTEGRATED PLC-BASED PREDICTIVE MULTI-ZONE IRRIGATION SYSTEM
126	ICSIEOCT260763	FROM ELECTROCARDIOGRAPHY TO ACTION: A REAL-TIME HEART ATTACK RISK ALERTING FRAMEWORK
127	ICSIEOCT260921	PERSONALIZED PHISHING AWARENESS AND ADAPTIVE EMPLOYEE RISK MANAGEMENT FRAMEWORK USING SALESFORCE
128	ICSIEOCT2600379A	DESIGN AND DEVELOPMENT OF A SOLAR-POWERED SMART AQUAPONIC SYSTEM WITH REAL-TIME MONITORING AND AUTOMATIC WATER CIRCULATION
129	ICSIEOCT260742	AN MRI IMAGE ENHANCEMENT AND AFFECTED REGION LOCALIZATION FRAMEWORK FOR ALZHEIMER DISEASE
130	ICSIEOCT260913	GRAPHENE-ASSISTED TERAHERTZ DIELECTRIC RESONATOR BIOSENSOR FOR QUANTITATIVE DETECTION OF VITAMIN B2 CONCENTRATION USING RESONANCE SHIFT ANALYSIS
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132	ICSIEOCT260865	REAL-TIME SEQUENCE MODELLING VIA LIGHTWEIGHT KEYPOINT NETWORKS FOR ENHANCED SIGN LANGUAGE INTERPRETATION
133	ICSIEOCT260907	BRAIN STROKE DETECTION, LESION SEGMENTATION AND BRAIN SWELLING RISK PREDICTION USING BIOMECHANICAL MODELLING
134	ICSIEOCT260767	THE DIGITAL COMMERCE MATRIX: A PLATFORM SELECTION FRAMEWORK FOR ALIGNING E-COMMERCE ARCHITECTURE WITH BUSINESS MODEL
135	ICSIEOCT2601033	AI-BASED PREDICTIVE MAINTENANCE SYSTEM FOR INDUSTRIAL EQUIPMENT USING MULTISENSOR AND VISION-BASED CONDITION ANALYSIS
136	ICSIEOCT260941	MACHINE LEARNING-BASED MULTI-CLASS BRAIN DISORDER PREDICTION: A COMPREHENSIVE REVIEW
137	ICSIEOCT2601067	AI AND IOT BASED SMART WASTE SEGREGATION SYSTEM AND BIN-LEVEL INDICATOR
138	ICSIEOCT260873	EOPS: A MULTI-AGENT AI ENGINEERING OPERATIONS PLATFORM FOR AUTOMATED INCIDENT INVESTIGATION AND ROOT CAUSE ANALYSIS
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141	ICSIEOCT2601087	CRASH SHIELD: AN ADVANCED AIRBAG SYSTEM FOR ENHANCED RIDER SAFETY
142	ICSIEOCT2601169	MICROSTRUCTURE AND HIGH-TEMPERATURE EROSION BEHAVIOUR OF PLASMA-SPRAYED NICRBSI COMPOSITE COATINGS WITH WASTE GRANITE AS REINFORCEMENT
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144	ICSIEOCT260765	SANSKRIT–MARATHI NEURAL MACHINE TRANSLATION: A REVIEW OF TRANSFORMER-BASED APPROACHES, CHALLENGES, AND FUTURE DIRECTIONS

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147	ICSIEOCT260796	ASSESSMENT OF MINIMUM MUSCULAR FITNESS IN MIDDLE AGED WOMEN WITH AND WITHOUT LOW BACK PAIN
148	ICSIEOCT260815	MACHINE LEARNING FOR UNIVERSITY STUDENT GRADUATION PREDICTION: A SYSTEMATIC REVIEW AND A GENERATIVE AI-DRIVEN FRAMEWORK FOR PERSONALIZED INTERVENTION
149	ICSIEOCT260946	A NEURO-SYMBOLIC FRAMEWORK FOR ADVERSARIAL PHISHING DETECTION AND ZERO-DAY MITIGATION
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155	ICSIEOCT260708	ANN-BASED CHARGING AND DISCHARGING CONTROL IN DC MICROGRID
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160	ICSIEOCT260740	ARTIFICIAL INTELLIGENCE IN RECRUITMENT AND INTERVIEW EVALUATION:A SYSTEMATIC REVIEW AND THE NEXUS FRAMEWORK
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178	ICSIEOCT2601052	AN INTELLIGENT CLINICAL DECISION SUPPORT PLATFORM FOR EARLY CARDIOVASCULAR DISEASE PREDICTION USING DEEP LEARNING
179	ICSIEOCT2601187	A CHARNES – COOPER FRAMEWORK FOR TRIANGULAR Q-RUNG ORTHOPAIR FUZZY LINEAR FRACTIONAL PROGRAMMING WITH AN APPLICATION TO PROFIT- TO – COST RATIO TRANSPORTATION PLANNING
180	ICSIEOCT260130	AN AI-DRIVEN KNOWLEDGE GRAPH FRAMEWORK FOR HERB-DRUG INTERACTION PREDICTION AND SAFE ALTERNATIVE HERB RECOMMENDATION USING AYURVEDIC KNOWLEDGE
181	ICSIEOCT2601259	A SURVEY ON AI-POWERED ENTERPRISE FINANCIAL DECISION INTELLIGENCE: KNOWLEDGE GRAPHS, DIGITAL TWINS, NEURAL-SYMBOLIC REASONING, AND CONVERSATIONAL ANALYTICS
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183	ICSIEOCT260975	A SAFETY-AWARE PLC–ROS 2 ARCHITECTURE FOR INDUSTRIAL ROBOTIC PICK-AND-PLACE AUTOMATION
184	ICSIEOCT260801	Q-TRANSPLANT: PRIVACY-AWARE, EXPLAINABLE ORGAN DONOR–RECIPIENT MATCHING WITH QUANTUM-INSPIRED CANDIDATE SEARCH
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239	ICSIEOCT2601114	AUTISM CATEGORIZATION USING CNN-BASED EFFICIENTNET AND ANT COLONY OPTIMIZATION
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249	ICSIEOCT2601290	MACHINE LEARNING AND DEEP LEARNING FOR SEWAGE TREATMENT PLANT PERFORMANCE PREDICTION AND OPERATIONAL OPTIMIZATION: A CRITICAL REVIEW
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252	ICSIEOCT2601250	A SPATIOTEMPORAL DEEP LEARNING FRAMEWORK FOR BRAIN TUMOR SEGMENTATION AND GROWTH PREDICTION USING MULTIMODAL MAGNETIC RESONANCE IMAGING
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255	ICSIEOCT2601189	DEEP LEARNING BASED PLANT DISEASE DETECTION FOR ECO FRIENDLY AGRICULTURE AND RECOMMENDATION SYSTEM
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258	ICSIEOCT260355	AI - BASED BLOCKCHAIN - SECURED CHRONIC DISEASE PREDICTION SYSTEM
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260	ICSIEOCT260364	ADAPTIVE TIMING-BASED MORSE CODE DECODER FOR REAL-TIME TEXT RECONSTRUCTION
261	ICSIEOCT2601509	INTELLIGENT INFRASTRUCTURE ASSESSMENT AND PREDICTIVE MAINTENANCE USING AI
262	ICSIEOCT2601573	ADAPTIVE MEMORY DESIGN FOR LOW POWER VLSI ARCHITECTURE
263	ICSIEOCT2601587	MEDICARE: AN INTEGRATED MEDICINE REMINDER AND DIGITAL HEALTH MANAGEMENT SYSTEM
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265	ICSIEOCT2601586	AN ENHANCED MACHINE LEARNING FRAMEWORK FOR CROP AND FERTILISER RECOMMENDATION WITH SOIL NUTRIENT DEFICIENCY ANALYSIS
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267	ICSIEOCT2601522	A SURVEY ON AN AGENTIC RAG AND LLM-BASED FRAMEWORK FOR INTELLIGENT HEALTHCARE INFORMATION RETRIEVAL AND SKIN DISEASE PREDICTION
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269	ICSIEOCT2601607	AI-BASED FOOD DEMAND, COOKING RESOURCE OPTIMIZATION AND SMART DONATION SYSTEM
270	ICSIEOCT2601280	PORTABLE SPIROMETER FOR LUNG FUNCTION TESTING AND EARLY DISEASE DETECTION
271	ICSIEOCT2601577	EARLY BRIDGE DAMAGE DETECTION
272	ICSIEOCT2601497	BUGSENSE AI: AN INTELLIGENT GITHUB BUG TRIAGE AND REPOSITORY-AWARE ROOT CAUSE ANALYSIS FRAMEWORK USING LOCAL LARGE LANGUAGE MODELS
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1. KSHETRA: A DECENTRALIZED DIGITAL INFRASTRUCTURE FOR SECURE LAND OWNERSHIP USING BLOCKCHAIN TECHNOLOGY

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Management of land ownership is vital for proper governance and socio-economic development. The traditional method of land registration depends on centralized information management system, written documentation, and manual process of authentication. Many times, these methods face issues like fraud, data manipulation, dispute among property owners, lack of transparency, delay, and corruption. In the absence of a trustworthy method for verification, there is no guarantee about any property deal.In order to resolve these issues, we have proposed KSHETRA, which is based on digital land ownership management system using Blockchain technology. It allows users to perform functions including property registration, government authentication, verification by other co-owner, document verification, and transferring property ownership in a transparent manner. Once the properties are verified, it gets registered to the Blockchain. KSHETRA employs a role-based architecture with the participation of users and government authorities. The user can register a property, upload documents, maintain the ownership database, and even request a property transfer. The government authority, on its end, validates the land record and approves or rejects such a request prior to recording it in the blockchain. In addition, KSHETRA can accommodate co-owner approval and secure communication among buyer and seller.The suggested architecture leverages the state-of-the-art web technology, blockchain technology framework, decentralized storage, and smart contracts in order to increase the security and efficiency of the process. The proposed system will ensure better transparency, decrease

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the incidence of fraudulent activities, increase the efficiency of verification, and ensure a tamper-proof environment for maintaining land records.

2. A PREDICTIVE ANALYTICS FRAMEWORK FOR ASSESSING CAREER PROGRESSION AND PROFESSIONAL EMPOWERMENT OF WOMEN IN IT ORGANIZATIONS

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The rapid digitalization of the IT sector has intensified the need for equitable career growth mechanisms; however, female professionals continue to encounter significant obstacles to career progression and professional empowerment. Traditional assessment methods for career advancement predominantly rely on retrospective qualitative evaluations, which fail to anticipate attrition risk, skill-gap trajectories, or leadership readiness in a timely manner. Thisstudy proposes a Predictive Analytics Framework designed to assess and forecast careerprogression pathways and professional empowerment indicators for women employed in ITorganizations. The framework integrates organizational data, such as tenure, skill acquisition,mentorship exposure, promotion velocity, and work-life balance indicators, alongsidemachine learning classification and regression techniques to generate empowerment scoresand progression forecasts at both the individual and cohort levels. Feature selection wasguided by established constructs from the career aspiration and empowerment literature,ensuring that the model's predictive variables were theoretically grounded rather than purelydata-driven. The proposed framework was validated using a dataset derived from ITprofessionals in Bengaluru, India, and assessed against standard performance metrics,including accuracy, precision, and F1-score. This study contributes a data-driven, scalabletool that integrates organizational behavior research with predictive analytics, providing bothacademic and practical value for advancing gender-inclusive career development strategies inthe IT industry.

3. MULTIGRADE DIABETIC RETINOPATHY CLASSIFICATION USING HYBRID AI MODEL ON RETINAL FUNDUS IMAGES

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Diabetic Retinopathy (DR) is a very dangerouscondition in the eye that may result in blindness in case it is nottreated and it mostly occurs to diabetics. Timely screening of DRis important to avoid blindness but the conventional screeningtechniques are time-consuming, laborious, and involve the use ofhighly qualified experts. To overcome such limitations, ArtificialIntelligence (AI) and Deep Learning (DL) tools have become oneof the prospective answers to automated diabetic retinopathydetection. In this

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project, it is suggested to develop a state-of-the-art system to analyze retinal fundus images, where the deeplearning models, including EfficientNet and U-Net, are used toclassify the DR and identify the retinal lesions. The explainableAI methods used in the system such as Grad-CAM or LIMEallow clinicians to gain a better understanding of the modelpredictions, which contributes to the clinical decision-makingprocess. The system proposed is a system that receives digitalcolor fundus photographs as an input and then classifies theseverity of DR into five grades (No DR, mild, moderate, severe,proliferative DR), and separates critical lesions, such asmicroaneurysms, hemorrhages, hard exudates, and soft exudates.Also, the system localizes anatomical structures like the optic discand the fovea. Outputs of the AI model are a severity grade witha confidence score, lesion segmentation masks, anatomicallocalization markers, and a Grad-CAM heatmap to explain themodel. A detailed medical report, with details of the patient,visualization and diagnostic information is produced to bereviewed by the clinicians and timely action is taken, which mayprevent the threat of vision loss in diabetic patients.

4. ELECTRICAL PERFORMANCE EVALUATION OF A SOLAR PVT COLLECTOR USING LIQUID BASED SNO_2 NANOFLUID AS COOLANT

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In this study, the electrical performance of a photovoltaic thermal (PVT) collector with liquid based SnO_2 nanofluid as coolant is assessed. SnO_2 nanoparticles prepared by sol-gel technique were dispersed in water at 0.05% and used in the experimental setup at flow rates of 0.5 and 1.0 litre per minute which were tested outdoors in Chennai from May to July 2024. Results show that the electrical efficiency of the SnO_2 nanofluid at 1.0 lpm is 14.80% which is 35.8% greater than that of pure water at 1.0 lpm with electrical efficiency of 10.90%. Because of the higher thermal conductivity and capacity of heat removal of the nanofluid, the temperatures of the glazing surface on the peak radiation hours were significantly lower, which reduced the temperature effects on the electrical losses. This work shows that SnO_2 nanofluid can be used as a promising coolant for increasing the electrical output of PVT system and it is a promising way of sustainable solar energy conversion.

5 .RELIABILITY-ENHANCED SENTIMENT CLASSIFICATION VIA UNCERTAINTY- AWARE HYBRID ENSEMBLE LEARNING ON LARGE-SCALE SOCIAL MEDIACORPORA

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Automated sentiment analysis of social media corpora presents considerable challenges owing to the prevalence of noisy text, informal register, and lexically ambiguous expressions. This study introduces a hybrid ensemble framework engineered to deliver reliable sentiment classification over a large-scale Twitter corpus. Raw tweets undergo a systematic cleaning pipeline before being encoded as numerical feature vectors through Term Frequency–Inverse Document Frequency (TF-IDF) representation incorporating unigram and bigram configurations. We train three different models: Logistic Regression, a calibrated Linear SVM, and an SGD classifier. Each one learns separately. After that, we combine their outputs using soft voting. This makes the final prediction more stable. We also check the confidence score for each prediction. If the score is low, we treat that result as uncertain. This step helps improve reliability. The model achieves an accuracy of 0.7853 and an F1-score of 0.7901. The ROC–AUC is 0.866. The confusion matrix shows that the model performs evenly across all sentiment classes. We remove low-confidence results to increase accuracy to 0.9178. In this process, 179,833 uncertain predictions are identified and separated. Cross-validation and statistical tests show that the method is stable and dependable. Overall, combining multiple models with confidence-based filtering improves sentiment analysis, especially when the data is noisy.

6.A COMPARATIVE EVALUATION OF SUPERVISED MACHINE LEARNING CLASSIFIERS FOR ANOMALY DETECTION IN IOT NETWORKS

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Securing Internet of Things deployments is difficult because authentication and encryption alone do not stop an adversary who has already reached the network, so intrusion detection is a necessary second layer and supervised machine learning is a common way to build it. This paper reports a controlled comparison of thirteen supervised classifiers from scikit-learn on the DS2OS traffic-trace dataset, 357,952 samples across one normal class and seven attack classes, trained with default hyperparameters on an identical 80–20 stratified split so that differences are attributable to the algorithm rather than to tuning. Twelve of the thirteen classifiers reached test accuracy between 96.0 and 99.4 per cent, led by K-Nearest Neighbours at 99.44 per cent with Decision Tree, Extra Trees, Random Forest and Multilayer Perceptron at 99.42 per cent. The central finding is that these figures are misleading. The dataset is 97.20 per cent normal traffic, so a classifier that predicts normal unconditionally scores 97.20 per cent without learning anything, and the best model exceeds that trivial baseline by only 2.24 points. Per-class analysis shows that every high- accuracy model misclassifies about a third of all Denial-of-Service traffic as normal — DoS recall near 65 per cent against a headline accuracy of 99.4 — and that Bernoulli Naive Bayes detects no DoS traffic at all. Accuracy should therefore not be the primary metric on this dataset; macro-averaged and per-class results are reported alongside it, and the apparent saturation of performance on DS2OS is shown to be an artefact of class imbalance rather than a solved problem

7. DESIGN AND DEVELOPMENT OF A LOW-COST 6-DOF ROBOTIC ARM WITH SOFTWARE-BASED CONTROL FOR PICK-AND-PLACE APPLICATION

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This project presents the design and development of a low-cost 6-DOF robotic arm for small-scale industrial and laboratory applications. Industrial robotic arms are usually expensive, which makes them difficult for small industries and laboratories that need a simple and affordable automation system. The proposed robotic arm is designed with six degrees of freedom to provide flexible movement and positioning of objects. A two-jaw gripper is developed for gripping and handling small objects during pick-and-place operations. This makes the system suitable for applications such as pick-and-place operations and material handling. A dedicated software application is also developed to control and program the robotic arm. The software allows the user to control individual joints, set the required movements, and create movement sequences for different tasks. The system combines mechanical design, motor control, kinematic modelling, gripper control, and software integration. The developed robotic arm is physically tested through gripper operation and pick-and-place tasks using different small objects. The proposed system aims to provide a simple, flexible, and affordable automation solution for small industries, laboratories, and educational applications where conventional industrial robotic arms may not be suitable.

8.SMART INTERACTIVE PHYSIOTHERAPY AND SPEECH THERAPY MAT [SIPST]

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Early childhood development plays a crucial role in shaping a child's cognitive, social, and communication abilities, with speech and motor skills serving as fundamental components of overall growth. Delays in these developmental domains can significantly affect a child's capacity to interact, learn, and perform daily activities, making timely and effective intervention essential for achieving optimal long-term outcomes. This paper presents a Smart Interactive Therapy Mat designed to simultaneously support speech development and motor skill rehabilitation through an integrated and engaging therapeutic platform. The proposed system combines touch- and force detecting button embedded within a mat, and an Espressif Systems ESP32-based wireless communication module to create a responsive and intelligent environment for therapy. User interactions are detected in real time and transmitted to a mobile application, where the sensor inputs are mapped to gamified physiotherapy exercises and audio-visual vocabulary activities. This integrated approach enables children to perform guided physical movements while participating in communication training tasks, thereby promoting concurrent development of motor coordination and speech skills. The system provides immediate visual and auditory feedback to enhance motivation, reinforce learning, and ensure accurate exercise performance. Experimental implementation demonstrates that the gamified therapeutic framework significantly improves children's engagement, encourages repetitive motor practice essential for rehabilitation, and strengthens vocabulary acquisition and communication abilities. Designed as a low-cost, user-friendly, and portable solution, the Smart Interactive Therapy Mat supports both clinical and at-home use, reducing dependence on frequent therapy visits while delivering an effective and accessible early intervention tool for comprehensive child development.

9.INTELLIGENT CLOUD-BASED VIDEO ANALYTICS USING YOLO AND DEEPSORT

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The research presents an intelligent cloud-assisted video analytics solution which utilizes the YOLO object detector and DeepSORT multi-object tracker. The proposed system uses the idea of tracking by detecting and is evaluated within three different modes: pure edge, pure cloud, and adaptive orchestration of edge and cloud environments. The experiments conducted with 300 artificial images yielded 2,352 total detections and the mean confidence score of 0.77. In edge-only mode, the system had the fastest performance rate with 291.3 FPS and mean latency of 3.43 ms, and there were no identity switch events. In cloud-only mode, the latency was 60.91 ms and FPS was 16.4 with 39 identity switches. Meanwhile, adaptive orchestration reduced latency by 92.3% in comparison with the cloud-only one. The cross-domain evaluation yielded MOTA scores of 0.96, 0.85, and 0.677 for sparse, moderate, and dense domains respectively, and the value of MOTP was equal to 1.0 in all cases. In addition to that, this research solves six challenges of the current literature including end-to- end latency benchmarking, lack of NMS in detection algorithms, thermal domain applicability, adaptive orchestration, and cross-domain evaluation.

10.HYBRID PHARMACHAIN: A SOFT-COMPUTING EXTENSION FOR INTELLIGENT AND AUDITABLE PHARMACEUTICAL COLD-CHAIN PROVENANCE

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Counterfeit and substandard medicines remain a persistent threat to public health, and blockchain-based provenance systems have emerged as a leading defence by making supply-chain records tamper-evident. However, existing pharmaceutical blockchain frameworks treat shipment acceptance as a rigid, threshold-based decision, discarding large numbers of consignments that remain fit for use while offering no mechanism to reason about graded, interacting cold-chain risk factors. This paper proposes Hybrid PharmaChain, a soft-computing extension that augments a provenance ledger with an intelligent decision layer. A Mamdani fuzzy inference system converts continuous cold-chain telemetry (temperature deviation and delivery delay) into a graded risk score that drives a smart-contract acceptance rule, while an Ant Colony Optimization router selects distribution paths and reinforces routes in proportion to assessed risk and cost. Every decision is committed to a SHA-256 hashlinked ledger, preserving auditability. Across ten independent trials of two hundred simulated shipments each, the proposed fuzzy-intelligent contract retains 72.2%±2.3% of consignments versus 24.6%±2.0% for a conventional rigid contract, recovering 47.6%±2.8% of provably-safe shipments that would otherwise be destroyed. The narrow variance confirms that the improvement is a stable property of the decision model rather than an artifact of sampling. Index Terms—pharmaceutical supply chain, drug counterfeiting, blockchain, fuzzy logic, ant colony optimization, cold chain, provenance, smart contracts

11. PREDICTION OF SIDE EFFECTS OF DRUGS USING GRAPH NEURAL NETWORKS

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Adverse Drug Reactions (ADRs) are a major problem in healthcare because some medicines may cause unexpected side effects. Detecting these effects early is important to reduce health risks and avoid failure during clinical trials. Traditional experimental methods require high cost and lot of time, and they may not identify rare side effects in the early stage. In this work, we use a Graph Neural Network (GNN) approach to predict potential drug side effects based on biomedical data. A knowledge graph is created by combining information about drugs, diseases, and known side-effects. Five datasets, namely drug names data, meddra all se, drug action labels data, drug disease labels data, and drug symptom labels data, are combined to build a knowledge graph The Graph Convolutional Network (GCN) learns relationships between these entities and predicts new drug–side effect connections. The model also provides confidence scores to show how strongly a side effect is related to a drug. In addition, the system identifies the biological action type of drugs such as virus, bacteria, fungi, or human cell targets. The results show that graph-based learning can capture complex biomedical relationships effectively. This approach may help researchers identify risky drugs earlier and support safer drug development.

12. A MULTI-MODAL AI FRAMEWORK FOR SAFETY INTELLIGENCE IN HEAVY-DUTY TRUCKS USING COMPUTER VISION, IOT SENSOR FUSION, AND EXPLAINABLE AI

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Heavy-duty trucks face big safety challenges due to trailer blockages, long braking distances and driver fatigue during long trips. The MAN TruckScenes dataset [1] the first multimodal dataset for autonomous trucking shows that no single sensor can give stable detection in all conditions. Camera-only methods got only 0.02 mAP whereas LiDAR-only methods got 0.27 mAP. We present a multimodal AI system that combines cameras, 4D radar, LiDAR, IoT telematics, and driver-state information for improved safety. The system uses YOLOv8x for camera-based object detection, CenterPoint for LiDAR 3D detection, 4D radar feature extraction, driver monitoring and telematics data. These features are combined using a cross- attention Transformer. To show the lack of aligned external and driver-state sensor data a feature-level Gaussian Mixture Model (GMM) sampling method is used to enhance and align the MAN TruckScenes and manD 1.0 datasets [2]. The full combination model got 48.6% NDS and 31.2% mAP for 3D detection, with a risk prediction AUC of 0.87 using GMM-sampled driver features. The complete system runs at 14.2 FPS with 70.2 ms end-to-end response time on an NVIDIA Jetson Orin NX. Grad-CAM and SHAP analyses also show that the model focuses on important objects and driving conditions. Index Terms—Heavy-duty trucks, multi-modal fusion, Trans- former, safety intelligence, driver monitoring, explainable AI, edge computing.

13 .SMART OBSTACLE DETECTION SYSTEM FOR VISUALLY IMPAIRED NAVIGATION USING ULTRASONIC SENSING

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This paper presents a low-cost device that helps visually impaired people to walk safely by detecting difficulties with an ultrasonic sensor. The system uses an HC-SR04 sensor and an Arduino Nano to measure the distance of nearby objects and gives sound alerts by a piezoelectric buzzer which beeps quickly when a barrier is 50 cm or closer it beeps slowly when it is between 50 and 100 cm away and stays silent when it is more than 100 cm . Tests were done at distances from 20 to 300 cm with 10 readings at each distance. The results showed high accuracy with a Mean Absolute Error (MAE) of 0.32 cm and a Root Mean Square Error (RMSE) of 0.41 cm. The system gives an alert in 30 ms which is faster than the normal human reaction time which is 200–300 ms. The total cost is less than $15 which make it cheaper than AI-based systems. The paper also tell about the limits of using only one sensor and adding AI in the future for better barrier detection and smarter feedback. Index Terms— Arduino, ATmega328 Microcontroller, Blind walking stick, Mobility aid, Smart cane, Ultrasonic sensor

14. A SHAP-DRIVEN EXPLAINABLE AI APPROACH FOR PREDICTING AND INTERPRETING RIVER POLLUTION RISK USING ENSEMBLE LEARNING

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Most times, water pollution is only found late. There are still places that collect water samples by hand and send them to a lab for testing. It can take a few days to a couple of weeks for a result. By the time the report is ready, the polluted water has already run further downstream, sometimes overflowing water intakes in nearby towns, harming fish and other aquatic life, and in many cases, it's too late. In this paper, we attempt to solve this problem with machine learning. We use two Random Forest models to predict the level of pollution (low, moderate, high, critical) and one that gives an actual risk score of 100. We take in ten commonly checked parameters such as pH, turbidity, dissolved oxygen, temperature, conductivity, TDS, BOD, COD, rain and flow rate. Since Random Forest models don't explain how each parameter is pushing the risk score up or down, we produce our own version of Tree SHAP to find Intelligent Cloud-Based Video Analytics using YOLO and DeepSORTout whether each parameter is pushing the risk score up or down. When tested, the classifier returned a 94.58% accuracy and an F1 score of 94.28% and our regression model returned an R2 of 0.9967, so both parts perform well. Since numbers alone aren't very useful for someone in the field, we turn the SHAP output into simple explanations and suggest what to do next, such as aeration when oxygen is low.

15.SILENT ASPIRATION DETECTOR

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Silent aspiration is a critical medical condition in which food or liquid enters the airway without producing noticeable symptoms such as coughing or choking, making early detection difficult. It is commonly observed in patients with neurological disorders, dysphagia, and in elderly individuals, and can lead to severe complications such as aspiration pneumonia and respiratory infections if left undetected. Conventional diagnostic techniques require specialized equipment and clinical settings,limiting their use for continuous monitoring. This project aims to develop a portable, low-cost, and non-invasive silent aspiration detection system capable of real-time monitoring outside hospitalenvironments.The proposed system utilizes acoustic and vibration-basedsensing to monitor swallowing activity. A MAX9814 MicrophoneAmplifier Module is used to capture swallowing sounds, while a MakerHawk Piezoelectric Sensor detects throat vibrations.These signals are processed using the ESP32-WROOM-32, which performs signal conditioning and feature extraction. Important parameters such as amplitude, frequency components, and signal/energy are analyzed to characterize swallowing behavior. Data were collected from multiple individuals using a stethoscope-based recording method, and a threshold value was determined by calculating the mean of the extracted features to represent normal/swallowing patterns.The system compares incoming signals with the established threshold to identify deviations that may indicate aspiration.Upon detection of abnormal patterns, the system provides immediate alerts through an RGB LED and a buzzer, enabling timely intervention. The device is powered by a rechargeable battery,ensuring portability and ease of use for continuous monitoring.The results demonstrate that the system can effectively differentiate between normal and abnormal swallowing patterns using a simple threshold based approach.Although the system shows promising performance, certain limitations such as sensitivity to noise, variations in sensor placement, and individual differences in swallowing behavior may affect accuracy. Future improvements can include the use of larger datasets, advanced signal processing methods,and integration of machine learning techniques for enhanced reliability. Overall, the proposed system highlights the potential of combining embedded systems and biomedical signal analysis to develop practical and accessible healthcare monitoring solutions for early detection of silent aspiration.Index Terms—Index Terms- Silent Aspiration Detection, Swallowing Analysis, ESP32, Piezoelectric Sensor, FFT, RMS, Wearable Device, Real- Time Monitoring.

16 .PROTECTING FOREST-EDGE COMMUNITIES THROUGH REAL-TIME WILDLIFE DETECTION: A YOLOV8M-BASED FRAMEWORK FOR THE WESTERN GHATS

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Human – environment conflict remains one of the several issues which is rapidly growing in the upcoming years. Faster developments, globalization remains one of the major reasons which causes a conflict in the balance between the human and wildlife. Over a span of 10 years, more than 100000+ displacements have been seen in the forest areas which includes killing and trafficking of the wildlife as well. As the result most of the wildlife population are being forced to have an impact on human civilization, more animals sighting has been seen on the roads causing more accidents and a loss of over 1000+ human lives caused by the wildlife. To overcome problem like we have introduced a solution collaborated with AI animal detection system and using YOLO model to increase the feasibility of the model which will not only help humans to take the best prevention for animal's lives and will help in preventing more accidents. We have preferably done this research on the Western Ghats area which uses Karnataka Forest department dataset helping in the rapid development of model's confidence using the dataset. The dataset covers different species like Leopard, Bengal Tiger, Elephant, Gaur and sloth bear. To improve the reliability of the system under real world scenarios, most of the training has been done on the outside forests along with the dataset. The model processes 640 * 640 pixels images and with 100 epochs uses f1-score and (mAP@0.5). The model achieves (mAP@0.5) 92.7%, precision of 89.7%, f1-score of 89.0%. Using TensorRT optimization, an inference speed of 26 frames per second was obtained on NVIDIA RTX 4050, which shows the uses of the real-world development and really

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helps in boosting the confidence of the model. We have trained this model on low light conditions which will help in night sightings as well as the areas which has low visibility and will preferably help in the prevention. The developed model demonstrates the model for real-time wildlife monitoring and can be further developed through multi-object tracking and community-facing alert techniques to support safer relationship between wildlife, forest-edge communities, and road users.

17. A COMPUTER-IMPLEMENTED METHOD FOR EXPLAINABLE WORKFORCE SKILL EVOLUTION FORECASTING USING RECENCY-WEIGHTED MULTI- SOURCE INTELLIGENCE

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The rapid evolution of technologies and changing workforce requirements make it difficult for individuals and organizations to determine which skills will remain relevant, emerge, or decline over time. Existing career recommendation and workforce management systems primarily depend on current job-market demand and static skill profiles, providing limited insight into future skill evolution. This project proposes a computer-implemented method for explainable workforce skill evolution forecasting using recency-weighted multi- source intelligence. The proposed method collects and integrates workforce and technology- market signals from multiple sources, including job postings, developer activity, technical discussion platforms, and technology ecosystems. A recency-weighted processing mechanism assigns greater importance to recent observations while retaining relevant historical trends. These signals are analyzed to determine skill growth, stability, and decline. A forecasting framework combines current demand, temporal trends, technology velocity, skill recency, and predicted skill decay to generate a Future Skill Relevance Score. The system also provides explanations identifying the factors contributing to each prediction. Furthermore, individual and organizational skill profiles are compared with forecasted market requirements to identify skill gaps and recommend targeted upskilling or reskilling pathways. The proposed approach enables proactive, explainable, and data-driven workforce planning and career development in rapidly changing technology markets.

18. COMPARATIVE ASSESSMENT OF UNTREATED AND NAOH -TREATED HEMP FIBRE-REINFORCED CONCRETE: EFFECTS OF FIBRE VOLUME FRACTION ON WORKABILITY AND MECHANICAL PERFORMANCE

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Cement and concrete production contribute substantially to resource depletion and greenhouse-gas emissions, creating a need for lower-impact reinforcement strategies that maintain structural performance. This study investigates the mechanical performance of hemp fibre- reinforced concrete (HFRC) incorporating untreated and NaOH-treated hemp fibres at an aspect ratio of 10 and volume fractions of 0.1–0.5%. M25-grade concrete was designed in accordance with IS 10262:2019, and compressive, split tensile, and flexural strengths were evaluated at 7, 14, and 28 days in accordance with the relevant provisions of IS 516. Workability and fibre-surface morphology were also examined to assess the influence of fibre incorporation and alkaline treatment. The results identified 0.3% fibre volume fraction as the optimum dosage within the investigated range. At 28 days, NaOH-treated fibres at 0.3% Vf increased compressive, split tensile, and flexural strengths to 46.33, 3.74, and 7.00 N/mm2, corresponding to improvements of 23.28%, 12.31%, and 20.40%, respectively, relative to conventional concrete. The corresponding untreated- fibre mixture achieved 39.61, 3.38, and 6.38 N/mm2,

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representing improvements of 5.05%, 1.50%, and 9.68%, respectively. Increasing the fibre volume fraction beyond 0.3% reduced workability and mechanical performance, indicating the importance of balancing fibre dosage with fresh-state workability and effective fibre reinforcement. SEM observations revealed distinct modifications in fibre-surface morphology following NaOH treatment, supporting the enhanced reinforcing efficiency of the treated fibres. Overall, NaOH-treated hemp fibres at 0.3% Vf and an aspect ratio of 10 provided the most favourable mechanical performance within the investigated range, demonstrating the potential of treated hemp fibres as a natural reinforcement for concrete.

19. SMARTSPOT: A CONTEXT-AWARE GEOFENCED REMINDER SYSTEM WITH ADAPTIVE LOCATION SENSING AND ON-DEVICE LEARNING

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Location-based reminders are only useful if they are timely, relevant and non-disruptive. Traditional geofencing applications rely on a simple static proximity condition and therefore suffer from false triggers when a user just passes by a location, context blindness that ignores weather, time and activity, and a complete lack of adaptation to user behaviour or battery constraints. In this paper we introduce SmartSpot, a cross-platform mobile application that enhances a base geofenced reminder system by adding an on-device, rule-based context-fusion layer. SmartSpot employs a single trigger- evaluation pipeline that fuses movement direction, detected physical activity, weather, temporal rules and historical visit patterns, and adapts both the geofence radius and GPS polling strategy to real-time context. A bearing-comparison route- intelligence mechanism differentiates between approaching a destination and being in proximity; an activity-adaptive radius policy mitigates the geofence blowthrough problem at driving speeds; a three-tier battery optimisation policy reduces the cost of location queries; and frequency-based place clustering surfaces predictive reminder suggestions and history-informed rescheduling. All logic is deterministic, transparent and fully on-device, with no trained model, paid API or backend server. Evaluations on synthetic movement traces and a short field trial demonstrate that SmartSpot reduces false-positive and missed triggers over a fixed-radius baseline while reducing estimated GPS query cost, proving significant context-awareness can be achieved under tight mobile engineering constraints.

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20.IOT-BASED SMART HELMET FOR REAL-TIME ACCIDENT DETECTION AND EMERGENCY ALERT

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Road traffic accidents remain one of the leading causes of preventable death, and delayed emergency response is a major factor contributing to fatality rates, particularly among two-wheeler riders. This paper proposes an IoT-based Smart Helmet system capable of detecting accidents in real time and automatically alerting emergency contacts and nearby medical services. The proposed helmet integrates an accelerometer-gyroscope module for impact and fall detection, a GPS module for real-time location tracking, and a GSM/IoT communication module for transmitting alerts. An alcohol sensor is embedded to restrict vehicle ignition when the rider is intoxicated, and a helmet-wear detection mechanism ensures the vehicle starts only when the helmet is properly worn. Upon detecting a sudden impact or fall beyond a defined threshold, the system triggers a short countdown during which the rider may cancel a false alarm; if not cancelled, it automatically sends the rider's GPS coordinates and accident alert via SMS and a cloud dashboard to registered emergency contacts and, where available, nearby hospitals. The proposed system aims to reduce the golden-hour delay in emergency response and improve rider safety through low-cost, real-time accident detection and communication.

21. SAFEGAURDHER: REAL-TIME PERSONAL SAFETY APPLICATION FOR WOMEN

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Personal safety remains a critical concern for women in public and private spaces, with existing safety applications often falling short in three key areas: speed of response during an emergency, discretion when an attacker is in physical control of the victim's phone, and the ability to preserve tamper-evident proof of an incident for later use. This paper presents Safe Guard Her, a cross-platform mobile application built using Flutter and backed by Supabase, designed to address these gaps through a unified, low-friction safety toolkit. The core of the system is a one-tap panic button that simultaneously triggers three actions: automatic video and audio evidence capture, continuous live GPS location broadcasting to a user-defined list of trusted emergency contacts at fixed intervals, and instant SOS alert dispatch via SMS. To ensure the evidentiary integrity of captured media, each recording is hashed using SHA-256 at the point of capture, allowing later verification that the file has not been altered. For

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situations where visibly reaching for a safety app could escalate danger, the application includes a stealth mode that disguises the entire interface as a standard calculator, requiring a discreet numeric sequence to unlock the real application. A shake-to-SOS gesture detector further allows the panic sequence to be triggered without unlocking the phone or navigating any screen, which is critical when a user's hands or attention are not free. Beyond the emergency-response core, Safe Guard Her incorporates several supporting safety features: a Safe Walk timer that monitors a user during a set journey duration and automatically alerts contacts if the timer lapses without check-in, a breadcrumb trail module that logs and visualizes location history during active journeys, a fake incoming call generator to help users exit uncomfortable situations discreetly, and a centralized evidence vault where all recorded media, timestamps, and location metadata are securely stored for later retrieval by the user or, where required, law enforcement. Access to the application itself is protected via a PIN-lock and account authentication layer, while a curated national and regional helplines directory gives users immediate access to verified support services. The backend is implemented using Supabase, providing authentication, a PostgreSQL-backed data layer with row-level security policies for contact and profile data, and cloud storage for evidence files, allowing the system to scale without a dedicated backend server. The application's architecture emphasizes low- latency alerting, offline resilience during intermittent connectivity, and privacy-by- design principles, ensuring that sensitive location and media data are only accessible to the user and their designated contacts. This work contributes a practical, deployable system that combines rapid emergency response, discreet operation under duress, and forensically sound evidence capture within a single mobile application. We discuss the design rationale behind each safety module, implementation challenges encountered in building real-time location sharing and stealth-mode UI switching on Flutter, and directions for future work including AI-based threat detection, wearable device integration, and community-based safe-route recommendation.

22. AI -POWERED TOURISM RECOMMENDATION AND INTELLIGENT PLATFORM FOR SAFE AND PERSONALIZED TRAVEL

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Tourism planning often requires travelers to gather information from multiple sources regarding destinations, attractions, accommodation, transportation, safety, and suitable activities. This paper proposes an AI-Powered Tourism Intelligent Platform for Safe and Personalized Travel that integrates these requirements into a unified intelligent system. The proposed platform incorporates an AI-based recommendation system that analyzes user preferences, interests, budget, travel duration, and preferred travel period to provide personalized recommendations for destinations, attractions, activities, and travel itineraries. An AI-powered conversational assistant enables users to interact with the system using natural language and receive destination-specific recommendations and travel guidance. The platform also incorporates safety-oriented features to provide relevant alerts and assist users in making informed travel decisions. Tourism information covering heritage, spiritual, nature, and beach destinations, along with accommodation and local travel resources, is organized within the system. By integrating an intelligent recommendation system, itinerary generation, conversational AI, and travel safety support, the proposed platform aims to enhance travel decision-making and provide a convenient, personalized, and safer tourism experience. Index Terms—Artificial Intelligence, Recommendation System, Personalized Tourism, Travel Recommendation, Intelligent Itinerary Planning, Conversational AI, Travel Safety, Smart Tourism.

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23.ROUTEIQ: ADAPTIVE COST–QUALITY ROUTING FOR LARGE LANGUAGE MODELS USING QUERY COMPLEXITY ESTIMATION

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The rapid proliferation of large language models (LLMs) with heterogeneous capability, latency, and cost profiles has made static, single-model deployment an inefficient default for production AI applications: always invoking the strongest available model guarantees quality at the expense of unnecessary computational and monetary cost, while always invoking a low-cost model risks unacceptable quality loss on difficult queries. This paper presents RouteIQ, an adaptive query routing framework that estimates the complexity of an incoming query prior to inference and dynamically selects the least expensive model tier expected to satisfy a target quality level. RouteIQ combines multiple normalized query-level features — including reasoning depth, task and domain complexity, linguistic ambiguity, contextual requirement, and expected token load — into a single bounded Query Complexity Score, C(q), which is mapped to a discrete set of model-capability tiers through a threshold-based decision rule. We formalize the routing problem as a constrained multi-objective optimization over response quality, inference cost, and latency, and we present a complete algorithmic specification, an experimental methodology with defined baselines and metrics, and an ablation design for isolating the contribution of individual complexity features. Because no production-scale measurements have yet been collected for this manuscript, all quantitative outcomes are represented as explicitly marked placeholders rather than invented values; the paper instead specifies, in full reproducible detail, how each reported number is to be obtained. The intended contribution is therefore twofold: a concrete, mathematically specified routing methodology, and a rigorous evaluation protocol against which that methodology can be measured once real experimental data are collected. Index Terms—large language models, query routing, model selection, inference cost optimization, query complexity estimation, cascaded inference, mixture of experts, cost–quality trade-off.

24. LIFEVAULT: A PRIVACY-PRESERVING, CONTEXT-AWARE FRAMEWORK FOR PERSONAL INFORMATION MANAGEMENT AND RETRIEVAL

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Personal records are not usually in one file. A receipt could contain the details of a purchase, a warranty document the terms of coverage, and a service record the history of a subsequent repair. Keeping them separate makes it difficult to answer any question that requires more than one document. Privacy

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compounds the problem because a single personal archive can have financial, transactional and identifying data, all of which can be easily mishandled. Existing work includes personalized anonymization for knowledge graphs [1], membership-inference risk in federated knowledge graph embeddings [2], and sensitivity-aware differential privacy for structured data [3]. Each provides a real privacy mechanism, but each was designed for graphs, learned embeddings, or tabular data — not for the more difficult job of fishing answers out of a pile of heterogeneous personal documents. LifeVault is an effort to close that gap. It extracts entities and attributes from personal records, links matching entities across documents, builds a Personal Document Knowledge Graph, answers queries by traversing relationships in that graph, and checks privacy rules before any sensitive information is returned. We assess the design across five axes: entity extraction, entity linking, retrieval quality, privacy protection and processing efficiency. Index Terms—Personal information management, privacy preserving retrieval, entity extraction, entity linking, knowledge graphs, relationship-aware retrieval, access control, heterogeneous documents.

25. DEEP FEATURE LEARNING FOR REAL-TIME BRAIN–COMPUTER INTERFACE APPLICATIONS USING MOTOR IMAGERY EEG SIGNALS

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Brain–Computer Interface (BCI) systems enable direct communication between the human brain and external devices by translating electroencephalography (EEG) signals into control commands. Motor imagery (MI)-based EEG is widely used in BCI applications; however, accurate multi-class classification remains challenging due to the non-stationary nature of EEG signals, low signal-to-noise ratio, and inter-subject variability. This paper presents a lightweight and computationally efficient CNN–LSTM-based framework for real-time multi-class motor imagery classification. Unlike conventional approaches that rely on complex deep learning architectures, the proposed work focuses on the development of a practical and deployable BCI system by integrating a structured end- to-end processing pipeline with an efficient spatio-temporal learning model. The framework incorporates motor imagery segmentation, band-pass filtering (8–30 Hz), and normalization to enhance signal quality, while a compact CNN–LSTM architecture enables effective feature extraction with reduced computational overhead. The model is evaluated on the BCI Competition IV Dataset 2a across nine subjects, achieving a mean classification accuracy of 89.6% ± 2.35%, with the highest accuracy of 92.3% for Subject A06. To assess real- world applicability, the framework is further validated using experimentally acquired EEG data from three subjects, achieving an average accuracy of 68.33% under realistic conditions. A graphical user interface (GUI) is developed to enable real-time visualization and classification of EEG signals. The results demonstrate that the proposed framework provides an effective balance between performance and computational efficiency, making it suitable for real-time BCI applications.

26. SMART PORTABLE KNEE ASSIST

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Knee weakness can make walking and other routine movements difficult, particularly for individuals who require additional support during lower-limb movement. Wearable assistive devices have therefore been developed to provide mechanical support at the joints. However, many powered systems require multiple actuators, complex control methods, and comparatively large mechanical structures. This work presents a Smart Portable Knee Assist that combines passive mechanical assistance from a spring mechanism with controlled assistance from a servo motor. A pressure sensor is used to monitor the loading condition associated with knee movement. The sensor signal is processed by an Arduino Nano, which compares the measured value with a predefined threshold. When the pressure exceeds the threshold, the Arduino Nano activates the servo motor. The servo assists the spring mechanism to provide additional support to the knee. When the pressure remains below the threshold, the servo assistance is not activated and the system remains in its normal state. A buck converter is used for regulated power supply to the electronic components. The prototype demonstrates a simple sensor-controlled approach for combining passive and active assistance in a portable knee-support device. Further development can focus on movement monitoring and adaptive assistance.

27.THE EFFECTS OF SOCIAL MEDIA ALGORITHMS ON THE BEHAVIOR OF NEPALESE VOTERS DURING ELECTION CAMPAIGN

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The rapid expansion of social media has transformed political communication in Nepal by changing how voters obtain information, encounter political candidates, evaluate competing narratives, and participate in election campaigns. Unlike traditional media, social media platforms do not simply transmit political messages; algorithmic systems select, rank, recommend, and repeatedly circulate content according to signals such as previous engagement, viewing behavior, network connections, and predicted user interest. This study examines how such algorithmic environments affect the political behavior of Nepalese voters during election campaigns. The study uses a secondary- data, such as Election Commission documents, election-monitoring reports, recent studies of Nepal's digital electorate, Facebook political content, misinformation, political polarization, and electoral behavior. Evidence from the 2026 parliamentary election campaign also showed strong concentration of political content around particular personalities and parties on high-reach Facebook pages, while monitoring research documented misinformation, disinformation, hate speech, identity-based attacks, and election-related manipulation. The study finds that algorithmic exposure can influence voter attention, emotional responses, candidate perceptions, political discussion, selective exposure, and trust in information. However, the evidence does not justify claiming that algorithms independently determine voting decisions. Rather, algorithmic effects interact with prior political preferences, social networks, candidate popularity, media literacy, socioeconomic conditions, and offline political experiences. The

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article concludes that algorithmic transparency, stronger election monitoring, digital literacy, platform accountability, and accessible fact-checking are necessary to protect informed electoral choice in Nepal.

28.SOCIAL MEDIA CONSUMPTION, PLATFORM DOMINANCE, AND NEWS DISTRIBUTION DURING NEPAL'S GEN-Z PROTESTS

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The widespread use of smartphones and social media has transformed how political movements are documented, disseminated, and consumed in real time. During Nepal's youth-led Gen-Z protests, traditional broadcast media faced growing competition from decentralised, user-generated content shared directly from the streets. This quantitative survey study examines changes in news consumption and distribution during the protests, focusing on the shift from television toward mobile-first social media platforms. The study compares platform use, perceptions of credibility, exposure to misinformation, verification practices, and the use of VPNs and alternative communication channels during social-media restrictions. The findings show that social media became the dominant source of real-time protest information, with Facebook emerging as the leading platform and citizen-generated smartphone footage playing a central role. However, concerns about misinformation and credibility sustained the relevance of traditional media for verification. Social-media reliance also extended beyond Gen-Z, indicating a broader transformation in news consumption. The study demonstrates that social media has become a major infrastructure for real-time political communication and news distribution while traditional journalism continues to perform complementary verification functions.

29.REHEARSING MORAL JUDGMENT THROUGH STRUCTURED ETHICAL DELIBERATION : A QUASI EXPERIMENTAL STUDY AMONG CHINESE UNIVERSITY STUDENTS

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Structured opportunities to examine ethical problems may strengthen how university students approach moral decisions. This study examined an active learning based moral education Intervention organized around ethical deliberation. A non-equivalent control- group pretest–posttest design involved 280 first-year undergraduates at a Chinese university, with 140 students in each of two intact classes. The experimental class received a ten-week Intervention incorporating case analysis, role-play, group deliberation, and reflective writing; the comparison class continued a conventional teaching approach. Moral judgment was measured using the 39-item Moral Judgment Scale within the Comprehensive Instrument Measuring Morality. Total scores increased by 14.89 points in the experimental group and 3.48 points in the control group. Analysis of covariance adjusting for baseline scores yielded an adjusted difference of 11.19 points, 95% CI [8.71, 13.67], F(1, 277) = 78.98, p < .001, partial η^2 = .222. Exploratory analyses found positive adjusted differences across all four subscales. Reflecting Outcome required an interaction model because its association with instructional condition varied with baseline scores. The findings are consistent with deliberative rehearsal as a pedagogical explanation, but do not establish a cognitive mechanism. Self-report measurement, non-random allocation, and complete confounding of instructional condition with class and teacher constrain causal inference. The study provides context- specific evidence supporting further evaluation of structured ethical deliberation using multiple classes and independent measures of moral reasoning.

30. MOODTUNE AI: SONG EMOTION ANALYSIS AND EMOTION-BASED MUSIC RECOMMENDATION USING DISTILBERT

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Music is an important medium for expressing emotions, and users often prefer songs that match their emotional state. Most existing music recommendation systems primarily rely on listening history, song genre, artist preference, or popularity, while giving limited importance to the emotional meaning of song lyrics. This project proposes an AI-based Song Emotion Analysis and Emotion-Based Music Recommendation System that utilizes Natural Language Processing (NLP) and a pre-trained DistilBERT model to analyze the emotional content of song lyrics. The system accepts a song name as input, retrieves the corresponding lyrics from a lyrics dataset, and processes the text using DistilBERT to classify the dominant emotion into categories such as Happy, Sad, Love, Anger, Fear, and Motivational. Based on the predicted emotion, the system recommends songs with similar emotional characteristics, providing a more meaningful and personalized music recommendation experience. The proposed approach leverages the contextual understanding capability of DistilBERT to improve emotion classification compared to traditional keyword-based methods. The system is designed using open-source datasets and modern NLP techniques, making it scalable and suitable for real-world applications. Future enhancements include multilingual lyric analysis, integration of audio features, and compatibility with online music streaming platforms.

31. DESIGN AND IMPLEMENTATION OF A REAL-TIME CONTINUOUS EEG STREAMING AND VISUALIZATION SYSTEM WITH STRESS DETECTION

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Electroencephalograph (EEG) is a non-invasive method for measuring the electrical activity of the brain and it also evaluate the neurological behavior and mental state of the person. This research implies the continuous monitoring of real time EEG data streaming and visualization associated with stress detection. The Proposed system has scalp EEG electrode, ADS 1299 analog front end, microcontroller, Bluetooth communication module for continuous acquisition and wireless transmission of EEG signals to a laptop. A Python based application has been developed to acquire multi-channel EEG signals in real time. The resulting signals are then processed using signal processing and frequency band analysis and a real time assessment of stress is done using statistical and frequency components. It continuously monitors the EEG waveform and a stress gage to differentiate the stress and calm state. The experimental data shows the real time EEG collections, wireless data transmission, continuous display of the waveform, and basic evaluation of the stress state can be determined with minimum delay function. The proposed system provides a low cost portable EEG monitoring prototype with further improvement Machine learning technique to reduce artifact for stress detection.

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32. FIRE PERFORMANCE OF REINFORCED CONCRETE FRAME BUILDINGS: A COMPREHENSIVE REVIEW OF EXPERIMENTAL AND NUMERICAL STUDIES

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Fire is one of the most critical hazards affecting the safety, stability and serviceability of reinforced concrete (RC) frame buildings. Elevated temperatures significantly alter the mechanical properties of concrete and reinforcing steel, resulting in a reduction in strength, stiffness and overall structural performance. This review paper critically examines recent experimental and numerical research on behavior of RC frame building subjected to fire. The review discusses the performance of concrete, reinforcing steel, RC beams, columns and complete frame structures under fire exposure, together with commonly adopted standard fire curves and numerical modelling techniques. A critical analysis of the available literature reveals that most existing studies concentrate on material behavior, individual structural components, or standalone numerical simulations. However, limited research has integrated experimentally obtained temperature-dependent material properties with nonlinear analysis of complete RC frame buildings to evaluate their overall structural response. Furthermore, the effects of fire on load redistribution, plastic hinge formation and collapse mechanisms at the building level remain inadequately explored. This review identifies the major research gaps, summarizes current advancements and outlines future research directions for improving the fire-resistant design, performance assessment and structural safety of reinforced concrete frame buildings.

33. VISION-GUIDED PRECISION PESTICIDE SPRAYING SYSTEM FOR SMART AGRICULTURE USING ESP32-CAM AND EXCESS GREEN INDEX CLASSIFICATION

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This paper presents a low-cost vision-based precision spraying system for smart agriculture using an ESP32-CAM and Excess Green Index (ExG). The system captures leaf images and analyses RGB values to identify plant health conditions. Based on the calculated ExG value, plants are classified into healthy, mild, moderate, or severe stress levels. A MOSFET-controlled pump automatically adjusts the

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spraying duration according to the detected severity, while healthy plants receive no spray. The proposed system performs image processing and decision-making locally without requiring cloud connectivity or a trained machine-learning model. Experimental results demonstrate consistent plant stress classification and controlled spraying, providing an affordable approach for reducing unnecessary pesticide usage in small-scale agricultural applications.

34. A SYSTEMATIC EMPIRICAL STUDY OF LIGHTWEIGHT REAL-ONLY ANOMALY-BASED FACE MORPHING DETECTION

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Face morphing attack detection (MAD) increasingly relies on supervised classifiers trained on labelled morph images, which are comparatively scarce and morphing- method-specific. This paper studies an alternative: a lightweight, REAL-only (bona-fide-only fitted) anomaly- detection pipeline built from handcrafted texture features (Local Binary Patterns, LBP, and Gray-Level Co-occurrence Matrix, GLCM) and classical anomaly mechanisms (K-Means and Local Outlier Factor, LOF), aiming to characterize — not resolve — its sensitivity(morph-recall)/specificity(genuine- image pass-rate) trade-off and its generalization behavior. We progressively evaluate a frozen LBP+K-Means baseline (V1), a controlled preprocessing-harmonization study (E0–E4), LOF alone and fused with K-Means, a further V1+E4 combination, and a controlled REAL-reference-pool experiment (V1-LFW), each assessed across a development split, a same-domain locked test scored once (hash-integrity- verified), and cross-dataset/external domains (LFW, SSMD, FairFace). Development-time gains do not reliably transfer: fusion's morph recall falls from 0.987 (development) to 0.227 (locked), concentrated in non-GAN morphing methods, while broadening V1's REAL reference pool to include LFW moves locked specificity from an already-poor 0.200 to exactly 0.000 rather than improving it. Domain-harmonized preprocessing (E4) reaches the most balanced same-domain operating point, but specificity still declines monotonically from FRLL through LFW and SSMD to the only formally independence-audited external domain, FairFace, for every model tested. A separate confound investigation shows that an apparently near-perfect configuration (locked AUC ≈ 1.0) reflects a dataset-construction confound and a complete failure to generalize to external bona- fide faces, and is rejected rather than reported as a working detector. To the best of our targeted literature search, no prior study combines this specific set of controlled interventions for REAL-only MAD; we make no claim of state-of-the-art performance, algorithmic novelty, or resolved generalization.

35.LEGAL NLP IN THE INDIAN JUDICIARY: A REVIEW OF EXISTING SYSTEMS, RESEARCH GAPS, AND FUTURE DIRECTIONS

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There are around 47 million unresolved cases pending in Indian courts. A large volume of these documents are produced each year. It includes the decisions of the court, legal filings, statutes, and case journals, far surpassing what can be managed manually. Natural language processing (NLP) offers a way out of this burden. These past five years a lot of systems have been developed to process legal text for works such as information retrieval, summarization, prediction of outcomes, and question answering. However, these systems are typically built for performing a single task at a time. Most of them depend on data from Chinese or Western courts, leaving very little support for the Indian legal texts and legal systems, which handles an enormous case volume across multiple languages and jurisdictional approaches. In this paper, we review fourteen studies on Indian court-ruling prediction, legal hallucination, legal reasoning benchmarks, and contract review. These studies are brief in a comparative table indicating each study's task, method, dataset, contribution, and limitation. The review finds that existing systems address only isolated parts of the legal-NLP pipeline, that very few studies target India specifically, and that even the best-performing models demand large computing resources. Based on these findings, the paper identifies seven open challenges. It includes data scarcity, multilingual and code-mixed text, and cross-domain generalization, It proposes an Intelligent Legal Document Analysis and Case Recommendation System that integrates classification, retrieval, summarization, and retrieval-augmented question answering for the Indian legal domain.

36. AGENTIC AI-BASED TENDER COMPLIANCE MANAGEMENT SYSTEM FOR MSMES: A REVIEW

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Artificial Intelligence AI has become an important research area in public procurement so mainly for tender document analysis and also for this compliance checking then supplier selection, recommendation and decision support. Recent studies have explxored Machine Learning ML, Natural Language Processing NLP, Retrieval Augmenteds Generation RAG, Large Languzage Models LLMs, explainable AI and agentic architectures to handle the challenges in complex and semi srtructured procurement doccuments. This literature survey reviews 14 exiscting research works by studuyying their apxproaches, architecztures, retrieval and analysis techniques, datasets and evaluation methods. The reviewed studies show the potential of AI in tender discovery, document intelligence, requirement analysis, compliance validation, historicfal tender retrieval, supplier recommendation and bid decision support. However, limitations like restricted datasets, retrieval inaccurcacies, hallucinaztion, limited explainability, fragmented functioxnality and security concernsh still remain. Basetd on these gaps, this review identifies the need for an integrated procurement intelligence platform which combines RAG, knowledge based retrieval, predictive analytixcs and agentic AI to support tendedr discovery, eligibilxity assessment, complianzce analysis and explainable decision making, mainly for Micro Small and Medium Enterprisexs MSMEs.

37.AI-DRIVEN AUTONOMOUS SYSTEMS: SURVEY OF ROBOTICS, DRONES, AND AUTOMATED VEHICLES FOR MODERN INDUSTRY

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AI-driven autonomous systems-including robotics, drones, and automated vehicles-are revolutionizing industrial operations, logistics, agriculture, and mobility. This paper presents a comprehensive survey of recent developments from 2020 to 2025, highlighting the technological enablers that drive autonomy, such as artificial intelligence, machine learning, computer vision, sensor fusion, and edge computing. Emphasis is placed on real-world deployments that demonstrate measurable improvements in operational efficiency, safety, and sustainability. A focused case study on warehouse automation illustrates the integration of collaborative robots and delivery drones, revealing tangible benefits in productivity, error reduction, and resource optimization. Key challenges-including regulatory compliance, cybersecurity threats, human-machine interaction, and ethical considerations-are critically analyzed to provide a holistic view of the current landscape. The paper also proposes a structured taxonomy of AI-driven autonomous systems, outlining emerging trends and research opportunities. By synthesizing technological advances, practical applications, and implementation insights, this survey offers guidance for researchers, engineers, and industry practitioners seeking to harness autonomous systems effectively. The findings underscore the transformative potential of AI-enabled autonomy, bridging the gap between cutting-edge research and safe, scalable, and sustainable deployment in modern industrial environments.

38. ARTIFICIAL INTELLIGENCE IN 3D PRINTING: PAVING THE WAY FOR SMARTER MANUFACTURING

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Additive Manufacturing (AM), commonly known as 3D printing, has become a transformative technology across various industries due to its ability to produce highly complex geometries with reduced material waste and lower production costs. Despite its advantages, optimizing AM processes remains a significant challenge. Several factors, including material characteristics, machine parameters, and environmental conditions, directly influence the quality and reliability of printed components. Managing these variables effectively is essential to ensure consistent performance and reduce defects. Artificial Intelligence (AI) has emerged as a powerful tool to address these challenges. This paper explores the integration of AI techniques such as Machine Learning (ML), Deep Learning (DL), and Computer Vision to enhance AM process optimization. It reviews recent advancements in AI-driven material selection, defect detection, predictive modeling, and real-time process monitoring, with a focused case study on metal additive manufacturing. These intelligent approaches enable improved accuracy, reduced production costs, enhanced efficiency, and better product quality. However, the adoption of AI in AM is not without limitations. Challenges such as high computational requirements, data dependency, and integration complexity must be addressed. The paper concludes by highlighting future research directions and emphasizing AI's critical role in advancing smart, autonomous, and sustainable manufacturing systems.

39. AN AI-DRIVEN, QUANTUM-RESILIENT BLOCKCHAIN FRAMEWORK FOR MITIGATING ZERO-DAY VULNERABILITIES IN DECENTRALIZED IOT NETWORKS

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The use of blockchain technology in the Internet of Things (IoT) has become one of the most promising paradigms for ensuring decentralized integrity of data and immutable access control. Nevertheless, according to conventional literature, classical blockchain architectures are inherently passive in nature, thus resulting in latency issues as well as high computation costs during standard cryptographic validation on heterogeneous IoT devices with limited energy resources. Emerging trends have been directed towards implementing ZKPs and encryption techniques that can be used to obscure identities of devices as well as protect transactional telemetry data from any form of unauthorized access. Although such cryptographic measures have been implemented effectively, there exists a fundamental limitation in the form of inability to detect and isolate active anomalies in edge layer attacks, data poisoning, and network partitions. This paper highlights the technical shortcomings of conventional

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cryptographic-based systems and proposes an innovative algorithm for securing and protecting the privacy of data using machine intelligence. Through the incorporation of decentralized blockchain consensus along with an adaptive federated learning module at the edge layer, this solution can provide post-quantum resilience through dynamic isolation of malicious nodes.

40.DIGITAL MEDIA AND GOVERNMENT LAWS AGAINST CHHAUPADI SYSTEM IN WESTERN NEPAL

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Chhaupadi is a menstrual taboo prevailing in some western regions of Nepal that requires women and girls to undergo isolation during their menstrual period. This practice, which is based on centuries-old cultural beliefs, exposes females to psychological torment, physical abuse, sexual assault, and lack of access to sanitary napkins. The Government of Nepal has adopted legal measures to stop this practice. However, the law alone is not enough to change the attitude of centuries-old cultural practices. This paper discusses how digital media can help the Government of Nepal in reducing the prevalence of Chhaupadi among women in western Nepal. The paper uses a qualitative documentary research design to examine scholarly articles, policies, government laws, reports, surveys, and recent studies on menstruation and digital media use. The paper demonstrates that digital media can help in legal awareness, menstrual health education, and public forum for discussion, youth engagement, female empowerment, and advocacy against the practice in the region. The findings further show that poor internet penetration, lack of ownership of digital devices by many locals, digital illiteracy, disinformation, privacy and security risks, language, and low levels of enforcement of the law have an impact on the effectiveness of digital media in reducing Chhaupadi. Overall, the paper argues that digital media should not be used as a stand-alone intervention in eliminating Chhaupadi. Instead, it should be used to encourage interactive communication between the government agencies, health workers, schools, youth, women, non-governmental organizations (NGOs), and communities. Effective implementation and enforcement of the law, legal awareness, health education, youth engagement, community involvement, and empowering women and girls through digital media can help eliminate Chhaupadi in the western part of Nepal.

41.MACHINE LEARNING-ASSISTED VALIDATION OF BOLLINGER BAND BREAKOUT TRADING SIGNALS ACROSS MULTIPLE TIMEFRAMES

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Financial markets exhibit changing volatility, trend structure, and market regimes, making the reliability of rule-based breakout signals difficult to assess across different timeframes. This study investigates a machine-learning-assisted validation framework for a Bollinger Band Breakout trading strategy, where the deterministic Bollinger strategy generates candidate trading signals and machine learning is applied as an additional validation layer. The experimental study evaluates 35 relevant configurations across five timeframes—1-minute, 3-minute, 5-minute, 15-minute, and 1-hour—comprising five TRAIN configurations, five unfiltered TEST configurations, and 25 ML-TEST configurations. The ML-TEST experiments use five confidence settings of 0.50, 0.53, 0.55, 0.57, and 0.60. The supplied experimental registry records 124,826 orders, including 43,579 wins and 81,235 losses, with an aggregate descriptive win percentage of 34.91% across the independent configurations. The results show a clear selectivity

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effect, with higher ML confidence settings reducing the number of recorded trades across the tested timeframes. However, the relationship between ML confidence and win percentage varies by timeframe and is not uniformly monotonic. The unfiltered TEST configurations provide a higher observed win percentage than the aggregate ML-TEST configurations, although this comparison does not establish economic superiority because the supplied data masks Total Profit, Total Loss, Net Realized, and Maximum Drawdown. Therefore, the study focuses on the effect of ML-based signal validation on trade activity and observed win characteristics rather than claiming improved profitability. The findings provide a structured basis for evaluating machine-learning-assisted filtering of Bollinger Band Breakout signals and motivate further analysis using complete profit, loss, drawdown, and risk-adjusted performance metrics.

42. AI-BASED EARLY DETECTION OF DIGITAL BURNOUT USING BEHAVIORAL PATTERNS

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The increasing dependence on smartphones, digital communication platforms, online learning, and continuously connected work environments has created new challenges related to prolonged digital engagement and psychological well-being. Conventional approaches for identifying burnout and excessive stress primarily depend on questionnaires and periodic self-reports, which may fail to capture gradual changes in everyday behavior. This paper proposes an AI-based early detection framework for digital burnout risk using behavioural patterns collected from personal digital devices. The proposed framework applies digital phenotyping to analyze behavioral indicators such as screen-use duration, frequency of device interaction, sleep patterns, physical activity, mobility, application usage, interaction frequency, and changes in daily routines. These behavioral features are processed through preprocessing, temporal feature extraction, normalization, and machine-learning-based risk prediction. The framework can employ classification models such as Logistic Regression, Random Forest, Support Vector Machine, and gradient-boosting methods to estimate a user's changing burnout-risk level. Importantly, the system is designed as an early-warning and self- monitoring tool rather than a clinical diagnostic system. Existing digital-phenotyping research demonstrates that smartphone-derived behavioral signals can be associated with stress, sleep, activity, social interaction, and other aspects of well-being. The proposed approach therefore aims to identify meaningful behavioral deviations from an individual's normal baseline and provide timely, privacy-aware feedback before prolonged patterns become more severe. The study also emphasizes privacy, informed consent, data minimization, explainability, and individual behavioral baselines as essential requirements for responsible AI-based well-being monitoring.

43.IOT-ENABLED AMBULANCE PATIENT MONITORING

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Continuous access to patient information during ambulance transport is essential for timely clinical decision-making and effective pre-hospital care. However, patient information available during transportation may not always be accessible to the receiving hospital in real time, creating a communication gap between the ambulance team and hospital staff. This paper presents an IoT-enabled ambulance patient monitoring system focused on the real-time transmission of existing patient vital data from an ambulance-side system to a hospital-side monitoring interface. The system uses an ESP32-S3 microcontroller to process and transmit the available patient data through wireless communication. LoRa communication is used to establish ambulance-to-hospital data transmission, while the transmitted information is made available through a remote monitoring interface for hospital- side observation. The developed prototype demonstrates continuous data transmission and real-time monitoring of patient information under laboratory conditions. The proposed system provides a low-cost and scalable approach for improving information sharing between ambulances and healthcare facilities, enabling hospitals to access patient information before ambulance arrival.

44. MULTI-AGENT SMART CAMPUS HELPDESK FOR STUDENT, FACULTY AND ADMINISTRATIVE QUERIES

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University campuses generate a continuous stream of heterogeneous queries from students, faculty, and administrative staff, spanning academics, examinations, hostel and facilities, finance and fee payments, library services, and IT support. Most institutions still route these queries through a single generic chatbot, a shared mailbox, or in-person helpdesk counters, producing slow, inconsistent, and unverifiable answers, with no clear path to a human when the system is uncertain. This paper proposes a Multi-Agent Smart Campus Helpdesk that decomposes the helpdesk into an orchestrator/router agent and a set of modular, domain- specialized agents, each grounded in an institutional knowledge base through Retrieval-Augmented Generation (RAG). Incoming queries are authenticated and scoped through role-based access control (Student, Faculty, Administrator), classified and routed to one or more relevant agents by a transparent, weighted routing score, and their retrieved evidence is checked for grounding before a conflict- detection stage reconciles disagreements across agents. A context-aware response synthesis step then produces a single coherent answer, or escalates the query to a human agent when confidence is low or an unresolved conflict remains. The paper details the layered multi-agent architecture, the routing and escalation formulation, the knowledge base and module design, security considerations, and a testing methodology, defining evaluation metrics for a future empirical study rather than reporting fabricated results. The proposed ecosystem is expected to reduce response latency and staff workload while keeping every answer traceable to a verified institutional source.

45. A SMART PLATFORM FOR SKILL GAP DETECTION, LEARNING ROADMAP GENERATION AND MENTOR MATCHING

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The rapid evolution of technology and shifting industry requirements have widened the gap between the skills students and professionals possess and the skills that employers actually demand. Traditional career counselling and generic online courses fail to address this gap in a personalized manner, leaving learners uncertain about which competencies to develop and how to develop them efficiently. This paper proposes a Smart Platform for Skill Gap Detection, Learning Roadmap Generation and Mentor Matching that leverages machine learning and natural language processing to bridge this divide. The platform evaluates a user's existing skill set against the requirements of a target job role, automatically identifies missing or weak competencies, and generates a personalized, milestone-based learning roadmap using curated resources. In addition, the system incorporates a mentor-matching module that connects learners with suitable mentors based on domain expertise, availability and mentoring style, using a similarity- based recommendation approach. Experimental evaluation on a prototype implementation demonstrates that the proposed system improves the relevance of learning recommendations and reduces the time learners spend searching for appropriate resources and guidance. The results indicate that an integrated, data-driven approach to skill assessment, roadmap generation and mentorship can significantly enhance employability outcomes and self-directed learning.

46. MEDITWIN: INTELLIGENT DIGITAL TWIN FOR MEDICATION AND PATIENT HEALTH MONITORING

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The increasing prevalence of chronic diseases such as hypertension, diabetes, and cardiovascular disorders has highlighted the need for intelligent healthcare systems that go beyond traditional medicine reminder applications. Most existing medication management systems provide only time-based reminders and basic tracking, lacking the ability to continuously monitor a patient's health or predict potential medical risks. To address these limitations, this project proposes MediTwin, an Intelligent Digital Twin- based Medication and Patient Health Monitoring System that creates a virtual representation of a patient by integrating real-time health data, medication history, lifestyle information, and behavioral patterns into a unified healthcare platform. The proposed system continuously synchronizes patient information, including blood pressure, heart rate, oxygen saturation, medication adherence, physical activity, sleep patterns, and medical history, to maintain an up-to-date Digital Twin. Data collected from wearable devices, IoT sensors, and manual user inputs are securely stored in the cloud and analyzed using Artificial Intelligence. Machine learning models such as Random Forest, XGBoost, and LSTM are employed to predict medication non-adherence, detect abnormal health conditions, and forecast disease progression. Additionally, Explainable Artificial Intelligence (XAI) techniques provide transparent explanations for each prediction, enabling users and healthcare professionals to understand the reasons behind AI-generated decisions. The system also incorporates several intelligent healthcare features, including adaptive medication reminders, prescription digitization using Optical Character Recognition (OCR), medicine refill prediction, emergency caregiver alerts with livelocation sharing, and personalized health recommendations. Dedicated dashboards for patients, caregivers, and doctors present real-time health trends, medication adherence statistics, AI-generated risk scores, and comprehensive health reports, facilitating informed decision-making and timely medical intervention. The proposed Digital Twin framework offers a comprehensive

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solution for personalized medication management and continuous health monitoring by combining IoT, cloud computing, Artificial Intelligence, and Explainable AI into a single platform. The system aims to improve medication adherence, enable early detection of health deterioration, reduce emergency hospitalizations, and enhance communication among patients, caregivers, and healthcare professionals.

47. DYNAMIC MULTI-DISCIPLINARY CAPSTONE PROJECT COLLABORATION AND MENTORSHIP MATCHING ECOSYSTEM FOR UNIVERSITIES

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Capstone projects are central to undergraduate engineering curricula, requiring students to form teams, secure mentorship, and manage collaborative work over a term. Most universities still coordinate this informally, through personal networks, notice boards, and generic chat apps, leading to mismatched teams, uneven skill distribution, and inefficient mentor allocation. This paper presents a web-based Dynamic Multi-Disciplinary Capstone Project Collaboration and Mentorship Matching Ecosystem that centralizes profile management, project discovery, skill-based team formation, mentor matching, and progress tracking in one platform. The system follows a three-tier architecture: a React 18 and TypeScript front end built with Vite and Tailwind CSS, a Node.js/Express.js REST API layer with Socket.io for real-time updates, and a PostgreSQL database (hosted on Supabase) managed through the Prisma ORM, secured with JWT authentication, Google OAuth 2.0, and role-based access control. A weighted, rule-based scoring model, combining skill similarity, interest alignment, discipline diversity, and availability, is proposed for matching, avoiding machine learning where a transparent formula is sufficient for a first, single-institution deployment. The paper covers functional and non-functional requirements, database schema, module decomposition, security, and a testing methodology, and defines the metrics, such as matching relevance, response time, and task completion rate, for a future empirical evaluation rather than reporting fabricated results. The proposed ecosystem is expected to reduce coordination overhead and improve interdisciplinary collaboration in capstone programs.

48. POST-OPERATIVE PATIENT REMOTE RECOVERY TRACKER AND WOUND HEALING EVALUATION PORTAL

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Patients are often discharged a few days after surgery even though problems such as surgical site infection (SSI), delayed wound healing and reduced mobility may become noticeable later at home. Conventional follow-up usually depends on scheduled hospital visits or telephone calls, so early changes can be missed. This paper presents the Post-Operative Patient Remote Recovery Tracker and Wound Healing Evaluation Portal, a full stack web application through which patients can record daily recovery information and upload wound photographs at regular intervals. The proposed system uses image processing and deep-learning methods to estimate wound size, tissue composition and changes in healing over time using the patient's earlier images as a baseline. Clinicians can view these

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records through a dashboard and identify patients whose recovery trend may require attention. A review of ten studies published between 2021 and 2026 found, within the reviewed set, that recovery tracking, AI- assisted wound evaluation and a clinician dashboard are not combined in a single platform. The paper therefore describes the proposed architecture, methodology and modules. Since the project is still at the design stage, no experimental performance results are claimed; such figures will be added after implementation and evaluation.

49. SMART COLLEGE ADMISSION COUNSELOR: AN AI AGENT FOR PERSONALIZED COLLEG AND COURSE RECOMMENDATION AND ADMISSION GUIDANCE

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Picking a college and a course is a big decision, but most students work it out from scattered material: what a counsellor says, what friends heard about cut-offs, and prospectus PDFs that are hard to compare. This paper describes the Smart College Admission Counselor, an AI agent that brings college and course search, admission-probability estimates and application guidance into one chat-based platform. Two parts do the main work. A weighted rule-based engine ranks colleges and courses by admission probability, interest match, budget, location and placement record. A retrieval-augmented generation (RAG) agent answers questions only from verified institutional documents, which limits made-up admission advice. The system has three tiers: a React/TypeScript front end, a Node.js/Express layer for the REST API and agent logic, and a PostgreSQL database, with JWT authentication and role-based access control. We set out the requirements, scoring formula, database schema, security design and test plan. No results are reported, since the system has not been deployed; the paper lists the metrics a later study would use instead.

50. AI-ENABLED PREDICTIVE MAINTENANCE DASHBOARD FOR SMART STREETLIGHTS

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Municipal streetlights are often repaired only after citizen complaints or manual inspections, which can cause delayed maintenance and longer outage periods. This paper presents a software-only AI-Enabled Predictive Maintenance Dashboard for Smart Streetlights that combines streetlight monitoring, complaint management, and failure-risk prediction in a single web platform. The system uses React/Next.js, REST API, real-time updates, and a relational database to manage streetlight information and maintenance records. Since physical IoT hardware is not required, simulated telemetry is used to generate voltage, current, lumen output, and temperature readings. A rule- based risk scoring model analyzes operating hours, fault history, complaint density, and sensor deviations to identify

51. ALGORITHMIC TRADING SYSTEMS EMPLOYING MACHINE LEARNING FOR FINANCIAL MARKET PREDICTION AND AUTOMATED EXECUTION: METHODS, ARCHITECTURE AND FEASIBILITY IN INDIAN EQUITY MARKET.

 ISBN 978-81-69206-38-9

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The volume of structured price data and unstructured text that are being created on electronic exchanges has become immense, and this has thrust machine learning (ML) into the forefront of modern algorithmic-trading research. Instead of suggesting another predictive model independent of the trading system, this paper discusses what such a predictive model must offer a defensible trading system for an equity market like the National Stock Exchange (NSE) or the Bombay Stock Exchange (BSE) from the predictive stage to the order execution and risk control stage. Five approaches are explored and compared: statistical and tree-based learners, recurrent and convolutional deep networks, transformer architecture, reinforcement learning (RL) agents, and text-driven sentiment models. That review in turn informs a methodology of data collection and cleaning, feature building, a multi-layered architecture for prediction and execution, signal generation on thresholds within explicit risk controls, and an event-driven and bias-conscious backtesting methodology. A comparison is made between rule-based technical trading and classical ML, deep sequence models, transformers, RL and sentiment-augmented hybrids, in terms of data requirements, interpretability and effective horizon, concluding in a hybrid-ensemble recommendation, instead of a single winning architecture. The paper presents a series of examples throughout to demonstrate that classification accuracy alone is not a good measure of trading viability and that, in fact, risk-adjusted measures and walk-forward (cost-aware) validation are the only true measures of whether a result generalizes. The latter part of the book presents open directions including refined reinforcement learning, multimodal and order-book inputs, explainability, and deploying in a regulation-aware manner. No architecture is guaranteed to be profitable, nor is any claim made that any of them are, in this paper they are merely prototypes that have been used to educate in the classroom, but they are studied with discipline and in the light of regulation in order to distinguish them from systems capable of trading real capital.

52.EFFECTS OF STRUCTURED ACTIVE LEARNING ON STUDENT ENGAGEMENT IN A MODERN CHINESE HISTORY COURSE: A QUASI-EXPERIMENTAL STUDY

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This study examined whether structured active learning was associated with stronger student engagement in a compulsory Modern Chinese History course at a private university in Zhejiang Province, China. A quasi-experimental pre- and post-intervention design involved two intact classes comprising 260 first- year undergraduates, with 130 students in each condition. The experimental class completed ten course-aligned active learning units organised around analysis, interaction, feedback, learning output, and reflection, while the comparison class received regular instruction. Engagement was measured using the 23-item Student Course Engagement Questionnaire. Although engagement increased significantly in both groups, baseline-adjusted post-intervention engagement was higher in the structured active learning condition, $F(1, 257) = 17.12$, p < .001, partial $\eta^2 = .062$. Significant adjusted differences were also found across all four engagement dimensions. The results suggest that structured active learning may be particularly valuable when it provides sustained opportunities for students to engage intellectually, interactively, and reflectively with disciplinary content.

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53. ELECTRIC VEHICLE ROUTE OPTIMIZATION AND ENERGY CONSUMPTION PREDICTION SYSTEM USING GOOGLE MAPS API AND AI

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This paper aims to create an EV's system which optimizes route, predict the energy consumption and show the best less energy consuming route. It basically optimizes routes for electric vehicle by considering the data of real-time traffic, terrain, weather, and the availability of charging stations through various sources. Google maps APIs gives route planning, live traffic updates, and elevation data etc. A machine learning model scrutinizes vehicle behavior, road conditions, and past driving data of the vehicle and according to that it predict energy usage. A web-based UI gives real- time navigation and suggests energy-efficient routes. This project improves electric vehicle efficiency, reduces range anxiety, and enhances the driving experience for the user.

54. AI-DRIVEN GOVERNMENT WELFARE SCHEME RECOMMENDATION AND EXPLAINABLE ELIGIBILITY CHECKING WITH CONTINUOUS ELIGIBILITY MONITORING

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Government welfare schemes provide essential financial, healthcare, educational, employment, housing, and social-security support to citizens. However, identifying suitable schemes remains challenging because of complex eligibility conditions, numerous schemes, changing beneficiary circumstances, and limited awareness. Existing welfare portals primarily depend on manual scheme searches and one-time eligibility checks, which may cause citizens to miss benefits for which they become eligible later. This paper proposes an AI-driven Government Welfare Scheme Recommendation and Explainable Eligibility Checking System that provides personalized and continuous welfare assistance. The proposed framework maintains a dynamic citizen profile containing attributes such as age, income, education, occupation, family composition, and relevant documents. An intelligent Scheme Recommendation Engine analyzes these attributes against scheme requirements to identify and recommend relevant government benefits. To improve transparency, the proposed Explainable Eligibility Checker does not merely classify citizens as eligible or ineligible. Instead, it identifies the specific eligibility conditions satisfied or violated, highlights missing requirements or documents, and provides actionable guidance. A novel Lifetime Eligibility Radar continuously re-evaluates eligibility when significant changes occur in a citizen's profile or when new welfare schemes are introduced. This enables proactive discovery of benefits without requiring repeated manual

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searches. The framework further supports benefit tracking, household-level welfare analysis, document verification, and personalized welfare planning through a unified platform. The system combines rule-based eligibility validation with AI-assisted recommendation and explainability mechanisms, implemented using a modern web architecture with React.js, Node.js, Express.js, MySQL, and OCR technologies. The proposed approach aims to transform welfare discovery from a reactive, one-time eligibility-checking process into a proactive, personalized, and continuously updated welfare intelligence system, improving scheme discoverability, eligibility transparency, and access to government benefits.

55. AI-BASED SMART ENERGY MONITORING AND OPTIMIZATION SYSTEM FOR CAMPUS BUILDINGS

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Campuses consume a huge amount of electricity in classrooms, laboratories, libraries, hostels, and administrative blocks, but it is still largely manually monitored or limited through fixed-alarm systems that neglect the context of use within the moment of the room. We present an AI-based solution for energy auditing in campus buildings that utilizes current and previous meter readings combined with four contextual dimensions of occupancy, schedules, room characteristics, and appliance usage patterns to identify normal and thus anomalous consumption behaviors specific for each building and room. The machine learned a demand pattern for each building and the room and estimated an anticipatory load for a given time instance that showed a significant deviation as a potential anomaly candidate. Before flagging, the candidates were filtered according to holidays, examinations, seminars, placement drives, room changes, or maintenance periods to avoid false positives that would otherwise penalize legitimate higher consumption areas such as laboratories. The anomaly reports were then converted to action items, including switching off idle appliances such as fans and lights or suggesting reduced air-conditioning usage and electricity consumption outside core working hours and were visualized in a web dashboard with an aggregated view on the building and room levels, including trends, alerts, and saving analytics, for administrators. By incorporating the contextual information, the system aimed to reduce false positives while providing relevant suggestions for administrators to make data-driven decisions for energy auditing in campuses.

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56. SENTINEL: A REAL-TIME YOLO-BASED DESKTOP COMMAND CENTER FOR AUTOMATED PERSONAL PROTECTIVE EQUIPMENT COMPLIANCE MONITORING

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Non-compliance with Personal Protective Equipment (PPE) regulations remains a persistent safety concern in construction and industrial environments, while many vision-based PPE monitoring systems depend on browser-based interfaces, server infrastructure, or continuous network connectivity. This paper presents SENTINEL, a network-independent, on-device desktop command center designed for automated PPE compliance monitoring using local hardware. The system employs a YOLO-based object detector, a greedy Intersection-over-Union (IoU) object tracker, and a sustained-violation alerting rule consistent with an existing open-source browser- based reference system. These components are integrated into a new multi-threaded PyQt5 architecture that separates camera capture and model inference from the graphical event loop. SENTINEL further provides a thread-safe live-monitoring interface, auto-refreshing analytics, a searchable and CSV-exportable detection log, and locally persisted SQLite records. The paper defines an evaluation methodology covering detection accuracy, real-time performance, alert latency, tracking robustness, and operation under degraded or absent network conditions. Preliminary measurements report an overall detector mAP@50 of 0.619 associated with the inherited model weights, a model size of 17.6 MB, approximately 3.2 frames per second on a CPU live-camera run (range 2.6–4.8 FPS), and an indicative alert latency of approximately 4 s from a replayed logged session. A one-minute CPU-only timing run is also reported. The remaining evaluations, including independent held-out detection validation, quantitative tracking assessment, and network-degradation comparison, have not yet been completed. Therefore, the network- independent reliability advantage of SENTINEL is presented as an architectural and systems-level contribution and is not claimed as experimentally demonstrated in the current study.

57 .ZERO HUNGER: AN AI-DRIVEN SMART FOOD REDISTRIBUTION PLATFORM TO COMBAT HUNGER AND REDUCE WASTE

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In India, nearly 40% of food produced never reaches those who need it, while a large section of the population faces chronic hunger; this reflects a distribution and coordination gap rather than an actual shortage of food. Restaurants, supermarkets, caterers, and households routinely discard edible surplus food because nearby NGOs, shelters, and community kitchens are informed too late, have no reliable way to mobilise volunteers, and cannot confirm whether donated food is safe to consume. This paper presents Zero Hunger, a proposed AI-driven smart food redistribution platform intended to connect food donors with verified NGOs, shelters, and available volunteers through a real-time digital system. The platform is designed to use AI-powered matching to pair each donation with the nearest verified recipient and volunteer, reducing reliance on manual, phone-based coordination. An AI-based Food Safety Scoring model is intended to assess each donation using factors such as food type, storage

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duration, and temperature before a pickup is confirmed. IoT sensors are planned for commercial storage facilities to continuously monitor temperature and humidity and alert donors when food approaches the end of its safe storage period. Data analytics is intended to support route optimization for pickups and deliveries, along with demand forecasting to help NGOs plan in advance. The platform also incorporates NGO verification to support recipient accountability, together with volunteer coordination and impact tracking to sustain engagement. This paper further situates Zero Hunger against comparable food-redistribution and food- safety-monitoring efforts reported in the literature, describes the intended matching and scoring algorithms in greater technical detail, and outlines the feasibility, limitations, and future development roadmap of the proposed system. Through this integration, Zero Hunger aims to reduce food wastage, improve the timeliness and safety of surplus food redistribution, and support more consistent access to food for underserved communities.

58 .AN AI-DRIVEN UNIFIED TRUST SCORE FRAMEWORK INTEGRATING CIVIC TAX COMPLIANCE AND OPEN-SOURCE CONTRIBUTION ANALYTICS FOR HOLISTIC DIGITAL CITIZEN ASSESSMENT

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Government tax records and personal software-development activity are, today, assessed by completely disconnected institutions, so a citizen's financial responsibility and their technical or community contribution are never viewed together, even though both are meaningful signals of reliability and civic value. This paper proposes a Unified Trust Score (UTS) framework that links a single identity — an Aadhaar number optionally paired with a GitHub username — to two independent data pipelines: a civic-compliance pipeline that aggregates property-tax payment records from linked bank accounts, and a community-contribution pipeline that queries the GitHub REST API for public repository, pull-request and issue activity. A four-layer architecture (identity resolution, data aggregation, AI intelligence, and application) feeds both streams into a shared reasoning layer built on a large-language-model-backed scoring engine (Claude 3.5 Sonnet) with a deterministic rule-based fallback, producing a composite trust tier — HIGH, MEDIUM, or LOW — together with a natural-language justification for each citizen profile. The framework is evaluated on a simulated cohort of 500 synthetic citizen profiles compiled from representative tax-payment distributions and public GitHub activity patterns, with the automated classification checked against manually labelled ground truth. The system reaches 91.4% overall agreement with manual assessment, an average end-to- end scoring latency of 640 ms, and cuts the average per-profile assessment time by roughly 87% compared with manual review. We discuss applications in scholarship screening, government incentive schemes and civic-recognition programs, along with the privacy and fairness safeguards that would be required before any real deployment.

59 .AN EXPLAINABLE DEEP LEARNING FRAMEWORK FOR BRAIN TUMOR DETECTION, CLASSIFICATION, AND GRADING USING EFFICIENTNETV2B0

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Brain tumors are among the most critical neurological diseases in the world and the prognosis of the treatment can significantly rely on early and precise diagnosis. The imaging modality most commonly used for this is Magnetic Resonance Imaging (MRI) but manual interpretation of MRI images by the radiologist is still time-consuming, relies on the experience of the individual reader and can be prone to inter-observer variability especially where the grade of the tumor also needs to be determined. In this work, we design a complete, learning-based system to assist radiologists in the analysis of brain MRI that is accurate, efficient, and explainable. The framework will be expanded to include tumor grading, a web-based interface for automated report generation, and visualization of model predictions by Gradient- weighted Class Activation Mapping (Grad-CAM) and will be categorized into four types: glioma, meningioma, pituitary tumor, and no tumor. In order to develop the model, the Br35H dataset (7,023 labelled MRI images) was preprocessed, which involved resizing, normalization, denoising, contrast enhancement, and augmentation of the MRI images. To examine the various candidate architectures, as a first step, four different architectures were trained with the same settings and evaluated on a held-out test set: a baseline CNN, EfficientNetB2, EfficientNetV2B0 and EfficientNetB3. Based on the overall performance of the model, EfficientNetV2B0 was chosen as the backbone since it achieved the highest test accuracy (90.56 %) with about 6.25 million parameters.

60 .ANCESTRA: A SELF-EVOLVING STRUCTURAL CONSENSUS INTELLIGENCE FRAMEWORK FOR PREDICTIVE HERITAGE CONSERVATION USING OPPORTUNISTIC CROWDSOURCED VISUAL DATA

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Heritage monuments deteriorate continuously under environmental exposure, biological growth, and visitor pressure, yet conservation authorities rely on infrequent manual inspection or costly sensor-based structural health monitoring (SHM) that is difficult to justify for the majority of protected sites. Meanwhile, tourists opportunistically capture and upload large volumes of monument photographs that remain unexploited as a structural sensing resource. This paper presents Ancestra, a Self-Evolving Structural Consensus Intelligence (SESCI) framework that repurposes such crowdsourced tourist photographs into a continuously evolving, reliability-weighted structural intelligence system. An ingestion module geometrically registers and segments incoming images into architectural regions. A Reliability- Adaptive Evidence Fusion Engine computes a per-image reliability coefficient from image quality, geometric consistency, viewpoint diversity, temporal relevance, environmental similarity, and inter-observation agreement, and uses it to govern a weighted running update of a persistent, region-wise Dynamic Structural Consensus Memory. A Consensus-Based Structural Validation module confirms candidate anomalies only when corroborated by multiple independent, sufficiently reliable observations, suppressing false positives from shadow, foliage, scaffolding, and glare. A Cross-Temporal Structural Evolution Intelligence module models deterioration trajectories, and an Autonomous Predictive Conservation Orchestrator computes a conservation urgency score, auto-generates work orders, and notifies heritage authorities. We describe the architecture and algorithms,

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work through illustrative consensus-update and urgency-scoring examples, and discuss an evaluation plan and open challenges.

61. AI-POWERED SMART HOSPITAL BED UTILIZATION AND PATIENT FLOW DECISION SUPPORT SYSTEM

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Efficient hospital bed utilization and smooth patient flow are critical for timely healthcare delivery and lower operational costs. This paper presents an AI-powered Smart Hospital Bed Utilization and Patient Flow Decision Support System that automates patient registration, ward and bed recommendation, transfer tracking, and discharge management, while applying machine learning to support administrative decisions. Patient movement from admission to discharge is recorded in a MySQL database and shown on a real-time dashboard of occupancy, admissions, and discharges. A Decision Tree combined with rule-based logic recommends the most suitable ward and bed based on disease, severity, and resource availability. A Random Forest Regressor predicts each patient's length of stay, an XGBoost Regressor forecasts future bed occupancy from historical admission and discharge patterns, an Isolation Forest detects abnormal, overcrowded wards, and a Logistic Regression model estimates the probability of 30-day readmission. Rule- based thresholds generate real-time alerts, and a Large Language Model converts the analytical results into SSSS readable summaries for hospital administrators. Because each module uses an algorithm suited to its problem type (classification, regression, or anomaly detection), the system improves both predictive accuracy and explainability, helping hospitals anticipate demand, reduce waiting times, and prevent ward overcrowding.

62. A CARBON-AWARE ADAPTIVE ROUTING AND ORDER-CONSOLIDATION FRAMEWORK FOR SUSTAINABLE E-COMMERCE LAST-MILE DELIVERY

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Last-mile delivery is one of the most carbonintensive parts of e-commerce logistics, yet conventional route planners optimize purely for distance or travel time and have no visibility into the emissions a route actually produces. This paper presents EcoRouteX, a multi-objective routing and order-consolidation framework that models a delivery network as a weighted road graph, scores candidate routes with a blended distance-time- carbon cost function, and uses a density-and-time-window clustering stage to identify orders that can be merged into a single trip. We benchmark EcoRouteX against three baselines shortest-distance routing, fastest time routing, and carbon- weighted routing without consolidation on a synthetic but realistically parameterized urban road network at order volumes of 50, 150, and 300, on a road network held fixed in size across all three scales so that order density is not confounded with network topology. Across all three scales, EcoRouteX cuts total CO_2 emissions by 9.7% to 56.8% relative to the individual- delivery baselines. The experiments also isolate where this reduction actually comes from: carbon-weighted edge routing on its own accounts for only a 1.6-2.2% emissions reduction over shortest-path routing, while order consolidation is responsible for nearly all of the remaining gain. This shifts the effective lever for emissions reduction in lastmile delivery from route-level path selection to trip level demand aggregation, and we present it here as a specific, reproducible, and falsifiable result rather than a general claim about carbon-aware routing. We describe the mathematical formulation, the consolidation algorithm, the experimental protocol, and outline how learned components — graph neural traffic prediction and reinforcement-learning- based dynamic re-routing — could extend the framework in future work.

63.SENTINELX AI: AN EXPLAINABLE MULTI-AGENT FRAMEWORK FOR DETECTING FAKE RECRUITMENT AND ASSESSING ONLINE JOB TRUST

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The rise of online recruitment marketplaces has led to an explosion of job scams, impersonation by recruiters, and fake offer letters resulting in significant loss to job seekers. Anti-fraud measures currently are limited in their capability, often acting as black-boxes, only considering a uni-model input, and not being able to ingest the various signals that appear as part of a real job application. This paper introduces SentinelX AI-an explainable multi-agent AI platform for detecting recruitment fraud and analyzing digital trust. The system takes in various modalities including the job poster, screenshots, PDFs, offer letters, recruiter information, email, website URL, and QR codes as input and directs them to eight different agents, including OCR, NLP, Computer Vision, and various verification modules for recruiters, companies, websites/URLs, QR codes, and threat intelligence. Evidence collected by these modules is then combined by an Adaptive Trust Scoring Engine (ATSE) that outputs a Trust Score, Fraud Score, Risk, and Confidence. An XAI layer provides natural language explanation rather than a binary real/fake prediction, while a Digital Trust Knowledge Graph (DTKG) connects related entities such as recruiters, companies, domains, emails, phone numbers, and QR codes across different fraudulent activities to uncover fraud rings and patterns at scale. SentinelX AI demonstrate the system architecture and explain how different agents contribute to the final Trust Scores and explainable evidence. SentinelX AI also describe the design principles for XAI and provide an overview of the evaluation methodology for detection accuracy, explanation performance, and network level analysis of fraud patterns. The proposed approach can potentially be generalized to other areas such as phishing, identity verification, document forgery detection, and other cyber-security related tasks.

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64. NLP-DRIVEN KNOWLEDGE GRAPH CONSTRUCTION FROM SCIENTIFIC RESEARCH PAPERS: A COMPARATIVE ANALYSIS OF EXTRACTION, REPRESENTATION, AND STORAGE TECHNIQUES

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The exponential growth of scientific literature has made it increasingly difficult for researchers to manually track concepts, methods, datasets, and their inter-relationships across thousands of publications. Knowledge graphs (KGs) offer a structured, queryable representation of scientific knowledge, but constructing them from unstructured research papers requires a pipeline of Natural Language Processing (NLP) tasks, including PDF parsing, named entity recognition, coreference resolution, relation extraction, and entity linking. This paper presents a comparative study of NLP-based techniques for building knowledge graphs from scientific research papers using Python, spaCy, transformer-based language models such as SciBERT, and the Neo4j graph database. Representative entity and relation extraction approaches are compared on the SciERC scientific benchmark, and key challenges such as domain adaptation, cross-document entity resolution, scalability, and evaluation are identified, along with directions for future research.

65. AI-DRIVEN INTRUSION DETECTION IN IOMT: A COMPARATIVE ANALYSIS OF DETECTION PERFORMANCE, PRIVACY, AND EXPLAINABILITY

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The growing trend of the Internet of Medical Things (IoMT) has made intelligent and connected healthcare solutions available, but at the same time has created vulnerabilities to advanced cyber-attacks. AI-based Intrusion Detection Systems (IDS) have become the key security technique that allows recognizing malicious behavior in heterogeneous IoMT environments. This paper proposes a comprehensive comparison of recent intrusion detection techniques employing ML, DL, FL, blockchain-based intelligence, and XAI. Recent approaches are analyzed taking into consideration learning technique architecture, used datasets, attacks covered, reported detection performance, privacy-preserving capabilities, computational complexity, explainability, and deployment limitations. The results of the comparative study reveal that advanced learning-based and hybrid IDS demonstrate impressive reported performance. Nevertheless, the direct comparison of performance is restricted due to the lack of consistent benchmarking across datasets, distribution of attacks, evaluation settings, and classification problems. Moreover, there are such issues as computational costs, limited cross-dataset generalization, communication overheads in federated environments, privacy concerns, and poor model interpretability that restrict practical adoption of current techniques. On this basis, research challenges and future directions related to the development of lightweight, privacy-preserving, explainable, and adaptable IDS are revealed. The proposed study gives a narrow-focused vision of trade-offs between performance, privacy, and explainability.

66. AN AI-BASED RECOMMENDATION AND STABLE ALLOCATION FRAMEWORK FOR THE PRIME MINISTER INTERNSHIP SCHEME

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The Prime Minister Internship Scheme needs a system that can match a large number of students with suitable internships in a fair and transparent manner. Most of the existing internship systems mainly recommend opportunities using keywords, skill similarity, or weighted scores. They do not fully deal with the next step, which is stability-aware allocation under seat capacity, student preferences, company preferences, policy rules, verification, and reallocation. This paper presents an AI-based recommendation and stable allocation framework for the Prime Minister Internship Scheme. The framework brings together structured candidate profiles, a skill knowledge graph, Sentence-BERT based semantic similarity, machine learning models for internship success, opportunity gap, and offer acceptance prediction, LightGBM LambdaRank for learning-to-rank, and a policy-aware Modified Gale- Shapley algorithm for allocation. A calibrated synthetic dataset was generated using public PMIS-related workflow information, AICTE internship records, PMIS partner company information, eligibility rules, and realistic student- internship distributions. Experimental results on held-out students show that the full AI pipeline improves NDCG@10 from 0.3975 for keyword matching to 0.4076. The Modified Gale-Shapley stage reduces the blocking pairs from 376 to 20 and increases the stable allocated candidates from 0 to 94. Dynamic reallocation helps recover 15 vacant seats after simulated verification or offer failure. Overall, the work displays how recommendation, opportunity-aware ranking, stable matching, and explainable AI can be grouped together into a single, practical allocation pipeline for public internship schemes.

67. A SECURE FEDERATED LEARNING FRAMEWORK FOR DIABETES AND CARDIOVASCULAR RISK ASSESSMENT

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Diabetes mellitus and cardiovascular diseases remain two of the leading causes of mortality worldwide, making early disease prediction an important component of modern healthcare systems. Conventional machine learning approaches generally require centralized aggregation of sensitive patient data collected from multiple healthcare institutions. Although centralized learning achieves good predictive performance, it raises significant concerns regarding patient privacy, data confidentiality, regulatory compliance, and security. Federated Learning (FL) has recently emerged as a promising distributed learning paradigm that enables collaborative model training while ensuring that raw patient data never leaves local healthcare institutions. This paper proposes a privacy-preserving healthcare prediction framework based on Federated Learning integrated with Weighted Conglomeration Optimization (FLWCO). Multiple healthcare institutions independently train local machine learning models using their own diabetes and heart disease datasets. Instead of transmitting patient records, only encrypted model parameters are communicated to a central aggregation server. A novel Weighted Conglomeration Optimization mechanism dynamically assigns aggregation weights according to the performance of individual local models, thereby improving the quality of the global model over conventional FedAvg aggregation. The framework incorporates Random Forest, Gradient Boosting, AdaBoost, XGBoost, and Multilayer Perceptron classifiers. Data preprocessing includes missing value handling, feature scaling, duplicate removal, feature selection, and Synthetic Minority Oversampling Technique (SMOTE) for addressing class imbalance. Experimental evaluation demonstrates improvements in prediction accuracy, precision, recall, F1-score, and robustness while maintaining strict privacy preservation. The proposed FLWCO framework provides an effective solution for collaborative healthcare analytics suitablefor multi-hospital environments and future smart healthcaresystems.

68. REAL-TIME SHIP ENERGY OPTIMIZATION USING INTEGRATED WEATHER, WAVE, AND ENGINE-PERFORMANCE DATA: A FRAMEWORK FOR FUEL-EFFICIENT VOYAGE PLANNING

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The maritime sector is currently witnessing a paradigm shift in its approach towards reduced fuel consumption, cost, and emissions with enhanced safety and efficiency of voyage operations. Ship energy performance is highly dependent on the interaction between weather, sea state, engine performance, ship speed, and route. These variables are often considered individually which does not allow the ships' operator to adjust the ship's energy performance in response to the actual operating conditions. Therefore, this study presents a proposal for a ship energy optimization model based on real-time analysis of weather conditions, wave state, engine performance, and ship operation. The framework will consider these aspects to provide an insight into the vessel's performance in relation to

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fuel consumption and other energy performance indicators. The data can be utilized to develop predictive models that can help estimate engine performance and fuel consumption. The framework can also provide dynamic optimization of the ship's route and speed considering the current and predicted sea conditions. By continuously analyzing the data, the framework can also update the suggested course and speed instead of depending on initial voyage planning. In this regard, it can be argued that the proposed framework can enhance optimized fuel consumption and reduced emissions while supporting improved voyage planning and decision-making for enhanced fuel economy and emission control.

69. DESIGN AND DEVELOPMENT OF A SOLAR-POWERED SMART AQUAPONIC SYSTEM WITH REAL-TIME MONITORING AND AUTOMATIC WATER CIRCULATION

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This paper introduces a smart aquaponic system using solar photovoltaic (PV) source for sustainable fish and plant production with real-time monitoring and automation of water circulation. The proposed system consists of solar PV source, aquaculture, plant growing through hydroponics, water circulation pump, sensors, microcontroller, and display unit. In the proposed system, fish excreta act as a nutrient for the plants, while plants and biological filtration improve water quality before its circulation to fish tanks. The control unit continuously monitors various parameters including water level, water temperature, pH, etc. The data from sensors is then fed into the microcontroller, which is shown in real time to monitor the system. Based on the monitored parameters, the controller controls the operation of water circulation pump to regulate the water flow between the fish tanks and plant-growing tanks. The solar PV source provides the needed electric energy to the pumping and control devices, hence reducing dependency on conventional electrical energy. Various operating conditions and irradiance values of solar sources may be considered to test the proposed system.

70.INTELLIGENT SOLAR PV-BASED BATTERY CHARGING SYSTEM USING BOOST AND SEPIC CONVERTERS WITH ANN AND ANFIS CONTROLLERS

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This paper proposes an intelligent solar photovoltaic (PV) based battery charging system using cascaded Boost and Single-Ended Primary-Inductor Converter (SEPIC) stages along with conventional and intelligent controllers. The proposed intelligent charging system consists of a variable solar PV source, boost converter to condition the PV side voltage and to form the DC-link and finally a SEPIC converter to control the battery charging. The battery voltage, charging current and state of charge (SOC) will be measured to regulate the charging process. Proportional Integral (PI), Artificial Neural Network (ANN)

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and Adaptive Neuro Fuzzy Inference System (ANFIS) controllers will be used for the purpose of generating the command to the SEPIC converter duty- cycle. Five input variables such as SOC, Battery voltage, Battery Current, PV Voltage and PV current will be supplied to the intelligent controller. The ANN model learns the non-linear relationship between the input parameters and the output duty cycle, whereas ANFIS uses the technique of combining fuzzy logic and neural learning for the purpose of the same. Different levels of solar irradiation will be considered to test the system. The training MSE and RMSE of the ANN and ANFIS models are 0.00042802 and 0.02068857, 0.00000704, 0.00265254 and 0.00133460 respectively. The outcomes reveal a lower prediction error in training of ANFIS than ANN. However, independent validation and experimentation is required in order to verify the generalization of the models.

71. REAL-TIME AUTOMATED VULNERABILITY DETECTION AND ITERATIVE AI-ASSISTED PATCH VALIDATION FOR WEB APPLICATIONS

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Web applications are released continuously, yet vulnerability scanning, triage, and remediation remain largely separate and manual, leaving exploitable flaws exposed for long periods, and patches proposed by large language models (LLMs) can look plausible while leaving the flaw in place or breaking existing functionality. This paper presents an AI-assisted framework that integrates real-time vulnerability detection with iterative patch validation for web applications. A change-triggered pipeline combines static, dynamic, and dependency scanning, correlates the findings, and applies AI-based triage to suppress false positives. Confirmed findings enter a patch pipeline in which an LLM proposes context-aware fixes that are validated in an isolated sandbox through build checks, regression tests, exploit replay, and targeted re-scanning, with structured failure feedback returned to the model until a patch passes every gate or the iteration budget is exhausted. A central orchestrator links both pipelines, delivers each validated patch as a pull request for human approval, and escalates unresolved cases to a security engineer. The framework aims to shorten the time between detection and verified remediation while maintaining human oversight,and the study also addresses patch overfitting, data privacy,and the security of the pipeline itself.

72.AGROCARBON AI: MOBILE-BASED CARBON CREDIT ESTIMATIONSYSTEM FOR SMALL FARMERS

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AgroCarbon AI is an image-based tree carbon assessment system designed to estimate tree carbon sequestration and its equivalent monetary carbon-credit value using two photographs: a whole-tree image and a close-up image of a leaf. The proposed system consists of a six-stage pipeline comprising species classification, health classification, tree detection, canopysegmentation, biomass estimation, and carbon-credit valuation. The first four stages employlearned computer-vision models, while the final two stages use deterministic mathematicalcalculations based on forestry allometric relationships and greenhouse-gas accountingstandards. Trunk diameter at breast height (DBH) is estimated from leaf length using a species-specific power-law relationship, followed by tree height estimation using a Chapman–Richardsheight–diameter model. Above-ground biomass is calculated using the Chave et al. allometricequation, while carbon stock is estimated using the IPCC default carbon fraction. The storedcarbon is then converted to carbon-dioxide equivalent and assigned an approximate monetaryvalue using a configurable carbon-credit reference price. The deterministic design providestraceability and interpretability for the final estimates. However, field calibration and ground-truth validation are required to establish the absolute accuracy of the proposed carbonestimates.

73. EDUSPHERE AI: INTELLIGENT ACADEMIC DECISION PLATFORM

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Conventional academic management systems primarily focus on storing and managing student records, while academic decision-making and personalized study planning often remain dependent on manual analysis and individual judgment. To address this limitation, this paper proposes EduSphere AI, an intelligent, cloud-enabled academic decision platform that integrates an ERP-based Student Management System with an AI-driven decision-support layer. The proposed platform consolidates academic information including attendance, marks, examinations, syllabus, academic performance, study progress, and revision history across Student, Faculty, and Admin modules. The AI layer processes these academic data along with Previous Year Questions (PYQs), examination requirements, topic importance, student performance, and revision patterns to generate personalized academic insights. A PYQ Trend Analysis module identifies frequently occurring and examination-relevant topics, while a Hyper-Dynamic Priority Score dynamically prioritizes topics by considering multiple academic factors. In addition, a Personalized Study Planning mechanism recommends topics and revision activities according to individual learning requirements. A cognitive forgetting-curve-based revision mechanism further identifies topics that require timely revision to support knowledge retention. The platform is implemented using the MERN stack and integrated with AWS cloud services to provide

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centralized data management, scalability, and reliable deployment. An AI chatbot is also incorporated to provide an interactive academic assistance interface. By transforming distributed academic records into actionable and personalized recommendations, EduSphere AI aims to enhance academic planning, support data-driven decision-making, and provide a unified intelligent environment for students, faculty, and administrators.

74.EVALUATING COMMAND VALIDATION IN AI AGENTS: CONFIGURING AND TESTING THE NANOBOT FRAMEWORK WITH OPENCODE'S DEEPSEEK-V4-FLASH-FREE MODEL

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Large Language Models (LLMs) are increasingly being integrated into AI agent frameworks capable of performing system-level tasks such as file management, command execution, software automation, and operating system administration. AI agents combine the reasoning ability of language models with software tools, allowing users to perform complex computer operations using natural language instructions. This research work investigates the configuration, testing, and command execution behaviour of the Nanobot AI agent framework running on a Windows 11 environment. Unlike traditional local deployments that require dedicated GPU hardware, this research work uses OpenCode's hosted DeepSeek-V4-Flash-Free language model through an OpenAI-compatible API using an API key. OpenCode was selected because it provides free hosted access to modern language models while maintaining compatibility with OpenAI-style APIs, making experimentation possible without requiring expensive cloud subscriptions or high-end local hardware. Nanobot was selected because of its lightweight architecture, configurable tool-calling system, WebSocket gateway, and flexible execution framework. The study focuses on how Nanobot's safety guard validates commands generated by the language model before execution. During testing, the DeepSeek- V4-Flash-Free model generated an alternative PowerShell command that assembled the destination drive letter and path dynamically at runtime instead of referencing it as a literal string. Because the framework's safety guard relied primarily on simple string-matching against literal drive paths, this generated command was not recognized by the filter before execution. Gateway log analysis confirmed that PowerShell reconstructed the full path internally and completed the requested file operation. The findings demonstrate that language models can produce syntactically valid but structurally varied commands when solving a requested task, and emphasize the importance of behaviour- based security mechanisms instead of simple text-based filtering when designing AI agent frameworks.

75. AI-ASSISTED PRODUCE AGGREGATION AND MARKET LINKAGE PORTAL FOR AGRICULTURAL COOPERATIVES WITH TRANSPARENT FARMER SETTLEMENT

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Smallholder farmers who sell through cooperatives still face fragmented small lots, inconsistent grading, limited visibility of verified supply for wholesale buyers, and opaque payment settlement. This paper presents an AI-assisted, role-based web portal for agricultural cooperatives that manages member farmers, verifies produce quality, consolidates approved submissions into aggregation batches, links batches to wholesale buyer demand, and settles farmer payments transparently. Farmers submit harvests through a form, an AI quality scanner that estimates ripeness, surface defects, and size uniformity, or a bilingual English–Tamil voice assistant. Cooperative staff verify quantities and assign Grade A, B, or C before batches are formed. A weighted matching engine scores each available batch against a buyer requirement on quantity (40%), quality grade (30%), price (20%), and location (10%) and returns a score explanation. On each order, revenue is allocated to contributing farmers in proportion to their contribution, a 5% cooperative fee is deducted, and a net settlement is recorded for every farmer. A public QR-based page exposes batch origin and contributor breakdown. The system is implemented on the MERN stack with JWT-based role-based access control. We describe the architecture, data model, and algorithms, work through an illustrative matching and settlement example, and discuss limitations and a plan for field evaluation.

76 .AI-BASED PRESCRIPTION IMAGE PROCESSING AND INTELLIGENT INVENTORY MANAGEMENT FOR AUTOMATED PHARMACY DISPENSING SYSTEMS

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Illegible prescriptions and manual stock management can cause inefficiency and errors in pharmacy operations. This paper presents an AI-based framework that integrates prescription image processing with intelligent inventory management for automated pharmacy dispensing. Prescription images are pre-processed and analysed using Optical Character Recognition (OCR) and deep learning techniques to extract and validate medicine names, dosages, and instructions. The inventory module monitors medicine availability using computer vision and applies demand forecasting to support timely stock replenishment. A central dispensing controller connects prescription processing with inventory management to verify and reserve medicines before dispensing, while exceptions are referred to a pharmacist for validation. The proposed framework aims to improve prescription processing, inventory accuracy, and dispensing efficiency while maintaining human oversight. The study also addresses challenges related to prescription variability, data privacy, and integration with existing pharmacy systems.

77. FROM CLASSICAL VISION TO DEEP REINFORCEMENT LEARNING: A THREE-PHASE AUTONOMOUS DRIVINGSYSTEM USING UDACITY, CARLA, AND PPO

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Self-driving cars need to be able to see well, control things perfectly, and change quickly. Canny edge detection, Hough Transform, and PID control are examples of classical computer vision (CV) methods that are fast but don't work well in environments that are dynamic and complicated. This paper presents a three-phase progressive research framework that enhances an autonomous driving pipeline, transitioning fromrule-based classical vision to a fully trained deep reinforcement learning (RL) agent. In Phase 1, The Udacity self- driving car simulator has a traditional CV lane-following system that uses HSV color filtering, Region of Interest (ROI) masking, Hough line detection, and a PID steering controller. In Phase 2, The system is transferred to the high-fidelity CARLA simulator. In that place, the Traffic Manager and map-based waypoint navigation work together with a YOLOv8 perception module to make a strong rule-based baseline. In Phase 3, A Deep Reinforcement Learning agent that only uses a camera and is trained with Proximal Policy Optimization (PPO) takes the place of the control paradigm. The agent learns how to steer by looking at four black-and-white camera frames. It does this without using any map data or ground truth from the CARLA API. The reward function is based on the output of semantic segmentation. It encourages covering the road, centering the lane, moving quickly, and steering smoothly. Experimental results show that the RL agent can follow lanes stably in a dynamic simulation environment. It does better than the classical CV baseline when it comes to generalization and robustness. The proposed framework emphasizes the transition from manually created rules to learned policies, establishing a reproducible benchmark for autonomous driving research in simulation.
Index Terms—Autonomous Driving, Lane Detection, Reinforcement Learning, Proximal Policy Optimisation, CARLA Simulator, Computer Vision, Canny Edge Detection, Hough Transform, PID Control, Semantic Segmentation, YOLOv8

78. EXPERIMENTAL INVESTIGATION ON PVC CONFINED CONCRETE WITH FRP WRAPPING

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In strengthening column, confinement is used as restraint and applies as a preventive measures to avoid deterioration of concrete structure in future. Lateral confinement by PVC, HDPE and UPVC increases strength, stiffness and durability of concrete structure .Nowadays it is necessary to increase strength, ductility and energy absorption capacity of column which is achieved by the application of various confining material wrapped with thin, flexible and high strength composite strip around column. Confinement of concrete around PVC wrapped FRP improves the maximum loading capacity of

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concrete structure, decreases deformation, does not cause buckling and improves the flexural capacity of compressive member .In this study the mechanical behavior and durability of concrete filled PVC wrapped with FRP, with PVC thickness 3mm, 5mm and 7mm are focused .The result presented will contain the following: Confinement effect, Maximum loading capacity and mode of failure. The result of the axial compressive test indicate that the CFRP wrapped PVC reaches its maximum load of 650 KN in full wrapped specimen in which the thickness of PVC is 7mm .Brittle failure will be delayed and capacity of crack arresting and deformation is improved as a result of confinement achieved by PVC and FRP wrapping .The PVC-FRP system combines the good features of both PVC and FRP namely the long term performance and chemical resistance with high strength and stiffness of FRP.

79 .DESIGN REVIEW AND FRAMEWORK FOR SIGN LANGUAGE TO TEXT AND SPEECH TRANSLATION VIA SMART GLOVES

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Sign language is a way that many Deaf and Hard-of-Hearing people talk to each other; however most people do not know sign language which makes it hard to communicate. This paper shows a smart-glove system that turns hand movements into words and sounds using sensors that can tell what the hands are doing. The system uses sensors that measure how much each finger bends and a device that checks the way the hand is moving and where it is. A small computer takes the information from the sensors. Figures out what the hand is doing using simple rules or by learning. The hand movements are then turned into words. Sounds through a device that shows or says them. Compared to using cameras this way of sensing is not affected much by light background noise, where the camera is or if part of the hand is hidden. Some smart-glove systems already work fast and can be right between 85% to more than 94% depending on the sensors the kinds of signs and how they are recognized. This system offers an cheap way to help people talk and future plans include recognizing more signs making it better, for different people and letting people talk back and forth.

80. A COMPREHENSIVE REVIEW OF DEMAND FORECASTING TECHNIQUES FOR URBAN FOOD SUPPLY CHAIN

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For urban food supply chains to be resilient and sustainable, accurate demand forecasting is essential, particularly for perishable foods and dairy products that are impacted by seasonality, short shelf life, and externally driven demand variability. This review paper provides a analysis of demand forecasting methods, including statistical time-series models, machine learning techniques, deep learning architectures, and hybrid approaches, that have been published in IEEE and other reputable journals. We compare the reviewed techniques based on data characteristics, evaluation metrics, and application domains such as retail supply chains, urban food, and dairy. The review identifies significant research gaps in model explainability and real-time adaptability due to culturally driven and festival-induced demand variations. The study also propose a framework aims to bridge these gaps by integrating cultural calendars with hybrid forecasting and decision-support mechanisms.

81.SHOP DROP: AN AI-POWERED SMART MULTIVENDOR CAMPUS FOOD ORDERING AND LOAD-BALANCING SYSTEM

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Campus food courts routinely face hundreds of students placing orders within a 10–15 minute break window, producing crowded stalls, disorganized pickups, and vague preparation-time estimates. Existing solutions — canteen ordering apps, single-restaurant delivery platforms, and token-based queue systems — digitize payment and menu browsing but leave the underlying kitchen-capacity problem untouched. This paper presents Shop Drop, a real-time scheduling layer that treats each vendor's kitchen as a live capacity timeline rather than a first-in-first-out queue. The system integrates four cooperating mechanisms: demand forecasting from historical order data, dynamic slot allocation that assigns a calculated pickup time to every order, adaptive prep-time learning through an exponential moving average, and cross-vendor pickup synchronization combined with walk-in fairness interleaving. An optional natural-language order parser and a vendor-facing analytics dashboard extend the core engine. We describe the system architecture, the scheduling and learning methodology, and a comparative evaluation against six categories of existing systems, showing that no current solution combines these mechanisms into a single real-time engine for multivendor, mixed app-and-walk- in campus environments.

82.SMART RURAL HEALTHCARE, APPOINTMENT SCHEDULING AND PHARMACY MANAGEMENT SYSTEM

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Rural communities rely on health centres and sub- centres that serve many people with very few doctors, inconsistent network connections and long distances to travel. Appointments are managed by people coming in person and using paper registers. Medicine stock is tracked by hand, which causes waits, unnecessary trips, medicines running out and medicines that have expired. This paper talks about a rural healthcare system that brings together appointment scheduling with pharmacy management at village, primary health centre and district levels. The system works without needing a network. Facilities can still book appointments give prescriptions and give medicines even when there is no internet. Their records update when the network comes back. Patients and community health workers can book appointments using an app, SMS or a phone call. A rule makes sure that there are slots for booking at the facility level so that offline and remote bookings do not cause problems. Reminders help patients plan for the time they need to travel. The pharmacy part of the system keeps track of stock by batch. It gives the batch that will expire first. It calculates when to order medicines taking into account different times it takes to get supplies. It also suggests moving stock between nearby facilities. A dashboard, for the district shows how many appointments are being used, how much stock is available and the risk of medicines expiring. The paper explains the structure, data setup, methods and how to check how well the system works.

83.UNIVERSITY BUDDY (UB): AN INTEGRATED ACADEMICSUPPORT AND ADAPTIVE STUDENT MENTORING PLATFORMUSING MACHINE LEARNING

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Most universities today run on a patchwork of disconnected tools, and students are left managing the chaos. University Buddy (UB) is a platform we designed to replace that patchwork with a single system that handles course materials, grades, attendance, deadlines, and faculty communications in one place. It also runs a machine learning layer in the background that flags students who are falling behind, ideally before they know it themselves. The architecture has five tiers: User Inter- face, Business Logic, Data Persistence, Machine Learning, and Communications. Unlike standard learning management systems that merely store information, UB acts on data by generating personalized recommendations, early warnings, and facilitating a peer-to-peer knowledge-sharing layer. Index Terms—learning system

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optimization, machine learning educational outcomes, unified academic platforms, intelligent tutoring systems, student risk prediction, adaptive learning per- sonalization, educational technology infrastructure

84. DIGITAL CONSENT AND PATIENT DOCUMENT MANAGEMENT PORTALWITH A RETRIEVAL-AUGMENTED ASSISTANT FOR MULTI-SPECIALTYHOSPITALS

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Hospitals continue to run a large share of their consent and record-keeping on paper. Forms are misplaced, approvals wait on signatures collected in person, and patients are asked for the same identity and insurance documents at every visit. This paper describes a Digital Consent and Patient Document Management Portal built for multi-specialty hospitals on the MERN stack, together with a Retrieval-Augmented Generation (RAG) assistant assembled entirely from openly licensed components. The portal issues role-scoped dashboards to patients, doctors, receptionists and administrators behind JSON Web Token authentication, keeps structured records in MongoDB and uploaded files in managed object storage, and writes an audit entry for every action that touches patient data. Consent is captured electronically once the patient clears a one-time- password check and supplies a drawn signature. The assistant addresses a problem that digitization by itself does not solve: consent forms and laboratory reports are written for clinicians, and a patient who signs a document they cannot read has not given informed consent in any useful sense. Uploaded files are segmented, embedded with a locally hosted sentence encoder and indexed in an open-source vector store; questions are answered only from the passages retrieved for that patient, and the model is instructed to state plainly when the document does not contain an answer. Because the encoder and the language model both execute on hospital-controlled hardware, document content never leaves the institution.

85.EARLY DETECTION AND RAG-BASED RECOMMENDATIONSYSTEM FOR INTELLIGENT MANAGEMENT OF DIABETICMELLITUS: A SYSTEMATIC REVIEW

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One of the most prevalent metabolic disorders that poses challenge to healthcare systems due to its rising frequency and accelerating prevalence is Type2 Diabetes Mellitus (T2DM) and consequently prediction and intelligent management is necessary to forbid the emergence and progression of disease at an early stage. The disordered physiology of T2DM is characterized by the interrelated coupling of genetic predisposition, clinical characteristics, behavioral patterns, and environment exposures and thus requires the implementation of advanced predictive modeling frameworks that support early disease identification, patient risk assessment, and individualized therapeutic interventions. Although accuracy

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and reliability in diabetic prediction and management have been enhanced through legacy technologies such as Artificial Intelligence (AI), exceptionally Machine Learning (ML) and Deep Learning (DL), they are still prone to limitations that compromise intelligent clinical decision support framework. For early prediction and personalized management of T2DM, the proposed conceptual framework includes AI-driven architecture that synergizes Generative AI, Explainable AI, a personalized management of T2DM and Retrieval- Augmented Generation (RAG)-enabled Large Language Models (LLMs) into a unified decision support system.

86. A SMART AGRICULTURAL EQUIPMENT RENTAL PLATFORM WITH INTELLIGENT SCHEDULING AND PREDICTIVE MAINTENANCE

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Agricultural equipment rental operations in small- scale and local farming environments are often managed through manual communication and recordkeeping, making it difficult to coordinate equipment availability, rental schedules, payments, and maintenance activities. Conventional digital rental systems primarily provide equipment listing and booking functionalities, with limited support for weather-dependent scheduling and equipment maintenance decisions. This paper presents a Smart Agricultural Rental Platform that integrates equipment rental management with intelligent decision-support features in a unified web-based system. The proposed platform enables equipment owners to manage equipment, customers, bookings, rentals, billing, payments, and maintenance, while allowing farmers to search for equipment, request rentals, track bookings, and manage their rental information. A constraint-based smart scheduling mechanism analyzes booking conflicts, equipment availability, maintenance restrictions, rental duration, and weather conditions to identify suitable rental slots and recommend alternatives when the requested slot is unsuitable. Weather-aware booking uses forecast information to classify rental conditions as best, usable, or not recommended, providing users with actionable scheduling information. A predictive maintenance module evaluates equipment usage, maintenance intervals, service history, and overdue conditions to generate equipment risk levels and maintenance recommendations. In addition, an AI-assisted equipment recommendation module

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suggests suitable equipment based on user requirements and current inventory, while a multilingual farmer assistant provides contextual support for equipment, booking, weather, and maintenance-related queries. The platform is implemented using the MERN stack with external services for weather information, artificial intelligence, online payment processing, and notifications. Functional testing demonstrates the integration of these components into a unified rental workflow, providing a practical approach toward more efficient equipment utilization, informed scheduling, and digitalized agricultural rental operations.

87 .DESIGN AND DEVELOPMENT OF A LOW-COST EMG-CONTROLLED PROSTHETIC HAND USING ESP32-BASED SIGNAL PROCESSING

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The loss of an upper limb can greatly affect a person's independence and make everyday tasks more difficult. Myoelectric prostheses offer a more intuitive way to control an artificial limb, but their cost and technical complexity can make them difficult to access. This work presents the design and development of a low-cost prosthetic hand controlled through surface electromyography (EMG). The system combines EMG signal acquisition, signal conditioning, embedded processing, and servo-based actuation. Surface EMG electrodes capture muscle activity, a Muscle Sensor v3 conditions the signal, and an ESP32 performs threshold-based processing. A PCA9685 PWM driver controls multiple MG996R servo motors used for finger and gripping movements. When the user contracts the selected forearm muscles, the resulting EMG activity is detected and translated into the corresponding prosthetic movement. Prototype testing demonstrated EMG-triggered gripping and releasing, coordinated servo operation, and real-time response during the tested movements. The 3D-printed mechanical structure also provides a modular platform that can be modified and improved in future iterations. At this stage, the system remains a proof-of- concept based on basic threshold control; quantitative performance evaluation, multi-channel control, adaptive grip control, and more advanced gesture recognition are left for future work. Keywords—Electromyography (EMG), Prosthetic Hand, ESP32, Rehabilitation Engineering, Biomedical Signal Processing, Embedded Systems.

88.PROMPT INJECTION ATTACK DETECTION AND DEFENSE FRAMEWORK FOR PRODUCTION LLM SYSTEMS

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Large Language Model (LLM) applications increasingly integrate retrieval systems, external documents, web content, and software tools. This integration exposes them to prompt injection attacks, in which malicious instructions manipulate model behavior, either directly through user prompts or indirectly through retrieved or uploaded content. Existing detection methods often depend on static rules or isolated prompt analysis and may fail against paraphrased, obfuscated, indirect, or multi-stage attacks. This paper presents a context-aware prompt injection detection and defense framework for production-oriented LLM applications. The framework combines input preprocessing, rule-based detection, lexical similarity analysis, hybrid risk assessment, attack classification, provenance analysis, policy enforcement, and controlled LLM and tool execution. It also proposes a cumulative security state that tracks repeated suspicious behavior across interactions to help expose multi-stage attacks. A preliminary proof of concept was implemented in Python and FastAPI using open- source components. In an initial evaluation of 14 manually prepared cases, the hybrid detector achieved an F1-score of 0.933 and a recall of 0.875 with zero false positives. These results are preliminary, rest on a small dataset, and are not intended to represent production- level performance. The framework aims to provide a reusable security layer for LLM applications and agent-based systems.

89.FINANCIAL INSTRUMENTS FOR SUSTAINABILITY, IMPACT INVESTING, GREEN BONDS AN CLIMATE FINANCE, SOCIAL RESPONSIBLE INVESTMENT(SRI) STRATEGIES

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The success of an enterprise depends upon its sustainability in the market. Businesses constantly seek out potential investors who may provide funds to ensure the long-term viability of the enterprise. This chapter demonstrates how the growth of sustainable financial literacy has altered investors' attitude. Specifically, those businesses that promote global sustainability goals are more likely to attract investors. Hence, adopting sustainable financing practices is crucial for businesses to secure funding. The ability to diversify the possible risks involved with investments is the driving force behind investing in green bonds and other securities. It is recommended that businesses should utilize credible indicators such as GRI, IIRC and CDP to influence investors who prioritize Environmental, Social, and Governance (ESG) criteria while investing. Global warming and carbon dioxide emissions have a profound impact on the environment. Investing in renewable energy sources can make a significant positive difference to the environment and enhance the quality of life for people. Public-private partnerships (PPPs) can be used to secure funding for infrastructure projects like solar power plants, waste management, water supply and sanitation. These collaborations have the potential to help developing countries overcome financial challenges to sustainable development. In addition, the government can encourage these projects by offering lower corporate income tax rates, various subsidies, and support for research and development in this field. Banks and other financial institutions can play a significant role in supporting Socially Responsible Investment (SRI). They can promote environmentally conscious business practices by using tools like impact investing and microfinance. Impact investing is a strategic method that supports the goals of a circular economy by balancing financial returns with environmental objectives, whereas Microfinance can assist Small and Medium-sized enterprises (SMEs) by providing funding for their eco-friendly initiatives. This chapter also explores a transformative approach to climate financing, focusing not just on generating new funds for environmental protection, but on redirecting existing resources. It suggests disinvesting in harmful projects and introducing taxes to gain better control over environmental outcomes. Additionally, it emphasizes the need for national and international regulations to reduce funding for fossil fuel production, which contributes significantly to harmful emissions. In this book chapter we will also examine the various factors driving Socially Responsible Investments (SRIs). While profit is an

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important consideration, it is not the only factor influencing investors, especially when profits may be steady but not immediate. It explores the elements that can influence individuals to invest in such entities, showing how emotional, psychological, past experiences and knowledge can affect investors' decisions in these projects. It also mentions that behavioral biases and demographic characteristics influence investors' decision-making. Those with a higher risk tolerance may view these investments as opportunities, while those with lower risk tolerance may be harder to persuade. Based on these individual differences, investors shape their portfolios accordingly. This chapter concludes by outlining the benefits of green financing as well as the difficulties faced by investors in this field. It recognizes the sector's growth potential, particularly with the support of several stakeholders. Key support pillars for promoting green entrepreneurship include financial literacy, regulatory authorities, government assistance, and strategic collaborations. It is an attempt to analyze the existing and prospective resources for the development of green projects, with the objective of establishing a sustainable environment for all species on the planet.

90 .A CLOUD-NATIVE SCALABLE TEXT DETECTION AND OCR EXTRACTION SERVICE USING CRAFT AND DEEP LEARNING

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Text detection is a fundamental computer vision task that underlies document digitization, automated data extraction, and archival systems. While numerous deep learning models achieve high detection accuracy, most academic implementations remain confined to local, single-user environments and are never engineered into deployable, multi-tenant services. This paper presents the design, implementation, and evaluation of a Cloud-Based Scalable Text Detection Service that converts a text-detection model into a production-oriented web API. The system is developed in two stages: a baseline Flask-based detection service that accepts an uploaded image and returns a structured JSON response indicating whether text was detected, and an enhanced CRAFT (Character Region Awareness for Text Detection) and OCR extraction pipeline augmented with OpenCV preprocessing (grayscale conversion, Gaussian denoising, Contrast-Limited Adaptive Histogram Equalization, and Laplacian sharpening) to improve character-region localization and layout-preserving text extraction. Experimental evaluation on representative inputs a plain-text image, a structured retail receipt, and a text-free photograph — shows that the baseline service correctly distinguishes text-bearing from text- free images with sub-second processing times (1.50 s-2.56 s), while the enhanced CRAFT + OCR pipeline localizes 24 discrete text regions on a structured receipt image and reconstructs the original line-by-line layout with high fidelity in 20.93 s on CPU-only hardware. The full architecture is builtfor being deployed as containers on Docker, and orchestrated on Kubernetes on a cloud platform, allowing for easy horizontal scaling to handle any number of concurrent users. The results illustrate a viable approach to making research-level text-detection models usable in a cloud-based service that is reliable, scalable, and functional for document processing in the real world.

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91. MATLAB-BASED COMPARATIVE PERFORMANCE AND COST ANALYSIS OF SOLID-STATE AND LITHIUM IRON PHOSPHATE BATTERY PACKS FOR ELECTRIC VEHICLE APPLICATIONS

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Electric vehicles (EVs) are becoming central to sustainable transportation, and their performance depends critically on the underlying battery chemistry. This paper presents a MATLAB-based comparative simulation of a 100-cell (10S×10P) Lithium Iron Phosphate (LFP) battery pack against a Solid-State Battery (SSB) pack under a WLTP-style drive cycle spanning urban, suburban, rural, and highway phases. State of Charge (SoC), cumulative energy consumption, power demand, driving range, and 15-year total cost of ownership were evaluated for both chemistries using an energy-balance vehicle model. Results indicate that the SSB configuration achieves approximately 15.9% higher driving range than the LFP configuration under the modeled conditions, along with lower pack mass and higher nominal pack energy, while the LFP pack maintains a substantially lower upfront cost, with cost break-even reached in Year 4. The comparison indicates SSB offers superior technical performance for range, safety, and cycle life, while LFP remains the pragmatic near-term choice on cost.

92.HUMAN-AI-MACHINE CO-INTELLIGENCE FOR PHYSICAL DECISION-MAKING IN INDUSTRY 5.0 MANUFACTURING: A CONSTRAINT-AWARE FRAMEWORK

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Industry 5.0 is commonly described as a human-centric development of digital manufacturing, but the engineering question of how human expertise, artificial intelligence (AI), and physical machines should jointly contribute to industrial decisions remains insufficiently specified. Existing research already provides substantial work on human-AI collaboration, human- cyber-physical systems, hybrid intelligence, and human-centric manufacturing. The narrower problem addressed here is how physical machine state and constraints can be treated as decision inputs alongside human context and AI inference, while keeping recommendation, feasibility, and authorization distinct. This paper develops a Human-AI-Machine Co- Intelligence (HAMCI) framework as a constraint-aware decision architecture. In HAMCI, human contextual knowledge, AI-based computational reasoning, and machine-based physical sensing and constraints enter a shared decision space, followed by an explicit decision-authority gate before execution. The framework also introduces a six-level co-intelligence maturity model, seven design principles, and a worked automotive manufacturing scenario. The contribution is conceptual: it does not claim experimental performance improvements. Instead, it specifies a bounded systems-engineering mechanism and propositions that can be empirically tested against adjacent human-AI and human-cyber-physical approaches. The framework is intended to support future research on decision quality, physical feasibility, role allocation, uncertainty communication, safety, and human agency in Industry 5.0 manufacturing.

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93.ADAPTIVE MACHINE LEARNING-BASED INTRUSION DETECTION IN SMART IOT NETWORKS

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The rapid growth of smart Internet of Things (IoT) networks has intensified security challenges due to heterogeneous devices, dynamic traffic patterns, and limited computational resources. Conventional intrusion detection techniques are often ineffective against evolving and sophisticated cyberattacks in such environments. Adaptive machine learning-based intrusion detection has therefore gained significant attention as an effective solution for enhancing network security in smart IoT ecosystems. This comprehensive review examines recent advancements in machine learning-driven intrusion detection approaches, including supervised, unsupervised, semi-supervised, and deep learning models. The review highlights how adaptive learning mechanisms improve detection accuracy and resilience under changing network conditions and concept drift. Key challenges such as data imbalance, scalability, real-time detection, and deployment constraints in edge- and cloud-assisted IoT architectures are critically analyzed. Furthermore, commonly used datasets, evaluation metrics, and comparative performance trends are discussed. Finally, open research issues and future directions are outlined to guide the development of robust, scalable, and intelligent intrusion detection frameworks for next- generation smart IoT networks.

94. VOLTAMMETRIC APPROACH FOR CAFFEIC ACID DETECTION IN OREGANO USING CARBON PASTE ELECTRODE

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Caffeic acid (CA) is an important phenolic constituent of many plant-derived products and is frequently investigated because of its chemical and biological properties. In this study, a bare carbon paste electrode (CPE) was developed as a simple and economical platform for voltammetric determination of CA. Cyclic voltammetry (CV) was employed to characterize the electrochemical response and optimize the measurement conditions, whereas differential pulse voltammetry (DPV) was used for quantitative analysis. Among the supporting electrolytes examined, 0.1 M acetate buffer solution (ABS) at pH 7 produced the most favourable response. Under the selected conditions, the oxidation current increased with increasing CA concentration over the range of 10–600 μM, giving a detection limit of 3.72 μM. The analytical performance of the electrode was further examined through repeatability, reproducibility, and stability tests, with RSD values of 1.35%, 1.95%, and 1.77%, respectively. The applicability of the method was demonstrated using an oregano sample, where the standard addition approach was employed to compensate for matrix effects. The results demonstrate that the proposed CPE-based method provides a convenient and low-cost alternative for voltammetric CA determination in plant-derived food and spice samples.

95. VISUAL CULTURE AND BUILT SPACE: ART, ARCHITECTURE, AND SOCIAL IDENTITY IN GEORGE ELIOT'S MIDDLEMARCH AND ELIZABETH GASKELL'S NORTH AND SOUTH

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This article examines how material architecture and visual culture actively shape social identity in George Eliot's Middlemarch and Elizabeth Gaskell's North and South. Going beyond metaphorical understandings of space, the analysis treats architecture as a material and lived space consisting of houses, streets, factories and interiors that control movement and visibility and bodies. Placing both of the novels in the context of the mid-Victorian discussions about the reform of architecture, the industrial design, and the visual order, the paper integrates the close textual analysis with the studies of the spatial theory and material culture. In Middlemarch, the architecture of the provinces creates modes of enclosure and moral panopticism which constitute gendered identity, especially by means of restrictive interior spaces. Conversely, North and South construct industrial architecture as a field of visibility, in which factories and housing of workers influence the relations of classes based on sensory and corporeal experience. Through creating a dialogue between these spatial regimes, the article shows that the Victorian realism is very concerned with the material organization of built space as a composition of social life. The study ultimately argues that Eliot and Gaskell imagine architecture not merely as a symbolic backdrop but as an active social force, capable of sustaining hierarchy while also enabling ethical reform.

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96.PHARMACY PRESCRIPTION INVENTORY MANAGEMENT AND DISPENSING AUDIT SYSTEM

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Pharmacy operations increasingly depend on the correct alignment of three activities that are often engineered as separate systems: reading and validating a prescription, verifying that the prescribed medicine is actually available in stock, and confirming that what is dispensed matches what was prescribed. Weak coupling between these activities leaves pharmacies unable to reconstruct, after the fact, why a particular medicine reached a particular patient, in what quantity, and under whose authorization. This paper proposes an integrated architecture for a Pharmacy Prescription Inventory Management and Dispensing Audit System that links prescription processing, medicine identification and validation, real-time inventory verifi- cation, and dispensing confirmation through a central workflow controller, with a dispensing audit module treated as a first- class component rather than an incidental log. The audit module is designed to record, for every transaction, the prescription and medicine identifiers, the acting user, the quantity dispensed, pre- and post-transaction stock levels, verification and exception status, and any manual override, so that discrepancies can be traced to a specific step and a specific actor. Confidence thresholds and pharmacist review are retained throughout the pipeline for low-confidence recognition, stock mismatches, and flagged exceptions. Drawing on literature in prescription recognition, medicine and pill identification, pharmacy inventory and demand forecasting, automated dispensing safety, and electronic health record audit-log analysis, the paper identifies a research gap in the joint treatment of dispensing audit and inventory verification, and proposes a methodology and an evaluation framework built around recognition, inventory, dispensing, and audit-level metrics. No experimental results are reported; expected outcomes are framed against results already documented in the cited literature, and the framework is presented as a basis for future empirical evaluation.

97.EARTHNSOL: A SUSTAINABLE E-COMMERCE PLATFORM FOR SOIL HEALTH AND DESI COW CONSERVATION

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Earthnsol: A Sustainable E-Commerce Platform for Soil Health and Desi Cow Conservation. India's agricultural sector faces persistent challenges in connecting farmers to fair markets, conserving indigenous cattle (Desi Govansh) breeds, and promoting chemical-free soil restoration. Existing E-commerce platforms are largely profit-centric and provide limited support for sustainability-focused agricultural stakeholders. This paper presents Earthnsol, a full- stack agri-commerce platform that integrates farmers, consumers, home gardeners, nurseries, and Govansh supporters within a unified digital ecosystem. The platform provides product catalogue management, secure authentication, cart and order workflows, and farmer-connect services, implemented using Node.js and Express with Supabase for relational data management. Earthnsol integrates five functional domains: soil conservation, farmer market linkage, Govansh support, natural Govansh by-products, and home gardening. An AI-enabled advisory chatbot, Mitti, further supports real-time soil health and farming guidance. The developed system was validated in a staging environment through functional testing of its major modules using API response time, order success rate, product search accuracy, authentication reliability, and cross-device compatibility as evaluation measures. The system achieved a 95% order success rate, 92% product search accuracy, 210 ms average API latency, and 96% mobile responsiveness, demonstrating reliable system performance. The proposed platform differentiates itself from conventional E-commerce systems by combining sustainable agri-commerce, Govansh conservation, soil-health awareness, and AI- assisted advisory support within a single platform. Future enhancements include AI-based crop recommendation, Hindi and Marathi language support, and a Flutter-based mobile application.

98. PWM-BASED FAULT-TOLERANT ELECTRONICDIFFERENTIAL FOR LOW-COST ELECTRIC VEHICLESUSING GYROSCOPE FEEDBACK

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Electric vehicles (EVs) commonly use mechanical differentials to accommodate the difference in wheel speeds during cornering. However, conventional differential systems contain several mechanical components that increase vehicle weight, frictional losses, maintenance requirements, and drivetrain complexity. This work presents an optimized electronic differential system for low-cost electric vehicles that replaces the conventional mechanical differential with independent control of the left and right drive wheels. The proposed system uses a NodeMCU ESP8266 microcontroller, an MPU6050 gyroscope, and a steering potentiometer to determine the steering condition and vehicle motion. Based on these inputs, the controller generates independent Pulse Width Modulation (PWM) signals to regulate the speeds of two DC geared motors. During straight-line motion, equal PWM signals are applied to both motors, while during cornering, the motor speeds are varied according to the steering direction to provide appropriate differential action. The control algorithm is developed and evaluated using MATLAB/Simulink under forward, left-turn, and right-turn conditions. To evaluate fault tolerance, gyroscope freeze, bias, and noise conditions are also introduced into the simulation. The controller detects discrepancies between the steering command and gyroscope feedback and adjusts the motor PWM signals to maintain stable vehicle motion. A hardware prototype incorporating the NodeMCU ESP8266, MPU6050 sensor, motor driver, potentiometer, DC geared motors, and display unit is developed to validate the simulation results. The experimental results demonstrate that the proposed system can independently control wheel speeds and reproduce the required differential

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behaviour. The comparison between simulation and hardware results shows a maximum deviation of less than 8% with an RMSE of approximately 1.08 rad/s. The proposed electronic differential therefore provides a compact, low-cost, and fault- tolerant approach for independent wheel control in small electric vehicles, while reducing mechanical complexity and improving drivetrain efficiency.

99.SMART WEARABLE MOBILITY AND HEALTH MONITORING ASSISTANT FOR THE ELDERLY

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Older adults are more vulnerable to falls and sudden health issues, especially when living alone, making continuous monitoring essential. This work presents a smart wearable system using an ESP32 with motion and biomedical sensors to track movement, heart rate, and oxygen levels in real time.To improve detection accuracy, a lightweight machine learning model is integrated to identify fall events and abnormal health conditions, reducing false alerts compared to traditional methods. When a critical event is detected, the system sends real-time alerts to caregivers through a backend server.Experimental results show reliable performance with minimal delay. The system's low cost, compact design, and ease of use make it a practical solution for enhancing elderly safety and supporting independent living. Index Terms—Elderly health monitoring, wearable IoT, fall detection, ESP32, vital signs, machine learning, heart rate, SpO2, remote monitoring

100 .QUEST-AI: A SECURE RETRIEVAL-AUGMENTED FRAMEWORK FOR INTELLIGENT QUESTION PAPER GENERATION WITH SEMANTIC ANALYSIS AND BLOOM'S TAXONOMY

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Generation of automated question papers faces persistent challenges in educational technology, including question repetition, non-uniform syllabus coverage, limited cognitive diversity, and weak control over paper composition. This paper proposes Quest-AI, a framework that combines Optical Character Recognition (OCR), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), semantic analysis, and Bloom's Taxonomy to automate

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question paper generation. Quest-AI builds a curriculum-rooted knowledge base from syllabi, lecture notes, and reference materials, using OCR for scanned content. Sentence Transformer embeddings support similarity-based redundancy elimination, while a Multi-Constraint Intelligent Question Selection Engine selects questions based on syllabus coverage, Bloom's cognitive levels, difficulty, marks, and semantic uniqueness. Gemini Flash Lite generates context-aware questions, while embedding-based modules perform syllabus and semantic auditing. The framework further supports template- based paper generation, three equivalent question sets, and role- based access secured through JSON Web Token (JWT) authentication with hashed passwords. Evaluation on 8 subjects and 12 generated papers shows 100% syllabus coverage, 100% no-repetition rate, 79% semantic matching and a 93% overall quality score, demonstrating the framework's effectiveness in generating diverse, curriculum-aligned, and institution-ready examination papers. Quest-AI offers an integrated, secure solution that reduces manual effort while improving the consistency and reliability of academic assessment generation.

101 .A MACHINE LEARNING BASED APPROACH TO PROFIT- OPTIMIZED COMPANION CROP RECOMMENDATION IN TAMIL NADU AGRICULTURE

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Intercropping and companion planting are well-established practices that improve soil health, reduce pest pressure and diversify farm income, yet smallholder farmers in Tamil Nadu largely rely on tradition and word- of-mouth rather than data when choosing which crops to grow together. Generic, pan-India crop advisories rarely account for the specific soil types, seasons and market conditions of a given Tamil Nadu district, and none translate companion planting choices into an expected monetary return. AgriPair addresses this gap with a machine-learning based decision-support system that predicts the profitability of companion crop pairs for Tamil Nadu farmland. A synthetic dataset of 750 records was constructed from agronomic and market references, capturing soil type, season, district, crop pair and cost/yield information. Random Forest, XGBoost and LightGBM regressors were trained and compared, achieving R2 scores of 0.97, 0.99 and 0.83 respectively; XGBoost also gave the lowest profit prediction error (MAE of 16,392) and is automatically selected by the pipeline for deployment. The trained model is served through a Streamlit application with soil-aware dropdown inputs that returns a season-by-season profit breakdown for candidate companion crop pairs, giving farmers an accessible, quantitative basis for intercropping decisions.

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102. SCAMSHIELD: A WEIGHTED MULTI-SIGNAL FUSION ARCHITECTURE FOR EXPLAINABLE SCAM, PHISHING, AND MALWARE DETECTION ON MOBILE DEVICES

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Modern mobile devices are exposed to threats via scam messages, social engineering, malicious APKs, and breaches, while current solutions address just one threat. Here, we propose ScamShield – a mobile security application that offers a comprehensible assessment of various risks based on text messages, phone calls, screenshots, APKs, and potential breach risk. Texts are analyzed through a mixture of large language model analysis, deterministic rule engine, and OSINT search where the last mentioned one serves as a backup in case external services cannot be applied. APks are evaluated using static analysis and signature matching with a limited on-device backup procedure. Our offline rule engine had 89.5% accuracy, 89.3% F1 score, and processed 200 messages within 18 milliseconds on average, although the evaluation of our entire pipeline will take place in the future. In general, ScamShield provides a practical approach to achieving comprehensive mobile security in a single consumer app.

103 .DESIGN AND ANALYSIS OF A PROSTHETIC LIMB WITH SHOCK ABSORPTION AND LIGHT WEIGHT COMPOSITE MATERIALS

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Transtibial prosthesis comfort and functional adaptation are strongly influenced by load transfer at the socket–residual limb interface, while prosthesis mechanical properties also affect gait mechanics, energy expenditure, and contralateral-limb loading. This study designs, analytically models, and computationally validates a transtibial prosthetic limb that integrates a helical compression spring within a lightweight carbon-Fiber-reinforced polymer (CFRP) structural tube. A Chrome Vanadium helical spring was sized analytically using the Wahl-corrected shear-stress equation and a Goodman fatigue criterion, then housed axially within a CFRP T700 pylon tube. Finite element analysis (FEA) in ANSYS Mechanical was used for quasi-static structural assessment under three representative stance-phase loading cases informed by ISO 10328 structural-testing principles. Analytical and FEA predictions were cross-validated for maximum spring shear stress. The spring stores 1.06–1.53 J of elastic strain energy per simulated gait phase. Static safety factors of 1.89–2.46 (FEA) and 2.05–2.46 (analytical) were obtained, with a Goodman fatigue safety factor of 1.79. FEA-predicted maximum shear stresses were 8.8–8.9% higher than the corresponding analytical predictions at all three phases. Peak stress and deformation occurred in the spring coil rather than at the socket-adapter region. The proposed Prosthetic Leg spring-integrated CFRP pylon demonstrates computational structural feasibility and close agreement between analytical and FEA predictions. The calculated elastic-energy storage supports further experimental investigation of the concept as a potential load-attenuating pylon architecture.

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104. DERMOSCOPIC IMAGE ANALYSIS FOR SKIN CANCER ASSESSMENT: VISUAL FEATURE INTERPRETATION AND AN ENHANCED PREPROCESSING FRAMEWORK

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Skin cancer assessment from dermoscopic images is strongly influenced by the quality of the image presented for analysis. Dermoscopic acquisition can introduce hair, shadows, uneven illumination, reflections, blur, compression effects and other disturbances that may overlap with clinically meaningful visual structures. This paper addresses the first two objectives of the proposed doctoral research: examination and interpretation of skin-cancer forms and dermoscopic visual features, and development of an effective preprocessing pipeline involving noise detection, image normalization and image enhancement. A structured review of the supplied literature is combined with a secondary-data-oriented preprocessing framework suitable for public dermoscopic image collections. The study organizes relevant image information into colour, lesion geometry, border characteristics, internal pigment structures and acquisition artifacts. Based on these observations, a sequential pipeline is proposed in which images are screened, artifacts are identified, spatial and intensity characteristics are normalized, enhancement is applied conservatively, and the processed output is checked for structural preservation and traceability. The work intentionally stops before feature extraction and diagnostic model evaluation. Accordingly, no unsupported accuracy, sensitivity or classification values are introduced. The contribution is a clear and reproducible foundation for subsequent computational analysis of dermoscopic images. Index Terms—dermoscopy, skin cancer, dermoscopic image analysis, image preprocessing, noise detection, image normalization, image enhancement, computer-aided diagnosis.

105. ANALYSIS OF DIABETIC RETINOPATHY DETECTION USING DEEP LEARNING MODELS

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Diabetic retinopathy (DR) is a microvascular diabetic complication and one of preventable vision blindness in the world. When retinal abnormalities are picked up early, screening and treatment can significantly minimise the risk of significant loss of visual function. Classification of Diabetic

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retinopathy are comprehensively reviewed in this paper based on evidence. CNNs architecture, transfer learning methods, image processing and augmentation techniques, segmentation of lesions, and severity grading of a lesion into various classes are discussed in this review, while microaneurysms, hemorrhages, hard exudates, soft exudates etc. are regarded as the main visual markers of disease progression in the retina. The architectures analyzed are from existing literature and report successful results in automatic retinal image analysis, without the need to conduct fresh experiments. Some current and possible future lines of research are summarized, including explainable artificial intelligence (XAI) and Bayesian/uncertainty-aware deep learning, which offer visualizations and confidence levels for model predictions. From the literature studies have been reviewed that deep learning shows high performance at analyzing DR, yet the problem of class imbalance, image quality, dataset variations, interpretability, computational demand and applicability to different clinical populations are major open issues. The review ends with an envisioned evaluation framework and future research directions in the direction of dependable, understandable, and clinically implementable automated DR screeners.

106. AI-BASED SMART LAPTOP STUDY MONITORING SYSTEM USING WEBCAM FACE-DISTANCE ESTIMATION, AUTOMATIC BRIGHTNESS CONTROL AND BREAK ALERTS

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Students now spend many hours each day studying on laptops. Sitting too close to the screen and studying for long periods without rest are linked to eye strain, headache, neck pain and tiredness, yet students seldom notice these habits, and adjusting screen brightness by hand interrupts their focus. This paper presents an AI-based smart laptop study monitoring system that uses the built-in webcam to look after the student automatically. The system captures live video, detects the student's face with OpenCV and MediaPipe, and estimates the viewing distance from the size of the face. The student is classified as Too Close (below 40 cm), Comfortable (40–70 cm) or Far (above 70 cm). Screen brightness is lowered and a warning is shown when the student is too close, kept normal at a comfortable distance and raised when the student moves away. A study- time tracker counts continuous study time and shows a "Take a Break" alert after a set period, for example 45 minutes. Video is processed on the laptop only and is not saved. Testing of face detection, the too-close alert, brightness adjustment and the break alert gave the expected output in all four test cases. Unlike earlier tools that only warn about distance or remind about breaks, the proposed system combines distance monitoring, automatic brightness control and break reminders in one lightweight application.

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107. DESIGN AND SIMULATION-BASED EVALUATION OF A SMART, IOT-ENABLED AUTOMATION SYSTEM FOR SMALL-SCALE INDUSTRIAL OVENS

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Small-scale industrial ovens used in bakeries, food processing, drying and curing units are mostly regulated by mechanical thermostats or basic ON/OFF controllers. These give large temperature swings, no process records and limited safety. This paper presents a low-cost smart automation system for a small-scale industrial oven, built around an ESP32 microcontroller, a K-type thermocouple with a MAX31855 converter and a zero-cross solid-state relay (SSR). The controller combines a PID algorithm with anti-windup and integral separation, model-based feed-forward for multi-stageramp–soak recipes, a supervisory safety state machine backed by a hardware thermal cut-off, and MQTT-based remote monitoring with energy metering. A lumped parameterthermal model of a 2 kW oven, including sensor noise, quantisation, transport delay and a door-opening disturbance, is used to compare the design with a conventional ON/OFF thermostat and a naive PID. For a 150 °C set-point step the proposed controller gives 0.3 °C overshoot and 0.5 °C peak-to- peak steady-state ripple against 11.5 °C for ON/OFF, recovers from a 30 s door opening in 87 s, and tracks a two-stage ramp–soak profile with 0.37 °C RMS error. A 100-run Monte- Carlo test with ±20 % plant variation confirms the robustness of the tuning. The results indicate that the proposed control strategy is a viable, low-cost alternative to ON/OFF regulation for small-scale industrial ovens; experimental validation on a physical prototype is identified as future work.

108 . DEVELOPMENT OF A PLC-BASED INTELLIGENT INTERLOCKING AND PROCESS CONTROL SYSTEM FOR INDUSTRIAL HOIST CRANE AUTOMATION

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Surface treatment of metal components—degreasing, phosphating, corrosion protection and similar chemical processes—requires each part to be immersed in a succession of chemical baths for a controlled length of time, a task that is conventionally handled by a central overhead hoist crane operated by a human. Relying on the operator's judgement introduces three recurring problems: the crane can be moved to the next station before the required dipping time has elapsed, a basket can be lowered into a heated bath before that bath has reached its process temperature, and a tank in the sequence can be skipped. Any of these mistakes reduces coating quality, raises the rejection rate and can create an unsafe crane movement. This paper presents the design, implementation and testing of a programmable logic controller (PLC) based interlocking system that eliminates all three problems for a seven-tank chemical dipping line used in automobile-component manufacture. The controller—a Delta DVP40ES2 with 24 digital inputs and 16 relay outputs—receives crane-position feedback from seven inductive proximity sensors (PX0–PX6) and seven lever-type limit switches (LS1–LS7), a key-

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operated manual bypass switch, and temperature-status contacts from three THEM AI-7982 digital controllers, each reading a PT100 resistance temperature probe mounted in one of the three heated tanks (Tank 1, Tank 5 and Tank 7). Through two eight-channel Omron relay modules the PLC drives the hoist motor contactors, one 24 V indicator lamp placed above each of the seven tanks, and an industrial hooter. The ladder logic enforces three conditions on every basket. First, a basket may be lowered into a temperature-controlled tank only after the corresponding THEM AI-7982 controller reports that the bath has reached its set-point, so the crane is held until the chemistry is ready. Second, once the basket is at the dipping position, a per-tank timer starts and the lift is locked until that timer expires, giving every basket exactly the same dip time regardless of how busy the operator is. Third, a basket can be placed only in the next tank of the predefined seven-tank order, which prevents tanks from being skipped and still permits several baskets to progress through the line simultaneously under operator direction. While a tank timer is running, the indicator lamp above that tank blinks; when the timer ends the lamp stays on continuously and the hooter sounds for ten seconds, alerting the operator that the basket is ready to move on. The system was assembled in a steel control panel, wired with shielded multicore cable and commissioned on a working prototype. Each interlock condition, the parallel handling of multiple baskets, the LED indication and the hooter were exercised and confirmed. Eleven practical faults were encountered during commissioning—covering sensor mounting, cable continuity, relay contacts, limit-switch adjustment and ladder-logic corrections—and all were identified and cleared; they are reported here because they show where a PLC interlocking system of this type is most likely to fail in service. The crane movement between tanks remains under operator command, so the system provides interlocked, semi-automatic operation rather than full automation.

109. A HYBRID ANN-LSTM FRAMEWORK FOR ACCURATE CARDIOVASCULAR DISEASE DETECTION

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Cardiovascular diseases (CVDs) are still one of the major causes of death in the world, calling for more accurate means of diagnosis. This paper introduces a Hybrid Artificial Neural Network- Long Short-Term Memory (ANN-LSTM) method that is beneficial in detecting CVD with increased efficiency due to the ability of the hybrid model to recognize temporal patterns and learn nonlinear features. The hybrid model is systematically tested using two datasets: cardio.csv (static clinical measures) and Cardiovascular Disease Dataset.csv (patient record data). In cardio.csv, the hybrid model performs outstandingly well, achieving 99% accuracy, 0.99 recall and 0.99 F1-score. In Cardiovascular Disease Dataset.csv, the hybrid model performs equally well with 97.67% accuracy, 0.98 recall, and 0.98 F1-score. It outperforms ANN alone (94.2% accuracy) and LSTM (95.8% accuracy) models. A comparative analysis with conventional machine learning approaches indicates higher accuracy than Random Forest (92.3%) and SVM (91.5%), especially in minimizing false negatives among high-risk groups. Index Terms—Cardiovascular Disease Detection, Hybrid ANN-LSTM, Medical Diagnostics, Static and Longitudinal Data

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110 .MULTIMODAL ARTIFICIAL INTELLIGENCE FOR PLANT DISEASE DETECTION: INTEGRATING CNNS, VISION TRANSFORMERS, BERT, GENERATIVE MODELS, AND AGENTIC AI

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Accurate identification of plant diseases at an early stage is key to ensuring agricultural production and food security. This work proposes a modular pipeline that leverages CNN models for local texture and lesion features, transformer models as backbones for global context, text- aware transformers (BERT) to fuse metadata and textual symptom descriptions, generative models for realistic data augmentation, and Agentic AI as an orchestration layer for continuous supervision and triage. We evaluate our framework on cash crop datasets such as cotton and soybean leaves and discuss deployment considerations such as latency, explainability, drift, and human-in-the-loop escalation. We include a literature review to place our work in context with CNN-only baselines, ViT backbone-based approaches, GAN-based data augmentation, and recent hybrid models. We share results from one ablation study, showing that ViT + CNN improves model generalisation in the wild, GAN/diffusion augmentation brings significant improvements in imbalanced settings, and a lightweight agent that uses both uncertainty estimation and active learning reduces annotation costs.

111. YOLOV8N-BASED DETECTION OF MARINE DEBRIS AND BIOLOGICAL OBJECTS IN UNDERWATER IMAGES

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Marine debris detection is important for understanding the condition of underwater environments and reducing the impact of waste on marine ecosystems. However, underwater images can be difficult to process due to poor visibility, color distortion, light attenuation and complex backgrounds, which may affect the accuracy of object detection. In this study, a lightweight YOLOv8n-based model is used to detect and classify seven types of underwater object categories: Plastic, Glass, Metal, Net, Rope, Rubber, Marine Life. A combined dataset was prepared using images from the SeaClear dataset along with additional marine debris images for model development. Bounding boxes were used to annotate the images and then converted into the YOLO format. The proposed model was trained for 50 epochs using an input resolution of 416 × 416 pixels. The model achieved 70.2% precision, 60.9% recall, 67.1% mAP@50, and 43.1% mAP@50–95 on the combined test dataset. These results show that the model can effectively detect underwater objects under different environmental conditions.

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112. A LEAKAGE-FREE MATERIALS INFORMATICS FRAMEWORK FOR AUTOMATED ASTM E112 GRAIN SIZE CLASSIFICATION AND HARDNESS PREDICTION IN TI-6AL-4V ALLOYS

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The rapid development of additive manufacturing technologies, most notably selective laser melting, has enabled the fabrication of complex near-net shape titanium alloy components for aerospace and biomedical applications. However, the macroscopic mechanical properties of these dual-phase Ti- 6Al-4V alloys are heavily dictated by their underlying hierarchical microstructures, which exhibit high morphological complexity. Traditional quantitative metallography relies on manual ocular comparison against standard charts or human-guided linear intercept methods, which are labor- intensive, require up to 5 hours of expert annotation per micrograph, and suffer from subjective operator bias. Furthermore, purely data-driven frameworks frequently introduce severe data leakage during random image-slicing partitions, artificially inflating validation accuracy while failing to generalize to unseen specimen pedigrees. To overcome these bottlenecks, this paper proposes an end-to-end, leakage-free materials informatics framework for automated ASTM E112 grain size classification and physical Vickers hardness (HV) prediction of Ti-6Al-4V alloys. Sourced from a public dataset comprising 1,225 high-resolution SEM micrographs, our binarization and direct least-squares ellipse-fitting pipeline determines average grain sizes (d avg) in milliseconds. Hardness targets are generated using the physical Hall-Petch scaling law. To ensure validation integrity, a strict parent-level partition ($D_{train} \cap D_{val} = \emptyset$) is enforced prior to feature extraction. Based on a 5-dimensional statistical and spatial texture descriptor matrix, an ensemble Random Forest regressor and classifier are trained. On a testing fold of 245 completely unseen specimens, our regressor achieves a hardness prediction R^2 score of 0.9102 and a Mean Absolute Error of 7.83 HV, while our classifier achieves a test classification accuracy of 91.43% across Coarse, Medium, and Fine classes. The final models are connected and deployed via an interactive Streamlit web application, establishing an objective, scalable, and high-throughput tool for advanced structural alloy design.

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113. MONITORING SPATIOTEMPORAL DROUGHT DYNAMICS USING AN ENTROPY-BASED SENTINEL-1 SAR INDEX AND GOOGLE EARTH ENGINE (GEE): A CASE STUDY OF PATNA DISTRICT, INDIA

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Drought monitoring in the Indo-Gangetic Plain remains challenging due to strong seasonal variability, persistent cloud cover, and rapid surface moisture dynamics. This study proposes an entropy-based approach for drought assessment over Patna District, India, using multi-temporal Sentinel-1 SAR VV polarization data acquired from 2017 to 2024. Monthly median composites were generated after masking per- manent water bodies, followed by min–max normalization of backscatter values. Shannon entropy was computed to characterize surface heterogeneity, and a Rela- tive Entropy Anomaly Index (REAI) was derived using percentile-based normali- zation to identify drought severity. The REAI time series reveals pronounced seasonal and interannual variability, with recurrent drought signals primarily observed during the post-monsoon period. Se- vere drought conditions were detected in 2018 and 2024, while other years exhibited moderate to low drought intensity with rapid post-event recovery. The results indi- cate that most drought events were short-duration meteorological droughts, rather than prolonged hydrological deficits. The study demonstrates that entropy-based SAR metrics provide a robust, cloud-independent framework for monitoring drought dynamics in data-scarce regions. The proposed methodology is well suited for operational drought assessment and early warning applications when integrated with rainfall and vegetation indicators

114. ROBUST MULTICLASS LEUKEMIA CLASSIFICATION USING HYBRID FEATURE REDUCTION AND ENSEMBLE LEARNING

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Multiclass classification of leukemia based on gene expression microarray analysis is still problematic due to the reasons of high dimensional space, correlation among probe sets, unbalanced class distribution, and complicated subtype structures. This study introduces a leakage-controlled methodology that integrates nonlinear SMOTE, PCA, LDA, and a heterogeneous ensemble. The main

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experiment is performed using GSE13159 MILE Stage-1 dataset and relates diagnoses with AML, ALL, CLL, and CML. Following an untouched stratified 80:20 split, standardization, nonlinear SMOTE, PCA retaining 95% variance, and LDA are fitted only within training folds. Random Forest (RF), XGBoost, and LightGBM are combined by weighted soft voting. The proposed weighted ensemble achieved an accuracy of 97.2%, macro precision of 96.8%, macro recall of 96.5%, macro F1-score of 96.6%, and an MCC of approximately 0.96, outperforming the individual RF, XGBoost, and LightGBM models, which achieved accuracies of 93.4%, 94.8%, and 95.1%, respectively. Independent verification is formulated separately on GSE9476 as AML-versus-normal classification. The design emphasizes imbalance handling, discriminative compression, classifier diversity, systematic ablation, and independent verification while explicitly preventing preprocessing leakage.

115. REAL-TIME SEQUENCE MODELLING VIA LIGHTWEIGHT KEYPOINT NETWORKS FOR ENHANCED SIGN LANGUAGE INTERPRETATION

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The Real-Time Keypoint Sequence Network (RKPS-Net) represents a strategic engineering advancement in the deployment of efficient sign language recognition systems. By pivoting away from the high computational demands of raw video processing toward a lightweight 63- dimensional feature vector derived from the MediaPipe Landmarker, and integrating a robust, multi-layer LSTM core, RKPS-Net achieves a high classification accuracy of 94.1% while fundamentally prioritizing the low latency required for real-world, edge-device deployment. The system's real-time inference pipeline, which uses a 30-frame sliding window, a 0.8 probability threshold, and a 10-frame temporal consistency check, ensures the output is both fast and temporally stable, overcoming typical challenges associated with sequential prediction jitter in live video streams.

116.AN EXPLAINABLE AI FRAMEWORK FOR ADAPTIVE PERSONAL BUDGET OPTIMIZATION USING MULTIMODAL TRANSACTION FUSION

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Most personal budgeting tools ask people to set category limits by hand and then report overspending after it has already happened. The few tools that automate the process usually depend on one data source, typically bank narrations, and say little about why a transaction was placed in a category or why a particular budget figure was suggested. This paper proposes an explainable AI framework for adaptive personal budget optimization that fuses several kinds of transaction evidence: structured payment records, bank and UPI SMS alerts, and scanned or photographed receipts. A confidence-aware gated fusion network assigns each transaction to a spending category even when one of these

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sources is missing or noisy. Category-level spending is then forecast with uncertainty estimates, and a constrained optimization step turns the forecasts, the expected income, and the savings goal into a budget that is re-solved as new transactions arrive. An explanation layer combines SHAP attributions, modality-level Shapley scores, and shadow-price counterfactuals, so that every classification and every budget figure can be traced back to its cause. The framework is designed to operate with user consent and minimal data retention. The paper describes the architecture, the methods, the system workflow, and a proposed evaluation protocol, and it discusses the limitations that need to be addressed before deployment.

117.PREDICTING E-COMMERCE PURCHASE INTENTIONS USING ENSEMBLE LEARNING AND EXPLAINABLE AI

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Online shopping has become increasingly prevalent, yet predicting customer purchase intentions remains challenging due to the complexity of user behavior and class imbalance in conversion data. This study presents a comprehensive approach for predicting purchase intentions using ensemble machine learning methods applied to the Online Shoppers Purchasing Intention dataset. Five ensemble classifiers—Random Forest, Extra Trees, XGBoost, LightGBM, and CatBoost—were evaluated using stratified train-test split and 5-fold cross-validation. Feature standardization and label encoding were performed during preprocessing to ensure model compatibility. Experimental results demonstrate that gradient boosting methods consistently outperform bagging-based approaches, with LightGBM achieving the highest F1- score of 0.644 and XGBoost attaining the best ROC-AUC of 0.929. Feature importance analysis reveals that ExitRates (importance score: 390), ProductRelated (260), and ProductRelated_Duration (240) are the most influential predictors, while SHAP analysis provides interpretable explanations for individual predictions. The confusion matrix analysis shows the model correctly classifies 1,996 non-purchase sessions and 159 purchase sessions, with a false positive rate of 7.4%. Correlation analysis confirms expected behavioral patterns, with ProductRelated_Duration positively correlated with PageValues ($\rho = 0.38$) and negatively correlated with BounceRates ($\rho = -0.39$). The findings demonstrate that ensemble learning, combined with explainable AI techniques, can effectively predict purchase intentions while providing actionable insights for e-commerce optimization strategies, including targeted marketing campaigns and user experience improvements.

118.DESIGN AND DEVELOPMENT OF A SOLAR-POWERED AUTOMATIC IRRIGATION SYSTEM USING SOIL MOISTURE-BASED CONTROL

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This paper One of the main problems in current agricultural practices is water scarcity along with poor irrigation methods. In this paper, a solar powered automatic irrigation system using soil moisture sensors for monitoring the moisture level of the soil in agriculture fields is introduced. The suggested design includes solar energy which is used as the energy source for such systems in remote

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agricultural places where there is no easy way to provide regular electricity. Soil moisture sensors measure the moisture level of the soil and then give feedback to a controller unit. Whenever the level of moisture in the soil becomes less than a predetermined value, the controller turns on the water pump to irrigate the crops and when the necessary amount of moisture level is reached, the pump is turned off by the controller. In this manner, waste of water and unnecessary use of energy are minimized.

119.DEEP LEARNING TECHNIQUES FOR MULTI-CLASS CLASSIFICATIONS OF DIABETIC RETINOPATHY SEVERITY

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Various eye diseases include diabetic retinopathy, diabetic macular edema, early age-related macular degeneration, and late age-related macular degeneration. Among these, diabetic retinopathy is identified as a leading cause of blindness in diabetic individuals if not detected early. This condition arises in patients with long-term diabetes and occurs due to damage to the blood vessels of the retina, the light-sensitive tissue at the back of the eye. The disease progresses through four stages, which are characterized by structural changes in the retinal blood vessels Conventional methods often utilize deep learning with convolutional neural network models for classifying retinopathy images. However, these approaches face challenges such as difficulties in processing large datasets, complex training procedures, and extensive computation time. In the proposed research, a hybrid approach is undertaken, combining multi-class Support Vector Machine learning with lesion and vessel analysis of retinopathy images. Additionally, deep learning techniques are\ employed to detect and classify diabetic retinopathy images, leveraging multi-layer performance to distinguish between normal and abnormal diabetic retinopathy image data. The research aims to develop a multi-class algorithm using both machine learning and deep learning techniques for the prediction and classification of diabetic retinopathy. This algorithm was applied to an open-source diabetic retinopathy image dataset.

120.SMART COMMUNITY TOOL & EQUIPMENT SHARING LIBRARY USING MOBILE APP

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The proposed project presents a Smart Community Tool and Equipment Sharing System designed to facilitate efficient sharing and utilization of commonly used tools and equipment within a community. Many tools, such as drills, hammers, ladders, and other household or maintenance equipment, are purchased for occasional use and remain unused for extended periods. Users can communicate through natural language or voice- based interaction to enquire about the availability of tools, obtain information about their usage, identify suitable equipment, and initiate borrowing or rental requests. The system can maintain information about available tools, their owners, rental charges, and availability periods, enabling users to efficiently identify and access required resources.The system provides a user-friendly interface that reduces the complexity of conventional application-based interaction and improves accessibility through interactive communication. It can assist users in searching for suitable equipment, providing tool- related information, checking availability, and guiding them through the borrowing and return process. The system promotes a peer-to-peer resource-sharing model, allowing community members to share underutilized tools with others who require them.By encouraging collaborative resource utilization, the proposed system aims to reduce unnecessary purchases, improve equipment utilization, minimize waste, and promote sustainable community practices. The integration of interactive communication with a smart tool-sharing platform provides an engaging and convenient approach to community resource management and supports the development of a more connected and resource- efficient community.

121.ANALYZING AND IDENTIFYING SOCIAL MEDIA ADDICTION THROUGH COMPUTATIONAL MODELS AND DATA-DRIVEN APPROACHES-A SYSTEMATIC LITERATURE REVIEW

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The explosive development of social media networks has qualitatively changed human communication and interaction, increasing the risks of social media addiction. The phenomenon is characterized by dependence, loss of control, and negative consequences for the quality of life and has attracted significant research interest. However, the problem is still under the stage of analysis, and various studies use different methodologies, from traditional psychological assessment to modern data science techniques. This article provides a qualitative systematic review of computational methods and tools for analyzing and detecting social media addiction. The focus is on machine learning and data mining approaches, including classification and deep learning models, natural language processing, and explainable artificial intelligence. The review is based on the analysis of scholarly articles retrieved from Scopus, Web of Science, and PubMed databases. It explores the effectiveness of different algorithms, including Random Forest, XGBoost, CatBoost[4], Support Vector Machines[7], and neural networks, in social media addiction detection[9] andprediction. The review also aims to discuss the role of different features, including behavioral, textual, and physiological data, in the algorithms' performance. The theoretical background of social media addiction detection using computational methods is also at the center of this review. The key concepts explored include reward circuitry–related mechanisms and various risk factors, such as the fear of missing out and the need for self-validation. Finally, the review provides a summary and future research prospects related to the use of algorithmic approaches in social media addiction research. The discussion highlights the opportunities for further research directions, including the development of an integrated and explainable classification system applicable to various social media networks, which is currently under investigation.

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122.A BINARY PATTERN BASED MATRIX FRAMEWORK FOR MULTI PARTICIPANT COMMUNICATION SYSTEMS

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This paper develops a binary pattern based framework for discrete commu- nication activity in multi participant systems. The Even-Odd (EO) method and simple shortcuts for special numbers like 2n, 2n 1, and 2pM are used to convert participant states intoa binary communication matrix C. C = [cij]r×m. Participant workloads Ri and positional channel loads Lj satisfy the conser- vation law ∑r i=1 Ri = ∑m j=1 Lj . In addition, communication activity density ρ and peak concurrency load Lmax are formalized under the capacity bound 0 ≤ Lj ≤ q, providing a compact algebraic model for concurrency analysis and capacity planning.

123.DEFENDING LARGE LANGUAGE MODELS AGAINST ADVERSARIAL PROMPT ATTACKS VIA A LIGHTWEIGHT SECURITY GATEWAY

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Large language models are now being used in conversational products, enterprise tools, retrieval pipelines, and autonomous agents.Their ability to follow instructions given in language creates a security problem. An untrusted prompt might try to break through the application's rules. It could access data it should not see. It might also make the model do things that go against its policies. We introduce an Adversarial Prompt Defense Gateway. This is an fast layer that sits in front of the main large language model. It checks the prompts before they reach the model. The gateway uses signals, like capitalization ratio, special character patterns, token-level b details prompt length and signs of Base64-style encoding. It also uses embeddings from a small transformer model—either DistilBERT or DeBERTa-v3. These signals work together to spot inputs. Both feature streams are concatenated and passed through a multilayer perceptron (MLP) resulting in a single threat probability. Prompts scoring below 0.85 are forwarded and prompts scoring 0.85 or above are blocked or sanitized. When benchmarked against TensorTrust, AdvGLUE and JailbreakBench, the system achieves 99.2% training accuracy, 98.4% validation accuracy and sub-5 ms processing time. We want to find a ground, between cheap but shallow static filters and powerful but expensive model-based judges. This creates a defense layer that works together with the target model's own safeguards not as a replacement.

124.AEGIS SHIELD: A REAL-TIME UPI FRAUD DETECTIONFRAMEWORK USING ENSEMBLE MACHINE LEARNING

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The convenience of digital banking and UPI payments has made everyday payments faster and easier, but the rise in their usage has also brought about concerns over transaction fraud. In this paper, we present Aegis Shield, a framework for analyzing digital transactions and detecting suspicious activity before confirming potentially fraudulent payments. Aegis Shield combines Random Forest and XGBoost classifiers to analyze transaction data, including the payment amount, the sender's previous payment patterns, receiver account information, previous fraud complaint history, recent transaction activity, unusual payment times, and trusted-contact history. The system does not rely solely on model predictions but also considers transaction patterns and previously logged reports to determine the final risk score. Transactions are then classified as low, medium or high risk. The system includes a mobile application based on Flutter, a backend based on FastAPI to detect fraud and a MongoDB database to store transaction records, receiver details, scam reports and system logs. Users can view QR-based payment details, check transaction history, receive explanations for risk warnings, and report suspicious accounts. Transaction and fraud information can also be monitored from the administrative interface. The performance of the individual models and their weighted combination is analyzed using performance metrics and ROC-AUC scores. Aegis Shield demonstrates a successful integration of a mobile payment interface, backend services, database management, and ensemble machine learning models to provide a practical framework for real-time transaction risk assessment

125. DESIGN AND DEVELOPMENT OF IOT-INTEGRATED PLC-BASED PREDICTIVE MULTI-ZONE IRRIGATION SYSTEM

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Agricultural irrigation is often performed using fixed schedules or manual decisions, resulting in over-irrigation, under-irrigation, water wastage, and unnecessary energy consumption. This paper proposes an IoT-enabled PLC-based intelligent multi-zone irrigation system designed to provide crop-specific water management. A Delta PLC serves as the primary industrial controller, while soil- moisture, temperature, humidity, pH, water-level, and rainfall sensors provide real-time field information. The agricultural field is divided into independent irrigation zones, allowing each zone to receive water according to its crop requirements and soil condition. Solenoid valves and a water pump are controlled automatically by the PLC using ladder logic. An ESP32-based IoT gateway communicates with the PLC through RS-485/Modbus RTU for remote monitoring, data logging, and dashboard visualization. Historical sensor data and weather information can subsequently be used to develop predictive irrigation algorithms. This proposed system aims to reduce unnecessary water consumption, improve irrigation

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efficiency, reduce manual intervention, and demonstrate the application of industrial PLC automation in precision agriculture. Experimental evaluation focuses on water savings, energy reduction, irrigation accuracy, and system reliability.

126.FROM ELECTROCARDIOGRAPHY TO ACTION: A REAL-TIME HEART ATTACK RISK ALERTING FRAMEWORK

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Myocardial infarction is a time-critical emergency, yet routine heart surveillance remains episodic, so early warning signs that appear between clinical visits often go unrecorded. Recent low-cost monitoring prototypes rely on fixed alarm thresholds or on a small set of basic vital signs, and the studies reviewed here do not report validation metrics for their complete pipelines. This paper proposes an integrated framework for continuous heart attack risk estimation and alerting. A single lead electrocardiogram front end connected to a wireless microcontroller streams samples to a Python service, which applies band-pass filtering, R-peak detection and extraction of heart-rate and heart-rate-variability features. These features are combined with a patient profile and classified by a Random Forest into Normal, Moderate or High risk, and a rule layer triggers notifications and updates a clinician dashboard backed by a relational database. The contribution is an architecture-level integration of existing techniques rather than a new algorithm, and the framework is at the proposed stage: no experimental results are reported. A validation protocol is specified, covering public annotated electrocardiogram datasets, patient-wise cross-validation, ablation of feature groups, latency measurement and a hardware bench test, with accuracy, sensitivity, specificity, F1-score and end-to-end alert delay as the planned metrics.

127.PERSONALIZED PHISHING AWARENESS AND ADAPTIVE EMPLOYEE RISK MANAGEMENT FRAMEWORK USING SALESFORCE

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Phishing remains a persistent cybersecurity concern because technical controls alone cannot completely prevent attacks that exploit human decision-making. Conventional awareness programs commonly provide uniform and periodic training without considering individual knowledge, previous failures,

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topic-specific weaknesses, training completion, or measurable improvement. This paper proposes a Personalized Phishing Awareness and Adaptive Employee Risk Management Framework implemented using Salesforce. The framework combines an employee-specific dashboard, interactive phishing assessment, question-level response capture, weak-category identification, configurable multifactor risk scoring, automatic personalized training assignment, scenario-based learning, training-attempt evaluation, and improvement tracking. Nine related Salesforce custom objects maintain employee, assessment, response, risk, module, assignment, scenario, and attempt information. Apex services coordinate data retrieval, validation, scoring, assignment creation, and profile updates, whereas Lightning Web Components provide a responsive user experience. The system converts assessment evidence into a continuous intervention cycle: an employee completes an assessment, the Risk Profile is updated, a relevant module is assigned, training performance is evaluated, and progress is reflected on the dashboard. Functional validation of the academic prototype confirms the expected employee-specific workflow, score calculation, category analysis, assignment logic, repeated attempts, and dashboard refresh. The framework demonstrates how a cloud application platform can support adaptive, explainable, and auditable phishing-awareness management.

128.DESIGN AND DEVELOPMENT OF A SOLAR-POWERED SMART AQUAPONIC SYSTEM WITH REAL-TIME MONITORING AND AUTOMATIC WATER CIRCULATION

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This paper introduces a smart aquaponic system using solar photovoltaic (PV) source for sustainable fish and plant production with real-time monitoring and automation of water circulation. The proposed system consists of solar PV source, aquaculture, plant growing through hydroponics, water circulation pump, sensors, microcontroller, and display unit. In the proposed system, fish excreta act as a nutrient for the plants, while plants and biological filtration improve water quality before its circulation to fish tanks. The control unit continuously monitors various parameters including water level, water temperature, pH, etc. The data from sensors is then fed into the microcontroller, which is shown in real time to monitor the system. Based on the monitored parameters, the controller controls the operation of water circulation pump to regulate the water flow between the fish tanks and plant-growing tanks. The solar PV source provides the needed electric energy to the pumping and control devices, hence reducing dependency on conventional electrical energy. Various operating conditions and irradiance values of solar sources may be considered to test the proposed system.

129 .AN MRI IMAGE ENHANCEMENT AND AFFECTED REGION LOCALIZATION FRAMEWORK FOR ALZHEIMER DISEASE

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Alzheimer's disease (AD) is a condition that gets worse over time. It affects the brain. Leads to a decline, in thinking skills and memory. The brain also changes in structure as the disease progresses. Magnetic resonance imaging (MRI) gives pictures of the brain's structure. These images can be studied using computer methods to find patterns. These patterns help in understanding Alzheimer's disease and how it changes at stages of dementia. [1], [2]. However, variations in image intensity and the visual

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similarity, between neighboring disease stages make automated MRI analysis a challenging task. In this paper, we propose an MRI framework, which consists of an image preprocessing procedure, contrast enhancement, affected-region localization and multiclass classification. The experiments were conducted using the Alzheimer's Disease Multiclass Images Dataset containing four categories: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented [3]. A stratified 80:20 division was used, resulting in 35,200 training images and 8,800 test images. The evaluated baseline achieved an accuracy of 62.26%, with weighted precision, recall, and F1-score of 62.44%, 62.26%, and 61.12%, respectively. The ModerateDemented category produced the highest class-wise F1-score of 80.77%, whereas VeryMildDemented produced an F1-score of 40.76%. A comparison with and without image enhancement showed that the tested enhancement configuration did not improve the classification performance of the evaluated baseline. Since the selected dataset contains image-level labels rather than expert pixel-level masks, the region- analysis component is treated as localization rather than supervised anatomical segmentation. The results provide a baseline for further investigation using advanced deep-learning architectures and datasets containing expert-annotated MRI regions.

130. GRAPHENE-ASSISTED TERAHERTZ DIELECTRIC RESONATOR BIOSENSOR FOR QUANTITATIVE DETECTION OF VITAMIN B2 CONCENTRATION USING RESONANCE SHIFT ANALYSIS

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Terahertz (THz) biosensing has emerged as a promising technology for rapid, non-invasive, and label-free identification of biomolecules due to its capability to detect unique molecular vibrational signatures. Riboflavin (Vitamin B2), an essential biomolecule widely used in biomedical diagnostics, pharmaceutical analysis, and food quality monitoring, exhibits characteristic absorption features within the terahertz spectrum. However, existing THz biosensors are predominantly based on metamaterial absorbers and plasmonic resonators, which often suffer from limited radiation efficiency, narrow operational bandwidth, low gain, and restricted sensing capability. To address these challenges, this paper proposes a graphene-assisted dual-band dielectric resonator antenna (DRA) integrated with a defected ground structure (DGS) for label-free terahertz riboflavin sensing. The proposed antenna employs a graphene-loaded rectangular dielectric resonator and a net-shaped DGS to improve impedance matching, radiation efficiency, and electromagnetic field confinement within the sensing region. A comprehensive full-wave electromagnetic analysis is performed using ANSYS HFSS/CST Microwave Studio. Parametric investigations are conducted to study the influence of graphene conductivity, defected ground dimensions, dielectric resonator geometry, and sensing layer properties on the antenna performance. The optimized structure exhibits dual resonances at 2.4 THz and 7 THz with reflection coefficients of −42.39 dB and −24.5 dB, respectively, while achieving radiation efficiencies exceeding 88% and realized gains above 4.5 dBi. Furthermore, riboflavin concentration variations are investigated through resonance frequency shift analysis, demonstrating the feasibility of quantitative biomolecule detection. The proposed biosensor offers high sensitivity, wide operational bandwidth, compact size, and low-loss performance, making it a promising candidate for next-generation THz biomedical sensing and pharmaceutical monitoring systems.

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131. SECURING DIGITAL IDENTITY FOR THE ADVANCED QUANTUM ERA: A MULTI-LAYERED FRAMEWORK FOR FORWARD SE- CURE AND ADAPTIVE MULTI-FACTOR AUTHENTICATION

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A scalable quantum computer using the Shor's algorithm will be capable of breaking the cryptosystems based on the RSA and elliptic curves which are currently used for securing digital identification systems, thus making it necessary to move towards lattice-based key encapsulation schemes like CRYSTALS-Kyber (NIST standard ML-KEM, FIPS 203). Nevertheless, a mathematical robust system does not ensure that the system would be secure;even a mathematically correct approach can leak its secret key via classical side channel attacks based on implementation. In this paper, we demonstrate an implementation of Kyber-512 which uses constant-time Barrett reduction, constant-time FO re-encryption check, fourth-order Boolean masking, and per query ciphertext rotation. In our experiments, the baseline is attacked and broken using just one power trace, but the enhanced version is not broken even after 10,000-traces representing a four order of magnitude im-provement in security strength. Additionally, we demonstrate, using the example of a password verification system, that the same vulnerability type (variable-time comparison) applies outside of the cryptographic context, and that the same solution (constant-time comparison) can be applied as an effective countermeasure. Lastly, we validate the threat model posed in the paper by running Shor's algorithm using a quantum circuit simulation tool (Google Cirq) to factor a toy RSA-style modulus. We then run Grover's algorithm to show that its speedup advantage over classical algorithms is only quadratic, and not exponential, validating the paper's central thesis that there is no quantum analog for the attack on lattice-based schemes such as Kyber

132.REAL-TIME SEQUENCE MODELLING VIA LIGHTWEIGHT KEYPOINT NETWORKS FOR ENHANCED SIGN LANGUAGE INTERPRETATION

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The Real-Time Keypoint Sequence Network (RKPS-Net) represents a strategic engineering advancement in the deployment of efficient sign language recognition systems. By pivoting away from the high computational demands of raw video processing toward a lightweight 63- dimensional feature vector derived from the MediaPipe Landmarker, and integrating a robust, multi-layer LSTM core, RKPS-Net achieves a high classification accuracy of 94.1% while fundamentally prioritizing the low latency required for real-world, edge-device deployment. The system's real-time inference pipeline, which uses a 30-frame sliding window, a 0.8 probability threshold, and a 10-frame temporal consistency check, ensures the output is both fast and temporally stable, overcoming typical challenges associated with sequential prediction jitter in live video streams.

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133. A SURVEY ON BRAIN STROKE DETECTION, LESION SEGMENTATION AND BRAIN SWELLING RISK PREDICTION USING BIOMECHANICAL MODELLING

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Stroke-oriented artificial intelligence has progressed from isolated image classification toward systems that detect abnormalities, delineate lesions, quantify tissue injury, estimate complications, and provide interpretable evidence for clinical review. It is based on ischemic and hemorrhagic stroke analysis using CT and MRI, with emphasis on lesion segmentation, multimodal learning, malignant cerebral edema and brain-swelling prediction, external validation, explainable artificial intelligence, and biomechanical modelling. The reviewed studies show that multimodal diffusion MRI and advanced U-Net/transformer architectures improve ischemic lesion delineation [2],[6], while CT-based deep learning can support acute ischemic stroke and intracranial hemorrhage detection [7],[10]. Large benchmark datasets such as ISLES 2022 strengthen reproducible evaluation and multimodal research [11],[12]. For post-stroke swelling, interpretable machine-learning and multi-region radiomics models show that edema risk can be estimated from clinical and imaging features, although external performance remains weaker than internal performance and outcome labels are less widely available than lesion masks [13], [14]. Recent XAI studies further demonstrate the value of SHAP, Grad-CAM, XRAI and uncertainty estimation [13], [15], [16]. Building on the biomechanical swelling framework [1]. It identifies a central research opportunity integrating patient-specific lesion segmentation, multimodal features, interpretable risk prediction, and biomechanical tissue modelling in one validated framework rather than treating these tasks independently.

134. THE DIGITAL COMMERCE MATRIX: A PLATFORM SELECTION FRAMEWORK FOR ALIGNING E-COMMERCE ARCHITECTURE WITH BUSINESS MODEL

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The shift in digital commerce has dramatically transformed the digital platform landscape, creating a complex web of e-commerce solutions, each with its own signature architecture (from API design patterns to micro-services compositions, data orchestration layers to third-party integration frameworks), and all influencing a business's ability to scale and go faster. However, while organizations are faced with the critical question of platform selection, the lack of a structured, criteria-based evaluation process leaves most teams in the dark when picking a platform, and when the platform’s architecture doesn't align with the functions and processes of a Direct-to-Consumer fulfillment model, a Business-to- Business procurement cycle, or a Marketplace vendor ecosystem, the misalignment is not only an annoyance - it tears the connection between technology infrastructure and strategic business goals apart, and the resulting costs are often subtle enough to be ignored. We propose a Platform Selection Framework (PSF), The Digital Commerce Matrix, that systematically classifies e-commerce platforms by their architectural properties. The idea behind the framework is to break down

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some of the platform characteristics that would be compatible with the operational needs of D2C, B2B and Marketplace business models. Also, the study offers practical recommendations that enable organizations to consider their digital strategy, platform choice, and scalability needs to support their growth and development in the ever-changing digital commerce landscape.

135. AI-BASED PREDICTIVE MAINTENANCE SYSTEM FOR INDUSTRIAL EQUIPMENT USING MULTISENSOR AND VISION-BASED CONDITION ANALYSIS

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Predictive maintenance enables early identification of equipment degradation by continuously monitoring operating conditions and detecting abnormal patterns. This paper presents an AI-based predictive maintenance system for industrial equipment using multisensor and vision-based condition analysis. The proposed system integrates a Raspberry Pi with an MPU6050 vibration sensor, DHT11 temperature and humidity sensor, ACS712 current sensor, RPM sensor, infrared (IR) sensor, and Raspberry Pi camera to monitor a motor under test. The acquired numerical parameters are processed and used by a Random Forest machine-learning model for motor-condition classification, while the camera provides visual monitoring of the motor. The system identifies motor operating conditions and presents the monitoring parameters and AI-based condition result through a Raspberry Pi TFT display. The proposed prototype demonstrates an integrated approach to real-time motor condition monitoring and predictive maintenance development.

136. MACHINE LEARNING-BASED MULTI-CLASS BRAIN DISORDER PREDICTION: A COMPREHENSIVE REVIEW

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Neurological and psychiatric disorders constitute a leading global health burden, with early and accurate diagnosis being critical for effective treatment. Machine learning (ML) and deep learning (DL) techniques have demonstrated remarkable potential in automating the classification of multiple brain disorders simultaneously from neuroimaging and physiological data. This review provides a comprehensive synthesis of ML-based approaches for multi-class brain disorder prediction, covering neuroimaging modalities, feature extraction techniques, model architectures, and public benchmark datasets. We critically evaluate convolutional neural networks (CNNs), recurrent architectures, graph neural networks (GNNs), and transformer based models applied to disorders including Alzheimer's disease,Schizophrenia, Autism Spectrum Disorder (ASD), Parkinson's disease, Major Depressive Disorder (MDD), brain tumors, and epilepsy. Key challenges such as data scarcity, class imbalance,interpretability, and multi-site variability are discussed along side emerging solutions

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including federated learning, domain adaptation, and explainability frameworks. This review identifies critical research gaps and outlines future directions for clinically deployable, interpretable, and equitable AI-based neurological diagnostic systems.

137.AI AND IOT-BASED SMART WASTE SEGREGATION SYSTEM AND BIN-LEVEL INDICATOR

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Waste Management is one of the most pressing logistical, environmental, and economic issues of the 21st century. With a growing population and rampant consumerism, the amount of waste that is produced in megacities is approaching an unsustainable level. While recycling programs and color-coded bins have been implemented worldwide, human psychology at the disposal station is a hurdle which causes an overall reduction in recyclable waste. Mixed waste with biodegradable matter tends to ruin other recyclable materials, and is consequently dumped into landfills. This paper proposes a human independent smart waste segregation system, which makes waste disposal easy for the common citizen, and has the potential to increase the amount of viable recyclable waste exponentially. Using deep-learning techniques for waste classification and IoT connectivity, we have created an automated smart bin, which can categorize the waste stream in terms of recyclables and biodegradable with an incredibly high degree of accuracy, as tested during our trials. When an item is placed on the bin's scanner, a lightweight Convolutional Neural Network identifies it as either Wet or Dry waste, and deposits it inside the appropriate chamber via a mechanized flap. An added advantage of the bin is its ability to calculate the fill level of its two compartments and alert the relevant authorities when it requires emptying, thus optimizing fuel consumption and helping to reduce unnecessary collection trips. This paper attempts to provide an in-and-out view of our prototype, ranging from the challenges faced in training the AI visual dictionary to the nuances of the mechanical structure, the working of the IoT network, and the psychological impact on the citizen. We present hard evidence of our test results, and discuss the potential economic and environmental impact that this system could have on modern day smart cities.

138. EOPS: A MULTI-AGENT AI ENGINEERING OPERATIONS PLATFORM FOR AUTOMATED INCIDENT INVESTIGATION AND ROOT CAUSE ANALYSIS

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The increasing complexity of modern distributed software systems has significantly elevated the challenges associated with production incident investigation and root cause analysis. Engineering teams typically rely on multiple disconnected tools, including version control platforms, application logs, monitoring dashboards, and internal documentation, resulting in time-consuming manual investigations and increased Mean Time to Resolution (MTTR). Although recent advances in Large Language Models (LLMs) and agentic AI have demonstrated strong capability in software engineering tasks, existing solutions primarily focus on code generation or conversational assistance rather than collaborative

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engineering investigation. This paper proposes the Engineering Operations Platform (EOPS), a supervisor-driven multi-agent AI framework designed to assist engineering teams during incident investigation. Specialized agents operate concurrently to analyze source-code repositories, application logs, and organizational knowledge bases, aggregating findings through a shared investigation state so that a dedicated Root Cause Analysis agent can generate evidence-based hypotheses and recommendations. The framework further integrates a Retrieval-Augmented Generation (RAG) based Engineering Memory Layer for contextual retrieval from structured engineering documentation. Designed as a decision- support system rather than an autonomous controller, EOPS preserves human oversight through an explicit approval stage. The proposed architecture provides a scalable foundation for future AI-assisted engineering operations and illustrates how collaborative multi-agent systems can improve contextual reasoning, knowledge utilization, and incident analysis in modern software environments.

139. PARTIALLY SET CONCRETE: A REVIEW ON RETEMPERING, DELAYED CASTING AND COLD JOINT

Sagar Arvind Agarwal, S. S Ajnadkar, and P.D. Jadhao

Delays between mixing and placing in concrete can cause concrete to set partially, reducing workability and increasing risk of cold joint. The review examines 20 studies from 1973 to 2026 for their relevance to retempering, remixing, blend ratio, controlled recovery, delayed casting and cold joint behavior. It distinguishes delayed concrete by elapsed time from partially set concrete, defined by material state. The literature shows that water retempering restores workability, but excess water beyond the designed water-cement ratio can weaken compressive strength. Controlled admixture dosage and blending with fresh concrete can limit losses. There are three gaps that emerge (i) real time concrete characterization, (ii) optimizing dosages to increase workability and mechanical properties of the concrete and (iii) surface treatment of cold joints.

140.AI AND CLOUD-ENABLED VIBRATION-ASSISTED SELF-CLEANING SYSTEM FOR PHOTOVOLTAIC PANELS

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Dust deposition on the photovoltaic (PV) panels is one of the major environmental impacts that can result in reduced solar energy conversion efficiency particularly in coastal, arid and industrial environments. Traditional cleaning methods can be very water intensive, require a lot of maintenance, be very mechanical and energy intensive in the cleaning of brushes and robotic cleaning devices. In order to tackle these issues, this paper introduces an AI and cloud based vibration self-cleaning system which can eliminate dust without rotating brushes, water or permanent man force, based on the vibration force. The prototype was designed to have a multi-layer vibration transmission structure, consisting of silicone protective layer, fiberglass sheet, PDMS layer and Kapton layer above the PV module. The ESP32 microcontroller is connected to dust sensor, voltage sensor, current sensor and the temperature sensor and the result of continuous measurement is displayed by the LCD. A controlled vibration is achieved by using two servo motors, symmetrically mounted on the fiberglass layer and applied to the multilayer structure which causes the separation of the dust deposited. Besides the existing threshold based cleaning mechanism there are two more components to support the proposed enhanced architecture: a cloud-based storage and analytics and an LSTM- based module for artificial intelligence

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based temporal fault and dust related performance prediction. The information from the sensors is sent via Wi-Fi to the cloud database and displayed on the web dashboard with calculated power information, AI- predicting power and cleaning status and cleaning history. The AI module analyzes series of temperature, dust, voltage, current and power data to calculate a probability that a cleaning is needed. A safety layer based on a threshold is left in place to allow for cleaning to still take place under abnormal electrical conditions when needed. Improvements in the electrical parameters of PV panel and effectiveness in cleanings are measured before and after each cleaning cycle. The integrated solution is a physical cleaning based on vibration and an IoT monitoring, cloud based data management and predictive intelligence in a single and modular solution, which provides low water, low maintenance PV operation.

141.CRASH SHIELD: AN ADVANCED AIRBAG SYSTEM FOR ENHANCED RIDER SAFETY

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Two-wheeler riders are exposed to severe injury because motorcycles and scooters lack the passive protection available in enclosed vehicles. This paper presents an integrated smart helmet that combines preventive, protective, and responsive safety functions. An Arduino Uno acts as the central controller. An MQ-3 alcohol sensor evaluates ethanol vapour, while an infrared sensor verifies helmet usage. A relay-based ignition interlock permits vehicle operation only when the helmet is worn and the alcohol reading is below a preset threshold. During riding, an MPU6050 inertial sensor measures three axis acceleration and angular velocity to identify a high-impact event followed by orientation change and post-impact inactivity. After accident confirmation, a second relay drives a DC geared motor that opens a compressed-gas cylinder valve to inflate a protective airbag. A GSM module simultaneously transmits an emergency SMS to predefined contacts. The proposed architecture therefore addresses prevention of unsafe riding, impact protection, and emergency response in one low-cost embedded platform.

142. MICROSTRUCTURE AND HIGH-TEMPERATURE EROSION BEHAVIOUR OF PLASMA-SPRAYED NICRBSI COMPOSITE COATINGS WITH WASTE GRANITE AS REINFORCEMENT

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Waste granite is an abundant mineral-processing residue whose incorporation into protective surface coatings offers an opportunity for both waste valorization and materials performance enhancement. In this study, NiCrBSi-based composite coatings containing 0, 10, and 20 wt.% waste granite powder were deposited on MDN321 steel by atmospheric plasma spraying. The influence of granite addition on the coating microstructure, phase constitution, hardness, and high-temperature solid-particle erosion behaviour was systematically investigated. X-ray diffraction, scanning electron microscopy, and energy-dispersive spectroscopy revealed a NiCrBSi matrix containing carbide phases, with SiO_2 and

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Al_2O_3 rich constituents introduced by the granite reinforcement. The average microhardness increased from 615 HV for NCB100 to 684 HV and 737 HV for NCB90G10 and NCB80G20, respectively. Solid particle erosion tests were performed at 200, 400, and 600 °C using Al_2O_3 particles at 30° and 90° impact angles. Granite reinforcement reduced the erosion rate at the 30° impact angle, with NCB80G20 exhibiting the highest erosion resistance. The effect of reinforcement was less pronounced at 90°, where impact-dominated deformation and coating fracture controlled material removal. Surface characterization revealed temperature and angle dependent degradation mechanisms involving micro-cutting, plowing, splat removal, crater formation, and micro-cracking. The results demonstrate that waste granite can function as an effective ceramic reinforcement for improving the mechanical and high-temperature erosion performance of plasma-sprayed NiCrBSi coatings.

143. ARTIFICIAL NEURAL NETWORK FOR FAULT DETECTION IN A PHOTOVOLTAIC SYSTEM

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A representation used for photovoltaic (PV) fault diagnosis must be invariant when a classifier is transferred from its training on a set of offline measurements to online fault detection. In this paper, a MATLAB/Simulink framework for seven-class fault detection of a 20-W PV system using a small-size artificial neural network (ANN) based on statistical waveform feature extraction is provided. An initial stage of our pipeline consists of generation of a single-diode PV system under five different levels of irradiance and four levels of temperature. Seven operational modes of the system: normal operation, partial shading, open circuit, short circuit, degradation, bypass diode fault, and ground fault are generated separately from the ANN decision. Voltage, current, and power waveforms are collected in a buffer and transformed into a fixed 31-feature vector comprised of eight voltage statistics, ten current statistics, ten power statistics, mean voltage-current product, irradiance, and temperature. Features are normalized based on the training set mean and standard deviation values and classified using a 31-20-10-7 pattern recognition network with tansig hidden layers and softmax output layer. A dataset of 140 samples, equally distributed among the seven classes, is split into 98 samples for training, 21 samples for validation, and 21 samples for testing. The resulting partitions yield 100% accuracy, while an independent forward-pass calculation of the network provides the same output as the one stored within it with the maximal difference around $1.27\times10{-}8$. The work highlights the potential of a transparent statistical feature ANN pipeline in PV fault classification based on simulations. In addition, a distinction between the offline ANN classification accuracy and the simulation implementation aspects is made and criteria for further experiments are defined.

144.SANSKRIT–MARATHI NEURAL MACHINE TRANSLATION: A REVIEW OF TRANSFORMER- BASED APPROACHES, CHALLENGES, AND FUTURE DIRECTIONS

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Translating Sanskrit into Marathi using Neural Machine Translation (NMT) is challenging because Sanskrit is a low-resource classical language with limited parallel data and a highly complex grammatical structure that differs considerably from modern Marathi. This survey examines how Transformer-based architectures address these challenges, with particular attention to multi-head attention, specialized tokenization for handling Sanskrit sandhi, and other techniques designed for morphologically rich languages. Since annotated Sanskrit–Marathi corpora are scarce, the review also considers the use of cross-lingual transfer from relatively high-resource Indic languages and multilingual pre-training to improve translation performance. Another major challenge is preserving meaning across the different grammatical systems of the two languages: Sanskrit uses the kāraka framework to express semantic roles through a flexible case system, whereas Marathi generally follows a more structured Subject–Object–Verb (SOV) pattern. A successful translation system must therefore capture these structural differences while preserving the cultural, religious, and contextual meaning of Sanskrit texts. Although Transformer models have shown clear advantages over earlier recurrent approaches, important research gaps remain, including the lack of standardized benchmarks, difficulties in evaluating domain- specific texts, and the need for better methods to assess philological accuracy along with fluency. This survey reviews existing research to summarize the current state of Sanskrit–Marathi NMT, identify its major challenges, and highlight promising directions for future work; it does not present new experimental results, architectures, or unpublished datasets.

145.FLOOR CLEANING ROBOT USING ARDUINO

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Cleaning large floor areas manually takes a lot of time and physical effort. To reduce this work, a floor cleaning robot has been developed using Arduino Uno. The robot can move on the floor and perform cleaning with the help of a brush and water spraying system. Ultrasonic sensors are used to detect obstacles during movement. DC motors provide the required movement, while the Arduino controls the different operations of the robot. The developed robot is intended for cleaning areas such as laboratories, offices, homes, and other indoor spaces. It provides a simple and low-cost method of floor cleaning with less human effort.

146.PHYSICOCHEMICAL ASSESSMENT OF SURFACE AND GROUNDWATER SOURCES IN THE DHARAMPUR REGION OF THE SHIVALIK HILLS, HIMACHAL PRADESH, INDIA

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This study evaluates the physicochemical quality of selected surface and groundwater sources in the Dharampur region of Solan district, Himachal Pradesh, India. Four water samples from a bore well, hand pump, river, and pond were analyzed using standard methods recommended by the American Public Health Association. Parameters assessed included pH, electrical conductivity, total dissolved solids, total suspended solids, total hardness, calcium hardness, and alkalinity. The results showed pH values of 6.44-7.52, electrical conductivity of 290-520 µS/cm, total dissolved solids of 200-600 ppm, and no detectable total suspended solids. Total hardness ranged from 260-625 ppm as CaCO 3 , while calcium hardness ranged from 515-625 ppm. Alkalinity varied from 65-125 ppm. The results showed variations in physicochemical parameters among the four water samples. Some measured values were outside the reference ranges, indicating the need for further assessment. These findings provide preliminary information on water quality; however, additional chemical and microbiological analyses are required before determining the suitability of these sources for drinking purposes.

147. ASSESSMENT OF MINIMUM MUSCULAR FITNESS IN MIDDLE AGED WOMEN WITH AND WITHOUT LOW BACK PAIN

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Being physically fit has been a fundamental human need. History demonstrates that physical fitness has always been essential to one's survival. The capacity to do everyday duties with energy and alertness, without experiencing excessive exhaustion, and with sufficient reserves to participate in recreational activities and respond to emergencies is known as physical fitness. (1) Too little attention has been paid to the fact that the dropping of physical fitness more specifically muscle fitness below a certain minimum actually jeopardizes well-being and health. (1)

148.MACHINE LEARNING FOR UNIVERSITY STUDENT GRADUATION PREDICTION: A SYSTEMATIC REVIEW AND A GENERATIVE AI-DRIVEN FRAMEWORK FOR PERSONALIZED INTERVENTION

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The growing demand for data-driven educational decision-making has fueled research into machine Learning (ML) techniques for forecasting university student graduation outcomes. Existing research has shown that ML algorithms such as Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting, and Artificial Neural Networks are effective at identifying students who are at risk of dropping out or delaying graduation based on academic performance markers. However, most of this research concentrates on predictive accuracy, providing little support for translating prediction results into individualized and practical academic interventions — a gap that limits the real-world impact of these models in academic advising. This paper presents a Systematic Literature Review (SLR) of recent research on ML-driven university student graduation prediction using academic performance data, following PRISMA 2020 guidelines to synthesize findings from 15 studies published between 2021 and 2026 across IEEE Xplore, Scopus, ScienceDirect, ACM Digital Library, and SpringerLink. The review synthesizes predictive models, commonly used academic attributes, evaluation metrics, and major findings from the literature. Based on the identified research gap, the paper proposes a Generative AI-

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driven conceptual framework that augments ML prediction by producing explainable insights and personalized intervention strategies, such as individualized study plans and early academic support recommendations. The proposed framework illustrates how Generative AI may help translate predictive analytics into an intelligent decision-support system that supports student engagement and academic success. This review offers scholars and higher education institutions a thorough understanding of current ML techniques, while highlighting the potential of integrating Generative AI to develop next-generation student success systems. Future research should focus on validating the proposed framework in real- world educational settings, particularly with respect to explainability, privacy, fairness, and ethical AI adoption.

149. A NEURO-SYMBOLIC FRAMEWORK FOR ADVERSARIAL PHISHING DETECTION AND ZERO-DAY MITIGATION

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Phishing attacks are becoming increasingly complicated with more clever use of misleading URLs, domain spoofing, and rapidly changing web infrastructure technology to evade old-fashioned detection methods. Existing phishing detection systems using machine learning need to be optimized and trained with many manually selected features, but these methods still can be affected by domain attacks, and there is no guarantee to explain how classification took place. In order to solve these disadvantages, we present a new neuro-symbolic phishing detection method called PhishGuard-X. PhishGuard-X has three main parts: Adversarial Grapheme & Phonetic Canonicalization Engine (A-GPCE) for Punycode normalization; Neuro-Symbolic Dual-Stream Inference Core (NS-DSIC), which combines deterministic security rules with machine learning predictions; and Edge Counterfactual Local Shapley Explainable AI (E-CXAI) for interpretive arguments of risks related to phishing. The PhishGuard-X algorithm generates 26 different features of URLs including lexical and syntactic, content and DOM, and adversarial brand features. During tests conducted on 65,000 URL samples, PhishGuard-X proved to achieve 98.74% accuracy and 0.9874 F1-score, in addition to 12.4 ms edge inference time. It has been demonstrated that the system has a false-positive rate for homoglyphs and Punycode mutations of less than 0.02%. Further research shows that it is possible to construct a system for real-time phishing detection and zero-day attack prevention, basing on the application of machine learning techniques with adversarial canonicalization, symbolic security proofs, and local explainability.

150. ENSEMBLE LEARNING FOR PULMONARY DISEASE DETECTION

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Lung disease refers to any condition that affects the proper functioning of the lungs. The World Health Organization reports that about 3.9 million people die each year from respiratory diseases, especially chronic obstructive pulmonary disease. Early and accurate diagnosis of pulmonary diseases such as pneumonia, tuberculosis, and lung cancer is very important. X-ray images are used to identify abnormalities in the lungs; however, manual analysis is time-consuming and may not be accurate due

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to image quality. The input image is preprocessed, and various texture features are extracted. This study presents an ensemble learning approach for the detection of pulmonary diseases using chest X-ray images. Multiple machine learning algorithms, including Random Forest, K-Nearest Neighbors (KNN), and Logistic Regression, are combined to improve the detection of pulmonary diseases. Various lung disease datasets, such as Kaggle, RSNA Pneumonia Detection Challenge (2018), NIAID TB, QU-MLG-TB, PadChest, the National Library of Medicine, and Belarus, are tested, and performance metrics are evaluated. The results show that the ensemble algorithm achieves better results than traditional algorithms.

151. AI – INTEGERATED SMART SAFETY HELMET SYSTEM

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Industrial workers are frequently exposed to workplace hazards, including accidental falls, improper use of personal protective equipment, and unauthorized entry into restricted areas. Conventional safety monitoring methods primarily depend on manual supervision, which may delay accident identification and emergency response. To address these challenges, this paper proposes an AI-Integrated Smart Safety Helmet System that combines embedded sensors, artificial intelligence (AI), computer vision, and the Internet of Things (IoT) to improve industrial worker safety through continuous monitoring and automated alerts.The proposed system incorporates a smart helmet equipped with motion and impact sensors to detect potential worker falls by analyzing acceleration, orientation changes, and inactivity patterns. A helmet-wearing detection mechanism identifies when a worker removes the helmet and generates an alert to notify the supervisor. In addition, an AI-based camera system is installed at restricted-area entrances to detect persons, identify safety helmets, and monitor movement across predefined boundaries. The system identifies unauthorized entry into restricted areas and detects situations in which a worker enters wearing a helmet but subsequently removes it. These functions enable continuous safety compliance monitoring beyond the entrance point. The detected events are processed and communicated through an IoT-based network to a monitoring platform, where supervisors can receive real-time notifications and review recorded safety incidents. The integrated system aims to reduce dependence on manual surveillance, improve hazard identification, and support timely intervention during emergencies. Its modular architecture allows the integration of additional sensors, worker identification mechanisms, and advanced safety analytics in future developments. The proposed system is suitable for manufacturing industries, construction sites, warehouses, power plants, and other industrial environments where worker protection and restricted-area monitoring are essential. The effectiveness of the system can be evaluated through fall detection accuracy, helmet detection performance, unauthorized-entry detection, notification delivery reliability, and overall response time under different operating conditions. This approach provides a scalable foundation for intelligent industrial safety management and proactive workplace hazard prevention.

152. A WEB-BASED INTELLIGENT SYSTEM FOR REAL-TIME EXERCISE MONITORING AND PERSONALIZED HEALTH GUIDANCE

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The proposed system is a web-based intelligent application designed to provide personalized fitness guidance, meal planning, and real-time exercise monitoring using Artificial Intelligence (AI) and Computer Vision (CV) technologies. Developed using Python, Flask, and MediaPipe, the system acts as a virtual fitness assistant that helps users perform exercises correctly, monitor their fitness progress, and receive immediate feedback through a webcam- based interface. The primary objective of this project is to promote healthy lifestyles by integrating exercise monitoring, nutritional recommendations, and intelligent assistance into a single accessible platform. Traditional fitness approaches often require professional trainers, separate applications, and continuous manual monitoring. The proposed system addresses these limitations by providing an integrated solution for home-based fitness management. OpenCV is used for real-time video capture and image processing, while MediaPipe performs human pose estimation and body landmark detection. The extracted body landmarks are used to calculate joint angles, analyze exercise posture, count repetitions, and identify potential errors in exercise movements. The system supports exercises such as squats, push-ups, and lunges, providing visual feedback to help users maintain proper exercise techniques and improve workout performance. In addition, the system includes an AI-powered chatbot that answers common fitness-related questions, recommends suitable workout routines, and provides motivational guidance based on user requirements. A personalized diet planning module generates meal recommendations by considering user information, fitness goals, activity levels, and body measurements. Body Mass Index (BMI) can also be used as a supporting parameter for general fitness guidance. User information, exercise records, dietary plans, and workout history are stored in an SQLite3 database for efficient data management and future reference. The interactive dashboard, developed using HTML, CSS, JavaScript, and Chart.js, presents exercise performance, workout history, and fitness progress through graphical visualizations. This enables users to monitor their activities, understand progress patterns, and maintain consistency in their fitness routines. The proposed system is designed to operate using a standard computer and webcam, reducing the need for specialized hardware and making fitness assistance more affordable and convenient. By combining computer vision, personalized nutrition, data visualization, and AI-based interaction, the system provides a unified digital fitness platform that supports exercise performance, healthy lifestyle practices, and continuous fitness tracking.

153. A COMPUTER VISION-BASED ADAPTIVE STERILIZATION SYSTEM FOR SURGICAL INSTRUMENTS USING UV-C AND CHEMICAL DISINFECTION

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This study presents a smart sterilization system that aims to make the disinfection of medical instruments safer and more efficient in healthcare facilities. Standard sterilization procedures often are error-prone due to human factors and lack of flexibility for different kinds of instruments, which in turn causes the increase of hospital-acquired infections (HAI). To address such problems, the system employs AI for both object detection and classification using YOLOv8n. It utilizes open datasets for

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training; after recognizing surgical instruments in real-time, it goes on to the automatic selection of sterilization protocols corresponding to the structural category of the instruments. Furthermore, the ESP32 microcontroller is responsible for the sterilization operations which are exposing to UV-C light and spraying of chemicals; the whole process is controlled via a web interface that is user-friendly. The results indicate that the system achieves a classification accuracy of 95.3% and a detection using YOLOv8n. The system that has been developed provides strong reliability for different types of surgical instruments reducing the involvement of humans directly and making the sterilization processes more efficient. Consequently, by automating the process management and guaranteeing full disinfection of the instruments, the chance of hospital-acquired infections becomes very low. The device is a great solution for highly developed hospitals and also for smaller clinics and it is built in a way that future upgrades, like connection with new sensor types and direct microbiological validation will be possible. The availability of such features helps to continue the transition to eco-friendly sterilization methods while still keeping the safety standards at a high level in the changing clinical environment.

154. ANALYTICAL METHOD DEVELOPMENT AND STRESS DEGRADATION STUDY OF SIMVASTATIN API BY ULTRA HIGH-PERFORMANCE LIQUID CHROMATOGRAPHY USING UV AND CHARGED AEROSOL DETECTOR

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Simvastatin is cholesterol-lowering drug which competitively inhibits cholesterol biosynthesis by HMG-CoA reductase [1]. Simvastatin is chemically modified 2, 2-dimethyl butyrate analogue of lovastatin [2]. Simvastatin is orally administered as inactive prodrug in the lactone form which is converted in the liver into its active metabolite β-hydroxy acid [3]. Currently, many method of analysis of simvastatin API has been published, but our approach for the identification of impurities in large scale involves the usage of instruments as UHPLC and CAD coupled with single quad mass spectrometry, gives opportunity to identify the impurities present during its formulation procedure. CAD being more sensitive than UHPLC covers the broad spectrum of impurities. Generating stress degradation profiling gives an additional edge for the identification of the impurities. Simvastatin API was subjected to forced degradation studies under various conditions, including Acidic, basic, neutral, oxidative stress. While UV detection targets the chromophoric core (238 nm), CAD provide mass sensitive, universal detection independent of chemical structure or absorption coefficients, successfully capturing non chromophoric degradation products (such as ring opened hydroxy acid forms or aliphatic fragments) that typically evade UV detection. Chromatographic separation on both systems was achieved using C18 column with the given dimension: L x i. d. (100 x 4.6mm), using volatile gradient mobile phase consisting of 0.05% formic acid in water and acetonitrile. This method has demonstrated specificity, linearity, precision, and making it highly suitable for routine quality control and comprehensive impurity profiling.

155.ANN-BASED CHARGING AND DISCHARGING CONTROL IN DC MICROGRID

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This paper, an artificial neural network (ANN)-based charging and discharging control algorithm is presented for the battery energy management of solar photovoltaic (PV)-powered DC microgrid. The control algorithm has been developed to attain adaptive energy management for the variations in the generation due to solar PV, load demand, battery state of charge (SoC) and bus conditions. An appropriate converter control signal for regulating the battery charging and discharging process has been generated with the help of the ANN controller with electrical parameters such as battery voltage, battery current, SoC, PV power, load demand and DC-bus voltage. The training of the control algorithm is based on typical operating data under various operating conditions of the microgrid. A model of the system has been created and analysed through simulation to evaluate its performance with varying levels of irradiance, load demand and battery SoC. The results reveal the adaptive control of battery power flow with the stability of DC- bus and efficient utilization of solar energy.

156.AI-BASED FAULT CLASSIFICATION AND REAL-TIME MONITORING OF MULTIPLE SOLAR PV PANELS

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This paper purpose that increasing deployment of solar photovoltaic (PV) systems has created a growing need for reliable fault detection and monitoring to maintain energy efficiency, system safety, and continuous power generation. This paper presents an AI-Based Fault Classification and Monitoring System for Multiple Solar PV Panels that enables automatic identification and monitoring of common PV panel faults. The proposed system collects electrical and environmental parameters such as panel voltage, current, power, irradiance, and temperature from multiple PV panels. These measured parameters are processed and provided to an artificial intelligence-based classification model for identifying fault conditions such as open-circuit faults, short-circuit faults, partial shading, abnormal temperature, and performance degradation. The classified fault information is continuously monitored through a centralized monitoring interface, enabling early detection and rapid maintenance decisions. The proposed approach reduces dependence on manual inspection and improves the reliability and operational efficiency of PV installations. Simulation and experimental analysis can be used to evaluate the classification accuracy, response time, and monitoring performance under normal and faulty operating conditions. The system provides a scalable and cost-effective solution for intelligent condition monitoring of multiple solar PV panels and can be extended to larger grid-connected and standalone PV installations.

157.FAIRTURN: A HEURISTIC SCHEDULER FOR MULTI-AGENT VOICE-BASED GROUP DISCUSSION SIMULATION

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Placement group discussions ask a lot of candidates following several viewpoints at once, knowing when to jump in versus hold back, and staying sharp under time pressure. Most prep tools don't actually train that. You get static Q&A banks, AI tools that simulate a single interviewer, or human-run mock GDs that depend on getting enough people in a room and end up wildly inconsistent. This paper builds something different: a real-time AI group discussion simulator where one human candidate sits with five separate AI participants plus a moderator AI, all talking through live voice. Each AI decides for itself when it wants to speak, based on how much it's already talked and how strongly it wants to contribute right now talk too much and you get penalized, show initiative and you get rewarded. There's also a mechanism that stops AI speakers from talking over the human the instant the person starts speaking, any AI response already in progress just gets cut. After each session, the system reports back on speaking time, how often the person spoke out of turn, and how much leadership they showed. We walk through how the whole thing works below, and close with where a learned turn-taking model could take this next.

158. AN EXPLAINABLE AI FRAMEWORK FOR PERSONALIZED INVESTMENT SUITABILITY AND BROKER RECOMMENDATION FOR BEGINNER INVESTORS

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Beginner investors often find it difficult to compare investment products, because suitability depends on how factors such as risk tolerance, investment horizon, liquidity needs, and expected return interact with an individual's financial situation rather than on any single product attribute. Many existing robo-advisory and recommendation tools present investment suggestions without an accessible account of the reasoning behind them, which can leave novice users unsure whether a suggestion actually reflects their circumstances. This paper proposes a web-based Explainable AI framework designed to support beginner investors, with an initial focus on the Indian retail investment context, through profile-driven and transparent investment and broker recommendations. The proposed framework is designed to collect investor profile information, assess risk characteristics, compute separate investment suitability and broker suitability scores, and combine these scores through a weighted recommendation approach. Rather than relying solely on a post-hoc, model- agnostic explainer, the framework is designed to generate a factor-based explanation that traces each recommendation back to the profile and product attributes that contributed to its score. The system is planned to be implemented using a React/Next.js front end, a FastAPI backend, PostgreSQL for storage, and Scikit-learn for the machine learning components, with Random Forest under consideration for risk classification. As of this writing, the dataset, backend implementation, and empirical evaluation are still in progress; this paper therefore documents the problem formulation, proposed architecture, and planned methodology, and reports the current evaluation status rather than final experimental results.

159. AI-DRIVEN SMART WHITE CANE USING RASPBERRY PI 5 FOR REAL-TIME NAVIGATION AND ENVIRONMENTAL AWARENESS

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Safe, independent navigation remains a major challenge for visually impaired individuals. The conventional white cane is inherently reactive and contact-dependent: it detects only obstacles its tip touches, cannot sense elevated obstacles or wet surfaces, and provides no situational awareness, constraining reaction time in unfamiliar or crowded environments. This paper presents an AI-driven Smart White Cane on the Raspberry Pi 5 that performs pre-contact obstacle detection with real time, multimodal audio–haptic feedback. The device integrates three complementary sensors an ultrasonic ranging sensor, an infrared (IR) proximity sensor, and a water-detection sensor under a priority-based arbitration scheme in which a wet-surface flag takes precedence over distance- threshold obstacle detection. A proximity-graded alerting module maps the obstacle distance and the wet-surface flag to multiple buzzer and vibration levels so that nearer hazards produce more urgent cues. On-device object recognition is performed by a MobileNetV2 classifier [17] deployed with TensorFlow Lite, trained by transfer learning as a frozen feature extractor on a 3,132-image custom dataset spanning ten obstacle classes (chair, door, fence, garbage bin, obstacle, plant, pothole, stairs, table, and vehicle); recognition is triggered when an obstacle is within 80 cm and is stabilized by a three- frame majority vote with a 0.60 confidence threshold. The prototype was evaluated over 62 controlled indoor and outdoor navigation trials. The obstacle-detection subsystem achieved an accuracy of 85.48%, a precision of 95.56%, a recall of 86.00%, and an F1-score of 90.53%, with a false- detection rate of 4.44% and a mean response time of 0.84 s (range 0.42–1.51 s). Multi-sensor operation outperformed either sensor alone, raising detection from 80% (ultrasonic only) and 76% (IR only) to 86% while cutting false alarms from five and six to two. The on-device recognizer attained a validation accuracy of 83.12% across the ten classes, and the prototype operated for 8–10 h on a 10,000 mAh battery. Across these controlled trials, multi-sensor operation combined with lightweight on-device vision produced higher detection and fewer false alarms than single-sensor operation while running entirely on low-cost, portable hardware; validation with blind and low-vision users remains future work.

160. ARTIFICIAL INTELLIGENCE IN RECRUITMENT AND INTERVIEW EVALUATION: A SYSTEMATIC REVIEW AND THE NEXUS FRAMEWORK

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The increased use of AI technology in recruitment makes clear the necessity of intelligent systems that will enable personalized interviewing, candidate assessment, and skill gap detection. Recent studies focus on the following issues: resume information extraction, resume-to-job matching, automatic questions generation, LLM-based interviewing, adaptive questioning, answer evaluation, coding assessment, and multimodal interviewing. Nevertheless, the current state of research is rather fragmented with regard to individual recruitment steps and suffers from the lack of integration, generalization, explainability, adaptive evaluation, fairness, and end-to-end evaluation. This paper aims at conducting a systematic literature review of 20 papers, which are analyzed in terms of their methodology, data sets, evaluation measures, and contribution to intelligent recruitment and interviewing. In particular, the following six gaps are revealed: fragmentation of the recruitment steps, lack of personalization of resumes to interview, insufficiency of adaptive questioning, heterogeneous evaluation approaches, lack of explainability and traceability, and fairness and privacy issues. Based on the results of the analysis, a literature-derived conceptual architecture of the proposed NEXUS (Next Generation Expert Interview and Evaluation System) is suggested, which includes Resume-JD matching, candidate skill profiling, personalized and adaptive interviewing, AI-based answer evaluation, coding assessment, performance assessment,and skill-gap based learning recommendations.

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161. MAGNETIC SMART NANO BRICKS

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The current invention relates to a construction material that includes magnetic nanoparticles for monitoring the structures condition. The construction material includes brick matrix along with Fe_3O_4 magnetic nanoparticles, where these magnetic nanoparticles are synthesized by employing appropriate iron material recovered from waste printed circuit boards. The iron material recovered from waste printed circuit boards is treated to get a suitable precursor material for magnetic nanoparticle synthesis, which is then included in the brick matrix in the proper proportion. The inclusion of magnetic nanoparticles ensures the provision of an extra magnetic response to the developed construction material. A magnetic scanning system is arranged to monitor any change in the magnetic response of the material upon application of mechanical stress or crack formation. These magnetic variations are further analyzed to find out whether there is any change in the materials structure and its approximate location.

162.AI-BASED LIVESTOCK BREED RECOGNITION SYSTEM USING MULTI-HEAD CNN

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Knowing which breed an animal belongs to affects its milk yield, its tolerance of heat and disease, and the breeding decisions a farmer makes. In most Indian villages, though, that call is still made by an experienced person looking at the animal, and two observers often reach different answers. We built a web-based tool that lets a farmer photograph a cow and receive a breed prediction on a phone or computer. The current version recognises five breeds found across the Indian subcontinent: Gir, Holstein, Jersey, Red Sindhi and Sahiwal. Its classifier starts from a MobileNetV2 network pre-trained on ImageNet. We first train only a small classification head, then unfreeze the last 30 layers of the base network and fine-tune them at a much lower learning rate, using class weighting to offset uneven image counts. Before any prediction is made, a Laplacian-variance test rejects blurred photographs and asks the user for a clearer one. When the model is torn between two breeds, the interface reports the two most likely breeds with confidence scores, which suits crossbred or poorly framed animals better than a single forced label. The classifier sits behind a FastAPI service, a React interface, Firebase sign-in and a MongoDB store of past predictions. This paper reports the completed first phase of the work: the system design, the data preparation pipeline and the end-to-end integration of the model. Full training, held-out testing with precision, recall, F1-score and confusion-matrix analysis, and field trials are planned for the second phase.

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163. ENERGY-EFFICIENT CRYPTOGRAPHIC FRAMEWORK FOR RESOURCE-CONSTRAINED IOT ENVIRONMENTS

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The rapid expansion of digital systems has significantly increased global energy consumption, raising critical concerns regarding the sustainability of computation- intensive technologies. Although cryptographic operations are indispensable for ensuring confidentiality, integrity, and authentication, their computational complexity introduces substantial energy overheads. Consequently, embedding energy efficiency as a first-order design constraint in cryptographic systems has become imperative. Green cryptography addresses this challenge by minimizing power consumption while preserving rigorous security guarantees. In this paper, we propose a VLSI-inspired optimization framework for sustainable cryptography that systematically evaluates workloads across algorithmic, architectural, and hardware layers. The framework leverages lightweight primitives, instruction-set architecture (ISA) enhancements, and parallel processing platforms to mitigate computational overhead. Evaluated across key application domains including the Internet of Things (IoT), cryptocurrency ecosystems, and quantum computing environments the proposed model demonstrates that coordinated, cross-layer software–hardware optimization substantially improves energy efficiency while maintaining robust security, making it highly suitable for future sustainable digital infrastructures.

164.INTELLIGENT AND SECURE TASK SCHEDULING FOR DIGITAL TWIN-ENABLED EDGE–CLOUD NETWORKS

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Secure task scheduling in digital twin enabled edge-cloud networks is challenging as latency sensitive workloads must be offloaded across heterogeneous user devices and nearby multi-access edge computing (MEC) servers while minimizing exposure of sensitive data to untrusted environments. This

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bidirectional physical-virtual mapping allows proactive risk detection and context aware scheduling decisions which are not achievable in conventional edge or IoT systems. Conventional schedulers which optimize only for completion time or energy ignore the dynamic threat state of the infrastructure which can leak data, incur security penalties, and miss deadlines. A Random Forest based AI risk engine classifies tasks and nodes into safe, trained, and critical risk levels with confidence scores which influence scheduling, optionally with a cloud large language model generating natural language explanations. The system is implemented as a Django and MySQL application with a real-time dashboard and is evaluated against random, greedy, and privacy aware baselines on reproducible seeded workloads with success rate, average latency, energy, privacy exposure, security risk metrics. Therefore, in this project we present SecureDT, a secure and privacy aware task scheduling framework which uses digital twin technology to maintain a continuously updated virtual representation of physical edge-cloud entities. The scheduler, based on a genetic algorithm, minimizes a weighted objective consisting of latency, estimated energy consumption, privacy loss, security risk, and a quadratic deadline penalty, with offline nodes never eligible for task assignment. A MEC agent with pairing-code registration, rotating bearer token authentication, heartbeat based lifecycle management, and automatic reconnection performs distributed task execution, with Google Drive as the object storage backend proxied through the server so cloud credentials never reach edge nodes. Results show that the security aware genetic algorithm improves reliability, privacy, and threat mitigation at a small latency cost, validating security embedded intelligent scheduling.

165.ATTENTION-GUIDED MULTI-SCALE EFFICIENTNET-B4 AND VISION TRANSFORMER FUSION FOR MULTI-CLASS SKIN LESION CLASSIFICATION

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Skin cancer is one of the most prevalent and life-threatening diseases worldwide, and early and accurate diagnosis is critical for improving patient survival. Automated skin lesion classification remains challenging because of high inter-class similarity, intra-class variation, and inconsistent illumination, color, and texture across dermoscopic images. This study proposes an attention-guided hybrid deep learning framework that integrates a multi-scale EfficientNet-B4 backbone, a projection-first convolutional block attention module (CBAM), a cross-scale attention mechanism, learnable scale weighting, a residual multi-scale fusion block, a ViT-Small transformer branch, and a learnable fusion gate followed by a residual multi-layer perceptron (MLP) classifier. EfficientNet-B4 extracts low-, mid-, and high-level feature maps (56/160/448 channels) that are projected to a common 256-dimensional space, refined by CBAM, exchanged through cross-scale attention, adaptively weighted, and fused with residual learning, while ViT-Small independently produces a 384-dimensional global representation; a learnable fusion gate combines both streams into a 640-dimensional feature for classification. The network is trained end-to-end with focal loss, label smoothing, class- balanced sampling, and a two-stage transfer-learning schedule (5-epoch backbone freeze followed by full fine-tuning) using AdamW and cosine-annealing with warm-up. On the balanced, seven-class HAM10000 dermoscopic dataset (7,049 test images), the proposed 43.9-million-parameter model achieved a test accuracy of 98.98%, macro precision of 98.98%, macro recall of 98.98%, macro F1-score of 98.98%, and macro-average AUC of 99.79%, with only minor confusion between melanoma and melanocytic nevi. These results indicate that the proposed attention-guided CNN–Transformer architecture is an effective, reproducible, and computationally efficient solution for automated skin lesion diagnosis—trainable end-to-end on a single 6 GB GPU—with strong potential to support dermatologists in clinical decision support.

166. AI-HEAL: AN INTELLIGENT AI-BASED HEALTHCARE MANAGEMENT AND DIAGNOSTIC SUPPORT SYSTEM

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AI-HEAL: An Intelligent AI-Based Healthcare Management and Diagnostic Support System is a comprehensive integrated healthcare management system using intelligent digital automation to manage Patient Care, consultation with Doctor Emergency Medical Records Diagnostics Hospital Billing, etc. The AI-HEAL has two distinct modules for approach Patient Login/ Registration & Doctor Login/ Registration. Each module brings safe access to the respective users and manages the respective organization's workflows. The patients can book their regular appointments and emergency appointments using date and time based appointment scheduling and queue management. The patient saves to a single centralized database the entire medical history (previous doctor consultants, medical details, laboratory reports, scan reports, medical prescriptions and other healthcare reports, etc.) which is securely accessed by the doctors during consultation to make clinical decisions. The AI-HEAL manages Identification of medical department. This ensures/induces patients being directed to the relevant medical department & Medical Consultant for their concerned treatments. The Emergency Slot & Queue Management Module: encourages the display of urgent cases, shortens waiting time by maintaining queue of patients based on appointment and emergency approval. Storage and display of Diagnostic and Scan Report Module: enables storage and display of investigation reports and scan reports and reference at the time of consultation. Linking the consultation data with medical shop/pharmacy and automated invoicing and billing module: prescription based processing of medicines and bill creation, and automate billing and invoicing can be efficiently generated. AI-HEAL overall offers a single digital health platform, which integrates all possible users like the patients and the physicians, through diagnostic channels, with various MD departments, pharmacy systems and billing systems. This integrated management healthcare architecture improved by AI-based diagnostic decision support seeks to increase access, decrease human effort, systematize miscellaneous medical knowledge and information, and make health care more rapid and efficient.

167. MEDSIMPLIFY AI: A FULL-STACK AI-POWERED ASSISTANT FOR PRESCRIPTION ANALYSIS, MEDICAL REPORT SIMPLIFICATION, AND EMERGENCY FIRST-AID GUIDANCE

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Prescriptions and diagnostic notes are delivered in clinical shorth and unintelligible to the average patient, resulting in poor adherence and slow response time to abnormal data. MedSimplify AI is a Cross-Platform(Full-Stack) MERN(MongoDB, Express, React, Node.js) application with optical character recognition(OCR), user-specified large language model(LLM) simplified conversion, and structured curated first-aid knowledge base behind an authenticated REST API. Report images are normalized through optical magic using a local machine-created library of medicine names, while report text is rephrased in plain language, and urgent hints retrieved through keywords without LLM use. User data protection is guaranteed using JWT sessions, and environment-based configurations. Architecture, steps, validation plan and constraints are discussed.

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168 .A PHASE-SYNCHRONY MODEL OF DÉJÀ VU GROUNDED IN DUAL PROCESS THEORY

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This paper presents a mathematical model for the phenomenon of déjà vu using existing equations which are modified with novel variables. While déjà vu has been studied extensively in psychology, no existing work offers a quantitative or mathematical framework for modelling it. This paper addresses that gap by using the Mamba-Hawkes equation as its base. We model déjà vu using the "tape-recorder" or "dual-processing" scientific theory which states that two different memory related processes that normally work synchronously become asynchronous leading to a false sense of familiarity. A simulation is run over 50 people whose data is synthetically generated to randomize the dataset. Graphs are plotted for the dual process for different scenarios to conclude which factor contributes more significantly. The results from this graph prove what we are already aware through psychology, as familiarity increases and recall decreases, confidence increases leading to higher probability of déjà vu.

169.BOREHOLE AND SPT-BASED CHARACTERIZATION OF SOIL–ROCK STRATIGRAPHY ALONG A LONG ECRF DAM CUT-OFF WALL ALIGNMENT

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The foundation records along a long dam cut-off-wall alignment show substantial changes in overburden thickness and rock condition. This paper synthesizes the reported borehole descriptions, soil-layer Standard Penetration Test (SPT) values, total core recovery and Rock Quality Designation (RQD) across chainages 0–1750 m. In the 0–450 m stretch, reported soil thickness is 12–49.5 m; it increases to 58–84 m between Ch. 450 and 600 m. Farther along the alignment, bedrock depths vary from about 3 to 37 m between Ch. 1085 and 1510 m. The Ch. 1005–1085 m stretch has silty sand over stiff to very stiff clay and medium-dense to dense sand, with rock at about 39–41 m. In the final stretch, Ch. 1510–1750 m, two boreholes record rock from ground level. The standard penetration number ranges are N = 3–61 for Ch. 0–450 m and N = 3–93 for Ch. 450–600 m. RQD and core recovery are evaluated separately. The results highlight why soil penetration resistance, refusal observations and rock-core indices must not be combined into a single numerical series.

170 . Q-ELASTICITY: A CAPABILITY-CENTRIC FRAMEWORK FOR MEASURING QUANTISATION SENSITIVITY IN LARGE LANGUAGE MODELS

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Post-training quantisation reduces the deployment cost of large language models (LLMs), but its impact on model capability is measured using aggregate benchmark scores that obscure the degradation of individual capabilities as quantisation increases. In this paper, we introduce Q-Elasticity, a unified metric to measure per-capability behavioural degradation across quantisation levels. Q-Elasticity is defined as the average absolute change in performance per bit across successive quantisation levels. We evaluate four open-weight LLMs—Llama-3-8B, Qwen2.5-7B, Qwen2.5-3B, and Mistral-7B-v0.3—across four quantisation tiers (FP16 to Q2_K) on four capability benchmarks: coding (HumanEval), general knowledge (MMLU), truthfulness (TruthfulQA), and safety (SafetyBench). Results show that capability degradation under quantisation is highly non-uniform across models. Safety is the most robust capability across all models, while coding is the most fragile for most models, exhibiting the sharpest decline at lower bit widths. Mistral-7B-v0.3 deviates from this trend, with truthfulness degrading more severely than coding. A chi-square analysis of answer-position distributions further reveals that positional heuristic bias in multiple-choice evaluations becomes more pronounced under aggressive quantisation. Q-Elasticity provides a model- and benchmark-agnostic framework for quantifying capability robustness across quantisation levels, enabling more informed deployment decisions than those based solely on perplexity or aggregate benchmark scores.

171.THE HUMAN-AI NEXUS IN HIGHER EDUCATION: GENZ ENGAGEMENT, IDENTITY AND ETHICAL AWARENESS

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Generative artificial intelligence (GenAI) is increasingly embedded in students' educational and everyday activities, raising questions about identity, human interaction, ethical awareness and well-being. This study examines these human-centred outcomes among Generation Z college students using a quantitative cross-sectional survey. Of 219 coded responses, 216 GenAI users were retained for inferential analysis. Six constructs were assessed: AI engagement, AI dependency, identity/authenticity, human interaction, ethical awareness and well-being. Cronbach's alpha values ranged from .799 to .905. Pearson correlations were positive and significant across all six constructs. Multiple regression showed that AI engagement significantly predicted all four outcomes, while AI dependency significantly predicted identity/authenticity, human interaction and ethical awareness, but not well- being. The findings support a human-centred approach to responsible GenAI use in higher education. shape ideas edit their writing solve problems and support their work. These uses can boost productivity and confidence.. Overuse may also reduce how much students think on their own or reach out to others for help. Recent studies are starting to look at AI dependency among university students. Evidence shows that unhealthy use of AI is linked to stress and high expectations about AI performance. Students themselves say they worry about creativity, misinformation and weaker independent thinking (Zhang et al. 2024). This study responds to this growing line of research by focusing on four human outcomes: identity and authenticity human interaction, ethical awareness and well-being. Identity and authenticity matter because students often use AI suggestions while trying to keep their voice and ideas intact. Human interaction is important because learning is not about sharing facts. It also involves

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relationships, empathy and social connection. Ethical awareness is relevant because using AI responsibly means paying attention to checking facts protecting privacy and knowing where help should stop. Well-being is essential because technology use can affect students' lives in both good and bad ways. Some research has already linked ChatGPT use to student life satisfaction and academic experiences. The findings are limited to specific situations and need more solid evidence (Rehman et al. 2024). This study therefore. Tests a simple human-centered model. It looks at how AI engagement and AI dependency affect identity and authenticity human interaction, ethical awareness and well-being among college students, from Generation Z. Of assuming generative AI is automatically good or bad the study lets the data reveal which connections are actually supported

172. PERFORMANCE EVALUATION OF NANOMODIFIED VG-30 BITUMEN FOR FLEXIBLE PAVEMENT APPLICATIONS

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Bitumen is the key binding material in flexible pavement construction, and its performance directly affects the durability and service life of roads. Conventional bitumen, however, is vulnerable to rutting, temperature fluctuations, and aging, which reduce pavement quality under heavy traffic. This study investigates the use of Iron Oxide (Fe_2O_3), Aluminium Oxide (Al_2O_3), and Zinc Sulfate ($ZnSO_4$) as nanocomposite modifiers in VG30 bitumen to enhance its physical properties. Modified binders were prepared by adding 2%, 4%, and 6% of each nanocomposite by weight of bitumen under controlled mixing. Their performance was evaluated using penetration, softening point, ductility, and viscosity tests in accordance with Indian Standard specifications. Results showed that nanocomposite modification significantly improved binder properties. Penetration values decreased with higher modifier content, indicating greater stiffness. Softening point and viscosity increased, reflecting improved resistance to hightemperature deformation. Ductility also improved, suggesting better flexibility. Among the three modifiers, aluminium oxide (Al_2O_3) consistently provided the most balanced enhancement, followed by iron oxide (Fe_2O_3) and zinc sulfate ($ZnSO_4$). Overall, the findings confirm that nanocomposite modification is an effective approach to strengthening bitumen. This method offers a promising pathway for developing durable, highperformance, and sustainable flexible pavements capable of meeting modern traffic demands.

173.A MATHEMATICAL FRAMEWORK FOR QUANTIFYING AGITATION IN PALLIATIVE SEDATION THERAPY

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In this study, we present a novel mathematical model designed to analyze sedative drug concentrations within the central and peripheral nervous systems during palliative sedation therapy (PST). The aim of

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the model is to optimize dosing intervals to minimize patient agitation while ensuring effective sedation. By simulating drug dynamics and the agitation-sedation cycle, we provide insights into how different dosing regimens can impact patient comfort. This research addresses a significant gap in current literature regarding the quantitative assessment of sedative distribution and its correlation with patient agitation levels, ultimately contributing to improved practices in end-of-life care.

174.A GENERATIVE AI-BASED TEST CASE GENERATION AND INTELLIGENT SOFTWARE TESTING

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Software testing depends on the ability to interpret software requirements and convert them intosuitable test cases. As requirement sets become larger and more frequently modified, manual test-case preparation becomes repetitive and difficult to maintain. This paper describes a Generative AI-based system, which aims to transform natural- language Software Requirements Specifications (SRS) into structured test cases, while keeping validation and human verification within the loop. The system utilises Natural Language Processing (NLP) for extracting relevant requirement information and potentially ambiguous expressions, which are subsequently interpreted by prompt- and template-based Large Language Models (LLMs). Generated test cases are linked to their source requirements to support traceability and are further organised through prioritisation, ambiguity analysis, validation, and export functions. In a project demonstration, the system generated 106 test cases, identified 50 potential ambiguity observations, and prioritised 233 test cases. These results demonstrate the operation of the integrated workflow; however, they are demonstration-levelobservations and do not establish superiority overexisting testing approaches. Human validationremains necessary to ensure that generated casescorrectly reflect the intended software behaviour.

175.INFORMATION-PRESERVING ADAPTIVE CANDIDATE SELECTIONFOR UNDERWATER IMAGE ENHANCEMENT

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Underwater images are frequently degraded by wavelength-dependent light attenuation, scattering, nonuniform illumination, and color distortion, which reduce contrast, alter color distributions, and obscure scene details. Many existing enhancement methods produce a single output for every input, even though the severity of degradation varies considerably across scenes. This paper presents an information-preserving adaptive candidate-selection framework. Instead of predicting one enhanced

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image, the method generates three candidate responses of different enhancement strengths (Conservative, Balanced, and Aggressive) from a learned condition representation of the input image using candidate-specific residual branches. Each candidate is represented by a learned candidate feature and characterized by feature-based preservation and reconstruction-based consequence measures, while reference-image fidelity provides the training target for candidate selection. A condition-aware selector produces the candidate assignment. Experiments were conducted on 3,700 paired images from the Underwater ImageNet paired subset of the EUVP dataset with a fixed training, validation, and test split. On the held-out test set of 370 images, the final V7 model achieved a PSNR of 21.2027 dB, an SSIM of 0.7896, a UIQM of 1.6703, and a UCIQE of 5.6111, with a preservation score of 0.9651. It selected Conservative, Balanced, and Aggressive candidates for 13.78%, 16.76%, and 69.46% of the test images. The results demonstrate input-dependent candidate selection and indicate the potential of conditioning enhancement strength on the input image condition.

176.HYBRID MACHINE LEARNING TECHNIQUES FOR OPTIMIZED 5G NETWORK PERFORMANCE

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The rapid growth of devices in generation (5G) that connects the internet apps need a lot of data handling and different kinds of services that has to manage the network and make it work better and more complicated. Traditional ways are not efficient for managing such a large amount of data and predicting attacks. Machine Learning (ML) is a way to handle the data moving through the network and help to make smart predictions, sort things out and improve things. Quantum Machine Learning (QML) is a way of doing calculations to solve problems and improves in learning patterns. This study provides a way to evaluate the performance of algorithms. This method includes ML, QML learning without labels and deep learning to help understand 5G network data and make it work better. In this concept, we will see about how the algorithms of Machine Learning, Quantum Machine Learning and Deep Learning (DL) are going to estimate the attack ratio before any loss occurs in online services and monitor the traffic level.

177. EDUBRIDGE: A CENTRALIZED ACADEMIC RESOURCES AND MENTORSHIP PLATFORM

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Engineering students search for study material, project ideas, internship guidance and mentorship across scattered channels such as YouTube, Google Drive, WhatsApp/Telegram groups and institutional portals, while the experience of senior students is lost after graduation. Existing research and applications address resource sharing, peer recommendation or adaptive learning individually, but rarely combine curriculum-wise resources, senior mentorship, career guidance and analytics. This paper presents EduBridge, a web-based platform that integrates these services for three roles: students, mentors and administrators. The methodology follows a client-server architecture with a React.js front end, a Node.js/Express.js REST back end, a MySQL database, Firebase authentication and storage, a Python/Scikit-learn recommendation component and Power BI analytics. Resources are organised by course, semester and subject, mentors and uploads are moderated by administrators, and recommendations are produced by a hybrid content-based and collaborative method. A working prototype of the user- management module has been implemented, and a feature-level comparison shows that EduBridge is designed to cover all eleven compared features, whereas each compared

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application covers only a subset. A metrics-based evaluation protocol is defined for the recommendation and analytics modules. The work demonstrates a feasible integrated design and a foundation for deployment and large-scale evaluation.

178. AN INTELLIGENT CLINICAL DECISION SUPPORT PLATFORM FOR EARLY CARDIOVASCULAR DISEASE PREDICTION USING DEEP LEARNING

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Cardiovascular disease is a significant health care issue and early risk assessment can help facilitate earlier clinical intervention. In the conventional assessment, physicians need to assess patient history, demographic data, clinical measurements, laboratory parameters, lifestyle factors, sources can make the clinical assessment more complex and automated interpretation difficult. In this paper, HeartCare AI, an intelligent clinical decision support platform for early cardiovascular disease prediction with deep learning and Explainable Artificial Intelligence (XAI) is presented. A structured clinical information and ECG signals are integrated in one prediction workflow. The ECG characteristics are extracted by a Deep Convolutional Neural Network (DCNN), the sequential relationships are modeled by a Bidirectional Long Short-Term Memory (BiLSTM) network, and an attention mechanism highlights informative representations. The learned features are used along with structured clinical data for cardiovascular risk prediction. An important factor analysis of the model output is then performed using SHAP. The platform also supports patient management, clinical evaluation, clinical decision tools, reporting and monitoring capabilities. According to the project material, the system is not meant to replace the physician's diagnosis and is to be used as a support to the physician. The provided project presentation does not include any experimental data that is verified and has been used to build the train-test protocol or the performance results; therefore, this paper does not assign any unsupported numerical results, but concentrates on the proposed technical architecture and evaluation methodology.

179. A CHARNES – COOPER FRAMEWORK FOR TRIANGULAR Q-RUNG ORTHOPAIR FUZZY LINEAR FRACTIONAL PROGRAMMING WITH AN APPLICATION TO PROFIT- TO – COST RATIO TRANSPORTATION PLANNING

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Linear fractional programming models are mainly used to optimize ratio objectives such as profit to cost, but real data are usully imprecise. This work considers a linear fractional programming problem (LFPP) in which the objective and constraint coefficients are triangular q-rung orthopair fuzzy numbers. Using a score function, the fuzzy models is reduced to an equivalent crisp linear fractional problem, which is converted into a linear problem by the Charnes--Cooper transformation and solved exactly by the simplex method. The conditions under which the transformation is valid are stated and proved. The procedure is validated by replicating the crisp solution of a published transportation problem and by solving a published q-ROF example, in which a small numerical inconsistency in the reported solution

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is identified and corrected. A profit- to – cost ratio transportation problem with triangular fuzzy data extended by q-ROF information is then solved, and sensitivity analyses over the rung parameter q, the membership and non- membership degrees, and capacity are described. For the balanced data examined, the optimal shipment pattern is the same for all tested q and degree pairs, while the score- based objective index changes with both the pattern changes only when capacity slack is introduced

180.AN AI-DRIVEN KNOWLEDGE GRAPH FRAMEWORK FOR HERB-DRUG INTERACTION PREDICTION AND SAFEALTERNATIVE HERB RECOMMENDATION USING AYURVEDIC KNOWLEDGE

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Herb–drug interactions (HDIs) are an important concern when Ayurvedic herbal medicines are used along- side conventional pharmaceutical drugs. Existing approaches for HDI analysis often focus on interaction prediction alone and provide limited integration of Ayurvedic medicinal knowl- edge, interaction-risk interpretation, and alternative herb selection. This study presents an AI-driven framework that combines Ayurvedic knowledge extraction, machine learning, and knowledge-graph representation for herb–drug interaction analysis. BioBERT is fine-tuned to identify medicinal entities from the Charaka Samhita, including herbs, diseases, therapeutic actions, and medicinal properties. The extracted information is integrated with structured herb–drug data to construct an expanded dataset of 1,200 interaction records. An XGBoost classifier is then used to categorize interactions into Low-, Medium-, and High- risk classes. For Medium- and High-risk cases, an alternative herb recommendation module identifies therapeutically similar herbs using shared disease associations, therapeutic actions, active compounds, and Ayurvedic properties. The resulting relationships are represented through an interactive knowledge graph to improve interpretation of the predicted interactions. Experimental evaluation achieved an accuracy of 72.5% with XGBoost, which outperformed the other conventional classifiers evaluated in this study. The framework was also deployed as a Streamlit application for interaction prediction, knowledge-graph visualization, and alternative herb recommendation. The system is intended as a research-oriented decision-support framework and not as a substitute for clinical judgment.

181. A SURVEY ON AI-POWERED ENTERPRISE FINANCIAL DECISION INTELLIGENCE: KNOWLEDGE GRAPHS, DIGITAL TWINS, NEURAL-SYMBOLIC REASONING, AND CONVERSATIONAL ANALYTICS

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The rapid convergence of Artificial Intelligence, Knowledge Graphs, Enterprise Digital Twins, and Large Language Models has created an unprecedented opportunity to re-architect corporate financial decision-making. Existing enterprise systems—including ERP, CRM, HRMS, core banking, and supply chain management platforms—generate vast volumes of heterogeneous data that remain fragmented

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across operational silos, rendering holistic, evidence-backed executive decision support computationally infeasible. This survey systematically reviews twenty-five peer-reviewed studies (2023–2026) spanning four thematic pillars: Digital Twin and Knowledge Graph foundations; Financial Knowledge Graph construction and intelligence; Natural language interfaces, Retrieval-Augmented Generation, and Explainable AI; and Enterprise integration, risk analytics, and financial digital twin extensions. The survey critically analyzes the foundational base paper [1], a grain-enterprise financial shared-service digital twin achieving 94.7% prediction accuracy, identifying eight structural limitations that collectively motivate the proposed Enterprise Financial Decision Making Platform. Crucially, this survey bridges the gap between theoretical multi-layer architectures and real-world deployment by detailing a concrete, lightweight Mixture-of-Experts (MoE) implementation stack (incorporating dlt, FalkorDB, GLiNER, SimPy, LangGraph, and LightRAG) orchestrated via localized Q4_K_M quantized expert models under Ollama. This architecture delivers sector-agnostic, non-invasive enterprise intelligence, multi-factor forecasting, neural-symbolic causal reasoning, and conversational executive decision briefs while eliminating multi-million-dollar infrastructure barriers.

182.MULTI-OBJECT TRACKING IN AN INTELLIGENT FACE RECOGNITION ATTENDANCE SYSTEM

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Manual roll calls, RFID cards, and fingerprint scanners Teachers still often use manual roll calls, RFID cards, and fingerprint scanners to record attendance in classrooms. These methods are slow, rely on contact, or are susceptible to proxy attendance, in which a student records presence on behalf of a peer who is not present. Although camera-based systems lessen this burden, they rarely preserve identity continuity across frames when multiple students are visible at once and often rely on static face matching, which is circumvented by printed images and screen replays. The intelligent, contactless attendance gateway presented in this work combines multi-face recognition in real-time with multi-object tracking and multi-cue LIVENESS verification. A refined AdaFace/ARCFACE encoder receives input from an YOLOV11-based face detector and a five-point landmark alignment stage. This encoder generates 512-dimensional identity vectors, which are then compared to an enrolled gallery using an FAISS FlatL2 index. While an LIVENESS layer combines Laplacian-variance texture analysis, HSV chroma statistics, eye-aspect-ratio blink detection, and three- dimensional head-pose variation into an LIVENESS probability, BYTETRACK provides temporal identity consensus. Attendance is only recorded when this probability hits 0.85. The entire pipeline analyzes a frame in an average of 31.4 ms, and the embedding model converges to 99.1% training and 98.6% validation accuracy within 30 epochs. Comparative trials demonstrate improvements over classical, MTCNN- based, and detector-plus-tracker baselines in terms of identification accuracy, spoof rejection, and identity-switch reduction. For automatic attendance in actual classrooms, the suggested architecture provides a workable, scalable, and spoof-resistant base.

183. A SAFETY-AWARE PLC–ROS 2 ARCHITECTURE FOR INDUSTRIAL ROBOTIC PICK-AND- PLACE AUTOMATION

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Industrial cells that combine conveyors with robotic arms usually rely on a Programmable Logic Controller (PLC) for sequencing and protective interlocks, while modern robot software is increasingly built on the Robot Operating System 2 (ROS 2). Joining the two in a small laboratory or factory cell raises practical questions of ownership, interface design, and failure behaviour. This paper proposes a safety-aware architecture for a conveyor-fed pick-and-place cell in which a Bosch Rexroth IndraControl XM21 PLC owns the conveyor, the sensing of parts and the cell state machine, and a ROS 2 system running on an edge computer owns vision- based colour classification and arm motion. The proposed design uses a direct Ethernet link with an OPC UA interface as the primary protocol, adeliberately minimal request–acknowledge signal set, bidirectional heartbeats, and an explicit fault state that stops the belt and withdraws permission for the arm to move. Protective stopping remains on a hardwired safety circuit and is never delegated to the network or to ROS 2. The paper presents the complete architecture, the signal table with detailed semantics, the PLC state machine with transition rationale, the fault-handling rules, the recommended ROS 2 node structure, and a concrete evaluation plan. It is a pure design proposal: the interface has not yet been validated experimentally, and no performance results are claimed. The work is intended as a reusable reference design for similar teaching and research cells that must integrate a conventional industrial controller with modern robot middleware.

184. Q-TRANSPLANT: PRIVACY-AWARE, EXPLAINABLE ORGAN DONOR–RECIPIENT MATCHING WITH QUANTUM-INSPIRED CANDIDATE SEARCH

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Organ transplantation depends on fast, accurate donor–recipient matching, yet coordination between hospitals is still largely manual, and digital systems that expose HLA data raise serious privacy concerns. This paper presents Q-Transplant, a research prototype of a decision-support platform for organ transplant coordination. The system pairs a deterministic, explainable scoring engine (blood group, organ type, HLA compatibility, medical eligibility and patient urgency) with a restricted- access design in which hospitals receive compatibility scores and marker counts, while raw donor or recipient alleles are not exposed during routine matching operations. A separate quantum-inspired candidate-search module, based on amplitude-amplification-style weighted sampling, runs on classical hardware and includes a benchmark harness for comparison with exhaustive search without claiming a quantum speedup. Role-based access control, validated case and emergency state machines, audit logging and an ESP32 emergency alert device complete the platform. Building on graph-based optimization for kidney exchange, the work extends optimization- oriented allocation into a privacy-aware, real- time coordination architecture. The prototype is evaluated using simulated data and implementation-level tests; quantitative benchmark results are not reported because a reproducible execution record was not available at the time of manuscript preparation.

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185. SKILLBRIDGE: AN INTELLIGENT WORKFORCE MANAGEMENTPLATFORM

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Existing familiar job-search portals facilitate vacancy searching and employer-candidate interaction but service marketplaces provide a parallel effort for search and booking of short-term skilled work. Such separation of processes results in duplicated registration effort, limited insight into workers' history, and inability to assess candidate fit or service dependability. SkillBridge is an integrated workforce management platform integrating workers, customers, companies, and admins in a single system. Workers manage professional profile and supply search and application process for jobs and service requests; customers search and discover skilled workers, view profiles and review ratings for selection and booking; companies has vacancy posting, candidate referencing, and recruitment support; admins monitor platform activity. A React.js frontend with FastAPI backend and Supabase PostgreSQL data layer serves the platform, using Supabase authentication and Google OAuth for identity management. An embedded intelligent recommendation component analyses job requirements against candidate attributes (skills, experience, education and certifications) and a Trust Score compiles activity and feedback into a structured dependability metric. The architecture design also focuses on role-based authorization, human verification, and traceability of recruitment decision-making. This delivers an architecture for a scalable workforce ecosystem in which workforce discovery and recruitment are integrated as coherent digital workflow.

186. DEVELOPMENT OF SALT-RESISTANT COASTAL BRICKS USING SALT-DAMAGED AGRICULTURAL WASTE AND DESALINATION BRINE

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Coastal areas have two waste-related issues that are rarely discussed together. Farmland is being destroyed by seawater intrusion, which also leaves behind agricultural biomass that is burnt outdoors because farmers cannot compost it and animals won't consume it. Concentrated brine is being sent back into the sea by desalination plants along the same beaches, silently endangering marine life. This review summarises the current knowledge regarding the use of desalination brine and salt-damaged crop residue as raw materials for bricks. It examines the effects of salt on agricultural waste fibres, the contributions of brine minerals when they are ground into powder, and the function of coconut coir as a reinforcing fibre in clay and cement matrices. Saline-alkali conditions actually remove the waxy layer on plant fibres, allowing them to connect with clay more successfully. This is one recurrent observation that sticks out. After evaporating into mineral powder, brine provides binding chemicals that can increase compressive strength. In several instances, coconut coir reinforcement has increased compressive strength by over 70%. However, it doesn't seem that anybody has mixed coir, brine minerals, and salt-damaged agricultural waste under a saline curing regime intended especially for coastal application. The conceptual framework put out here, which outlines how such a brick may be made and tested using reasonably priced, locally accessible materials, is based on that gap.

187 .A SURVEY ON MULTI-PERSPECTIVE LLM HALLUCINATION DETECTIONUSING BGE-SMALL EMBEDDINGS AND ADAPTIVE FUSION

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Large language models (LLMs) can write fluent, convincing answers to almost any prompt, yet they sometimes get the facts wrong while sounding completely sure of themselves. This is called hallucination, and in real applications it can push people toward poor decisions. Many detection methods depend on outside knowledge sources, extra supervised models, or access to a model's internals, and none of these is available for closed-source commercial LLMs. What is needed is a black-box, zero-resource approach that works from the model's own outputs. This survey reviews the research behind one such approach, multi-perspective consistency checking. It generates several forward responses and backward reconstructed queries, encodes them with SBERT, and blends their cosine similarities through a fixed linear fusion to decide whether an answer is factual or hallucinated. We point out two weaknesses in this design: SBERT may limit how richly meaning is captured, and a fixed fusion coefficient ignores how reliable each perspective is for a given sample. We then describe a proposed framework that replaces SBERT with BGE-small-en- v1.5 and uses confidence-based adaptive fusion to set the weights for each sample. Whether these changes actually improve detection will be tested in controlled experiments.

188. FORGEAI – AUTONOMOUS MACHINE LEARNING ENGINEERING PLATFORM

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ForgeAI – Autonomous Machine Learning Engineering Platform is an intelligent platform developed to automate the major stages of the machine learning development lifecycle. In conventional machine learning development, tasks such as understanding the dataset, identifying data types, handling missing values, selecting suitable algorithms, training models, evaluating performance, and improving models require considerable manual effort and technical knowledge. ForgeAI addresses these challenges by providing an integrated and automated workflow that accepts a user-defined machine learning requirement and an uploaded dataset through a web- based interface. The system dynamically analyzes the uploaded dataset to identify its structure, features, data types, missing values, and target variable without depending on fixed dataset- specific information. Based on the analysis, ForgeAI performs the required preprocessing and prepares the data for machine learning. The platform supports the training and comparison of multiple machine learning algorithms, including Logistic Regression, Random Forest, and XGBoost. The trained models are evaluated using Accuracy, Precision, Recall, and F1-score, while cross-validation is used to examine the consistency of model performance. The model selection process is driven by the generated evaluation results, enabling ForgeAI to identify an appropriate model for the given machine learning task. An important feature of ForgeAI is its autonomous model debugging and improvement workflow, which extends the system beyond basic automated model training. After model evaluation, ForgeAI analyzes the generated performance results to identify potential performance issues and determines whether additional improvement or retraining is required. When improvement is necessary, an appropriate improvement strategy can be applied and the model can be retrained and re-evaluated. When the model demonstrates stable and satisfactory performance, unnecessary retraining can be avoided. The final model, evaluation metrics, metadata, and generated results are stored for future prediction and analysis. The system is implemented using a Next.js and React-based frontend with a FastAPI and Python-based backend, integrating machine learning libraries such as Scikit- learn and XGBoost. Experimental evaluation using a real-world

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customer churn dataset demonstrated the complete workflow of dataset analysis, preprocessing, model training, cross- validation, performance evaluation, model selection, debugging, and prediction.

189.HYBRID RENEWABLE SMART STREET LIGHTING USING LORAWAN FOR ENERGY SAVING

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Street lighting is very important for safety and daily life in cities. Street lighting matters a lot. It keeps cities safe and life running smoothly after dark. But traditional systems still chew through electricity from the grid, racking up big energy bills and lots of maintenance. As cities try to get smarter and greener, there's a push for lighting that uses less energy and reacts on its own to what's happening outside. Everything connects with LoRaWAN, so it's simple to manage even over long distances. Sensors monitor light, motion, and battery power.

190.FROM VISION TO VOICE: A REVIEW OF AI-DRIVEN MULTIMODAL ASSISTIVE SYSTEMS

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Visual impairment can limit independent access to text, objects, and environmental information, creating a need for accessible assistive technologies. Recent developments in Artificial Intelligence (AI) have made possible the development of speech-driven systems that combine visual perception with natural interaction. This paper reviews 25 studies spanning object detection, computer vision, optical character recognition, speech recognition, wake-word detection, text-to-speech, and vision-language models for assistive applications. The technologies are considered with respect to computational efficiency, real-time performance, offline capability, privacy, and robustness. The challenges of multimodal integration, computational cost, latency, contextual understanding, and real-world reliability are discussed, highlighting the need for adaptive, low-latency, privacy- preserving, and context-aware assistive systems.

191.PERFORMANCE ASSESSMENT OF B20– METHANOL BLENDS IN A CI ENGINE UNDER VARIABLE LOAD CONDITIONS

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The present work experimentally evaluates the performance of a B20–methanol blend in compression ignition (CI) engine at different load conditions. This study analyses a CI engine fuelled by diesel, B20, B20 + 5% methanol, and B20 + 10% methanol. The engine performance was tested at 0%,25%, 50% and 100% load conditions as brake power, brake thermal efficiency, mechanical efficiency and specific fuel consumption. The results of the experiments demonstrate a significant impact of engine load on all

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tested fuels. Among the tested fuels, B20 with 5% methanol exhibited favourable performance at medium and high load performance. The brake power achieved at full load was 2.76 kW with B20 +5% methanol, while it was 2.60 kW and 2.57 kW for diesel and B20 respectively. This same blend also achieved the maximum mechanical efficiency of 51.81% at full load. The brake thermal efficiency improved as the load increased, where at full load 27.31% achieved for B20 + 5% methanol and the lowest specific fuel consumption of the methanol-containing blends was 0.29 kg/kWh for B20 + 5% methanol at full load The result indicates that the system appeared to have improved selected performance characteristics of a CI engine with methanol addition into B20 at higher-load operation. The research offers an experimental evaluation of the feasibility of B20–methanol blends as alternative fuels for potential use in CI engine applications.

192. DESIGN AND PERFORMANCE ENHANCEMENT OF A DENIM-BASED 2×2WEARABLE MIMO ANTENNA USING ORTHOGONAL PLACEMENT ANDDEFECTED GROUND STRUCTURE

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This paper presents a comparative study of a denim- based 2×2 wearable MIMO antenna designed to operate at the 2.45 GHz ISM band for wearable biomedical wireless communication applications. Four antenna configurations are investigated by considering two element orientations, parallel and orthogonal, along with two ground-plane configurations, with and without a defected ground structure (DGS). The same patch geometry and substrate characteristics are used for all configurations to provide a consistent basis for comparison. The study investigates how the element orientation and ground- plane modification affect the electromagnetic interaction and mutual coupling between the MIMO elements. The DGS is incorporated to alter the ground-current distribution and provide an additional approach for controlling mutualcoupling. The performance of the proposed configurations is evaluated using important MIMO parameters, including impedance matching, port isolation, envelope correlation coefficient, diversity gain, total active reflection coefficient, channel capacity loss, realized gain, and specific absorption rate. The study provides an understanding of the effects of antenna orientation and ground-plane engineering on the development of compact, low-coupling denim-based wearable MIMO antennas for biomedical wireless communication.

193. LEAN CONSTRUCTION AND CIRCULAR ECONOMY INTEGRATION FOR SUSTAINABLE CONSTRUCTION WASTE MANAGEMENT

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The construction industry remains one of the largest generators of solid waste worldwide, largely because it still relies on linear "take-make-dispose" processes. Lean Construction (LC) and the Circular Economy (CE) have each been proposed as remedies, but they are rarely combined into a single framework for reducing construction waste. This paper develops a conceptual, literature-based framework that integrates LC principles (waste elimination, pull planning, Just-in-Time delivery, standardised work and continuous improvement) with CE R-strategies. Using a narrative literature synthesis, the framework divides the project life cycle into five stages: design, procurement, construction, operation and end-of-life. For each stage it pairs LC principles with CE strategies and identifies the waste reduction mechanism of each pairing. A qualitative synergy map shows the strongest synergies between value stream mapping and design-stage circular strategies such as Design

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for Disassembly, and between standardised work and on-site reuse and refurbishment. An illustrative refurbishment example shows how the pairings translate into project decisions. The paper also discusses barriers to integration, including fragmented supply chains, weak material tracking and misaligned contractual incentives, together with possible enablers. The framework is not yet empirically validated. It offers researchers a basis for testing, and practitioners and policymakers a reference point for action.

194.CARELINK LEDGER: A DECENTRALIZED TRUST INFRASTRUCTURE FOR INTEROPERABLE HEALTHCARE DATA ECOSYSTEMS

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Modern healthcare systems generate vast volumes of electronic health records (EHRs), yet clinical data remains frag- mented across isolated institutional databases with heterogeneous schemas and incompatible ledger platforms. While blockchain frameworks have been explored to enhance integrity and access auditability, direct on-chain storage of high-volume medical files introduces severe scalability bottlenecks, and existing decentralized architectures frequently operate as isolated homogeneous networks. This study presents CareLink Ledger, a layered, de- centralized trust infrastructure engineered for verifiable, patient centric healthcare data exchange across heterogeneous healthcare environments. The architecture decouples heavy clinical payload storage from state verification by combining InterPlanetary File System (IPFS) content-addressed storage with on-chain SHA 256 cryptographic integrity anchoring. Access control and entity verification are governed through modular EVM smart contracts and decentralized identifiers (DIDs) via the Veramo framework, while a dedicated healthcare data-processing pipeline normalizes raw clinical datasets into canonical HL7 Fast Healthcare Interoperability Resources (FHIR) bundles. To bridge disparate institutional ledgers, CareLink Ledger incorporates an eventdriven, application-level cross-ledger relay that synchronizes verified state transitions across Ethereum, Polygon, and Hy- perledger Fabric, and demonstrate the technical feasibility of integrating off-chain storage, standardized clinical modeling, and multi-blockchain synchronization within a unified prototype architecture.

195.ARTIFICIAL INTELLIGENCE TECHNIQUE FOR AUTOMATEDDETECTION OF MULTIPLE EYE DISEASES : A REVIEW

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Eye diseases represent a major global health concern, affecting people in all age groups and often leading to visual impairment or blindness if left undiagnosed. Among the most prevalent conditions are refractive errors, glaucoma, diabetic retinopathy, and cataracts, which together account for a significant proportion of visual disorders worldwide. According to recent World Health Organization estimates, more than 2.2 billion people experience some degree of vision impairment, a considerable portion of which could be prevented through timely detection and intervention. In older adults, particularly those above 60 years of age, cataract, glaucoma, age-related macular degeneration, and diabetic retinopathy

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remain the predominant causes of vision loss, while in children, disorders such as amblyopia, congenital cataract, and refractive errors can severely impact visual development. Accurate diagnosis of these diseases relies on various clinical examinations and imaging techniques, and early identification is essential to preserve sight and improve treatment outcomes. With recent progress in machine learning and deep learning, automated diagnostic systems have emerged as powerful tools for early disease detection. The combination of advanced imaging modalities such as fundus photography, optical coherence tomography (OCT), and scanning laser ophthalmoscopy with artificial intelligence algorithms has markedly enhanced diagnostic precision and efficiency, offering a promising avenue for large-scale, accessible, and reliable eye disease screening.

196 .SUBJECT-INDEPENDENT DEEP LEARNING-BASED EEG MOTOR IMAGERY CLASSIFICATION OF RIGHT-HAND GRASP AND RELAX FOR UPPER-LIMB NEUROREHABILITATION

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Recognising motor imagery (MI) in electroencephalography (EEG) signals remains a core challenge for non-invasive brain-computer interface (BCI) research in rehabilitation. This paper compares three deep learning models, EEGNet-Strong, DeepConvNet-Ultra, and Hybrid-EEGNet- Fusion-V2-TL, for binary REST vs. RIGHT_HAND MI classification. The data comprise 108,276 four-second epochs from 208 subjects in seven public datasets (PhysioNet-MI, Stieger2021, Schirrmeister2017, Weibo2014, AlexMI, Beetl2021-A, and Beetl2021-B), recorded from seven sensorimotor channels at 160 Hz and partitioned 70/15/15 for training, validation, and testing. Each model was evaluated in terms of accuracy, precision, recall, F1-score, area under the receiver operating characteristic curve(AUC-ROC), and Cohen's Kappa, together with confusionmatrix, calibration, and overfitting analyses. DeepConvNet-Ultra achieved the highest performance, with 86.7% accuracy, 86.6% F1-score, 93.5% AUC-ROC, and a Cohen's Kappa of 73.4%. EEGNet-Strong performed weakest (73.7% accuracy, Kappa 47.4%) and frequently misclassified REST trials as RIGHT_HAND. The hybrid transfer learning model reached 81.4% accuracy but showed the largest train-validation gap. All three models were reasonably well calibrated. The results indicate that a deeper residual convolutional architecture outperforms both a compact multi-scale model and a transfer-learning hybrid for this task.

197.INTERACTIVE AUTISM THERAPY ASSISTANT USING ARTIFICIAL INTELLIGENCE AND REALITY

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Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder characterized by difficultieswith communication, social interaction and behavioural regulation. The Conventional Methods lack in providing an immersive interactive environment to develop the cognitive ability of the autistic individual .In this paper we present an adaptive and interactive intervention framework that combines Augmented Reality (AR) with AI systems for personalized therapy. The system gathers data in real time through the use of multimodal sensors . It analyzes facial and behavioral patterns to assess affective cognitive states. Based on the acquired data, the model is trained in a way to help the individual to interact with the AR environment, which aids in improving the responsiveness of the individual in a social environment.

198.AI–BASED MOTOR DETECTIONFOR SPINAL CORD INJURY PATIENT

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Recent developments in biomedical instrumentation, computational intelligence and physiological signal processing allow non-invasive systems to do much better in neuromuscular assessment and rehabilitation applications. Among various bio-signal acquisition methods, Electroencephalography (EEG) and Electromyography (EMG) are widely recognized as reliable techniques for monitoring cortical neural activity and skeletal muscle responses. Current approaches typically rely on uni-modal or multimodal bio-signals combined with machine learning algorithms for movement classification, motor imagery recognition, and rehabilitation support. However, limited research has examined the physiological interplay between cortical activation and peripheral muscular response during voluntary motor tasks. In this paper, we present a framework for multimodal bio-signal analysis that integrates EEG and EMG recordings with advanced signal processing techniques driven by artificial intelligence. The physiological signals obtained are pre-processed, features extracted, synchronized and computationally analyzed to assess functional relationships between neural and muscular activities. This framework aims to strengthen the understanding of neuromuscular communication by applying advanced pattern recognition and data- driven approaches. The methodology delivers a portable, non-invasive, and efficient system with applications in neurophysiological assessment, rehabilitation, and smart biomedical monitoring.

199.A COMPARATIVE STUDY OF CLASSICAL SPATIAL DENOISING FILTERS UNDER GAUSSIAN AND SALT-AND-PEPPER NOISE

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Noise removal is a fundamental pre-processing step in almost every image processing pipeline, yet the choice of denoising filter is often made without a quantitative basis. This paper presents a controlled comparative study of four widely used classical spatial filters, namely the Gaussian, median, bilateral

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and non-local means (NLM) filters, under two common noise models: additive white Gaussian noise (σ = 10 to 50) and salt-and-pepper impulse noise (5% to 30% density). Experiments are performed on six standard grayscale test images, and performance is measured using peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) and execution time. Results show that NLM gives the best restoration for Gaussian noise at every level, reaching a mean PSNR of 33.64 dB at $\sigma = 10$ and improving the noisy input by 11.2 dB at $\sigma = 50$, but it is roughly three orders of magnitude slower than the median filter. For salt-and-pepper noise, the 3×3 median filter is clearly superior, achieving 32.35 dB and SSIM of 0.909 at 5% density, while the bilateral filter performs worst because its range kernel preserves impulses as if they were edges. The study also shows that blind NLM, whose strength is set from a wavelet-based noise estimate, fails at low impulse densities because the estimator underestimates impulse noise. Practical guidelines for selecting a filter based on noise type, noise level and time budget are derived from the results.

200. A SMART CAMPUS RECRUITMENT FRAMEWORK FOR AUTOMATED ELIGIBILITY MATCHING, APPLICATION MANAGEMENT, AND PLACEMENT ANALYTICS

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Campus recruitment is an important process that connects students with employment opportunities through college placement cells. However, traditional placement management often involves manual eligibility verification, application tracking, communication between placement officers and students, and preparation of placement reports. These activities can be time-consuming and may lead to errors, missed opportunities, and difficulties in managing recruitment information. This paper proposes a Smart Campus Recruitment Framework that automates student eligibility matching, job application management, and placement analytics through a centralized web- based platform. The system evaluates student profiles against company recruitment criteria such as CGPA, department, academic backlogs, technical skills, and graduation year. Eligible students can view suitable opportunities and submit applications, while placement officers can manage recruitment drives, verify eligibility, monitor application progress, and analyse placement statistics. The framework also introduces data-driven placement insights to identify skill gaps, compare recruitment outcomes, and support better placement planning. By centralizing recruitment activities and automating repetitive tasks, the proposed system aims to improve transparency, efficiency, and decision- making in campus placements.

201. AI-POWERED IMAGE CAPTION GENERATION USING DEEP LEARNING AND NATURAL LANGUAGE PROCESSING

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Image caption generation is an important application of artificial intelligence that combines computer vision and natural language processing to describe the contents of an image using meaningful sentences. Understanding an image and converting its visual information into a natural-language description remains a challenging task because an image may contain multiple objects, activities, relationships and background details. This paper proposes an AI-based Image Caption Generator that automatically generates descriptive captions from uploaded images. The proposed system uses a deep learning-based image encoder to extract visual features and a language generation model to convert those features into a meaningful textual description. The application is designed to accept an image through a user-friendly interface, process its visual content, and display the generated caption. The generated description aims to identify important objects and their relationships while maintaining grammatical clarity and relevance to the input image. The proposed system may use a pretrained vision encoder and a transformer-based caption generation model to reduce the complexity of training a complete model from scratch. Potential applications include image accessibility, educational support, digital content management, and automatic image description. The system will be evaluated using suitable image-caption datasets and standard evaluation metrics, along with qualitative examination of caption relevance and fluency. As the implementation is proposed, actual model performance and caption quality remain to be established through experimentation. The project aims to provide an accessible image description tool that demonstrates the practical use of artificial intelligence in connecting visual information with human language.

202. AI AND PERMISSIONED BLOCKCHAIN BASED STUDENT CERTIFICATE VERIFICATION AND FORGERY DETECTION SYSTEM WITH LIFECYCLE AWARE EVIDENCE ANALYSIS

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This project proposes a certificate verification platform that combines issuer-controlled digital records with AI-assisted examination of uploaded certificate images. An institution will issue a signed certificate record and anchor its digest and lifecycle events on a local permissioned blockchain. A verifier will check issuer authorization, digital integrity and current status, while an OCR and image-analysis service will identify suspicious field inconsistencies for human review. The contribution is an evidence-based verification report that separates cryptographic verification from visual suspicion and correctly handles corrected or revoked certificates. The college prototype will use fictional student records, a limited certificate template set and a controlled tampering dataset. Evaluation will report verification correctness and false alarms on genuine scans separately.

203 .AI ASSISTED NATURAL LANGUAGE SQL QUERY GENERATION AND BUSINESS INTELLIGENCE SYSTEM WITH SEMANTIC VALIDATION AND TRACEABLE REPORTING

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This project proposes an analytics assistant that converts natural-language business questions into validated SQL and explains the resulting tables and charts. It will operate on a controlled sales database with approved metric definitions and role-specific views. The contribution is a pipeline that asks users to resolve ambiguous business terms, checks for aggregation errors and preserves a reproducible record

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of each answer. A language model will generate candidate SQL, while an independent validator and restricted database account will control execution. The final- year prototype will compare this pipeline with direct text-to- SQL generation using a reviewed question set. Results will be assessed by answer correctness, ambiguity handling, access control and traceability rather than syntactic validity alone.

204 .CLOUD BASED EMERGENCY BLOOD DONOR DISCOVERY AND APPOINTMENT COORDINATION SYSTEM WITH AVAILABILITY AWARE MATCHING

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This project proposes a web application that helps authorized hospital coordinators find and contact consenting blood donors during time-sensitive requests. It combines verified request creation, donor availability, location-based shortlisting, staged notifications and a traceable response workflow. The main contribution is a coordination method that considers the freshness of availability information and prevents the same donor from being committed to competing requests. A React interface, an API service and PostgreSQL will support the college prototype. The system will be evaluated with simulated requests and consenting test users by comparing its coordination performance with distance-only matching. It supports donor discovery and scheduling; clinical donor assessment and blood compatibility decisions remain with the blood centre.

205. DEEP LEARNING BASED CROP PEST DETECTION AND LEAF DAMAGE SEVERITY ESTIMATION SYSTEM WITH IMAGE QUALITY ASSESSMENT AND PERCENTAGE MEASUREMENT

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This project proposes a crop-image analysis system that identifies visible insect pests and estimates pest associated damage on supported leaf images. A detection model will identify pest classes and visible counts, while a separate segmentation model will estimate leaf area and annotated damage area. The reported percentage will mean visible damaged leaf area divided by visible leaf area, rather than model confidence or the percentage of pests in a field. The contribution is an evidence-linked report that presents pest identity, damage percentage, image quality and repeated observations together. The college prototype will cover one crop and a small number of pest classes with a manually annotated damage subset. Evaluation will assess detection, segmentation and severity agreement independently.

206.CYBER THREAT HUNTER: INTELLIGENT CYBERATTACK DETECTION AND PREVENTIVE RECOMMENDATION SYSTEM

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The increasing use of digital technologies, cloud platforms, and network-connected devices has increased the risk of cyberattacks such as unauthorized access, malware infections, brute- force attempts, suspicious network traffic, and data breaches. Traditional security mechanisms may struggle to identify unfamiliar attack patterns and provide timely guidance to users. This project proposes Cyber Threat Hunter, an intelligent cyberattack detection and preventive recommendation system that analyzes system logs, network traffic features, and security events to identify potentially malicious activities. Machine learning techniques are used to classify suspicious behavior, while rule-based threat analysis helps identify known attack indicators. The system evaluates detected threats based on their severity and provides preventive recommendations, such as blocking suspicious connections, strengthening authentication, reviewing access permissions, and isolating affected systems. A centralized dashboard displays detected threats, severity levels, analysis details, and recommended security measures. The proposed system aims to support early threat identification, improve security monitoring, and help administrators make informed decisions when responding to potential cyberattacks.

207 .AI-POWERED SMART PHISHING ATTACK DETECTION AND EARLY WARNING SYSTEM USING MACHINE LEARNING AND NATURAL LANGUAGE PROCESSING

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This project proposes an AI-powered Smart Phishing Attack Detection and Early Warning System designed to identify potentially malicious URLs, emails, and online messages. Phishing attacks have become a major cybersecurity concern because attackers use deceptive websites and convincing messages to obtain sensitive information from users. The proposed system combines machine learning, URL feature analysis, and Natural Language Processing (NLP) to examine suspicious links and message content. It identifies patterns such as unusual domain structures, suspicious characters, urgent requests, and attempts to obtain confidential information. Based on the extracted features, the system classifies submitted content and generates an appropriate risk warning. The platform also provides understandable reasons for its predictions, helping users recognize potential threats before interacting with suspicious content. The proposed system aims to improve phishing awareness, support safer online activities, and provide an additional layer of cybersecurity protection.

208.CLOUD BASED VIRTUAL SCIENCE FAIR MANAGEMENT PLATFORM WITH EVIDENCE LINKED ASSESSMENT AND CONFLICT AWARE JUDGE ALLOCATION

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This project proposes a web platform for conducting a college science fair through digital project booths, structured submissions and coordinated judging. Students will submit research questions, methods, evidence and demonstration material; judges will evaluate frozen submission versions using a published rubric. The contribution is an evidence-linked assessment workflow that connects scores to artefacts, schedules conflict-free judging and records responses to reviewer questions. The prototype

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will use React, an API service, PostgreSQL and object storage, with recorded presentations and asynchronous questions as its core communication mode. Evaluation will compare it with a basic upload-and-score portal using controlled event scenarios. The system supports human scientific assessment and does not assign automated originality or research-quality verdicts.

209. VISUALIZING MULTI-CROP DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORKS AND EXPLAINABLE ARTIFICIAL INTELLIGENCE

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Agriculture plays a significant role in food security and economic development. However, crop diseases can reduce agricultural productivity and affect the quality of agricultural products. Traditional disease identification methods often depend on manual inspection and agricultural expertise, which may delay the identification of diseases. This paper proposes a deep learning- based system titled Visualizing Crop Disease Detection, designed to classify selected crop diseases from leaf images and present the results through an interactive visualization interface. The proposed system uses a Convolutional Neural Network (CNN) to learn image features and classify healthy and diseased leaves across multiple crop categories. The system is designed to display the uploaded image, predicted disease category, classification confidence, and a visual heatmap generated using Gradient-weighted Class Activation Mapping (Grad-CAM). This combination enables users to inspect the image regions that influence the model's prediction rather than relying exclusively on a textual classification result. A web-based interface is proposed to make the system accessible to farmers, agricultural students, and researchers. The methodology includes dataset preparation, image preprocessing, CNN model training, visual explanation generation, and performance evaluation using standard classification metrics. As the implementation is planned, the actual classification performance and practical effectiveness remain to be established through experimentation. The proposed system aims to support preliminary crop disease screening and provide a foundation for interpretable, image- based agricultural applications.

210.AMBIENT-CURED GEOPOLYMER BRICKS FROM FLY ASH AND RICE HUSK ASH WITH PLASTIC WASTE AND COIR FIBRE.

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Fired-clay and cement-based bricks demand a lot of energy and carry a heavy carbon cost, yet fly ash, rice husk ash and plastic waste keep piling up with little useful reuse. This work proposes a different route: a masonry brick that uses no cement and no kiln firing. The brick is made from four waste materials, shaped by cold-pressing and left to cure at ambient temperature. Class C fly ash acts as the main binder source, rice husk ash adds reactive silica, recycled plastic powder is meant to cut the brick's weight, and treated coconut coir fibres provide reinforcement. A sodium silicate and sodium hydroxide solution activates the mix. We will prepare several mix proportions, vary the share of each waste material, and compare the results against conventional fired-clay bricks. Each mix will be checked for compressive strength, water absorption, surface hardness (nail scratch test), durability, efflorescence

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and dimensional tolerance, which together show whether a brick is fit for real masonry work. The goal is to find a mix that gives good strength, durability and dimensional accuracy without cement or firing. If it works, the brick could be a more sustainable option for affordable construction, particularly where agricultural and industrial residues are plentiful, while putting four waste streams to use.

211.ADAPTIVE DUAL-CHAOTIC ECC STEGANOGRAPHY FOR SECURE CLOUD DATA STORAGE WITH INTEGRITY VERIFICATION

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Secure cloud storage of sensitive data requires confidentiality, imperceptibility, and integrity at the same time. This paper presents ADCES, an adaptive dual-chaotic steganographic framework based on elliptic-curve cryptography (ECC) for secure, lossless cloud storage. The framework combines five layers: super-resolution deep neural network (SRDNN) preprocessing, saliency-guided adaptive multi-image steganography, 3D Lorenz chaotic diffusion, ECC encryption over the secp256k1 curve, and SHA-256 chunk-level integrity verification. A Sobel-gradient importance map identifies image regions best suited for embedding, so the process adapts to local image characteristics. The embedded data is then diffused with Lorenz-based pixel scrambling and XOR keystreams, encrypted block by block with ECC, and split into hashed chunks so unauthorized modification can be detected. Evaluation on the DIV2K dataset covers image quality, cryptographic security, steganographic performance, recovery accuracy, and integrity verification.

212. EXPLAINABLE MULTIMODAL DEEP LEARNING FOR AI-DRIVEN DERMATOLOGICAL DIAGNOSIS: A COMPREHENSIVE REVIEW OF CLINICAL DECISION SUPPORT SYSTEMS

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Skin conditions rank among the most common health complaints worldwide, yet diagnosis still depends heavily on visual judgment, a process that is slow, subjective, and hard to scale wherever dermatologists are scarce. This shortage, combined with heavy patient loads and inconsistent clinical assessment, often delays diagnosis or leads to error. Deep learning has begun to change this picture, showing real promise for automating skin disease recognition from images and clinical records alike. In this review we critically examine explainable multimodal deep learning approaches for AI- driven dermatological diagnosis and decision support, drawing on nineteen studies that span convolutional and ensemble networks, attention mechanisms, transformer- and graph- based multimodal models, and explainable AI (XAI) methods that combine visual, textual, and structured data, with particular attention to how explainability builds clinician trust and supports adoption. We compare the studies by data modality, model architecture, evaluation approach, and clinical relevance, and use this comparison to surface recurring gaps, especially around explainability, generalization across diverse patient populations, and real-time clinical integration. Building on these findings, we propose DermaVision AI, a modular, explainable multimodal framework meant to guide future work toward more reliable, deployment-ready dermatological decision support systems.

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213.A SURVEY ON REAL-TIME SIGN LANGUAGE & SPEECH TRANSLATION

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The integration of Artificial Intelligence and computer vision is creating an opportunities to avoid communication barriers between signer and a normal hearable individual. However, despite recent progress, communication between them is still a challenge. Translating continuous hand movements into meaningful and context-aware sentences in real time is technically very difficult to implement. Moreover, most existing systems are one-directional they only focus on recognizing signs and fail to provide a complete two-way conversation where a hearing person can also reply using speech. In this system, we translate the British Sign Language into Text for Normal Hearable users. For the sign language user the voice will be converted into text. It is a two-way communication. To overcome these issues, this paper presents Communication Bridge: An AI-Powered Real-Time Sign Language and Speech Translation System for seamless bidirectional communication. The proposed system uses a camera-based system with computer vision techniques to track hand and body movements. A Convolutional Neural Network (CNN) is used to classify individual gestures, and a Large Language Model (LLM) then converts the recognized sequence of signs into grammatically correct sentences with proper meaning. In the reverse direction, integrated speech-to-text converts the spoken language of a hearing person into text for the sign language user. This system also include numbers and improves real time analysation and provide strong two-way communication which helps a normal hearable users and signer to communicate without barrier.

214. NUMERICAL SIMULATION OF THE SPACE-TIME FRACTIONAL THERMOELASTIC PROBLEM ON A THICK HOLLOW SPHERE

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The present work investigates the thermoelastic response of a thick hollow sphere within the framework of the space-time fractional heat conduction equation. The formulation is developed by utilizing the Caputo fractional derivative α with respect to time and the spatial fractional derivative β with respect to space, enabling the incorporation of hereditary time effects and spatial non-locality within the thermoelastic framework. Initially, the sphere has a non-zero temperature distribution g(r). The inner surface is kept at zero temperature, while a certain amount of heat is supplied to the outer surface for t > 0. The heat conduction equation is solved using the Laplace transform technique. A mathematical formulation is constructed to simulate the thermoelastic response of titanium alloy. The effects of varying the fractional orders of space and time on thermoelasticity are investigated and the results are depicted graphically.

215. ARTIFICIAL INTELLIGENCE ADOPTION AND SUSTAINABLE SUPPLY CHAIN PERFORMANCE IN INDIAN MANUFACTURING: THE PARALLEL MEDIATING ROLES OF OPERATIONAL EFFICIENCY AND GREEN SUPPLY

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CHAIN CAPABILITY. A CONCEPTUAL FRAMEWORK AND RESEARCH AGENDA

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Manufacturers are investing in artificial intelligence (AI) for forecasting, planning, logistics and quality control, yet it remains unclear whether these investments improve sustainability or only lower costs. This conceptual paper examines how AI adoption capability affects sustainable supply chain performance (SSCP) in Indian manufacturing. Drawing on the dynamic capabilities view and the natural-resource-based view, together with an integrative review of empirical research on AI, analytics and sustainable supply chains, it develops a moderated parallel- mediation framework and seven propositions. AI adoption capability is expected to strengthen both operational efficiency and green supply chain capability, and each capability is expected to carry part of the effect of AI on SSCP. The green capability route is proposed to be the stronger of the two, because efficiency gains mainly serve economic outcomes and can be offset by higher output, whereas green capabilities act directly on environmental and social outcomes. Organisational digital readiness is proposed to strengthen both routes. The paper also specifies a 41-item measurement instrument, sample size requirements and a two-wave, multi-informant design using Business Responsibility and Sustainability Report (BRSR) data for testing the framework with partial least squares structural equation modelling. For managers, the framework suggests that AI projects justified on cost and service grounds will not deliver sustainability gains on their own unless environmental and social objectives are built into their design and supported by investment in data, systems and skills. By separating the efficiency and environmental routes and proposing how they compare, the paper moves the debate from whether AI improves sustainability to how it does so.

216. A SURVEY OF MACHINE LEARNING AND DEEP LEARNING TECHNIQUES FOR ORAL CANCER PREDICTION

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Oral cancer is a serious disease in which early and accurate detection is essential for improving treatment outcomes. Recent studies have explored artificial intelligence, machine learning, and deep learning techniques to improve automated oral cancer prediction and detection. This survey compares five approaches with different strategies for improving diagnostic performance. Patel and Kumar assessed various preprocessing techniques with CNN, SVM, and Random Forest models and found normalization to be the most effective method. Kulkarni et al. compared VGG16, ResNet50, LeNet-5, MobileNetV2, and InceptionV3 for oral squamous cell carcinoma classification, with MobileNetV2 demonstrating the strongest performance among the evaluated models. Firdaus and Raza introduced federated learning using CNN with FedAvg and FedProx to address data privacy and enable decentralized model training across multiple clients. Kumar et al. proposed Multi-Layer Visual Feature Fusion (MLVFF), combining Local Binary Pattern features with CNN to strengthen discriminative feature representation for oral cancer image classification. Dharani and Danesh developed an optimized

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ensemble combining Enhanced EfficientNet-B5 and ResNet50V2 with attention mechanisms and Tunicate Swarm Algorithm optimization for improved oral cancer classification.

217. SAFEBUY AI: AN INTELLIGENT ONLINE SHOPPING SCAM DETECTION SYSTEM

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With the rapid growth of ecommerce, the prevalence of phishing websites, fake products, and price manipulation has also increased. This paper presents SafeBuy AI - an intelligent scam detection system which utilizes supervised machine learning, natural language processing (NLP) and statistical anomaly detection techniques to provide real-time consumer protection for online shoppers. Logistic Regression, Decision Tree and Random Forest models are trained on curated datasets of labeled shopping to-end detection pipeline that includes URL-level phishing analysis, product-listing text classification, and market-price anomaly detection. The system automatically distinguishes fraudulent from legitimate online shopping experiences at both the URL and content levels while maintaining computational latency acceptable for browser-integrated deployment.

218. INTEGRATED CAD–CFD–FEA-BASED DESIGN OPTIMIZATION OF A QUADCOPTER UAV FOR AERODYNAMIC EFFICIENCY AND STRUCTURAL PERFORMANCE

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This paper shows a multiphysics simulation framework that helps design and improve aerial vehicles or drones. The method mixes computer-aided design (CAD) computational fluid dynamics (CFD) and finite element analysis (FEA) into one solid engineering workflow. I chose a four-arm quadcopter as the example. I built its shape with SolidWorks and ANSYS SpaceClaim. Using ANSYS Fluent I ran CFD simulations to measure drag, lift, rotor downwash interference and flow separation at speeds from 0 to 25 m/s and angles of attack from –10° to +20°. With ANSYS Mechanical I performed FEA to check the strength of the frame motor mounts and aluminium parts when they face gravitational and vibrational loads. The simulation results point out where the stress is highest, where the aerodynamics could be better and what the vibration looks like so I can tweak the shape. Pick better materials. The earlier study claimed a 17.4 % drop in drag a safety factor of about 238 and a 12.1 % rise, in hover efficiency. I need to verify those numbers with the quadcopter simulations. This workflow can help optimize UAV design cut down on prototypes and make engineering work faster for delivery, farming and surveillance missions.

219. AEROSHIELD: A SOFTWARE-BASED FRAMEWORK FOR CYBER THREAT DETECTION IN ADS-B-ENABLED CIVIL AVIATION SURVEILLANCE SYSTEMS

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The increasing dependence on digital communication and surveillance systems in civil aviation has heightened the need for effective protection against cybersecurity threats. Automatic Dependent Surveillance– Broadcast (ADS-B) is widely used for aircraft tracking and situational awareness by continuously broadcasting aircraft information. However, the open and broadcast nature of ADS- B communication makes it vulnerable to cybersecurity threats, including spoofing, message manipulation, false aircraft data, and other anomalous activities that can affect the reliability and integrity of aviation surveillance information. This paper presents AeroShield, a software-based cybersecurity framework designed to monitor and analyse ADS-B data for detecting potentially malicious or abnormal aircraft activity. The proposed system applies software-based data processing and intelligent anomaly detection techniques to analyze aircraft information, identify deviations from expected behavior, and generate alerts when suspicious patterns are detected. AeroShield is designed as a non- intrusive solution that does not require modifications to existing aircraft hardware or the installation of additional physical security devices. The proposed framework is motivated by existing research on ADS-B vulnerabilities, anomaly detection, and aviation surveillance security. AeroShield integrates these concepts into a practical monitoring framework for continuous analysis and threat identification in civil aviation environments. By providing automated data analysis and timely security alerts, the proposed system aims to improve the visibility of potential ADS-B-related cyber threats and support aviation security personnel in making informed decisions.

220.NANO-ENHANCED KARANJA-OIL CUTTING FLUID FOR SUSTAINABLE DRILLING OF EN8 STEEL UNDER MQL

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Machining performance is strongly influenced by heat generation, friction, tool wear and the quality of the machined surface. Conventional flood cutting fluids can provide effective cooling and lubrication, but their high consumption, handling and disposal requirements have encouraged the development of sustainable lubrication methods. Minimum Quantity Lubrication (MQL) reduces fluid consumption by delivering a small quantity of lubricant directly to the cutting zone. The incorporation of nanoparticles into a Karanja- oil-based lubricant can further modify thermal and tribological behaviour. This review focuses on TiO_2 nano-enhanced cutting fluids for drilling of EN8 steel under MQL conditions. The review summarizes the principles of MQL, Karanja-oil-based lubricants, nanofluids, TiO_2 mechanisms, drilling responses and the influence of spindle speed, feed and nanoparticle concentration. The literature indicates that TiO_2 concentration should be optimized because excessive concentration may cause agglomeration, increased viscosity, poor stability and inferior atomization. A major research need is the systematic evaluation of cutting temperature,

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surface roughness, tool wear and cylindricity together. Based on this gap, a compact Design of Experiments (DOE) framework using a three-factor, three-level Taguchi L9 design is proposed. The factors are spindle speed, feed rate and TiO_2 concentration. Taguchi signal-to- noise analysis and ANOVA are proposed for determining significant factors and optimum conditions, followed by a confirmation experiment. This framework provides a practical basis for the subsequent experimental investigation of sustainable TiO_2 nano-MQL drilling of EN8 steel.

221.AI-POWERED END-TO-END AUTOMATION OF THE DATA SCIENCE LIFECYCLE

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The increasing adoption of Artificial Intelligence (AI) and Machine Learning (ML) across diverse industries has significantly accelerated the demand for efficient and scalable data science workflows. However, traditional data science pipelines remain highly dependent on human expertise, requiring substantial manual effort during data preprocessing, exploratory data analysis (EDA), feature engineering, model selection, visualization, and interpretation of analytical results. Although existing Automated Machine Learning (AutoML) frameworks simplify model development, they often provide limited support for intelligent decision-making, interactive data exploration, and end-to-end workflow automation. This paper presents AI-Powered End-to-End Automation of the Data Science Lifecycle, an intelligent framework that integrates Large Language Models (LLMs), autonomous AI agents, and automated data analysis techniques to streamline the complete analytical workflow through a conversational interface. The proposed system enables users to upload datasets, perform automated data validation, generate exploratory analyses, detect anomalies, create interactive visualizations, execute natural language queries, and obtain AI-generated insights with minimal human intervention. By combining LLM-driven reasoning with automated analytical pipelines, the framework assists both technical and non-technical users in understanding complex datasets while significantly reducing analysis time and manual effort. Furthermore, the architecture promotes modularity, extensibility, and seamless integration of advanced AI services for future enhancements in deployment automation and intelligent monitoring. The proposed framework demonstrates how conversational AI and intelligent automation can transform conventional data science practices into an accessible, efficient, and user-centric analytical ecosystem.

222. NEXT-GENERATION BLOCKCHAIN SYSTEMS FOR INTELLIGENT FOOD TRACEABILITY

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Food supply chains have grown into long, multi- party networks in which a single product may pass through farms, processors, distributors, retailers and consumers across several countries. Conventional centralized traceability systems keep records in isolated databases that are slow to reconcile, easy to alter and difficult to audit, which delays recalls and leaves room for fraud. This paper proposes a next-generation framework for intelligent food traceability that combines a permissioned blockchain, Internet of Things (IoT) sensing, off-chain distributed storage and machine-learning- based anomaly detection. IoT devices capture temperature, humidity and location data at every stage; cryptographic hashes of these records are anchored on a Hyperledger Fabric ledger while bulky data are kept in IPFS. Smart contracts enforce cold-chain rules and custody transfer automatically, and an Isolation Forest model flags abnormal handling patterns that indicate spoilage risk or tampering. Consumers verify a product's complete history by scanning a QR code. The paper presents the design, architecture, implementation stack and an analytical evaluation against centralized systems, and discusses advantages, limitations and future directions.

223.INTELLIGENT INDUSTRIAL MAINTENANCE SYSTEM ARCHITECTURE USING AI AND PREDICTIVE ANALYTICS

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The paradigm of industrial maintenance is undergoing a major transformation with the advent of Industry 4.0, shifting from traditional reactive and scheduled preventive maintenance to data-driven predictive maintenance (PdM). This paper proposes a comprehensive, scalable Intelligent Industrial Maintenance System architecture that leverages Artificial Intelligence (AI), the Industrial Internet of Things (IIoT), and deep learning methodologies. The proposed framework integrates real-time sensor data acquisition, edge-to-cloud data processing, and sequence-based machine learning models, specifically Long Short-Term Memory (LSTM) networks, to accurately estimate the Remaining Useful Life (RUL) of critical machinery. Furthermore, the architecture incorporates mathematical cost-optimization models based on Weibull distribution reliability to prescribe the optimal time for maintenance interventions. Experimental simulations using the C-MAPSS turbofan engine dataset demonstrate that the proposed LSTM-based approach outperforms traditional machine learning models, achieving a high degree of predictive accuracy. By implementing this intelligent architecture, industrial plants can significantly reduce unexpected equipment downtime.

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224 .GENETIC ALGORITHM–TUNED GAUSSIAN–CLAHE PREPROCESSING FOR HIERARCHICAL HYBRID MOBILENETV2– EFFICIENTNETB0 DIAGNOSIS OF COTTON LEAF DISEASES

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On a cotton leaf, a fungal spot, a bacterial lesion, viral curling and the damage left by aphids can look surprisingly alike, and a single flat classifier often mixes them up. We built a two- level diagnosis system instead, working from a dataset of 19,808 cotton leaf images. A Level-1 network first decides the broad cause (fungal, bacterial, viral, pest or healthy), and a dedicated Level-2 network then names the disease, giving nine final labels. Each network in the cascade is a hybrid that joins pooled features from ImageNet-pretrained MobileNetV2 and EfficientNetB0. We also questioned the preprocessing. The 3×3, $\sigma = 1$ Gaussian filter and CLAHE settings we began with were defaults rather than deliberate choices, so a genetic algorithm (GA) searched seven parameters of the Gaussian–CLAHE chain. Each candidate was scored by a fast proxy classifier and an image-quality term that punishes over-smoothing. The search took 50.7 min and, because candidates are scored on frozen features, it did not add a single training epoch to the hybrid model. It raised proxy Level-1 accuracy from 85.14% to 87.43% and edge retention from 0.12 to 0.98 while leaving SSIM unchanged, and it settled on CLAHE followed by a light 7×7 blur ($\sigma = 0.556$) and unsharp masking. On 746 unseen test images the hierarchy reached 95.04% Level-1 accuracy and 94.50% end-to-end accuracy over nine classes (macro-F1 94.47%, $\kappa = 0.937$). Once the super-class was known, its Level-2 networks scored 98.8–99.3%, and training was quick: Level-1 passed 90% validation accuracy by epoch 4, and the Viral and Pest Attack networks passed 95% by epoch 3. Nine in ten of the remaining errors were Level-1 routing mistakes, mostly fungal leaves sent to the pest branch, which is where sharper preprocessing should help most.

225. SMART FERTILIZER MANAGEMENT IN AGRICULTURE: A SURVEY OF PRECISION TECHNOLOGIES, APPLICATIONS AND CHALLENGES

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Precision fertilization applies nutrients at the right place, time and rate by combining sensing, connectivity and data analytics, and it has become a core element of smart agriculture. This paper surveys the technologies behind it: geographic information systems (GIS), artificial intelligence and machine learning (AI/ML), the Internet of Things (IoT) and wireless sensor networks, variable rate technology (VRT), low-cost mobile and embedded platforms, and aerial and ground robots. We summarise how these tools are used in aerial crop monitoring, data-driven fertilizer recommendation,

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real- time soil nutrient tracking, automated application and organic nutrient management, and we compare their strengths, costs and suitability for different farm sizes. The survey follows and condenses the PRISMA-based review of Liu et al. [1], which screened 226 records and retained 122 studies published between 2020 and 2025. Persistent barriers are high upfront cost, fragmented ecosystems, data scarcity and limited access for smallholders. We close with research directions: multi-sensor feedback loops, open agronomic datasets, affordable hybrid systems and explainable, eco-friendly automation.

226.A SURVEY ON AI POWERED VISION ASSISTANT FOR VISUALLY IMPAIRED PEOPLE

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For a person who cannot see clearly, walking through a new place can be difficult. There may be people, vehicles, stairs, or other objects in the way, and knowing what is nearby is not always easy. Recognizing money is another common problem. Our project is a mobile application made to give some help in these situations. The phone camera is used to view the surroundings. The existing work uses YOLO for finding obstacles and gives alerts through sound and vibration. It uses 7,600 images for its work. However, that system mainly deals with obstacles. It does not include a way to recognize currency, and it does not give the user an idea about how far an object is or which side it is on. In our work, the camera view is given to YOLO26n, which is a lightweight object detection model. It can find objects such as people, vehicles, stairs, walls, and poles. Once an object is found, the application checks its approximate distance and direction. The result can then be spoken to the user through the phone. We have also included currency recognition so that a supported banknote can be identified from the camera image. If an obstacle is close, the phone can give a vibration alert as well. In this way, the application provides information about the user's surroundings and helps with a simple task like checking currency, using one mobile device.

227.INTEGRATED COMPUTATIONAL AND INVITRO VALIDATION OF MILADEMETAN AS POTENTIAL LEAD CANDIDATE FOR NON-SMALL CELL LUNG CANCER (NSCLC)

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Non-small cell lung cancer (NSCLC) is one of the leading causes of cancer-related mortality worldwide, highlighting the need for the identification of effective therapeutic targets and multitarget drug candidates. The present study aims to identify key hub genes associated with NSCLC and evaluate potential multitarget ligands through an integrated computational approach. Gene expression data were

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analysed to identify differentially expressed genes (DEGs) between NSCLC tumour and normal lung tissues, followed by protein–protein interaction (PPI) network analysis to identify hub genes based on their connectivity and centrality. The identified hub genes were further considered as potential therapeutic targets, and molecular docking was performed to investigate the binding interactions of candidate ligands with the selected target proteins. Based on the docking results, Milademetan was identified as a potential multitarget ligand showing favourable interactions with the selected NSCLC-associated targets. Molecular simulation studies will further be carried out to evaluate the stability and dynamic behaviour of the ligand–target complexes. The computational findings will subsequently be supported through in vitro studies to evaluate the biological activity of the identified candidate. Thus, the study integrates gene expression analysis, network biology, molecular docking, molecular simulation, and in vitro validation to explore Milademetan as a potential multitarget therapeutic candidate for NSCLC.

228.SHAP-KRUM: AN INTERPRETABILITY-GUIDED BYZANTINE-ROBUST FEDERATED LEARNING FRAMEWORK FOR ALZHEIMER'S DISEASE CLASSIFICATION FROM MRI IMAGES

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Alzheimer’s disease (AD) is a progressive neurodegenerative disorder that affects memory, cognition and daily functioning. Deep learning models applied to magnetic resonance imaging (MRI) can support the analysis of structural brain changes associated with AD. Federated learning (FL) allows models to be trained using distributed medical data without transferring the original images to central server. However, malicious clients can introduce backdoor behaviour into the global model. To investigate this problem, we propose SHAP-Krum, a client-selection approach that combines Krum- based model-parameter distances with SHapley Additive exPlanations (SHAP)-based attribution similarity. The approach was evaluated using five federated clients with non- IID data, including one malicious client performing a backdoor attack. Across 20 communication rounds, Krum selected the malicious client in 55% of rounds, while SHAP-Krum selected it in 40%. SHAP-Krum recorded an accuracy of 0.7500 and an ROC-AUC of 0.6953 whereas Krum recorded 0.8341 and 0.4868, respectively. The attack success rate remained high for bot methods reaching 95.41% for SHAP-Krum and 100% for Krum. Although combining parameter-space and explanation- space information reduced malicious-client selection in this experiment, it did not provide sufficient protection against the persistent backdoor attack.

229. ENHANCED BRAIN TUMOR SEGMENTATION FROM MRI IMAGES USING SMALL- TUMOR-AWARE ATTENTION RESIDUAL U-NET

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Brain tumours are one of the critical neurological abnormalities which are needed to be detected and diagnosed accurately and in due time. If a brain tumour remains undiagnosed, it can progress quickly and result in serious neurologic problems such as cognitive problems, movement disorders, and seizures. Moreover, traditional detection models have shortcomings in correctly identifying small tumors, regions of low contrast, and irregular tumor boundaries. Therefore, this research introduces an effective Modified Small-Tumor-Aware Attention Residual U-Net (STARA-UNet) for brain tumor segmentation. Initially, the Quality of the obtained MRI brain images is enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). Then, a median filter is used to reduce noise and enhance the image quality. After that, the improved images are normalized using the z-score normalization technique to make the intensity distributions comparable and eliminate the effect of high-intensity pixels on learning. Furthermore, the normalized images are processed by the proposed STARA-UNet, which accurately segments the brain tumor regions. In addition, the evaluation results reveal that the proposed technique achieved 93.01% Pixel Accuracy, , Average Surface Distance (ASD) of 0.041, Dice Similarity Coefficient (DSC) of 0.920, Hausdorff Distance of 0.033, and an Intersection over Union (IoU) value of 0.920.

230. AI-POWERED APPOINTMENT BOOKING PLATFORM FOR SMALL LOCAL BUSINESSES

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Small local businesses such as salons, beauty centers, clinics, repair services, fitness centers, tutoring centers, and personal service providers frequently rely on telephone calls, messaging applications, social media conversations, or manual registers for appointment management. These approaches can result in double booking, missed appointments, inefficient staff utilization, delayed responses, and limited business analytics. This paper proposes an AI-Powered Appointment Booking Platform designed specifically for small local businesses. The proposed platform integrates conversational artificial intelligence, natural language processing, intelligent time-slot recommendation, appointment management, automated reminders, no-show prediction, and a business analytics dashboard within a unified framework. Customers can interact with the platform using natural language requests to identify services, select preferred dates and times, and receive suitable appointment recommendations. The recommendation module considers service duration, staff avail ability, customer preferences, historical booking patterns, and business constraints. A machine learning component can estimate the likelihood of appointment no-shows and support proactive reminder strategies. The platform provides small businesses with a centralized dashboard for managing services, employees, availability, bookings, cancellations, and performance indicators. The proposed architecture aims to reduce administrative effort, improve appointment-slot utilization, and provide a convenient digital booking experience without requiring complex enterprise- level infrastructure. A prototype-oriented evaluation framework is also defined using response time, booking accuracy, recommen- dation relevance, slot utilization, and no-show prediction metrics.

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231.FABRICATION AND CHARACTERISATION OF FREE-STANDING POROUS AND NON-POROUS PCL/PCL-PEO FILMS USING SPIN COATING TECHNIQUE

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Polycaprolactone (PCL) is one of the most extensively investigated biodegradable polyesters in biomedical engineering, valued for its mechanical flexibility, slow hydrolytic degradation over two to four years, and biocompatibility. However, its pronounced hydrophobicity characterised by water contact angles exceeding 80° substantially limits protein adsorption, cell adhesion, and biological integration. Blending PCL with the hydrophilic, water-soluble polymer polyethylene oxide (PEO) provides a straightforward and highly effective strategy to overcome these limitations without altering the fundamental biocompatibility of the system. This chapter presents a systematic and comprehensive investigation into the fabrication of free-standing porous and non-porous thin films from PCL and PCL-PEO blends using spin coating as the primary deposition technique. Four blend compositions (0, 10, 20, and 30 wt% PEO) were prepared and deposited at three spin speeds (500, 1000, and 1500 rpm), generating a 4 × 3 full factorial experimental design comprising 12 unique film conditions tested in triplicate (36 total samples). Surface wettability was quantified by static water contact angle goniometry, and in-vitro degradation was evaluated by monitoring mass loss over 12 weeks in phosphate-buffered saline (PBS, pH 7.4, 37 °C) with periodic SEM and FTIR-ATR analyses. Results confirm that increasing PEO content systematically reduces PCL crystallinity from 57.8% to 34.9%, decreases water contact angle from 82.4° to 47.8°, induces interconnected pore formation with diameters of 2–10 µm, and accelerates in-vitro degradation from 2.1% to 27.6% mass loss at 12 weeks. Film thickness is precisely governed by spin speed, decreasing from approximately 5 µm at 500 rpm to 2 µm at 1500 rpm. These tuneable and interrelated properties establish PCL-PEO spin-coated free-standing thin films as versatile, scalable candidates for wound dressings, drug delivery membranes, and tissue engineering scaffolds. The quantitative process–property–structure relationships presented provide a rational design framework for future biomedical thin film development.

232. HEALTHTWIN-AI: AN EXPLAINABLE LONGITUDINAL DIGITAL TWIN FRAMEWORK FOR PATIENT-SPECIFIC RISK PREDICTION AND WHAT-IF ANALYSIS

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Digital twin technology is gaining attention in personalized, predictive healthcare because it can keep a continuously evolving computational model of an individual built from patient- specific data. Many existing healthcare digital twins, however, are still conceptual, rely on a single data source, or focus on monitoring without offering a reproducible prediction workflow. This paper presents HealthTwin-AI, a patient-specific digital twin framework that merges electronic health record (EHR) attributes, physiological readings, laboratory results, wearable-sensor data, and lifestyle variables into a virtual patient profile that is updated continuously. The framework has five main layers: data acquisition and quality control, patient-state construction, temporal feature engineering, AI-driven risk prediction, and explainable clinical decision support. A scenario engine supports what-if analysis by estimating how

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predicted risk shifts when selected patient variables are changed. The pipeline is modular: gradient-boosted decision trees act as the main tabular predictor, and SHAP provides feature-level explanations. The proposed evaluation protocol covers discrimination, calibration, temporal prediction performance, explanation consistency, and computational latency on retrospective longitudinal patient data. In contrast to a static disease-prediction model, HealthTwin-AI works as a closed loop: each new observation updates the patient state and triggers a fresh risk estimate. The framework is meant as a research prototype for clinical decision support, not as an autonomous diagnostic or treatment tool. It addresses a clear research gap by turning the digital-twin concept into a reproducible, patient- centered AI pipeline with explicit updating, prediction, explanation, and evaluation stages.

233.SMART FOREST FIRE RESCUE SYSTEM USING EDGE AI, IOT AND UAV TECHNOLOGY

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Forest fires cause severe loss of biodiversity, property and human life, and their impact grows sharply when ignition is detected late. Conventional watchtowers, satellite imagery and manual patrols suffer from long revisit times, limited coverage and dependence on human observers. This paper presents a Smart Forest Fire Rescue System that combines Internet of Things (IoT) sensing, Edge Artificial Intelligence (Edge AI) and Unmanned Aerial Vehicle (UAV) technology to detect fires at an early stage, verify them autonomously and support rescue operations. Low-power sensor nodes distributed across the forest continuously measure temperature, humidity, smoke, carbon monoxide and flame intensity and transmit data over a long-range wireless network to an edge gateway. A lightweight machine learning model running at the edge fuses these readings to estimate fire probability without depending on cloud connectivity. When the probability crosses a threshold, a UAV fitted with thermal and RGB cameras is dispatched to the sensor location, where an onboard deep learning detector confirms the fire, locates trapped people or animals and estimates the fire boundary. Geo-tagged alerts, live imagery and suggested safe evacuation routes are delivered to a command dashboard used by forest and rescue teams. The proposed architecture reduces detection latency, lowers false alarms and works in areas with poor network coverage, making it a practical solution for proactive forest fire management.

234.DECOX AI MULTI AGENT COLLABORATIVE INTELLIGENCE PLATFORM

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Decox AI is a next generation, full-stack AI assistant, far more advanced than your typical single-model chatbot. Rather than a single AI trying to answer every question, a panel of many LLMS Gemini Claude GPT-4o, Groq et al are elicit to work totally independently, outvote by blind peer-review and cross-exanchge, and arrive at a marked final answer with trust- based weighting. Their query router intelligently recognizes whether your question needs: direct answer; multi-agent debate; web search; code execution; symbolic math validation; AI-image generation. Switching between Turbo Mode (ultra-fast 2-step speculative consensus) and Detailed Mode (slow but exhaustive 3 rounds of peer-review debate) is easy. Code answers are validated in a live Python subprocess sandbox. Math answers are validated with SymPy symbolic computation. Image prompts are also simultaneously rendered by 6 parallel diffusion engines. And each panelist's accuracy is quantified using a dynamic Elo trust-scoring system that makes debates smarter over time.

235 .A COMPUTATIONALLY LIGHTWEIGHT CONVOLUTIONAL NEURAL NETWORK FOR GENERALIZED CHEST X-RAY DIAGNOSIS

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Chest radiography is the most widely used and most affordable imaging modality for detecting thoracic disease, yet accurate reading is limited by a shortage of radiologists, especially in rural and resource-constrained clinics. Deep convolutional neural networks (CNNs) such as DenseNet-121 reach radiologist-level performance on several findings but need millions of parameters and billions of operations, which prevents deployment on low- cost hardware. This paper presents LiteCXR-Net, a computationally lightweight CNN for generalized, multi- label chest X-ray diagnosis. The network combines depthwise-separable inverted-residual blocks, selective squeeze-and-excitation attention and a single-channel input stem, giving only 0.59 million parameters and 0.09 GMACs per 224×224 image, which is about 13× fewer parameters and 32× fewer operations than DenseNet-121. Generalization is addressed through CLAHE-based intensity harmonization, class-balanced focal loss, strong augmentation and multi-source training. Grad-CAM heat- maps provide visual explanations, and post-training INT8 quantization targets real-time inference on CPU and mobile devices. The system is described together with its evaluation protocol.

236. ELECTROCHEMICAL ANALYSIS OF SALICYLIC ACID IN SHAMPOO USING A CARBON PASTE SENSING ELECTRODE

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Salicylic acid (SA), owing to its keratolytic, anti- inflammatory, and antimicrobial properties, is extensively incorporated into cosmetic and personal-care formulations, including acne treatments, scalp-care shampoos, facial toners, and exfoliating formulations. Therefore, accurate determination of SA in cosmetic formulations is essential for quality control and regulatory compliance. In this work, a carbon paste sensing electrode (CPE) was fabricated by blending graphite powder with paraffin oil and employed as a cost-effective electrochemical sensor for SA determination. Electrochemical analysis was carried out in phosphate- buffered saline solution adjusted to pH 3.0, using cyclic voltammetry (CV) and differential pulse voltammetry (DPV). After optimizing the experimental parameters, the proposed sensor showed a proportional electrochemical response to analyte concentrations between 1 and 100 μM. Its detection and quantification limits were calculated as 1.43 μM and 4.33 μM, respectively, while the analytical sensitivity reached 0.705 μA/μM. The practical applicability of the proposed method was confirmed by applying it to the analysis of a commercially available cosmetic product. Recovery studies based on the standard addition method yielded an average recovery of 95.43%, indicating satisfactory analytical accuracy. In addition, a partial least squares regression (PLSR) model developed from the DPV responses and validated using the manufacturer's declared SA concentration achieved an overall prediction accuracy of 95.24%, highlighting the suitability of the proposed CPE sensor as a simple, reliable, and economical platform for the quantitative determination of salicylic acid in cosmetic products.

237. DEVELOPMENT OF A LOW-COST MODULAR EARTHQUAKE-RESILIENT CONSTRUCTION SYSTEM INSPIRED BY TRADITIONAL TAQ AND DHAJJI TECHNIQUES

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This study explores how traditional Kashmiri building methods, Taq and Dhajji-Dewari, can be used to design affordable and earthquake-resistant homes. Both methods use timber and masonry to help buildings cope with ground movement. In Taq construction, horizontal timber beams are built into load-bearing masonry walls. Dhajji- Dewari uses a timber frame with diagonal bracing, with the gaps filled using brick or stone masonry. These timber elements help hold the structure together and allow it to handle earthquake forces more effectively. During the 2005 Kashmir earthquake, many well-built traditional structures performed well compared with more vulnerable construction. Learning from these methods, this study looks at their materials, structural details and construction techniques to develop a modular system that is easier to build, assemble and repair while keeping costs low.

238.CARDIOSENSE AI: REAL-TIME HEART DISEASE PREDICTION SYSTEM USING WEARABLE ECG SENSORS AND DEEP LEARNING

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Cardiovascular disease remains the leading cause of mortality worldwide, underscoring the need for continuous, early detection strategies that extend beyond episodic clinical assessment. This paper presents CardioSense AI, a real-time heart disease prediction system that integrates wearable ECG sensing with deep learning to enable non-invasive, continuous cardiac monitoring. A compact chest- or wrist-worn sensor acquires ECG signals, which are preprocessed to suppress noise and motion artifacts prior to analysis. Cardiac signal classification is performed using a hybrid CNN-LSTM architecture, in which convolutional layers extract spatial morphological features while LSTM layers capture temporal dependencies, enabling accurate detection of arrhythmias, ischemic events, and other cardiac abnormalities. The model is trained and validated on the MIT-BIH Arrhythmia and PTB-XL datasets, ensuring generalizability across diverse patient populations. CardioSense AI employs an edge-cloud architecture: on-device inference delivers instantaneous anomaly alerts, while cloud-based processing supports deeper diagnostic analysis and longitudinal health trend tracking. A companion mobile application delivers real-time risk assessments, alerts, and interpretable health reports to both patients and clinicians. Experimental evaluation demonstrates classification accuracy exceeding 95%, with latency low enough to support continuous ambulatory use. By coupling wearable sensor technology with deep learning-based analytics, CardioSense AI offers a scalable, proactive framework for cardiac care with particular relevance for early intervention in remote, elderly, and resource-limited populations.

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239. AUTISM CATEGORIZATION USING CNN-BASED EFFICIENTNET AND ANT COLONY OPTIMIZATION

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Reliable identification of Autism Spectrum Disorder (ASD) from image-based neuroimaging data remains difficult because the deep feature representations produced by modern convolutional networks are typically high-dimensional and contain redundant or weakly discriminative components. This paper presents EfficientNet-ACO, a two-stage categorization framework that couples EfficientNet-B0, a compact convolutional network adapted through transfer learning, with Ant Colony Optimization (ACO), a swarm-intelligence feature-selection metaheuristic. Images drawn from the Autism Brain Imaging Data Exchange (ABIDE) repository are preprocessed and augmented to normalize acquisition variability and enlarge the effective training set. EfficientNet-B0 converts every image into a compact deep feature vector, after which the ACO stage searches this feature space for the subset that best separates ASD from Control subjects while discouraging unnecessary dimensionality. A fully connected classification layer then assigns each subject to the ASD or Control category, with performance measured using Accuracy, Precision, Recall, F1-Score, Specificity, and ROC-AUC. Reported accuracies for comparable bio-inspired ASD-detection pipelines range from 73.86% to 94.49%; the proposed Efficient Net-ACO pipeline is designed to reach an accuracy of 95.8%, with corresponding gains in Precision, Recall, and F1-Score, while remaining architecturally simpler than competing multi-stage designs.

240.TCP CONGESTION CONTROL ALGORITHMS IN HIGH-SPEED NETWORKS: A COMPARATIVE CASE STUDY OF RENO, CUBIC, AND BBR

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TCP congestion control decides, at every instant, how fast a connection may send, using only acknowledgements, round-trip-time (RTT) samples, and loss or ECN signals. In high-speed networks with a large bandwidth- delay product (BDP), mixed deep- and shallowbuffered paths, and non-congestive loss, the classical loss-based answer becomes a serious limitation. This paper presents a comparative case study of three influential algorithms—TCP Reno, TCP CUBIC, and Google's BBR

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(Bottleneck Bandwidth and Round-trip propagation time). We describe the design principles, growth functions, strengths, and weaknesses of each, and compare them on throughput, fairness, latency (bufferbloat), and loss recovery. Worked examples on a 10 Gbps, 80 ms path (100 MB BDP) show that after a single loss Reno needs about 46 minutes to regain line rate while CUBIC needs about 37 seconds, and that BBR's probing adds only about 20 ms of queuing delay at most. We close with practical guidance for data centres, long-fat networks, and CDN or video traffic.

241. MULTIMODAL PREDICTIVE ANALYTICS FOR INTELLIGENT CARDIAC DISEASE DIAGNOSIS

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Heart disease remains one of the leading causes of mortality worldwide. Early detection and timely medical intervention can significantly reduce the risk associated with cardiovascular diseases. With the advancement of machine learning techniques, predictive systems can assist healthcare professionals in identifying potential heart disease risks at an early stage. This project presents a Machine Learning Driven Predictive System for Heart Disease Diagnosis, implemented as a web-based application. The system analyzes multiple health-related parameters such as age, gender, blood pressure, cholesterol level, blood sugar level, Alcohol consumption, Smoking habits. These medical attributes are processed using machine learning algorithms to predict the likelihood of heart disease occurrence. The proposed system implements two classification algorithms: Logistic Regression and Random Forest. Among these models, the Random Forest algorithm de monstrates superior performance with an approximate prediction accuracy of 95%, making it the primary model used for risk prediction. The system categorizes the prediction results into High Risk or Low Risk levels, while also providing explanations related to the contributing health factors. The application is developed using Python with the Flask framework, integrating machine learning libraries such as Scikit-learn, along with SQLite database management for storing user and prediction data. Additionally, the system provides graphical analysis and automated PDF report generation for prediction results.

242. HIGH PERFORMANCE CONCRETE: ADVANCEMENT, CHALLENGES AND

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FUTURE SCOPE IN SUSTAINABLE CONSTRUCTION- REVIEW

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Concrete is one of the foundational materials in Civil Engineering Infrastructure. Concrete with high strength, enhanced durability, better workability, and a longer service life is necessary as modern constructions are developed. Performance, which encompasses not only mechanical strength but also workability, aesthetics, finish, integrity, and durability, is defines High performance concrete. Effects of supplementary cementitious materials (SCMs), aggregate, fiber, curing method, Admixtures and water-to-binder (w/b) ratio on the strength of HPC are reviewed. The main innovation of this review paper is to highlight the potential for new materials that can be considered in the manufacture of special concretes. The review reports that SCM incorporation generally improves HPC strength, Dosage of fly ash varied from 10 to 40% in that at 40% optimum volume fraction reached the highest compressive strength about 120MPa for 28 days, by using advance curing process to obtain the greater strength. While silica fumes content ranges varied from 0%to 23%, at 3% attained compressive strength of 86 MPa and Flexural strength of 12.4 MPa. Metakaolin gained 72MPa an effective dosage 15% within the 0 to 25% range and GGBS of 30% volume fraction identified as the highest compressive strength about 72 MPa. LC3 produce 54MPa, at 45% replacing cement. The incorporation of LC3 in HPC provides a sustainable way to almost reducing 50% cement clinker consumption and reduces CO_2 emissions while simultaneously using industrial by-products to improving concrete durability and extending structural service life. In Fibre Reinforced Concrete, glass fibre at 1.5% attained higher compressive strength of 152MPa; subsequently PP fibres achieved 121MPa at 1.5% and recorded highest flexural strength of 11.11MPa between the content of 0 to 2%. Basalt fibres at 8% reached highest spilt tensile strength of 8.4Mpa with in the range of 4 to 16%. The optimised utilization of natural and synthetic fibers in hybrid to development of HPC is also emphasized, creating composites with better qualities than traditional concrete. Using Combination of SCM's and Hybrid fibres to development of HPC reduce the cement demand, Waste generation and reach the higher mechanical and Durability properties of concrete. This demonstrates its potential of efficient and low carbon construction.

243. AI-BASED INTELLIGENT VEHICLE DIAGNOSTICS AND PREDICTIVE MAINTENANCE: A REVIEW OF SENSOR ANALYTICS, FAULT DETECTION, FAILURE PREDICTION AND EXPLAINABLE AI

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Cars today produce a steady stream of electronic data through their On-Board Diagnostics (OBD) interface and embedded sensors, yet workshop diagnosis still leans mostly on stored trouble codes, fixed thresholds and the experience of the technician. Machine learning can automate fault detection, fault classification, failure forecasting and health assessment. This review examines machine learning, deep learning, anomaly detection, sensor analytics, predictive maintenance and explainable artificial intelligence (XAI) as applied to road vehicles, and discusses recurring obstacles: noisy and incomplete

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signals, scarce labelled faults, imbalanced classes, shifting operating conditions and opaque models. The survey shows that most published work addresses one diagnostic task at a time. To close this gap, we propose AutoSense AI, an integrated framework that chains anomaly detection, fault classification, root-cause analysis, failure prediction, health scoring, explanation and report generation into one workflow. AutoSense AI is at the design stage: no system has been built and no measured results are reported. Instead, we state the objectives, the planned methodology, an evaluation protocol and the outcomes we expect, clearly marked as predictions to be tested. We argue that such an integrated platform can shift vehicle maintenance from reactive repair to proactive, transparent and actionable decisions.

244.ECOFEED 2.0 - HYPERLOCAL FOOD REDISTRIBUTION PLATFORM

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The major objective of EcoFeed 2.0 is to build an AI-powered food circularity platform that ensures every food surplus gets the safest, most useful, and highest-impact second purpose instead of becoming waste. The system intelligently identifies surplus food, evaluates its safety, urgency, quantity, quality, location, demand, and remaining usable time, and then uses AI to determine the most suitable destination—prioritizing human consumption and community needs first, followed by food businesses, animal care, composting, biogas, mushroom cultivation, innovation, and finally authorized disposal as a last resort. EcoFeed connects food donors, NGOs, community kitchens, schools, volunteers, businesses, farmers, animal shelters, and organic-processing facilities into one coordinated network, while using real-time logistics and route optimization to move surplus efficiently before it expires. Beyond rescuing food, the platform focuses on prevention, using historical and real-time data to predict future demand, identify potential waste, optimize food distribution, and continuously improve decision-making. It also provides transparent AI reasoning so users can understand why a particular destination was selected, while measuring the resulting social and environmental impact through metrics such as meals rescued, people benefited, food diverted, CO_2 emissions avoided, waste prevented, and circular score. The ultimate goal is to transform food waste management from a simple donation process into a closed-loop circular ecosystem where surplus food is prevented, rescued, redistributed, reused, recycled, and regenerated, creating measurable social, economic, and environmental value.

245. AI-BASED SMART FINANCE TRACKER FOR INTELLIGENT

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PERSONAL FINANCIAL MANAGEMENT

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Managing personal finances manually can be difficult because users often record transactions inconsistently, struggle to identify spending patterns, and find it hard to maintain a realistic budget. This paper presents an AI-Based Smart Finance Tracker designed to support personal financial management through digital transaction recording, expense categorization, spending analysis, budget monitoring, and intelligent recommendations. The proposed system combines a web-based application architecture with a relational database and AI-assisted analytical functions. The application can collect income and expense information, organize transactions into meaningful categories, calculate financial summaries, and analyze historical spending behaviour. Based on the available transaction history, the intelligent layer can identify recurring patterns and provide budget-oriented recommendations. The system is designed using a Spring Boot backend, MySQL database, and web interface technologies, providing a structured and maintainable platform for personal finance tracking. The approach aims to reduce manual effort, improve financial awareness, and help users make better day-to-day financial decisions.

246.CAVIX: A LOW-COST AI-ASSISTED ROBOTIC CRAWLER FOR REMOTE CONFINED-SPACE INSPECTION

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Many small manufacturing units still rely on workers to enter tanks, silos, ducts, and chimneys for routine inspection, although confined spaces may contain oxygen- deficient air or hazardous gases. Existing remotely operated inspection robots can be costly for micro, small, and medium enterprises (MSMEs). This paper presents CAVIX, a tracked robotic crawler designed as a lower-cost alternative for confined-space inspection. The proposed platform integrates an HD camera, LED illumination, electrochemical sensors for carbon monoxide, hydrogen sulphide, and oxygen, onboard control, hybrid wireless/tethered communication, cloud storage, and an operator dashboard. A lightweight YOLO-based vision module is planned to detect cracks, corrosion, rust, and material buildup and to support an inspection-level health indicator. The project is currently at Technology Readiness Level 1; a physical prototype and field-tested AI model have not yet been completed. Accordingly, this paper presents the system design, hardware and sensing choices, projected cost, development roadmap, and planned evaluation rather than experimental results.

247. PROTEUS: ADAPTIVE AGENT TEAM FORMATION THROUGH RUNTIME COMPOSITION CONTROL FOR NICHE WEBSITEDEVELOPMENT

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Every widely used multi-agent framework traditionally fixes the team configuration at design time, leaving the team unchanged regardless of the task's complexity. While recent works have relaxed

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single dimensions—such as DyLAN pruning weak agents [cite: 1.2.2], AgentVerse recruiting new ones [cite: 1.3.3], or AgentInit selecting an initial team [cite: 1.1.2]—they remain partially static. This paper presents PROTEUS, a novel system that treats team composition as a closed control loop. PROTEUS dynamically decides the number of agents, required skills, leadership, merging, and removal at runtime, all constrained by a cost budget. By leveraging this adaptive runtime composition, PROTEUS provides a highly flexible infrastructure capable of tackling dynamic and multifaceted tasks, such as specific niche website development. We evaluate PROTEUS across standard benchmarks and simulated domain-specific environments, demonstrating significant improvements in cost-normalized accuracy compared to static architectures. The results validate that runtime compositional control allows autonomous agent teams to mirror human developmental paradigms in complex engineering workflows, significantly reducing token consumption while maintaining or increasing output fidelity.

248. RENEWABLE ASSET FAILURE & MAINTENANCE ASSISTANT

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The increasing adoption of renewable energy sources such as solar and wind power has created a growing need for intelligent monitoring and predictive maintenance of energy assets. Renewable energy equipment is continuously exposed to environmental conditions, mechanical stress, temperature variations, dust, corrosion, and other factors that can lead to component degradation and unexpected failures. Traditional inspection methods generally depend on manual monitoring, scheduled maintenance, and physical inspection, which can be time-consuming, costly, and prone to human error. This study proposes an AI-Based Renewable Energy Asset Failure Detection and Predictive Maintenance System Using OpenCV, Convolutional Neural Networks (CNN), and Vision Transformers (ViT) to provide an automated and intelligent solution for asset monitoring. OpenCV is utilized for image acquisition, preprocessing, enhancement, resizing, noise reduction, and extraction of relevant visual information from renewable energy asset images. The CNN model is employed to identify spatial patterns and detect visible defects such as cracks, surface damage, corrosion, overheating indications, panel abnormalities, and structural faults. To improve feature representation and classification capability, a Vision Transformer is incorporated to capture global relationships and complex visual patterns that may not be effectively recognized by conventional convolution-based models. The proposed system analyzes inspection images and classifies renewable energy assets into normal and different failure or defect categories. By combining CNN-based local feature extraction with Vision Transformer-based global feature analysis, the system aims to improve detection accuracy and reliability. The detected fault information can be used to estimate maintenance requirements and prioritize assets that require immediate attention. Predictive maintenance enables operators to identify potential failures at an early stage before they develop into severe operational problems.

249.MACHINE LEARNING AND DEEP LEARNING FOR SEWAGE TREATMENT PLANT PERFORMANCE PREDICTION AND OPERATIONAL OPTIMIZATION: A CRITICAL REVIEW

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Sewage treatment plants (STPs) experience operational variability arising from changes in influent composition, flow, weather, and biological processes. Machine learning (ML) and deep learning (DL) have been investigated to predict influent and effluent characteristics and support plant operation. This critical review examines applications of random forests, gradient boosting, support vector regression, artificial neural networks, long short-term memory networks, and gated recurrent units to wastewater prediction and decision support. It organizes the evidence around data requirements, forecast horizons, validation practice, model interpretability, and potential connections to oxygen demand, aeration, and chemical dosing. Attention is given to the distinction between predictive accuracy and demonstrated operational improvement. The review identifies gaps in temporal validation, cross-plant transferability, uncertainty assessment, and prospective testing, and proposes a staged prediction-to-decision framework for future STP research. Keywords—Sewage treatment plant, machine learning, deep learning, effluent prediction, influent forecasting, operational optimization.

250.HYBRID PARABOLIC TROUGH SOLAR AUTOCLAVE WITH BACKUP FOR RURAL HEALTHCARE STERILIZATION

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Reliable sterilization of medical instruments in rural healthcare facilities is constrained by unreliable grid electricity, which renders conventional electric autoclaves impractical and forces reliance on boiling-based disinfection that fails to destroy bacterial spores. Pure solar-thermal sterilizers address the grid- dependence problem but introduce a new one: cloud-induced temperature drops of up to 30°C within a minute can silently compromise a sterilization cycle. This paper proposes a hybrid parabolic trough solar autoclave that uses concentrated solar thermal energy as the primary heat source and an automatic, battery-powered electric heater as backup. A microcontroller- based closed-loop controller monitors chamber temperature via a K-type thermocouple and switches the backup heater through hysteresis logic (115°C ON, 120°C OFF), while a temperature- aware sterilization timer pauses whenever the chamber falls below 120°C, guaranteeing a full 40-minute exposure regardless of weather. A dual-battery architecture isolates the high-current heating load from the control electronics, so the controller, display, and alarm remain operational even if the heating battery is depleted. The complete system architecture, control logic, component specifications, and 3D design are presented, and design-based expected performance is reported as calculated projections rather than experimentally measured results, since this work constitutes a preliminary design study pending prototype fabrication and biological validation.

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251.FROM STUDENT FEEDBACK TO INSTITUTIONAL INTELLIGENCE: A HUMAN-CENTERED SURVEY OF NLP METHODS FOR HIGHER-EDUCATION ANALYTICS

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Universities regularly request feedback from students, even though a great deal of the most valuable information in that feedback is contained within brief comments, suggestions, complaints, and personal reflections. While numerical ratings are helpful for tracking general trends, they usually fail to explain the reasons for those trends. Natural Language Processing (NLP) provides a means of systematically examining large amounts of educational text and converting it into evidence regarding teaching, the curriculum, infrastructure, student support, employability, and institutional services. This survey looks at the human and analytical aspects of NLP in the field of educational analytics, focusing on sentiment analysis, aspect-based sentiment analysis, topic modelling, semantic representation, multilingual processing, transformer models, and large language models. Instead of seeing NLP as a substitute for human judgement, the article views it as a tool for organizing and interpreting student feedback on a large scale. Current research indicates that there is considerable activity in the area of sentiment analysis and increasing interest in the use of large language models for survey analysis. Nevertheless, there are still important gaps: educational datasets are scattered, many benchmark resources have been drawn from non- educational areas, feedback that is multilingual or code-mixed is not well represented, longitudinal student experiences are seldom linked together, and the results of the analysis are not always turned into clear institutional actions. The paper puts forward a human-centered conceptual approach in which quantitative indicators and textual evidence are analyzed together, and in which cohort context, privacy, human validation, and actionability are regarded as essential design requirements. The survey ends with a research programme for trustworthy educational analytics that can connect what students expect, what they experience, and what they later report about their education.

252 .A SPATIOTEMPORAL DEEP LEARNING FRAMEWORK FOR BRAIN TUMOR SEGMENTATION AND GROWTH PREDICTION USING MULTIMODAL MAGNETIC RESONANCE IMAGING

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Brain tumors, particularly gliomas, exhibit heterogeneous shape, intensity, and growth behaviour that make manual assessment from magnetic resonance imaging (MRI) time-consuming and prone to inter observer variability. Most existing deep learning solutions treat brain tumor analysis as a single static segmentation task and do not model how a tumor is likely to evolve over time. This paper proposes a unified spatiotemporal deep learning framework that jointly performs multimodal MRI denoising, voxel- level tumor segmentation, discriminative feature extraction, and future tumor-growth prediction. Multimodal MRI volumes (T1, T1- CE, T2, and FLAIR) are first denoised using a Denoising Autoencoder (DAE) that suppresses acquisition noise while preserving tumor boundaries. A 3D ResNet encoder extracts morphological, textural, and intensity descriptors from the segmented volumes. These descriptors condition a Conditional Generative Adversarial Network (cGAN) that synthesizes plausible longitudinal tumor states, while a Convolutional LSTM (ConvLSTM) models the spatiotemporal dependencies across the synthesized sequence to forecast future tumor morphology. The framework is evaluated using Dice similarity coefficient (DSC), Intersection-over-Union (IoU), precision, recall, Hausdorff Distance (HD95), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). Experimental results show module-wise accuracies of 92.4% (DAE), 94.1% (3D U-Net), 93.6% (3D ResNet), and 94.8% (cGAN-ConvLSTM growth prediction), with the fully integrated Attention U-Net pipeline achieving 95.8% overall accuracy and a mean Dice score of 0.914 across tumor sub-regions.

253.AI-BASED FLASH FLOOD PREDICTION FOR HILLY REGIONS USING MULTI-SOURCE DATA

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Flash floods in hilly and mountainous regions develop within minutes to a few hours after intense rainfall, leaving very little time for evacuation and emergency response. Steep slopes, shallow soils, narrow valleys and sparse gauging networks make conventional hydrological and hydraulic models difficult to calibrate and slow to run in such terrain. This paper proposes an artificial-intelligence-based framework for short-lead flash flood prediction in hilly regions that fuses multiple heterogeneous data sources: ground rain-gauge records, satellite precipitation estimates, reanalysis soil-moisture and weather variables, digital elevation model (DEM) derived terrain descriptors, land-use and vegetation indices from Sentinel imagery, and river-level or crowd-sourced observations. A dual-branch architecture is adopted, in which a Long Short-Term Memory (LSTM) network captures the temporal rainfall–runoff dynamics and a tree-based ensemble (Random Forest / XGBoost) encodes static terrain susceptibility. The two branches are combined by a fusion layer that outputs a four-level flood risk class together with a probability score. The paper describes the data pipeline, model design, training strategy, evaluation protocol and a lightweight alert-delivery layer suitable for deployment vin remote communities. The proposed approach aims to improve lead time, reduce false alarms and support disaster-management authorities in hill districts.

254.PROMPT SHIELD: REAL-TIME, REVERSIBLE, AND MULTIMODAL PII REDACTION FOR GENERATIVE AI INTERFACES USING FORMAT-PRESERVING SYNTHETIC MASKING

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The widespread use of online generative AI platforms, such as OpenAI ChatGPT, Google Gemini, Anthropic Claude, and Microsoft Copilot, has caused the emergence of a major concern that has gone mostly unresolved users sharing important personal or corporate information within their prompts. With every instance of a user pasting an actual production key into the chat, a medical worker using a model with a patient's data, or an analytical specialist attaching a screenshot with payment data, the information crosses an important line of trust and enters servers belonging to third parties where it can be stored, analyzed, or even utilized for training. Today's browser-based solutions try to tackle the problem by replacing the secrets with static placeholder tokens like [API_KEY] or [NAME_1]. While this results in actual values' protection, it destroys the context that is vital for language models to be able to produce meaningful responses. The paper describes a new system called PromptShield that solves this problem in a browser-native and zero-trust manner. Instead of removing any sensitive information identified, PromptShield simply replaces the identified secret with its Format Preserving Synthetic Mask (FP-SM) compatible version that is generated randomly but is structurally correct. For

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instance, a synthetic API key possesses the same vendor prefix and character length as the original one. Similar to this, a synthetic credit card number meets the requirements of Luhn checksum, while a synthetic email address has exactly the same domain structure. The generative AI model receives a prompt that appears entirely normal, produces a full-quality response referencing the synthetic values, and those values are swapped back to the real secrets inside the browser's Server-Sent Events stream before the user interface renders them. PromptShield further extends this protection to multimodal inputs through a novel OCR-guided pixel-level redaction pipeline. When a user uploads or pastes a screenshot, the system sends the image to its trusted backend, which calls Azure AI Vision to extract word level bounding boxes, runs the same detection cascade on the extracted text, and returns the pixel coordinates of every PI region together with a synthetic replacement value. The browser then draws the original image onto an HTML5 canvas, paints over each sensitive region using a background-sampled color for visual coherence, and renders the synthetic text directly onto the image before re-injecting it into the ChatGPT upload flow. Evaluation across 120 text prompts and 30 test images on ChatGPT (GPT-4o) demonstrates 99.8% semantic preservation, 100% restoration accuracy, and an average per request latency overhead of 140 milliseconds, making PromptShield a practical and deployable privacy layer for generative AI workflows.

255.DEEP LEARNING BASED PLANT DISEASE DETECTION FOR ECO FRIENDLY AGRICULTURE AND RECOMMENDATION SYSTEM

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Crop diseases pose a major threat to agricultural productivity, particularly in rural regions where timely expert guidance and reliable internet connectivity are limited. This project presents AI-Smart Agro Advisor, a hybrid artificial intelligence–based mobile application designed for real-time crop disease detection and intelligent crop recommendation. The system employs dual-mode architecture to ensure continuous operation under both offline and online conditions. In offline mode, a MobileNetV2-based Convolutional Neural Network optimized using TensorFlow Lite performs on-device inference to identify commonly occurring crop diseases from leaf images captured using a smartphone camera. In online mode, the application integrates a cloud-based deep learning model (ResNet50) accessed through a RESTful API to enable large-scale detection of crop diseases and pests with higher accuracy. Additionally, crop suitability predictions are generated using machine learning models trained on soil parameters and seasonal data. To enhance accessibility, the system incorporates offline Tamil voice-assisted interaction implemented using on-device Text-to- Speech and Speech Recognition modules. The proposed system aims to reduce crop losses, improve farmer decision making, and support sustainable agriculture through an efficient, scalable, and farmer-centric smart advisory solution.

256. AN EFFICIENT DEEP LEARNING APPROACH FOR DIABETIC FOOT ULCER DETECTION USING GENERATIVE AUGMENTATION AND ATTENTION MECHANISM

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Diabetic foot ulcer (DFU) is a serious complication of diabetes whose outcome depends heavily on how early it is recognized, yet routine screening still relies on visual inspection by trained clinicians and is limited by specialist availability and inter-observer variability. This paper proposes GAA-Net, a deep-learning framework that combines generative augmentation with an attention-enhanced EfficientNet-B3 backbone to classify foot images into ulcer and healthy-skin classes. A deep convolutional GAN (DCGAN), trained only on the training partition, synthesizes additional images to improve training-data diversity and reduce class imbalance; synthetic images pass a quality filter and are capped at a fixed share of the training set. A Convolutional Block Attention Module (CBAM) is attached to the final EfficientNet-B3 feature map so that the classifier emphasizes ulcer-relevant channels and regions rather than background artifacts, and Grad-CAM is used to inspect where the model looks. Training uses AdamW with cosine-annealing learning-rate scheduling. The GAN is used only at training time, so the deployed model adds roughly 0.3 million parameters to a 12.2-million-parameter backbone. Because implementation and empirical testing are at the design stage, the accuracy, precision, recall, F1-score, specificity, and ROC-AUC values reported here are design targets rather than measured outcomes; an ablation protocol is specified to test each component.

257.LIFE SKILLS IN THE MACHINE AGE: A SYSTEMS THINKING FRAMEWORK

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Today’s fast-moving world requires both self-reflection and flexibility. This study emphasises the need to harness technology as a force for good and ensure innovation aligns with humanity's highest ideals. It underscores the collective duty to ensure that a strong ethical framework guides progress and supports a future in which technology enriches, rather than undermines, our shared humanity. To make this understanding effective, one needs to learn not only technical skills but also how to approach complex problems such as sustainability, the relationship between people and machines, and existential concern. The present study aims to bring to the forefront a paradigm shift towards life skills education that will prepare people for both a mechanised world and a future that is both sustainable and based on

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strong morals. To better understand the concept, a researcher explores a system dynamics model to make a human-driven life.

258 . A SURVEY ON IOT, HYBRID ARTIFICIAL INTELLIGENCE, ENCRYPTION, AND BLOCKCHAIN- BASED SECURE CHRONIC DISEASE MANAGEMENT

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Chronic disease (CD) management poses significant challenges due to the long-term nature of illnesses such as kidney disease, diabetes, cardiovascular disease, and related conditions. Current chronic disease management (CDM) systems typically do not enable continuous monitoring of patients, nor do they accurately predict disease risks. They may additionally expose confidential patient health information to unauthorized users, and approaches that store patient records centrally raise the risk of data being altered or lost. Moreover, numerous current CDM systems lack efficient integration of live IoT sensor readings into AI predictive models, plus a provision for secure management of sensitive patient data. This paper surveys recent literature at the intersection of IoT, Hybrid AI (HAI), encryption, and Blockchain to construct a secure system for managing CD. We analyse deep learning frameworks, ensemble approaches, and blockchain secured data management systems, identifying key performance trends, research gaps, and future directions.

259. GUARDIANFALL AI: A POSE-ESTIMATION-BASED REAL-TIME FALL DETECTION AND EMERGENCY RESPONSE PLATFORM

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Falls remain one of the leading causes of accidental injury among elderly and independently living individuals, and delayed detection frequently turns a survivable incident into a medical emergency. This paper presents GuardianFall AI, a contactless, camera-based fall monitoring system that combines a YOLOv8-Pose keypoint detector with a lightweight rule engine to distinguish genuine falls from everyday activities such as sitting at a desk, bending down, or sleeping. Rather than relying on a single bounding-box aspect-ratio heuristic, the system tracks the spatial relationship between the shoulder and pelvic girdle to estimate torso inclination, and combines this with short-horizon motion history to separate a controlled recline from a sudden collapse. Two rule-based safeguards—an anatomical upright

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veto for near-field desk scenarios and a static-versus- dynamic disambiguation rule for resting postures—are introduced to reduce the false-alarm rate that commonly affects vision- only fall detectors. When a fall is confirmed, the system opens a ten-second cancellation window with an on-screen countdown and audible alert before autonomously notifying a designated contact over WhatsApp (via a Wi-Fi based webhook) and email, together with the location and a snapshot of the incident. The paper describes the system architecture, the detection and disambiguation logic, and the evaluation methodology. Experimental validation on 75 locally recorded trials across diverse indoor activity sequences demonstrates an overall detection accuracy of 93.33%, a precision of 94.59%, a recall of 92.11%, and sustained real-time throughput of 32 to 41 frames per second, while eliminating false triggers during seated desk work and recumbent resting.

260 .ADAPTIVE TIMING-BASED MORSE CODE DECODER FOR REAL-TIME TEXT RECONSTRUCTION

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Morse code is a temporal communication system in which information is represented through sequences of short and long signals separated by exact intervals. This paper deals with the design and implementation of an Arduino-based real time Morse code decoder that turns manually generated signals into readable text; A push-button is used to generate Morse input, while an Arduino UNO measures signal duration, identifies dots and dashes, reconstructs complete Morse codewords, and produces the corresponding characters through a serial interface. The implemented decoder extends a conventional fixed-threshold approach by incorporating adaptive timing estimation. Instead of depending exclusively on one absolute dot/dash threshold, the system estimates the operator's elementary Morse timing unit and evaluates the signal durations in relation to this value. A debounce mechanism suppresses short unwanted transitions, while confidence-aware timing classification prevents uncertain observations from excessively affecting the timing estimate. The system was tested using individual characters, multi-symbol sequences, the standard SOS sequence, variable operating speeds, and short unwanted transitions, and this implementation shows that the temporal structure of Morse code can be interpreted reliably using a low-cost microcontroller while retaining a simple and deterministic mechanism for character decoding. The study includes a compact embedded platform that combines the historical communication principle of Morse code using adaptive real-time signal processing.

261.INTELLIGENT INFRASTRUCTURE ASSESSMENT AND PREDICTIVE MAINTENANCE USING AI

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The rapid growth, increasing complexity, and aging of civil infrastructure have created a need for efficient, reliable, and intelligent approaches to infrastructure inspection and maintenance. Conventional inspection practices largely depend on periodic manual surveys, engineering reports, visual observations, and expert judgment. Although these methods remain important, they are time-consuming, labor-intensive, and difficult to scale when infrastructure assets generate large volumes of heterogeneous information. This paper proposes an Intelligent Infrastructure Assessment and Predictive

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Maintenance Framework using Artificial Intelligence (AI) based on a multi-modal and multi-agent architecture. The proposed framework integrates diverse infrastructure data sources, including site photographs, drone videos, engineering drawing PDFs, inspection reports, maps, satellite images, complaint text, and voice notes. AI techniques are used to process these data sources and extract relevant information related to infrastructure condition. A multi-agent architecture is proposed in which specialized agents perform visual damage detection, drawing interpretation, document analysis, GIS-based assessment, structural risk prediction, repair cost estimation, and maintenance planning. The vision-based agent identifies and classifies visible defects such as cracks and other forms of deterioration, while other agents interpret engineering and contextual information required for comprehensive assessment. The proposed system establishes an integrated workflow from data acquisition and defect identification to condition assessment, risk evaluation, remaining useful life estimation, repair-cost assessment, and maintenance prioritization. Unlike conventional approaches in which inspection information is often handled as separate reports, the proposed framework aims to establish a centralized infrastructure intelligence platform capable of combining multiple forms of evidence. The framework can support engineers, infrastructure managers, and authorities in making faster and more consistent maintenance decisions. The paper presents the conceptual architecture, workflow, AI- agent functions, implementation strategy, expected benefits, challenges, and validation framework for the proposed system. The approach has potential applications in buildings, bridges, roads, and other civil infrastructure systems and represents a step toward intelligent, predictive, and sustainable infrastructure management.

262.ADAPTIVE MEMORY DESIGN FOR LOW POWER VLSI ARCHITECTURE

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263. MEDICARE: AN INTEGRATED MEDICINE REMINDER AND DIGITAL HEALTH MANAGEMENT SYSTEM

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Medication adherence and proper management of personal health information are important aspects of effective healthcare. However, many individuals experience difficulties in following prescribed medication schedules, remembering medicine timings, maintaining medical records, and accessing essential health information when required. Existing healthcare applications often provide medication reminders or digital healthrecord management as separate services, requiring users to depend on multiple platforms. This paper proposes MediCare, an integrated Medicine Reminder and Digital Health Management System designed to simplify medication scheduling and personal health-record management through a unified digital platform. The proposed system combines medicine scheduling, automated reminders, medication adherence tracking, digital prescription storage, medical history management,appointment reminders, and emergency medical information into a single application.

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264.TIME-SPACE FRACTIONAL THERMOELASTIC ANALYSIS OF A THICK CIRCULAR PLATE WITH NON-HOMOGENEOUS ROBIN BOUNDARY CONDITIONS AND RELAXATION TIME

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A fractional-order thermoelastic model for a thick circular solid plate is developed in this study. The formulation employs the Caputo time-fractional derivative and the Riesz space- fractional derivative (fractional Laplacian) to simultaneously capture temporal memory effects and non-local spatial interactions, integrated with a single relaxation time and non- homogeneous Robin boundary conditions. The plate boundaries are assumed traction-free, and an axisymmetric heat source is applied externally. The governing boundary value problem is solved via Laplace and fractional Fourier integral transforms, reducing the system to an eigenvalue problem, with numerical inversion of the Laplace-domain solutions performed using the Gaver–Stehfest algorithm. The proposed framework is validated through a numerical case study on a copper plate subjected to a partially distributed heat flux under non-homogeneous Robin conditions at the curved surface. Distributions of temperature, displacement, and thermal stresses are computed and compared against predictions from classical and tempered fractional models. The results demonstrate that increasing the space fractional order β introduces pronounced non-local effects, producing heavy-tailed temperature profiles and significantly mitigating thermal stress concentrations. Additionally, the non-homogeneous Robin boundary condition is shown to provide effective active control over edge heating and cooling, offering enhanced design flexibility for advanced thermal management applications.

265. AN ENHANCED MACHINE LEARNING FRAMEWORK FOR CROP AND FERTILISER RECOMMENDATION WITH SOIL NUTRIENT DEFICIENCY ANALYSIS

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Agriculture plays a vital role in food security and economic development, yet inappropriate crop selection and unbalanced fertilizer application can reduce productivity, increase cultivation costs, and affect soil health. This paper presents an enhanced machine learning framework for crop and fertilizer recommendation with soil nutrient deficiency analysis. The proposed system uses an expanded dataset of 4,000 samples with ten input parameters representing soil and environmental conditions. The framework operates in two stages. First, a machine learning model recommends the most suitable crop based on the given conditions. Second, the system analyses the nutrient requirements of the selected crop, identifies deficiencies in nitrogen, phosphorus, and potassium, and recommends an appropriate fertilizer with an explanation of the nutrient deficiency. Random Forest, Decision Tree, K-Nearest Neighbours, Support Vector Machine, and XGBoost algorithms are trained and compared using accuracy, precision, recall, and F1-score. By integrating crop recommendation, nutrient deficiency

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analysis, and fertilizer recommendation, the proposed framework provides an interpretable decision-support system for agriculture. It aims to promote balanced fertilizer usage, reduce unnecessary input costs, and support sustainable and efficient farming practices.

266. FORENAI: AI – POWERED FORENSIC INVESTIGATION ASSISTANT

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Forensic examiners work with many kinds of evidence at a crime scene, including weapons, bloodstains, and wounds, and each is usually studied separately with its own tools. AI and computer vision can help with this analysis, but most existing work stays within one type of evidence. Current computer-vision approaches can detect weapons in crime-scene images, but they handle multiple weapons poorly and cover only a limited range of weapon categories. They focus mainly on weapon detection and reconstruction, and they do not include bloodstain or wound analysis. An examiner therefore has to move between separate tools to piece the evidence together. We propose ForenAI, an AI-assisted application that brings weapon, bloodstain, and wound analysis together for the preliminary examination of evidence. A YOLO-based module detects multiple weapons in a single image across a wider set of categories. A ResNet18-based module classifies wounds using region proposals. A fuzzy active-contour module segments bloodstains and fits ellipses to analyze their geometry. Grad-CAM shows which image regions drive each CNN prediction, and a report module combines all results into a structured forensic-style summary.ForenAI gives examiners an integrated and explainable way to screen evidence early. It is meant to assist forensic experts, not replace their judgment.

267. A SURVEY ON AN AGENTIC RAG AND LLM-BASED FRAMEWORK FOR INTELLIGENT HEALTHCARE INFORMATION RETRIEVAL AND SKIN DISEASE PREDICTION

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Traditional health care systems can be limited when it comes to comprehending complex medical searches, finding medically relevant information or producing dependable answers. Large Language Models (LLM), in combination with Retrieval-Augmented Generation (RAG), provide a viable solution in addressing some of the limitations associated with accessing reliable health care related information. Although most of the current RAG-based health care applications are able to retrieve relevant information and produce responses, their ability to perform multi-step reasoning, understanding context, verifying the produced information is limited. Applications involving identification of skin diseases also need to extract both texture based and deep-level visual features effectively in order to identify reliability. This project work addresses these limitations by introducing a hybrid agentic workflow to support both factual evidence verification and contextual understanding as well as a five-stage agentic workflow for providing responses: Think, Retrieve, Act, Reason and Answer. The proposed Agentic RAG and LLM- based framework for intelligent Healthcare Information Retrieval and skin disease prediction uses the Gemini model for reasoning. Hugging Face Sentence Transformers are used for generating embeddings for semantics and FAISS (Facebook AI Similarity Search) for performing similarity based retrieval from a knowledge base of health care data. Additionally, the proposed framework incorporates a Genetic Algorithm (GA) for selecting the most relevant features from the extracted skin-image features, such as Local Binary Pattern (LBP) and Gray-Level Co-occurrence Matrix (GLCM) features. The selected features are then given to a Multilayer Perceptron (MLP)-based Deep Neural Network (DNN) classifier for skin disease prediction. The MLP-DNN model is designed to classify the input into the corresponding skin disease category. The predicted result is further integrated with the Agentic RAG framework to retrieve relevant medical information and generate a contextual and conversational response for the user.

268. AGRIFUSION AI: MULTIMODAL INTELLIGENCE FOR CLIMATE-SMART AND PRECISION AGRICULTURE

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Smallholder farmers still make harvest, storage, and selling decisions on intuition, which exposes them to weather shocks, spoilage, and poor price realization. Existing agricultural tools usually address one concern at a time, such as crop monitoring, weather advisory, or market price reporting, and rarely turn these into a single actionable recommendation. This paper presents AgriFusion AI, a multimodal intelligence framework for climate-smart and precision agriculture that fuses satellite imagery, field-level crop images, soil and moisture readings, weather forecasts, historical crop cycles, and mandi price data into one decision pipeline. The framework has five layers: multimodal data acquisition and quality control, crop-state estimation, climate-risk assessment, price forecasting, and an explainable recommendation engine. YOLOv11-based vision models estimate crop maturity and expected yield, LSTM networks forecast market prices, and a fusion module combines these outputs with weather and soil context to recommend the optimal harvest window, the best market ranked by price and distance, and whether to store or sell immediately. Transport and buyer coordination alerts are delivered to the farmer's mobile device. The proposed evaluation protocol covers harvest-window accuracy, yield-estimation error, price- forecast error, net-profit gain over sell at harvest, and system latency, with a target prediction accuracy of 90% or above. AgriFusion AI is intended as a decision support prototype that reduces post-harvest loss, improves farm income, and supports sustainable, climate-resilient farming.

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269. AI-BASED FOOD DEMAND, COOKING RESOURCE OPTIMIZATION AND SMART DONATION SYSTEM

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Food wastage and improper meal planning are major challenges in colleges, hostels, hotels, and other food-serving organizations. Preparing excess food leads to the wastage of ingredients, water, energy, and money, while many people still struggle to access sufficient food. To address these problems, this project proposes an AI-Based Food Demand, Cooking Resource Optimization and Smart Donation System. The system uses machine learning to predict the required number of meal portions based on previous consumption, menu details, weekdays, and special events. Based on the predicted demand, it helps estimate the required ingredients and cooking resources, reducing unnecessary preparation. If surplus food is available, the system identifies suitable nearby NGOs and food donation centers and helps coordinate timely distribution. A centralized dashboard allows administrators to monitor food demand, resource usage, surplus quantities, and donation activities. By combining demand prediction, smart cooking planning, and food redistribution in a single platform, the proposed system aims to reduce food waste, save resources, and support people in need. This project provides a practical and innovative approach to building a more efficient, sustainable, and socially responsible food management system.

270. PORTABLE SPIROMETER FOR LUNG FUNCTION TESTING AND EARLY DISEASE DETECTION

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Peak expiratory flow rate (PEFR) is an important respiratory parameter used to monitor the maximum airflow during forceful expiration. Conventional pulmonary function testing equipment may be costly or less accessible in some healthcare settings. This paper presents the design of a low-cost portable PEFR monitoring prototype using an MPXV7002DP differential pressure sensor and an Arduino Uno microcontroller. The system consists of a mouthpiece, pressure sensor, LCD, and HC-05 Bluetooth module. During expiration, the sensor measures the pressure difference associated with airflow. The Arduino Uno processes the sensor output and estimates PEFR using a calibrated pressure-to-flow relationship. The estimated value is displayed on the LCD and can be transmitted wirelessly through Bluetooth. The proposed system aims to provide a compact platform for preliminary respiratory monitoring. Calibration and comparison with a reference peak flow meter are required to evaluate its accuracy and repeatability. The prototype is intended for educational and preliminary monitoring purposes and requires further validation before clinical application.

271.EARLY BRIDGE DAMAGE DETECTION

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Bridges are important components of transportation infrastructure, and their structural safety is essential for public safety and reliable transportation. Traditional bridge inspection methods mainly depend on manual observation and periodic maintenance, which can be time-consuming, expensive, and insufficient for detecting early structural abnormalities. To address these challenges, this paper presents an intelligent bridge damage detection framework that integrates the Internet of Things (IoT), Artificial Intelligence (AI), and Machine Learning (ML) for continuous structural health monitoring. The proposed system uses vibration sensors, sound sensors, and tilt sensors connected to an ESP32 microcontroller to collect structural data in real time. The collected data is transmitted to a Python-based monitoring system, where signal preprocessing, baseline calibration, and anomaly detection are performed. Isolation Forest is used to identify unusual sensor patterns, while Random Forest supports structural risk classification. A multi sensor fusion mechanism combines AI anomaly scores, damage accumulation, sound analysis, and tilt analysis to estimate the overall structural risk. A Streamlit-based dashboard displays sensor readings, damage percentage, structural health status, and warning notifications. The system classifies bridge conditions into Normal, Load, and Damage stages to support early warning generation and predictive maintenance. The proposed framework provides a low-cost and scalable approach to intelligent bridge monitoring. Quantitative performance claims require further validation using representative, labelled measurements.

272. BUGSENSE AI: AN INTELLIGENT GITHUB BUG TRIAGE AND REPOSITORY-AWARE ROOT CAUSE ANALYSIS FRAMEWORK USING LOCAL LARGE LANGUAGE MODELS

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In today's software world, timely issue report triage and accurate resolution of these reports in collaborative version-control repositories are crucial to modern software maintenance. Despite all this, today's development teams invest significant human effort in reading and understanding unstructured issue descriptions, attempting to replicate failures, pinpointing problematic modules, and writing suitable patches. Many issue tracking systems do not provide automatic tools to collaborate with others on reasoning with natural-language problem reports, repository abstract syntax trees (ASTs), call graphs and historical change context. In this paper, we introduce BUGSENSE AI, a repository-aware bug localization and root cause analysis framework that is autonomous and runs entirely using a set of locally deployed large language models (LLMs) with enterprise code privacy. BUGSENSE AI is orchestrated through a pipeline of specialised agents, synchronously triggered via GitHub Webhooks, through a FastAPI backend. Reports are normalized and categorized based on functional, performance, dependency, configuration, build an security categories and a vector database (ChromaDB) using dense semantic embeddings and static call graphs fetches candidate code snippets. A dedicated diagnostic agent pulls repository context to help isolate faults and generate candidate diffs which are presented to maintainers via an interactive dashboard where a developer is in the loop before pull request generation automatically takes place. To assess BUGSENSE AI, we test on a carefully selected set of 1,250 production defects from 15 production repositories. It not only delivers high multi-class classification accuracy (91.12%, macro F1 = 0.9117) and Top-5 method-level localization (87.60%,

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MRR = 0.792), but it also achieves a valid fix recommendation rate of 79.20%, with mean triage turnaround reduced from 3.1 hours to 8.6 minutes under strict execution on-premises.

273.HYBRID CONTACT–CONTACTLESS IOT WEARABLE FOR CONTINUOUS VITAL MONITORING IN INFECTIOUS ENVIRONMENTS

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Continuous patient monitoring is particularly important in infectious environments such as isolation wards, intensive care units, quarantine centres, and emergency-care areas, where repeated bedside interaction can increase workload and physical exposure [1][6]. The proposed Hybrid Contact–Contactless IoT Wearable combines contact-based and contactless sensing in a single compact platform. The MAX30102 is used for heart rate and peripheral oxygen saturation (SpO2), the MLX90614 provides contactless temperature measurement, and the HLK-LD2410 radar sensor is incorporated for respiratory activity and body- movement monitoring. An ESP32 microcontroller acts as the central processing and communication unit. Sensor information is acquired, filtered, normalized, analysed using threshold-based logic, and transmitted through Wi-Fi to a cloud platform using MQTT or HTTP [2]. The system is intended to provide local display, remote visualization, historical data storage, and abnormal-condition alerts. A compact ergonomic enclosure, skin-friendly contact materials, adjustable strap, rechargeable battery, and bamboo-based structural concept are included to support prolonged use. This paper presents the system concept, literature-based research gap, architecture, implementation methodology, ergonomic design, energy management, and proposed validation framework. Quantitative performance values are intentionally reserved for the experimental phase.

274. AI-BASED SMART TRAFFIC PREDICTION, ACCIDENT AND ROAD BLOCKAGE DETECTION, AND INTELLIGENT ROUTE OPTIMIZATION

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Urban road users lose a significant amount of time and safety to congestion, accidents, and unannounced road closures, yet most navigation tools react to conditions only after they have formed. This paper presents the design and prototype evaluation of an AI-based smart navigation system that integrates four capabilities in a single pipeline: (i) machine-learning-based short-term traffic prediction, (ii) vision-based accident detection using a YOLO object detector, (iii) road blockage detection that fuses camera analysis, user reports, and authority notices, and (iv) an intelligent route optimization engine that converts these signals into a dynamic edge cost for shortest-path search. The system is implemented with Python, Flask, MySQL, OpenCV, and OpenStreetMap. On a synthetic benchmark of 40 road segments over 120 days, a Random Forest predictor reduced mean absolute error by 56% against a persistence baseline. In a 200-trial routing simulation on a 12 × 12 road grid with random incidents, the proposed dynamic cost reduced mean travel time by 34% relative to shortest-distance routing and eliminated all routes passing through blocked segments. The results are obtained on synthetic data and demonstrate the design rationale; validation on real traffic feeds is identified as future work.

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275. HIREWAVE: A RECRUITMENT PLATFORM WITH RESUME PARSING, EXPLAINABLE CANDIDATE RANKING, AND REAL-TIME RECRUITMENT NOTIFICATIONS

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Online recruitment can require candidates to enter information that is already present in their resumes, while recruiters must review applications and communicate status changes. This paper presents HireWave, a recruitment platform that combines rule-based resume parsing, an interpretable weighted ranking model, and real-time notifications. We define the recruitment-efficiency setting and outline four measurable contributions: (C1) a structured resume-to-profile extraction pipeline with evaluated accuracy; (C2) an explainable weighted candidate-ranking model; (C3) an integrated recruitment work flow with Socket.IO-based real-time notifications, which is evaluated for latency; and (C4) a four part empirical evaluation covering resume parsing accuracy, application efficiency, recruiter filtering quality, and user satisfaction. On a set of 100 resumes, the parser achieved a macro-average F1-score of 0.920. In a study with 30 participants, mean completion time fell from 4 min 12 s to 2 min 26 s (42.1%; $p < 0.001$), while overall satisfaction with HireWave was 4.4/5.0. At 100 concurrent connections, mean notification latency was 156 ms and the 95th-percentile latency was 278 ms. These results suggest that combining profile ex- traction, transparent scoring, and status notifications can reduce effort in the evaluated recruitment workflow. The study does not establish that the ranking model outperforms alternative matching methods.

276.EMBEDDED WIRELESS COMMUNICATION INTERFACE FOR COCHLEAR IMPLANT DATA TRANSFER

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Cochlear implants restore hearing by electrically stimulating the auditory nerve, and increasingly rely on wireless links to transfer audio data between external and implanted components, replacing or supplementing wired and inductive connections to improve patient comfort and mobility. Continuous wireless transmission, however, can raise the local temperature of implanted or wearable electronics, which poses a safety risk to surrounding tissue if left unmanaged. This work presents an embedded wireless communication interface for cochlear implant data transfer that integrates real-time temperature monitoring with safety-driven transmission control. Audio signals are acquired by a microphone, conditioned and amplified, digitized by an embedded microcontroller, and transmitted

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wirelessly to a receiver module, while a dedicated thermal safety block continuously polls an analog temperature sensor and reduces transmission power whenever a defined safety threshold is exceeded. The system was implemented as a low-cost prototype using an Arduino Uno (ATmega328P), an LM35 analog temperature sensor, visual and audible alerting, and a liquid-crystal display for real-time status reporting, and was evaluated for its ability to detect elevated-temperature conditions and respond by throttling transmission before an unsafe condition is reached. The architecture and validation approach are designed to generalize to any embedded biomedical wireless blink where thermal safety must be supervised alongside data transfer.

277. LIFELINK AI: A HOSPITAL-FIRST, CONSENT-AWARE EMERGENCY RESPONSE PLATFORM WITH MULTILINGUAL AI ASSISTANCE

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Emergency care outcomes depend on how quickly a patient reaches a facility that can actually receive them. Existing mobile safety applications mostly forward a location and place a call; they rarely confirm that a destination hospital has capacity before an ambulance is sent. This paper presents LifeLink AI, a web and mobile platform that reorders the response sequence: the system first locates hospitals reporting a free emergency bed for the required speciality, obtains an explicit acceptance from one of them, and only then lets that hospital send its own ambulance to the patient's live location. The platform adds a keyword-based multilingual voice trigger (Tamil, English, Hindi), a consent-based Digital Health Card, time-limited location sharing with family, curated first-aid guidance, and an experimental severity classifier that is wrapped in a conservative rule-based safety layer. We describe the architecture, the dispatch protocol, the data model and the privacy controls, and we report a controlled experiment on a synthetic 5,000-record dataset in which a Random Forest reached 95.7% cross-validated accuracy and 97.4% recall on the critical class. Because the labels are synthetic, these figures measure model fit to a documented rule set, not clinical performance. The system is a prototype and is not a certified dispatch or diagnostic service.

278.INFRANEXUS AI: A ZERO-TRUST MULTIMODAL EDGE INTELLIGENCE FRAMEWORK FOR GEOSPATIAL URBAN HAZARD DETECTION AND AUTOMATED WORK-ORDER DISPATCH

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Rapid urbanization severely degrades municipal civil infrastructure through unmonitored road craters, water main pipe ruptures, structural bridge shear fissures, and electrical hazards. Conventional civic complaint portals suffer from heavy inspection backlogs, manual triage bottlenecks, and complete vulnerability to fraudulent submissions, yielding an average remediation turnaround exceeding 21 days. This paper introduces InfraNexus AI, an autonomous closed-loop urban infrastructure monitoring and rapid civic action framework integrating zero-trust edge computer vision, tactical geospatial intelligence, and automated contractor dispatch. The system features a sub-second client-side HTML5 Canvas pixel telemetry engine that analyzes optical Bayer sensor noise variance and chromatic tone profiles to detect and reject synthetic AI- generated deepfakes, domestic scenes, and non-hazards in under 140ms before network transmission. Valid civil defect images undergo multimodal edge inference to classify hazards across six municipal domains, calculate physical cavitation dimensions, estimate a dynamic severity index (1–100), and quantify daily carbon penalties. Incident coordinates are deduplicated on a tactical GIS grid using Haversine radius grouping, triggering automated work-order generation and an Official Memorandum email loop via Google SMTP with tamper-proof PDF dockets. Experimental validation demonstrates 96.8% defect detection accuracy, 99.2% anti-fraud precision, and a reduction in civic repair turnaround from 21.4 days down to 3.5 hours, establishing an actionable paradigm for sustainable smart city governance.

279. PHYSICAL RECREATION OF TAMIL PALM LEAF MANUSCRIPTS USING IMAGE-BASED DOT-PATTERN TRAJECTORIES

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Tamil literature spans over 3,000 years, holding a vast classical heritage which is recorded by ancient poets and scholars on palm leaf manuscripts. However, these fragile materials are highly prone to age-related degradation and it is difficult to preserve them. While modern preservation initiatives focus heavily on digitalization, digital formats do not retain the physical form, originality, and antique feel of

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manuscripts. To bridge this gap, this research introduces a method to physically reproduce ancient scripts onto new palm leaves. For this the images of the manuscripts are captured and the digital images are processed via AI and image-processing algorithms to remove background noise and to enhance tamil characters. Advanced image processing algorithms such as adaptive gaussian thresholding, stroke centerline extraction, polyline stoking, dot conversion are used to replicate the text in the manuscripts to dot pattern trajectories. The average deviation between the original and processed image is done using the F measure complemented by the modified Hausdorff distance. The results show a geometric similarity of 92.7% at a ±2-pixel tolerance, rising to 99.1% at ±3 pixel. A browser-based graphical user interface is developed to run the recreation workflow.

280. HARNESSING SOLAR ENERGY FOR ZERO CARBON EMISSION HYDROGEN PRODUCTION

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Green hydrogen is now-a-days promising way for clean energy that uses solar photovoltaic technology along with electrolysis process by eliminating carbon footprints. The ob- jective of the project proposed is to produce green hydrogen with zero carbon emission by using renewable solar energy with highly efficient solar driven water electrolysis process. Here, PV Panels convert sunlight into electricity powering mini electrolyzer setup to split Water (H2O) into Hydrogen (H2) and Oxygen (O2). Unlike grey and blue hydrogen, green hydrogen is an approach for enabling carbon free hydrogen production and the oxygen thus produced here is also a useful substance in hospitals in emergency purposes, as we have seen the necessity of oxygen in COVID-19 pandemic times. Overall, the main aim of the project is to design a Solar Driven Hydrogen Production System with better efficiency and reduced cost at a small-scale drive prototype for reducing carbon footprints and supporting clean fuel in Industrial and Transportation sectors. Index Terms—Green Hydrogen, Solar Photovoltaic panels, Solar driven Electrolysis, Decarbonization, Fossil Fuels

281. DECEPTION-AS-A-SERVICE (DAAS): REAL-TIME ACTIVE DEFENSE AGAINST AI VOICE CLONING USING AUDIO WATERMARKING AND AI HONEYPOTS

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Voice cloning has quietly become one of a fraudster's most effective tools. Given just a few seconds of an executive's recorded speech, an attacker can now produce a convincing imitation, opening the door to social engineering schemes that trick employees or partners into handing over money or sensitive information. Existing detectors such as AntiFake and RoVo can identify a cloned voice, but only after a call has already ended, by which point the damage is done; neither tool monitors the conversation as it unfolds, and neither retains any record of the attacker for later analysis. This paper introduces Deception-as-a-Service (DaaS), a framework designed to catch voice-cloning attempts while a call is still underway and respond to them immediately through an AI honeypot. DaaS watermarks an executive's genuine voice recordings in advance and combines that safeguard with deepfake-detection models that continuously classify a live call as genuine, suspicious, or cloned. Calls flagged as suspicious or cloned are quietly rerouted into an AI-driven honeypot conversation built to keep the attacker talking and revealing more, while administrators watch the exchange unfold on a live dashboard. Taken together, watermarking, real-time deepfake detection, honeypot engagement, and complete attack logging give an enterprise the ability to respond to a cloning attempt as it happens rather than after the fact. The present paper focuses on the design and methodology behind DaaS; building and evaluating a working prototype is left for follow-up work.

282 .A SMARANDACHE SEMIRING-BASED FRAMEWORK FOR POST-QUANTUM KEY ENCAPSULATION: ALGEBRAIC CONSTRUCTION AND SECURITY ANALYSIS

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This paper examines the structure of Smarandache semirings and some of their applications in public-key cryptography and computational hardness. Based on the structure that resulted, a candidate key-encapsulation mechanism is designed, which includes key generation, key encapsulation, key decapsulation algorithms and a formal correctness proof. A computational problem which attempts to express how hard it is to get secret information from public information. This proposed construction is subsequently considered and compared with classical algebraic attacks, structural decomposition techniques and to relevant models of quantum attacks. The purpose of this study is to see if Smarandache semirings is used as a practical basis for a new algebra for post-quantum cryptography. This analysis helps to elucidate the possible benefits and the mathematical and computational constraints of this method.

283. INTELLIGENT AI-BASED FINANCIAL FRAUD DETECTION AND RISK ASSESSMENT SYSTEM

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The Intelligent AI-Based Financial Fraud Detection and Risk Assessment System is an Artificial Intelligence-based system developed to identify suspicious financial transactions and assess their potential fraud risk. With the rapid growth of digital banking, online payments, and mobile transactions, financial fraud has become a major security concern. Traditional rule-based methods may not effectively detect new and changing fraud patterns. The proposed system analyzes various factors such as transaction amount, frequency, location, device information, transaction time, and previous transaction history to understand the user's normal transaction behavior. When a new
transaction occurs, it is compared with the user's behavioral profile. Unusual activities are identified and assigned a low, medium, or high risk level based on the detected behavior. An important feature of the system is Explainable Fraud Detection, which provides understandable reasons for identifying a transaction as suspicious. For example, the system may indicate that a transaction is risky because of a new device, unusual location, high transaction amount, or unusual transaction time. This makes the fraud detection process easier for users and administrators to understand. The system also includes Fraud Pattern Memory, which stores important characteristics of previously detected suspicious transactions. Future transactions are compared with these stored fraud patterns. If a similar pattern is identified, the system can generate an early warning and help detect repeated or related fraudulent activities. Another proposed feature is the What-if Fraud Simulator, which allows different transaction conditions to be tested and helps understand how they affect the fraud risk. For example, the system can compare scenarios involving OTP verification, known or new devices, and normal or unusual transaction locations. This provides a better understanding of how different conditions influence the overall risk. The system can support different actions based on the calculated risk score. Low-risk transactions can be processed normally, while medium-risk transactions can require additional verification. High-risk transactions can be flagged for further investigation. This approach helps reduce unnecessary alerts and allows attention to be focused on transactions that require immediate action. The system is designed to continuously improve its fraud detection capability by learning from newly identified suspicious transactions and changing user behavior. By combining historical transaction data, behavioral patterns, and multiple risk factors, it can adapt to emerging fraud techniques. This adaptive approach can help improve detection accuracy and reduce false alerts while maintaining a secure transaction monitoring process. Overall, the proposed system aims to provide an intelligent, explainable, adaptive, and behavior-based approach for early financial fraud detection and risk assessment. It can help users and financial institutions identify suspicious activities more effectively and support faster and informed decision-making.

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284.GENETIC ALGORITHM DRIVEN PORTFOLIO OPTIMIZATION IN INDIAN EQUITY MARKETS

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This paper presents a Genetic Algorithm (GA) combined with Modern Portfolio Theory (MPT) for optimizing portfolios of 15 NSE-listed Indian stocks using data from January 2022 to August 2026. The GA uses a population of 150 over 100 generations to maximize the Sharpe ratio without restricting long and short positions. The optimized portfolio achieved a Sharpe ratio of 1.9556, an expected annual return of 104.78%, and volatility of 49.90%, outperforming the equal-weighted benchmark. Backtesting of a ₹1,00,000 investment resulted in approximately ₹6.47 million, with a total return of 6,372%. The results demonstrate the potential of GA-based portfolio optimization, while highlighting the need to consider transaction costs, liquidity, risk management, and limitations of historical data.

285. METHODOLOGIES FOR CROP RECOMMENDATIONS

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Existing methodologies for crop recommendations depend largely on centralized cloud computing infrastructure, which introduces network latency of multiple seconds and makes existing methods impractical to operate without an Internet connection. Moreover, commercial nutrient probes are too expensive for resource-constrained systems, resulting in a usage of artificial soil nutrient representations in the baseline study. This project envisions an autonomous, low-latency, and edge- native precision agriculture system, which is not dependent on cloud APIs and remote servers. In our proposed architecture, a local field sensing node (Arduino Nano) is used to collect real-time physical measurements of soil moisture and temperature values through capacitive soil moisture and digital DS18B20 temperature probes, respectively. Due to the cost restrictions of industrial NPK probes, important soil macronutrient properties (Nitrogen, Phosphorus, and Potassium) will be artificially generated and added to the telemetry stream in accordance with the baseline study approach. Sensor readings are wirelessly communicated through a 2.4 GHz Nordic Enhanced ShockBurst (ESB) radio link using nRF24L01+ modules to the receiver gateway equipped with an ESP32 microcontroller. Instead of uploading telemetry to external cloud services, the key innovation point of this project is the

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utilization of an on-device TinyML classification engine. The pre-trained Enhanced Gaussian Naive Bayes (EGNB) classifier, trained using the default multi-class agronomic dataset of 2,200 samples, where feature scaling is done using StandardScaler and variance smoothing is incorporated, is ported in the native C++ code of ESP32 board.This application intends to make deterministic offline inference under 10 milliseconds, providing instant crop recommendations out of 22 possible classes to be displayed on the ESP32's on-board OLED screen. Through this research, we seek to validate a low-cost decision-support tool that works without internet connectivity using physical microclimates and simulated nutrients on the TinyML architecture.

286. MULTILINGUAL AI-BASED FORENSIC FACE SKETCH GENERATION AND SUSPECT IDENTIFICATION FROM EYEWITNESS DESCRIPTION

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Identification of suspects based on eyewitness descriptions is a crucial process in criminal cases. However, traditionally, the process of producing forensic sketches depends primarily on the work of artistic professionals. Varying interpretations of verbal descriptions lead to the inconsistencies in the final facial depictions produced. Language barriers and human memory limitations further complicate this process. This paper presents an innovative Multilingual AI-Based Forensic Face Sketch Generation and Suspect Identification System that will transform voice descriptions given by eyewitnesses into a visual representation of the suspect's face. The practical idea presented in this study consists of converting descriptions given by eyewitnesses into voice and then automatically making a search for the criminal in the database based on markings obtained from the eyewitness testimonies using the voice-inputting technology currently available. Voice-to-text conversion is done using Whisper system that provides excellent recognition of different languages. Further, extracted descriptions are analyzed in order to define important facial features like age, sex, hairstyle, type of face, etc., and used to input the information needed into AI-based model. The created sketch is also modified into a face representation that can be compared with images kept in the criminal database. A feature encoding method based on Vision Transformer (ViT) is utilized to obtain significant facial embeddings that can be differentiated while a cosine similarity process is applied in order to compute the likeness between the generated suspect representation and the images stored in the database. The system produces ranked similar suspects and presents their facial images along with relevant criminal information.

287.BLOCKWEAVE: A PRIVATE PERMISSIONED BLOCKCHAIN FRAMEWORK FOR GI TAGGED KOTA DORIYA HANDLOOM SUPPLY CHAIN AUTHENTICATION

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The Indian handloom sector, supporting more than 4.3 million weavers and bringing in Rs. 31,000 crores each year, faces an extremely severe problem of counterfeiting. One of the most impacted Geographical Indications (GI) tagged textiles is the Kota Doriya fabric from Rajasthan. With a large number of replicas entering the market, weavers are facing considerable economic losses, and customers are deceived with poor quality products. This work, therefore, introduces BlockWeave a private permissioned blockchain-based system for GI-tagged textile supply chain verification. It uses SHA-256 hashing along with block chaining, which effectively creates tamper-proof immutable records of each saree's path from manufacturer to customer. Each saree is tagged with a QR code which upon scanning queries the blockchain, providing real-time public verification of the product's authenticity without needing a login/authentication. The system has a role stratified access mechanism to distinguish roles like the manufacturer, distributor, retailer and the customer, with a governance layer in the system which is used to approve new participants. Evaluation on performance parameters of creation latency and tamper detection rate have given us promising results: blocks are created in less than 50 milliseconds and the chain integrity validator successfully detects 100% tampering by re-calculating the SHA-256 hash for every block in the blockchain.BlockWeave offers an affordable, zero-cost alternative compared to Ethereum-based solutions while providing the same level of cryptographic security, and has been field-tested with 3 manufacturers and 5 distributors over the entire supply chain.

288. A COMPREHENSIVE REVIEW OF DIGITAL TWIN- DRIVEN HUMAN-MACHINE COLLABORATION FOR PREDICTIVE AND SUSTAINABLE SMART MANUFACTURING

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Digital Twin is a key technology in Industry 4.0 and Industry 5.0, which is considered a tool for real-time synchronisation of intelligent virtual and physical assets. This review paper will discuss the latest research, summarizing and synthesizing the Digital Twin architectures, the enablers, predictive maintenance, human-machine collaboration, smart manufacturing, sustainability and future research directions. All the reviewed studies conclude that the use of the Internet of Things (IoT), Artificial Intelligence (AI), Machine Learning (ML), cloud computing, edge computing and data analytics all have a significant impact on operational efficiency, equipment reliability, production quality and intelligent decision making. In addition to manufacturing, Digital Twin technology also has potential applications in healthcare, smart cities, transportation, aerospace, energy management, and logistics, where its real-time monitoring and predictive capabilities can optimize system performance and resource utilization. The review also identifies several areas of concern, including interoperability, cyber security, scalability, implementation costs and lack of standardization. Based on the comparative analysis, this paper highlights research opportunities for developing integrated, human-centric, explainable, and sustainable Digital Twin frameworks capable of supporting next-generation intelligent manufacturing ecosystems.

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289.AN EFFICIENT MACHINE LEARNING FRAMEWORK FOR RABIES RISK PREDICTION

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Rabies is a serious viral disease, but it can be avoided if the infected person receives early treatment. The assessment of the factors associated with exposure and clinical conditions can help in making the right decision about the case. In this research, the author offers an efficient machine learning approach to address the problem based on decision trees and random forests. The algorithm uses the clinical symptoms and exposure factors, such as vaccination, animal bite severity, wound condition, and time to treatment. The method was tested on a dataset that included 1200 instances related to exposure and patient conditions. The experimental results were obtained using the three most common supervised machine learning algorithms. The Decision Tree and Random Forest methods demonstrated outstanding accuracy (98.75% and 98.33%) and precision (98.67 and 98.44), respectively. Logistic Regression showed lower efficiency with 90.00% accuracy and 90.67 precision. The study also revealed that the time to treatment, vaccination, and bite severity were the most determinant factors in terms of the model's performance. The research provides a new perspective on addressing the critical problem of rabies risk prediction using machine learning. It should be noted that the proposed methodologies have demonstrated high efficiency based on the experiments, but they still require further testing and verification before actual implementation.

290. LifeLink AI: A Hospital-First, Consent-Aware Emergency Response Platform with Multilingual AI Assistance

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Emergency care outcomes depend on how quickly a patient reaches a facility that can actually receive them. Existing mobile safety applications mostly forward a location and place a call; they rarely confirm that a destination hospital has capacity before an ambulance is sent. This paper presents LifeLink AI, a web and mobile platform that reorders the response sequence: the system first locates hospitals reporting

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a free emergency bed for the required speciality, obtains an explicit acceptance from one of them, and only then lets that hospital send its own ambulance to the patients live location. The platform adds a keyword-based multilingual voice trigger (Tamil, English, Hindi), a consent-based Digital Health Card, time-limited location sharing with family, curated first-aid guidance, and an experimental severity classifier that is wrapped in a conservative rule-based safety layer. We describe the architecture, the dispatch protocol, the data model and the privacy controls, and we report a controlled experiment on a synthetic 5,000-record dataset in which a Random Forest reached 95.7% cross-validated accuracy and 97.4% recall on the critical class. Because the labels are synthetic, these figures measure model fit to a documented rule set, not clinical performance. The system is a prototype and is not a certified dispatch or diagnostic service.

291.AI-ENABLED SELF-POWERED INDUSTRIAL VIBRATION MONITORING USING PIEZOELECTRIC MEMS ENERGY HARVESTING

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Battery replacement is the main obstacle to putting wireless vibration sensors on every motor, pump and gearbox in a plant. This paper describes a battery-less monitoring node in which a piezoelectric MEMS cantilever array supplies all of the electrical energy, and a small neural classifier on an ultra-low-power microcontroller decides locally whether the machine is healthy, unbalanced, or shows an outer- or inner-race bearing defect, so that only a one-byte health code is transmitted. A lumped electromechanical model of a four-beam PZT harvester tuned to 120 Hz predicts 5.6 μW at 0.2 g and 17.1 μW at 0.5 g. With an energy-aware scheduler, the node can send a diagnosis every 57 s at 0.5 g and every 6.4 min at 0.2 g, which is 4.4 times as often as a node that streams raw windows. On synthetic vibration data with a mean signal-to-noise ratio of 7.7 dB, an int8 multilayer perceptron of 0.36 kB reaches 93.9% four-class accuracy, 95.2% fault detection and 6.0% false alarms. All results come from simulation, and the paper closes by listing what a hardware prototype must confirm.

292. POSTURESENSE: A WEARABLE DEVICE FOR REAL-TIME POSTURE MONITORING AND DETECTION WITH CORRECTIVE FEEDBACK

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Poor posture is a major health concern due to the excessive and prolonged usage of digital devices, computers and smartphones leading to a number of health concerns such as back pain, neck strain,

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spinal discomfort and long-term musculoskeletal issues. PostureSense is a wearable device designedMfor real-time posturing and corrective feedback. The proposed device integrates an MPU6050 accelerometer and gyroscope sensor integrated with ESP32 Devkit microcontroller to continuously monitor the user's body orientation and detect deviation from desired posture in real time. The posture is evaluated based on a measured tilt angle and a predefined threshold. When incorrect posture is detected beyond a specified threshold angle for a given time, the system provides immediate corrective feedback using a buzzer or vibration motor, thereby alerting the user to correct theirposture. The proposed wearable device is compact, low cost, portable and suitable for everyday use. By providing continuous monitoring and immediate feedback, PostureSense aims to encourage proper posture habits and support the prevention of posture-related health complications.

293.SMART CYBER THREAT DETECTION AND FORENSIC INVESTIGATION USING AI

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Modern information systems generate continuous records of authentication, resource access, privilege changes, and other user activities. These logs are valuable sources of evidence, yet their volume and temporal complexity make manual inspection slow and inconsistent. This paper presents a real-time security monitoring framework that combines Long Short-Term Memory (LSTM) sequence modeling with Transformer-based contextual analysis to identify deviations from expected user and system behaviour. Incoming events are normalized and represented as ordered sequences. The LSTM component models temporal transitions, while the Transformer component captures dependencies among events over a wider context. Their outputs are combined into an anomaly score and mapped to a configurable risk level. Depending on policy and confidence, the system can raise an alert, create an incident, or temporarily restrict an account pending administrator review. Related events are grouped into a forensic timeline containing timestamps, identities, resource references, model scores, and response actions. A central dashboard supports investigation, access restoration, and report generation. The framework is designed to connect detection, response, and evidence review in a single workflow. Since measured experimental results are not supplied in the project description, this paper specifies an evaluation protocol rather than claiming unverified performance. The proposed design emphasizes auditability, controlled automation, and the preservation of evidence needed for incident investigation.

294. INTELLIGENT MULTI-AGENT CYBER THREAT DETECTION AND DIGITAL FORENSICS

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The increasing complexity of cyber threats has created a need for security systems that can analyze information from multiple sources and support timely incident response. Large Language Models (LLMs) have attracted attention in cybersecurity because of their ability to interpret text, summarize security information, and assist with threat analysis. Multi-agent systems extend these capabilities by assigning specialized tasks to different agents and combining their findings. This survey reviews research on LLM-based cybersecurity applications, multi-agent architectures, digital forensics, and the security challenges associated with intelligent agents. Five research papers are examined to understand their approaches, application areas, and limitations. The reviewed studies highlight the potential of LLMs for phishing detection, threat intelligence, security analysis, and collaborative problem-solving. However, challenges remain in ensuring reliable outputs, protecting sensitive information, coordinating agents, and validating evidence during digital investigations. The survey identifies the need for better integration between threat detection, event correlation, and forensic investigation. It also discusses future research opportunities for developing reliable systems that can organize security events, construct attack timelines, and support evidence-based incident reporting.

295.AI-POWERED MULTIMODAL DEEPFAKE DETECTION SYSTEM

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The rapid growth of generative artificial intelligence has made it possible to create realistic manipulated images, videos, and audio recordings. Such content can be misused for impersonation and misinformation. This paper describes a multimodal deepfake detection framework with separate pipelines for image, video, and audio analysis. The image pipeline uses a fine-tuned DenseNet121 model; the video pipeline extracts frames, optionally examines subtle temporal variations using Eulerian Video Magnification (EVM), extracts frame features with ResNeXt, and models temporal relationships using Long Short-Term Memory (LSTM); and the audio pipeline converts signals into Mel spectrograms for CNN-based classification. The outputs are presented as modality-specific predictions. A combined multimodal decision should only be reported if a fusion method is implemented and evaluated. The system is intended to support media verification, and its performance must be established through controlled experiments.

296.AI-BASED RANSOMWARE DETECTION AND REAL-TIME FILE ENCRYPTION ALERT SYSTEM

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Machine learning has transformed the detection of cyber threats by enabling automated monitoring, anomaly classification, and incident response across enterprise environments. Developing high-performing ransomware detectors, however, requires large labelled datasets of file-operation logs that are difficult to acquire because of privacy regulations, organizational sensitivity, and the cost of expert annotation of attack windows. Centralised collection is further restricted by legal and policy constraints, creating a clear need for lightweight, server-side detection frameworks that observe centralised file

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activity in quasi real time without requiring client-side instrumentation or invasive endpoint telemetry. In the existing system, Aranyi et al. proposed a server-side file-operation logging framework that aggregates file creation, rename, and deletion events into fixed one-second windows using Fluent Bit and InfluxDB, and classifies them with five machine learning models (Random Forest, SVM, Decision Tree, AdaBoost, and XGBoost), where XGBoost achieves a sensitivity of 91.87 % with sub-millisecond prediction latency across simulated SME file-server environments. However, that system relies on a single fixed one-second aggregation window, a static decision threshold of 0.5, and models trained on data from a single environment, which together reduce its ability to detect low-and-slow encryption, adapt to shifting class distributions, generalize to unseen ransomware families, and provide explainable alerts to security operators. To address these challenges, the proposed Adaptive Server-Side Ransomware Detection (ASSRD) framework integrates multi-scale temporal aggregation over trailing windows of 3 s, 5 s, 10 s, and 30 s with 26 derived temporal features — sum, mean, max, active fraction, absolute delta, coefficient of variation, active-run length, and burst ratio — applies five gradient-boosted and classical classifiers under a fixed asymmetric cost matrix with offline cost-sensitive threshold selection over a fixed 49-point probability grid, and incorporates permutation and decision-path feature attribution grouped into operator-readable families, together with Population Stability Index drift monitoring and offline batch model recovery. Attack traffic is produced by a behaviour-faithful emulator rather than executed malware, which bounds the external validity of all reported figures. Evaluation reports a sensitivity of 0.9630–0.9778 and a specificity of 0.9965–0.9990 across the five classifiers, with XGBoost and SVM tied at a total cost of 63, an inference latency ranging from 0.085 ms to 0.390 ms, 245–249 of 254 attack episodes detected at a median time-to-detect of zero windows, and a mean sensitivity drop of 0.015 under leave-one-family-out evaluation across the five modelled ransomware families.

297.TAI-SHACS: TRUSTWORTHY AI-BASED SECURE HEALTHCARE ACCESS CONTROL FOR CLOUD-BASED ELECTRONIC HEALTH RECORDS

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Cloud-based Electronic Health Record (EHR) systems enable healthcare professionals to access and manage sensitive patient information efficiently. However, protecting these records requires effective authentication, access control policies, and detection of abnormal user activities. Traditional security mechanisms, such as Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC), enforce predefined permissions but may not adequately identify misuse by authenticated users. Artificial Intelligence (AI)-based anomaly detection can identify unusual access behaviour, while Explainable AI (XAI) can provide understandable reasons for security decisions. This paper surveys existing research on healthcare cybersecurity, cloud-based EHR access control, anomaly detection, Explainable AI, and Trustworthy AI. Based on the identified research gaps, it presents the proposed TAI- SHACS framework, which extends the Secure Healthcare Access Control System (SHACS) by integrating anomaly detection, dynamic trust/risk evaluation, XAI, and adaptive access control. The framework classifies access requests into high-, medium-, and low-trust levels to support access approval, additional verification, or restriction. MIMIC-III is considered as the primary healthcare dataset; however, access-control evaluation requires suitable access-event data or a clearly defined event-generation procedure. Authentication time, access decision delay, processing time, computational overhead, anomaly-detection performance, and explanation quality are identified as evaluation metrics.

298.AMDL-TPNE: AN ADAPTIVE MULTI-MODEL DETECTION LAYER FOR MULTI-ROUND ADVERSARIAL PROMO FRAUD DETECTION

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Multi-round promo fraud is an adaptive problem. A detector is fitted on an initial graph of abusive accounts, and a motivated adversary then manufactures fresh fraudulent nodes that resemble the accounts the detector has already learned to reject. This paper presents AMDL-TPNE, a framework that couples a temporally pre-trained node embedder (TPNE) with an Adaptive Multi-Model Detection Layer (AMDL) and evaluates both against a reproducible multi-round adversarial scenario. TPNE maps every account to a 384-dimensional vector formed by concatenating a static half with a temporal half, optimised under three self-supervised objectives: neighbourhood reconstruction, temporal disentanglement and temporal maximisation. AMDL replaces the single gradient-boosted head of the reference implementation with three heterogeneous learners — XGBoost, histogram gradient boosting and extremely randomised trees — fused by an adaptive F1 proportional weighting with a relative floor, and all warm-started so that a round costs additional boosting rounds rather than a full refit. Measurements are reported on the tolokers bid interaction graph (11,758 nodes, 1,049,758 edges, 21.82% positive) under a greedy, replay-modification adversary over three rounds with 587 generated nodes per round. Under this protocol the matched ablation — the same embedding, the same scenario and the same budget — raises the average on-round F1 from 0.119 for a single XGBoost head to 0.263 for AMDL, a 2.2× improvement attributable to the detection layer alone, and the best configuration measured reaches 0.391 with a final-round recall of 0.523. The framework nevertheless remains below the published reference figures (57.09 ± 2.33 average on-round F1 over five trials), and two nominally identical runs of the ensemble arm differ by a factor of four at a single-trial setting. Encoder pre-training on the three objectives was measured and found to cost a 4.3× wall-clock increase without an F1 gain. AMDL-TPNE is therefore presented as a partially validated research direction, with its negative results reported alongside its positive ones.

299.A COMPREHENSIVE REVIEW OF NANOSILICA IN CEMENTITIOUS SYSTEMS: MECHANISMS, PERFORMANCE ENHANCEMENT, AND SUSTAINABLE APPLICATIONS

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Nanosilica (NS), typically produced through bottom-up synthesis, has shown strong potential in refining the microstructure of cement-based materials at the nanoscale. It functions in multiple roles—as a highly reactive pozzolan, a nucleation site for hydration products, and a fine filler—making it particularly effective in enhancing concrete performance. When used in optimized dosages (generally 2–3 wt.% of binder for conventional concrete and 0.5–1 wt.% for self-consolidating concrete), nanosilica contributes to noticeable improvements in both mechanical strength and durability. Studies consistently report gains in compressive strength (around 15–29%), alongside reductions in chloride diffusivity (18–24%) and apparent porosity (up to 70%). These improvements also support reduced cement consumption, which in turn helps lower the overall carbon footprint. The discussion extends beyond performance outcomes to address key practical considerations such as dosage sensitivity, dispersion challenges, and interactions with chemical admixtures. It also explores behavior in complex multiphase systems, long-term durability aspects, and applications across emerging areas including recycled concrete, self-consolidating mixes, and geopolymer systems. Recent advancements, particularly the integration of machine learning techniques and life cycle assessment (LCA), are also highlighted as part of ongoing research developments.

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300. CONFITICKETAI: INTELLIGENT AUTO-HANDLING OF SUPPORT TICKETS WITH CONFIDENCE-BASED HUMAN-IN-THE-LOOP

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Organizations handle a high volume of information technology (IT) support tickets daily, which makes the traditional manual processing of tickets challenging and time-consuming. Most current helpdesk ticketing systems are highly dependent on humans to triage, categorize, prioritize, assign, and resolve tickets, which often results in slow-processing rates, poor judgment, and increased workload. This paper proposes ConfiTicketAI: Intelligent Auto-Handling of Support Tickets with Confidence-based Human-in-the-loop (HITL), an artificial intelligence (AI)-powered support ticketing automation system that aims to improve the efficiency and effectiveness of the current IT support system. The proposed solution utilizes natural language processing (NLP), machine learning (ML), retrieval- augmented generation (RAG), ChromaDB semantic search, and large language models (LLMs) to process support tickets by classifying and predicting the priority of support tickets, retrieving similar past tickets and solutions, and generating context-aware responses to tickets automatically. In addition, the system has a confidence-based HITL framework that recommends tickets for review and provides explanations while enabling the escalation of complex and ambiguous tickets to human analysts for higher-level resolution, improving the ticketing system's overall performance. The proposed system also includes features such as explainable AI, duplicate detection, and security threat detection, as well as learning from past tickets to improve its responses to future tickets, thereby reducing the overall workload required to handle support tickets manually.

301. BIO-REINFORCED ECO WALL PUTTY USING WASTE-DERIVED MATERIALS

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Our project is about developing an eco-friendly wall putty using waste derived materials that are normally discarded. In our proposed putty,white cement is used as the main binder, waste sea shell powder is used as a calcium-carbonate-rich filler, marble dust is added as a fine filler,silica fume helps improve compactness and bonding, cassava starch improves workability, and chicken feather fibres act as natural reinforcement to help control small cracks. The materials are cleaned, dried, processed and mixed in suitable proportions to prepare the putty. Different mix proportions will be tested and compared with conventional wall putty for consistency, setting time, adhesion, compressive strength, water absorption and crack resistance. The main aim of our project is to convert waste materials into a useful construction product, reduce waste disposal and promote a more sustainable approach to wall finishing.

302.DEEP LEARNING-BASED FINGERPRINT RECOGNITION FOR UNKNOWN PERSON IDENTIFICATION

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DeepScan-ID is a deep learning-based fingerprint recognition system designed to identify unknown accident victims during medical emergencies. It processes uploaded fingerprint images using image enhancement, CNN feature extraction, and hybrid matching techniques. The system retrieves critical medical information, including blood group, allergies, chronic conditions, and emergency contacts, from a secure database. With reported 98.42% TAR and identification time below 1.5 seconds, it supports faster emergency treatment and Golden Hour care.

303.DISCRIMINATION OF FENUGREEK SAMPLES BASED ON VANILLIC ACID CONCENTRATION USING NIR SPECTROSCOPY

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Fenugreek (Trigonella foenum-graecum L.) is an important food and medicinal plant which contains several bioactive constituents including vanillic acid. The discrimination of fenugreek samples as a function of different vanillic acid concentrations is relevant for the characterization of their chemical composition. This study investigated the spectral differences among four fenugreek sample groups with varying vanillic acid concentrations by using near infrared (NIR) spectroscopy combined with multivariate statistical techniques. The reference values of vanillic acid concentration have been determined by high-performance liquid chromatography (HPLC) technique. After the successful capturing of the NIR spectral data in the wavelength range of 900-1700nm, it has been building a 600 x 458 dataset for further analysis. Principal component analysis (PCA) was applied to identify the major patterns of spectral variation, it has been demonstrated distinct clustering of the fenugreek samples, with PC1 and PC2 explaining 94.42% and 4.78% of the total variance, respectively.Another discrimination method, namely linear discriminant analysis (LDA),was subsequently applied to evaluate the separation among the groups based on the pre-defined sample labels. The differences in the spectral distribution of the four fenugreek samples were revealed with the PCA score plot, while the loading profiles identified the wavelength regions contributing to the dominant variation. The LDA1 versus LDA2 plot gave a complementary perspective of the groups' discrimination. Overall, these findings indicate the possibility of using NIR spectroscopy with PCA and LDA to discriminate among the fenugreek sample groups examined in relation to their vanillic acid concentrations.This approach is a basis for the spectroscopic characterization and multivariate discrimination of fenugreek samples with different vanillic acid concentration.

304. SUSTAINABLE CONCRETE WITH CO2 SAND:STRENGTH INSIGHTS FROM C &D WASTE REPLACEMENT

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Sand is a crucial material in modern construction and an essential component of concrete production. As demand for construction materials increases, there is a growing interest in replacing conventional sand with alternative materials that can match the strength of M-sand. In this research work, it is proposed to focus on recycled fine aggregates (CO2 Sand) obtained from construction and demolition waste, which includes brick, cement, and other fine aggregate debris. These recycled fine aggregates are produced through a carbonation process and are available in three colours: black, white, and grey. The study will investigate the partial replacement of M-sand with these new fine aggregate materials at various proportions (0%, 20%, 40%, 60%, 80%). The strength of the resulting concrete will be evaluated using tests for compressive strength, At a curing age of 28 days reached 50.37MPa with a 60 % RFA replacement. The highest splitting tensile strength of 4.5 MPa was recorded at a curing age of 28 days the highest flexural strength, of 6.01 reached at a curing age 28 days.

305. SUSTAINABLE PLANT-EXTRACT-BASED PROTECTIVE COATINGS FOR REINFORCING STEEL IN CONCRETE: AREVIEW

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Corrosion of reinforcing steel is one of the problems that affect how long reinforced concrete structures can last. The process of corrosion is mainly affected by chloride penetration, carbonation, moisture and electrochemical reactions. These things can slowly hurt the steel. Lead to cracking, spalling and losing the structures strength. Along with making things steel corrosion causes big economic and environmental losses because of more repair work, maintenance, replacement and using more materials. Traditional ways to protect steel like epoxy, galvanized, zinc-rich fusion-bonded and polymer coatings can work well. They also have problems like being expensive using harmful chemicals giving off VOCs and not being good for the environment. To fix these issues plant-extract- based corrosion inhibitors and coatings have become a choice because they are better for the environment. Plant materials taken from leaves, fruits, peels, flowers, seeds and other parts have chemicals like flavonoids, tannins, polyphenols, alkaloids, terpenoids and saponins. These chemicals can stick on steel surface. This creates a protective layer, these chemicals can stop corrosion by helping with adsorption making a passive layer creating a protective film and stopping electrochemical reactions. Many methods like weight-loss, immersion, polarization, EIS, LPR, OCP, SEM, FESEM, AFM, FTIR, Raman, XPS, XRD, TGA and DSC are used to check how well these materials work. There are still problems like differences in plant chemicals effects, from the season getting the same results when extracting, making standards making sure they last a long time breaking down from sunlight making larger amounts and selling them. So future research should try to create long-lasting, cheap and eco-friendly plant-based coatings. These coatings should be able to protect reinforcing steel in concrete for a time.

306. COMPARATIVE ANALYSIS OF STEEL FIBER AND GLASS FIBER REINFORCED CONCRETE: A REVIEW OF PROPERTIES, DURABILITY AND HIGH-TEMPERATURECHARACTERISTICS

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Discontinuous fibers have been widely used in concrete in order to increase tensile strength, control cracking and improve durability of the construction material because of its brittle and weak behavior under tension. The two most common fibers are steel fiber and glass fiber which are usually investigated separately without recommendations how to choose the most appropriate one under particular conditions. This paper is a comparative analysis of steel fiber reinforced concrete and glass fiber reinforced concrete according to 85 papers published in 2010-2026 on the issues of properties of fresh concrete, compression, tension, bending, shear and fracture, impact and dynamic loadings, hybrid fiber reinforced concrete, durability, corrosion resistance, high temperature characteristics, sustainability and modeling. Out of 85 papers which were reviewed in this study only 3 papers compared steel fiber and glass fiber in one experiment, hence the comparison of these fibers is conducted according to those papers and a lot of papers about each fiber separately. The most remarkable improvements in mechanical properties were found when steel fiber was used as it resulted in increasing strength together with increasing the content of fiber of 2-3%. The modest improvements in strength and other properties were found when glass fiber with the content of 0.5-1.5% was used; the best results appeared earlier than in case with steel fiber; further increase in strength led to decreasing of the performance due to poor workability and fiber clustering; post-cracking behavior was brittle. Glass fiber does not corrode, whereas steel fiber tends to corrosion near the surface. Glass fiber reduces water absorption and porosity, but steel fiber increases chloride transport in mortar, yet the fully embedded fibres remain uncorroded. At high temperature glass fiber increases residual compressive strength by 14.6-20.3% at 500°C and reduces thermal conductivity. It can be concluded that steel fiber is more preferable when tensile strength and ductility play an important role, and glass fibre – when corrosion resistance or low weight or thermal insulation are important.

307. A COMPREHENSIVE REVIEW OF LIP-VOICE SYNCHRONIZATION-BASED MULTIMODAL AUTHENTICATION SYSTEMS

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Biometric-based authentication has become one of the necessary components of secure solutions used in mobile phones, banking, healthcare, and access control. At the same time, the usage of unimodal speaker authentication based on voice or face identification is vulnerable to such issues as replay attacks, spoofing attacks, and attacks by artificial intelligencebased deepfakes. In order to solve the aforementioned issues, multimodal authentication, which uses lip movement and voice data, can be considered as one of the reliable approaches to enhance security and spoof-resistance. In the current paper, recent advances in the field of multimodal authentication using lip–voice synchronization are analyzed. In particular, recent advances in speaker verification, lip visual biometrics, multimodal fusion, deep learning architecture, and benchmarking dataset will be reviewed. The reviewed studies generally indicate that multimodal authentication can improve recognition performance and spoof resistance compared with unimodal systems. Besides that, several challenges will be discussed, including computation complexity, real-time capability, multilingual adaptation, and spoof attacks by deepfake.

308. AN INTELLIGENT UPI FRAUD DETECTION AND CYBERSECURITY SYSTEM

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The rapid growth of UPI and digital payment systems has significantly improved the convenience and speed of financial transactions. However, the increasing dependence on online payments has also resulted in a rise in cyber frauds, including phishing attacks, fake QR codes, identity theft, and unauthorized transactions. To address these challenges, an AI-powered cybersecurity system is proposed, where fraudulent UPI transactions are detected using Machine Learning techniques and user behavioral analysis. Transaction details such as amount, location, device information, transaction frequency, and historical patterns are analyzed to calculate a fraud risk score. Phishing URLs and malicious QR codes are also identified to prevent users from interacting with fraudulent payment sources. Whenever suspicious activity is detected, real-time alerts are generated before the transaction is completed. An administrative dashboard is provided for monitoring transactions, viewing fraud reports, and evaluating system performance. As a result, financial fraud is reduced, user security is strengthened, cyber threats are identified at an early stage, and the reliability of digital payment systems is enhanced through intelligent, AI-driven fraud detection and cybersecurity mechanisms.

309.SMART MEMORY COMPRESSION: A SALIENCE-WEIGHTED CURATION FRAMEWORK FOR LONG-HORIZON CONVERSATIONAL AGENTS

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As large language models (LLMs) are increasingly deployed in long-horizon tasks, managing conversational memory within finite context windows has become a critical bottleneck. Traditional methods such as naive summarization or retrieval-augmented generation often suffer from information loss or noise. This paper presents the Smart Memory Compression (SMC) framework, a technically grounded approach to memory management that combines hierarchical segmentation, salience-weighted curation, and semantic-aware deduplication. We formulate memory curation as a constrained combinatorial optimization problem, show that it is NP-hard by reduction from 0/1 knapsack, and derive a greedy approximation algorithm with a provable performance bound. We give the full SMC pipeline as pseudocode, analyze its computational complexity, and walk through a worked synthetic example that illustrates how the salience function resolves competing memory candidates. We position SMC relative to existing memory architectures (MemGPT, StreamingLLM, H2O, Mem0) and long-horizon benchmarks (LongMemEval, LoCoMo), and lay out a concrete, falsifiable evaluation protocol for future empirical validation. This paper is presented as a systems proposal: we make no empirical superiority claims and instead focus on the formal correctness, complexity, and evaluability of the framework.

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310.MORINGA FLOWER EXTRACT AS A CANDIDATE GREEN CORROSION INHIBITOR FOR STEEL REBAR: FTIR, UV-VIS AND XRD CHARACTERIZATION

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Corrosion of reinforcing steel is a leading cause of deterioration in reinforced concrete, and carbonation of the concrete cover removes the passive film that protects the rebar. Moringa leaf extract inhibits steel corrosion in acidic and chloride media, but the flower extract has received little attention. This paper characterizes an ethanolic Moringa flower extract as a candidate green corrosion inhibitor for steel rebar, using FTIR, UV-Vis spectroscopy and XRD. FTIR showed eleven bands, including O–H (3332 cm^{-1}), C=O (1637 cm^{-1}), aromatic C=C (1538 cm^{-1}) and phenolic C–O (1246 cm^{-1}). UV-Vis scans of two replicates gave maxima at 323 and 326 nm, within the flavonoid and phenolic range. XRD showed seven reflections, the strongest at $2\theta = 21.3°$, which indicates ordered domains. The extract therefore contains functional groups associated with adsorption-type inhibition. Corrosion performance was not measured.

311.CYBERDECEPTION AI: AN ADAPTIVE AI-POWERED HONEYPOT FRAMEWORK FOR ATTACKER BEHAVIORAL FINGERPRINTING, INTENT PREDICTION, AND DYNAMIC DECEPTION

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While traditional honeypots do gather valuable attack telemetry, the majority of them still rely on static service configurations and pre-scripted responses, which limits their ability to adjust if an attacker deviates from what the operator intended. Recently, LLM-based honeypots have advanced session length and interaction realism, but coming up with a plausible response is not the same as determining what an attacker should see next, how their risk level should change, or which decoy is disclosed as the conversation progresses.These signals are combined into a single dynamic risk score by a weighted risk engine, which the adaptive decision engine then uses to choose which deception resource the attacker

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will see next. The loop between observation and dishonesty is closed when each decision is recorded along with the supporting data. In addition to deception- effectiveness metrics like session duration, decoy engagement, and threat-intelligence coverage, a controlled experimental protocol is proposed to assess classification, anomaly detection, clustering, and intent-prediction performance, benchmarked against static and LLM-based honeypot baselines.

312.BEHAVIORLOCK: ZERO-TEXT CONTINUOUS WEB AUTHENTICATION USING HYBRID ANOMALY ENSEMBLES

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Session-based web authentication verifies a user once, at login, and then trusts the bearer of a token for the remainder of the session. This single- checkpoint model leaves every subsequent action exposed to cookie theft, cross-site scripting, and simple workstation abandonment. We present BehaviorLock, a continuous authentication layer that re-verifie the operator of a browser session using only involuntary behavioral signals keystroke inter-key timing, mouse kinetics, and scroll dynamics without ever recording which keys were pressed. Raw DOM events are reduced to a 28-dimensional feature vector inside a dedicated Web Worker, so the main UI thread never blocks. A hybrid anomaly ensemble combining an Isolation Forest with a One-Class SVM (RBF kernel) scores each vector, and the resulting stream is smoothed with an exponentially weighted moving average (EWMA) to suppress single-action noise before triggering an adaptive step-up re- authentication modal. On a corpus of 180,000 labeled interaction samples collected from 45 participants, BehaviorLock reaches an Equal Error Rate of 2.1%, a median end-to-en inference latency of 1.18 ms, and flags an impostor takeover within four post-hijack actions, while imposing no measurable UI jank. We describe the architecture, the feature and scoring mathematics, and the evaluation protocol in full.

313.DIGITAL CHAT ANALYTICS AND SENTIMENT ANALYZER USING AI

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The digital communication platforms have evolved very rapidly in the last few years and the amount of unstructured chat data has grown exponentially in personal, academic and professional settings. This is a manual, time consuming and qualitative biased method to manually analyze these huge volumes of conversational logs. Given the backdrop of these challenges, the paper proposes a novel solution, the Digital Chat Analytics and Sentiment Analyzer Using AI -powered Dashboard, using techniques of Natural Language Processing (NLP), Machine Learning and Data Analytics. During end-to-end data processing, the proposed framework automatically preprocesses the text, such as noise reduction, tokenization, lemmatization and stop-words removal. Can detect multi-class sentiment, six major mood classes with fine-grained emotion detection, word frequency, emoji frequency and modelling of user behaviour over time. The proposed framework is integrated with interactive dashboard using AI technology for Streamlit to turn the raw analytics into visual elements, such as time-series activity timelines, user interaction heatmaps, social interaction network graphs and contextual word clouds. The system enables to transform raw unstructured chat files to structured data that can be acted upon, and offers an intuitive manner of decision making for students, researchers and enterprise organizations.

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The future opportunities include utilizing real time data stream processing through live messaging APIs, and improving multilingual support with transformers.

314. A SURVEY OF MACHINE LEARNING AND DEEP LEARNING APPROACHES FOR ADAPTIVE CACHE FRESHNESSMANAGEMENT

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Caching is widely used to reduce data-access latency, communication overhead, and system load. However, maintaining the freshness of cached data under dynamic workloads remains challenging because a cache must determine when an object should be retained, refreshed, or invalidated. Conventional Time-to-Live (TTL) policies use fixed or configured expiration intervals, while sketch based approaches provide compact frequency estimates but limited temporal information. This paper surveys workload-aware approaches for adaptive cache-freshness man- agement, with particular emphasis on machine-learning (ML) and deep-learning (DL) techniques. The survey considers TTL and Count-Min Sketch as conventional baselines and examines Linear Regression, Random Forest, XGBoost, Multilayer Perceptron (MLP), and Long Short-Term Memory (LSTM) as learned workload estimators. A common decision-engine perspective is used to separate workload estimation from cache action selection, thereby providing a consistent basis for comparison. The paper discusses temporal, frequency-based, recency, read/write, and sequential workload characteristics; summarizes prediction and cache-level evaluation metrics; and identifies limitations and future research directions. The resulting framework highlights how workload prediction can complement conventional cache mechanisms while emphasizing that prediction accuracy must ultimately be evaluated in terms of cache-level cost, freshness, and operational overhead.

315. BOREHOLE AND SPT-BASED CHARACTERIZATION OF SOIL–ROCK STRATIGRAPHY ALONG A LONG ECRF DAM CUT-OFF WALL ALIGNMENT

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The foundation records along a long dam cut-off-wall alignment show substantial changes in overburden thickness and rock condition. This paper synthesizes the reported borehole descriptions, soil-layer Standard Penetration Test (SPT) values, total core recovery and Rock Quality Designation (RQD) across chainages 0–1750 m. In the 0–450 m stretch, reported soil thickness is 12–49.5 m; it increases to 58–84 m between Ch. 450 and 600 m. Farther along the alignment, bedrock depths vary from about 3 to 37 m between Ch. 1085 and 1510 m. The Ch. 1005–1085 m stretch has silty sand over stiff to very stiff clay and medium-dense to dense sand, with rock at about 39–41 m. In the final stretch, Ch. 1510–1750 m, two boreholes record rock from ground level. The standard penetration number

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ranges are N = 3–61 for Ch. 0–450 m and N = 3–93 for Ch. 450–600 m. RQD and core recovery are evaluated separately. The results highlight why soil penetration resistance, refusal observations and rock-core indices must not be combined into a single numerical series.

316. LOW-COST MULTI-VIEW VIDEO COMPRESSION USING ADAPTIVE VIEW SELECTION AND LIGHTWEIGHT INTER-VIEW PREDICTION

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Multi-view video systems capture the same three-dimensional scene simultaneously from multiple cameras and are increasingly important in immersive video, free- viewpoint television, virtual reality, surveillance, telemedicine, autonomous systems, and remote collaboration. However, transmitting and storing multiple synchronized video streams requires substantially higher bandwidth, computational resources, memory, and energy than conventional single-view video. Existing multi-view and immersive-video coding approaches improve compression efficiency by exploiting temporal and inter-view correlations, but their computational complexity and hardware requirements can limit deployment on low-cost edge devices and resource-constrained networks. This paper proposes a Low-Cost Multi-View Video Compression (LC-MVC) framework that combines adaptive view selection, lightweight inter-view prediction, region- based frame partitioning, residual-aware compression, and complexity-constrained encoding. The proposed framework first estimates redundancy among neighboring views and selects a compact set of reference views. A lightweight correlation model then identifies reusable spatial information between views, while temporal prediction is applied independently within each selected view. Regions with high inter-view similarity are represented using reference information, whereas view-specific regions are encoded as residual information. The framework is designed to remain compatible with conventional video codecs, enabling practical implementation without specialized high-end hardware. Performance is evaluated using bitrate, compression ratio, PSNR, SSIM, VMAF, encoding time, memory consumption, and energy-related computational cost. The proposed architecture provides a practical direction for reducing the cost of multi-view video transmission while maintaining acceptable reconstructed-video quality. The study also discusses the relationship between the proposed approach and contemporary MPEG Immersive Video (MIV) systems, which exploit multiple texture/depth views and existing 2D video codecs for immersive-video compression.

317.MACHINE LEARNING APPLICATIONS IN ELECTRICAL IMPEDANCE TOMOGRAPHY: A REVIEW OF RECONSTRUCTION METHODS,CHALLENGES, AND FUTURE DIRECTIONS

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Electrical Impedance Tomography (EIT) is a non-invasive imaging technique. It is used to estimate the internal conductivity distribution of an object from boundary voltage measurements. EIT has applications in medical imaging, industrial monitoring, and process control. Producing a clear and accurate image from these voltage measurements is not easy. Small errors in the measurements can sometimes cause large changes in the reconstructed image. Machine learning has become a useful

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approach because it can learn the relationship between measured voltages and the corresponding conductivity images. This review discusses different machine learning methods used for EIT image reconstruction. These methods include artificial neural networks, convolutional neural networks, autoencoders, recurrent neural networks, attention-based models, graph neural networks, and physics-informed neural networks. It also explains how Fourier Neural Operators and hybrid models can be used to improve reconstruction. The review highlights the importance of using good-quality datasets, proper data preparation, and fair model comparisons. It also discusses the need to test these methods using real EIT hardware. Despite the progress made so far, some problems remain. Experimental datasets are still limited, and models trained on simulated data may not perform equally well with real measurements. Models may also struggle when they encounter object shapes or measurement conditions that were not included in their training data. Future research should focus on improving reconstruction accuracy, including physical information in the learning process, estimating predictionuncertainty, and testing models carefully on real EIT systems.

318.AN AGENT-ASSISTED FRAMEWORK FOR SOLIDITY SMART CONTRACT SECURITY ANALYSIS

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Smart contract vulnerabilities remain a critical security concern in blockchain ecosystems, largely due to the immutable and permissionless nature of the deployed contracts. Although numerous tools exist for analyzing Solidity smart contracts, current approaches often lack reproducibility, standardized evaluation, and objective benchmarking, which limits consistent comparison and validation across tools and environments. An agent-assisted security analysis of Solidity smart contracts is addressed through a modular framework specifically designed for reproducibility and benchmark drive evaluation. Within this framework, vulnerabilities are formalized as standardized, executable security tasks, enabling consistent repeatable, and tool-agnostic evaluation across heterogeneous detection paradigms. The architecture integrates a task specification layer, a containerized audit execution interface, an agent-assisted vulnerability task generation module leveraging large language models (LLM), and automated validation for objective assessment. Experimental evaluation demonstrates that the framework effectively identifies smart contract vulnerabilities while improving the reliability of detection through LLM-assisted verification and standardized validation. The proposed framework supports reproducible experimentation, scalable task reuse, and systematic benchmarking of smart contract security analysis tools.

319. ARTIFICIAL INTELLIGENCE AND DEEP LEARNING IN LUNG CANCER DETECTION AND CLASSIFICATION

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Lung cancer remains a major cause of cancer related mortality, in part because many cases are identified after the disease has progressed. Low dose computed tomography (LDCT) enables screening of eligible high risk populations, but interpretation is complicated by small lesions, anatomical structures that resemble nodules, variation between readers, and the downstream burden of false positive findings.

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Artificial intelligence (AI), particularly deep learning, offers computational methods for assisting image review across lung field segmentation, pulmonary nodule detection, candidate characterization, malignancy risk estimation, and longitudinal assessment. This review examines the technical foundations and clinical considerations of AI assisted lung cancer detection and classification. It describes convolutional neural networks (CNNs), three dimensional CNNs, encoder decoder segmentation networks, detection architectures, attention mechanisms, transformers, transfer learning, radiomics, and ensemble methods. It also discusses image preprocessing, annotation quality, dataset partitioning, class imbalance, calibration, external validation, explainability, workflow integration, and reporting practices. Published meta analytic evidence has reported promising pooled diagnostic performance, although estimates differ according to study selection, task definition, threshold, reference standard, and validation design. High accuracy on a curated dataset should not be interpreted as proof of clinical benefit or generalizability. Robust evaluation requires patient level separation of training and test data, independent multi centre testing, confidence intervals, calibration analysis, and assessment of clinically relevant outcomes. The review concludes that AI is best positioned as a carefully validated decision support component within a multidisciplinary pathway rather than as an autonomous replacement for radiologists. Reproducible research, representative datasets, transparent reporting, prospective evaluation, and ongoing monitoring are essential for safe deployment.

320.LUMINA: A CITATION-MANDATED RETRIEVAL-AUGMENTED GUIDE

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Lumina is a browser-based conversational guide that answers everyday life questions from Islamic and Hindu scriptural passages and requires the language model to cite the retrieved passages, together with their original-script text. This paper describes the system as implemented, a React/TypeScript client with a hand-annotated knowledge base, a keyword-and-stem retriever, a citation- mandating prompt, and a Supabase persistence layer, and reports a small, fully reproducible pilot probe of its retrieval stage. The probe issues 12 queries (6 answerable from the corpus, 6 deliberately unanswerable) against both tradition-filtered retrievers, for 24 retriever invocations in total. In all 24 the retriever returned five passages, including for queries with no relation to the corpus, because a constant score bonus makes every passage score above zero and a score filter intended to drop non-matching passages therefore never fires. Combined with a prompt that requires at least three citations, the system cannot decline to answer. We also report an encoding error in one Arabic text field, a tokenizer that retains stopwords, and the absence of per-passage provenance. We do not claim retrieval-quality results: the system has no gold query set or evaluation harness yet. We state the evaluation protocol required to turn the observed failure mode into a measured one, and we scope the claims of this paper accordingly.

321. VISION TRANSFORMER-BASED INTELLIGENT PLANT DISEASE DETECTION AND SEVERITY ANALYSIS

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Plant diseases are a serious concern for agricultural production and impact yield, crop quality, food security, and farmers income. Early detection and precise identification of crop diseases are of great significance to minimize crop loss and provide a basis for timely crop decision management. This project aims to build a Vision Transformer (ViT)-Based Intelligent Plant Disease Detection and Severity Analysis System through Python and OpenCV environment. This system analyzes plant leaf images, classifies the plant disease condition, and determines the severity level of plant disease using a deep learning model built based on Vision Transformer. The OpenCV environment is used for image acquisition resizing smoothing, color space conversion, segmentation, and image preprocessing before classification. This system can help to identify plant disease types from leaf images and estimate the severity level of plant disease by the absented leaf region. Severity analysis categorizes the infection into severity levels like Healthy, Mild, Moderate, and Severe and helps farmers to understand the level or damage happened to the crop. It can display the disease related information and helps to make early decisions to grow healthy crop. The proposed system based on Python and OpenCV is a simple to use application where the user can either upload a photo of the plant or can capture the live image of plant, then can preprocess the image, detect the disease, analyze severity of the infection and then display the prediction results. The system can be extended to maintain disease history, can support multiple crop types, can generate alerts and advices for disease management.

322. HEALTH HUB: AN ONLINE PHR SYSTEM FOR FAMILIES

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Properly managing healthcare records is a huge challenge in modern healthcare systems. Quick access to healthcare is often slowed down by inefficient data storage systems. This paper describes the development of a standalone PHR (Personal Health Record) system: a web application designed to digitise, organise, and analyse medical records of an entire family as a unit. The system uses a client-server architecture using a React-based frontend, Node.js backend, and MongoDB database. Its features include user authentication, invitation-based family group management and storage of medical reports in PDF and image formats. Optical Character Recognition (OCR) using Tesseract.js has been used to extract health parameters such as hemoglobin, cholesterol, and blood pressure from uploaded records. Trend analysis and graph-based presentation of extracted data allow users to monitor their health parameters over time. The system does not provide medical diagnoses or health advice, focusing only on record management and health trend visualisation.

323.UNDERSTANDING THE IMPACT OF MICRO STEEL FIBERS ON CONCRETE CHARACTERISTICS: AN UP-TO-DATE REVIEW OF STRENGTH, CRACK RESISTANCE, AND DURABILITY

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This up-to-date review presents an extensive compilation of the improved mechanical properties, structure, and durability of concrete due to the incorporation of micro steel fibres (MSF) into composite materials. Conventional concrete is intrinsically limited by its low tensile strength, brittleness, and tendency to micro-crack under loading and deterioration conditions. The addition of micro steel fibers can overcome these drawbacks due to their high aspect ratio, tensile strength, and micro-crack bridging properties. In order to ensure peak efficiency and environmental sustainability, today's concrete formulations consist of multi- component systems that integrate MSF and additional cementitious components such as silica fume, fly ash, and GGBS in conjunction with hybrid synthetic fibres. This paper provides an assessment of the effect of MSF and multi-component synergy on compressive, tensile and flexural strength, post-peak ductility, crack mitigation, and various measures of durability (permeability, freeze-thaw, and chemical resistance). Moreover, a critical comparison of 15 most important studies reveals the densification process of the ITZ microstructure and mechanics of stress redistribution within these composite materials. This paper ends up with a summary of open research areas and relevant suggestions to improve MSF concrete formulations based on multi- component mixtures.

324.NANO-SILICA AND SODIUM-SILICATE BASED SELF-HEALING CONCRETE: A CRITICAL REVIEW OF STRENGTH, DURABILITY, CRACK CLOSURE, AND HEALING MECHANISMS

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Concrete is one of the most used materials in construction because of its good compressive strength, availability, and economy. However, cracking is a common problem during the service life of concrete structures. Cracks can allow air, water and aggressive substances to enter the concrete; this will lead to deterioration and corrosion. Therefore, self-healing concrete has attracted considerable attention for reducing cracks and extending the service life of concrete structures. This review focuses on the use of nano-silica and sodium silicate in self-healing concrete. Nano-silica will improve the hydration and pozzolanic reactions of cement. It promotes the formation of additional calcium silicate hydrate (C-S-H) and helps refine the concrete matrix. Adding nano-silica from 1% to 4% for 28 days of curing increased the compressive strength by 16% to 43.5%, but the optimal dose of nano-silica is 3% (43.5%). For split tensile strength, adding 1 to 4% nano-silica resulted in a 16% to 35% increase in strength, so the optimal strength is 35% for 4% nano-silica. Although the investigated nano-silica dosage ranges from 3% to 10%, the optimum flexural performance was reported at about 3%. It can heal cracks of up to 3 mm. Adding 0.3% to 0.5% nano-silica decreases the permeability and enhances the sulphate resistance. Water absorption decreases by 5.15%, 30.15%, and 35.66% for M30 concrete, for M40 1.47%, 30.40%, and 58.97% for 1% to 3% nano-silica, which improves the pores and results in less absorption of water and improved durability. Sodium silicate can increase the strength of concrete and decrease the absorption of water and permeability; it can be added at 0% to 8% in microcapsules. In capsules or as powder, it can be added at 0% to 6%. The optimum value for flexural strength recovers 45% for 16% addition. For tensile strength, 69% is recovered at 50%. When cracks develop, the available sodium silicate can react with calcium-containing compounds and seal the cracks around 250 μm to 450 μm, but if we add sodium silicate powder, it may heal crack widths up to 2 mm. Previous studies have shown that the healing depends on crack width, material dosage, and curing conditions. The review further highlights that considering crack closure is insufficient for evaluating efficiency. Improvements in compressive strength, split tensile strength, and flexural strength should also be considered. SEM and XRD are also useful for examining the healing products. Finally, the review discusses current limitations, testing methodologies, and future research needs.

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325. A SURVEY ON AI-POWERED AUTOMATED BUG REPORTING, CLASSIFICATION AND RESOLUTION SYSTEM

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Software testing produces useful evidence about application failures, but the resulting logs and issue descriptions can be difficult to review when many test cases fail. This paper surveys research on automated bug-report classification and presents a proposed workflow that connects Selenium-based functional testing with natural language processing (NLP) and deep learning. The base study, Testing Bug Report Generation–Knowledge Management (TBRG-KM), analyses existing textual bug reports and compares CNN, LSTM, GRU, and Bi-LSTM with Attention. Using 20,171 cleaned reports from Agile e-commerce projects, the study reports 95% accuracy for Bug Notification and 92% for both Severity and Responsible Side with Bi-LSTM and Attention [1]. Building on that direction, the proposed platform captures test failures, prepares textual descriptions, represents words using GloVe embeddings, predicts Bug/Non-Bug, severity, and responsible software side, and compiles the available evidence into a structured report. The proposed platform has not yet been experimentally validated; therefore, its performance is to be measured using classification and workflow metrics rather than assumed from the base paper.

326. A SURVEY ON RELIABILITY-AWARE ADAPTIVE MEMORY FOR HALLUCINATION-RESISTANT GENERATIVE AI AGENTS

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Generative Artificial Intelligence agents are nowadays considered to perform complex tasks that require reasoning, planning, tool interaction, and memory capabilities. Such tasks can be solved by means of these agents through a sequential decision-making process. Nevertheless, each wrong step performed by the agent can introduce errors that impact negatively all the subsequent steps. Therefore, the wrong decisions lead to hallucinated responses, wrong actions, and recurring failures. Previous works tried to address this issue by leveraging retrieval-augmented generation, Knowledge Graphs, and observability-based memories. However, an experience stored in memory that was helpful for some task may not be helpful for a different task anymore. This project, Reliability-Aware Adaptive Memory for Hallucination-Resistant Generative AI Agents, aims at finding out how to tackle this issue. The key idea is to make the agents aware that an experience previously stored in memory and retrieved for the current task resolution could be reliable or not based on its validation results. Each retrieved experience is assigned a reliability score updated according to the experience itself's execution outcomes. Therefore, the experiences' reliability is determined depending on whether they successfully completed the task or failed, how similar they are to the current task, and how fresh their context is. Then, when addressing a new task, the agent's planner considers both the relevance and reliability of retrieved experiences to reason and plan about the current task. Moreover, the most frequent successful experiences could be prioritized by the agent, whereas the often-failing ones would be penalized accordingly. Finally, the project will evaluate how efficiently the adaptive reliability memory approach can be applied by comparing it with the state-of-the-art memory mechanisms for generative agents. The evaluation will be performed using such metrics as task success rate, hallucination rate, successful

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experience reuse rate, failure recovery rate, and execution efficiency. Therefore, the project will investigate if adapting the experiences' reliability depending on their execution outcomes improves agents' memory usage efficiency and robustness against hallucination issues.

327.RECENT ADVANCES IN HISTOPATHOLOGICAL STRUCTURES SEGMENTATION: TOWARD CLINICAL DIAGNOSIS ANDPERSONALIZED TREATMENT

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Doctors diagnose cancer by studying thin slices of tissue under a microscope. Today these slices are scanned into huge digital images, and computers can learn to outline the important parts of them: the cell nuclei, the glands, the tumor, the supporting tissue, and the immune cells. This outlining task is called segmentation, and it is the first step toward turning a picture into measurements that can guide a diagnosis. This paper explains, in plain language, how segmentation methods have progressed from simple image rules to deep neural networks, attention-based "transformer" models, and large"foundation" models trained on millions of slide images. We describe the data these methods need, how their quality is judged, and how their outlines are converted into information such as tumor grade, immune-cell counts and survival risk. We then show how this information can help choose a treatment suited to an individual patient. Finally, we discuss what still prevents these tools from being used everywhere in hospital differences between laboratories, scarce expert labels, trust, and regulation and suggest directions for safe, useful progress.

328. CARDIOEDGE: A LOW-POWER TINYML WEARABLE FOR REAL-TIME CARDIAC HEALTH INTELLIGENCE

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Continuous cardiac monitoring using wearable sensors requires a balance between physiological information, computational efficiency, and interpretability. This paper presents CardioEdge, a TinyML-based wearable framework that integrates ECG and PPG sensing for local cardiac-status classification on an ESP32 microcontroller. Unlike approaches that directly process high-dimensional raw ECG or PPG waveforms, CardioEdge extracts ECG-derived heart rate and PPG-derived heart rate and combines them with age and gender to form a compact four-dimensional feature representation. These features are standardized and supplied to a lightweight 4–16–8–2 Tiny Neural Network (TinyNN) with ReLU hidden layers and Softmax output. The trained model is converted into a Float32 TensorFlow Lite model with a size of 3,028 bytes (approximately 2.96 KB) for embedded deployment. The implemented model achieved 99.88% accuracy, precision, recall, and F1-score on the evaluated dataset. The framework further incorporates an Explainable Artificial Intelligence (XAI) stage to provide feature level interpretation of the model prediction. By combining multimodal ECGPPG-derived features, demographic information, compact TinyML classification, local ESP32 inference, and explainability within a single wearable-oriented framework, CardioEdge provides an integrated approach to resource-constrained cardiac-status classification.

329.SOCIAL MEDIA MISINFORMATION: WHAT IT IS, WHY IT SPREADS, AND HOW TO DETECT IT

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Misinformation is defined as false or unsubstantiated information that is shared inadvertently whereas disinformation is defined as false or unsubstantiated information that is shared deliberately to deceive. Sharing can be incentivized by the normal human tendency to seek out novelty, network effects, and a sharing platform's distribution mechanism. Such incentive for sharing at scale may drive phenomena of interest such as the findings that false stories exhibit larger cascades and spread farther and faster than truth in large scale cascade analyses and that false beliefs can sometimes persist following the viewing of a correction in psychological studies. This research reviews the characteristics of the information landscape according to Wardle and Derakhshan's (2017) information disorder framework, explores motivations for sharing using uses and gratifications, social identity, dual process, and motivated reasoning accounts, and explores effectiveness of correction with continued influence and inoculation literatures and concludes that these bodies of evidence support 4 characteristics of a potential "prebunking" strategy: (1) preparation for processing the information (2) automatic triage, (3) not automatic truth assessment, (4) clear correction; assessment of impact across communities and platforms [3, 4].

330 .AN AI-DRIVEN CYBER-RESILIENT FRAMEWORK FOR MITIGATING RANSOMWARE- INDUCED PATIENT SAFETY RISKS IN SMART HOSPITALS

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The increasing adoption of Internet of Medical Things (IoMT) devices, Electronic Health Records (EHRs), cloud platforms, and interconnected clinical systems has transformed hospitals into highly connected smart healthcare environments. However, this connectivity increases exposure to cyber threats that can disrupt critical services and directly compromise patient safety. System encryption and service disruption may restrict access to medical records, laboratory results, medication systems, diagnostic services, and connected medical devices, leading to treatment delays, medication errors, and interruption of emergency and intensive care. This paper proposes a patient-safety-aware cyber-resilient framework for smart hospitals that integrates behavioral anomaly detection, clinical risk assessment, service prioritization, automated containment, and secure recovery. The framework evaluates malicious activity based on threat severity and the clinical criticality of affected services, enabling priority protection of essential functions such as emergency care, intensive care, pharmacy, laboratory, and medical-device infrastructure. A simulated smart-hospital environment will be used to evaluate detection performance, response time, critical-service availability, and recovery efficiency. The proposed approach aims to reduce clinical disruption and maintain continuity of essential healthcare services during cyber incidents, providing a practical foundation for resilient and patient-centered healthcare cybersecurity.

331.MODELLING AND OPTIMIZATION OF LM26 HYBRID COMPOSITES FOR RECIPROCATING FRICTION AND WEAR

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This study investigates the modelling and optimization of the friction and wear behaviour of LM26 aluminium alloy hybrid composites reinforced with almandine garnet and MoS_2 under linear reciprocating sliding. The hybrid composites were fabricated using a stir casting mechanical method with almandine garnet contents of 5–20 wt.% and MoS_2 contents of 1–4 wt.%. The tribological characteristics of the developed composites were evaluated using suitable experimental methods and reciprocating friction and wear tests were conducted using a linear reciprocating tribometer in the laboratory. Taguchi-based experimental design and optimization were used to investigate the effects of load, frequency, stroke length, temperature, and filler content on friction and wear of hybrid composite. The optimum parameter combination was identified as A4B4C4D1E1, corresponding to a load of 80 N, frequency of 50 Hz, stroke length of 9 mm, temperature of 50 °C, and filler content of 5 wt.%, with a predicted optimization accuracy of 95.8%. The results indicate that the LM26 hybrid composite containing 5 wt.% almandine garnet provides the best overall balance of friction and wear resistance. A1B3C1D1E2, i.e., Load = 20 N, Frequency = 40 Hz, Stroke length = 2 mm, Temperature = 50 °C, and Filler content = 10 wt %, for coefficient of friction properties. The developed modelling and optimization approach demonstrates the potential of almandine garnet–MoS_2 reinforced LM26 hybrid composites for improved tribological performance under reciprocating sliding conditions.

332. EVALUATING AXIAL LOAD-CARRYING CAPACITY OF NORMAL STRENGTH CONCRETE (M-30) RC COLUMNS: AN EXPERIMENTAL APPROACH WITH ANALYTICAL VALIDATION BY USING PYTHON SOFTWARE

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This study experimentally investigates the axial load behavior of short reinforced concrete columns under concentric loading. Twelve square specimens (125 mm × 125 mm × 1000 mm) were cast using M-30 grade normal strength concrete, with longitudinal reinforcement varied across four bar diameters (8, 10, 12, and 16 mm) and lateral confinement varied across three stirrup spacings (100, 200, and 300 mm). For each specimen, the cracking load, cracking deformation, ultimate load, and ultimate deformation were recorded experimentally. To extend this investigation, the experimental ultimate axial load capacities were further validated against analytical predictions obtained using Clause 39.3 of IS 456:2000 for short axially loaded columns with lateral ties. A Python-based computational framework was developed to automate the analytical calculations, tabulate the comparative results, and generate visualizations for both experimental and analytical datasets. The comparative analysis reveals that the code-based analytical predictions are conservative relative to the experimentally observed capacities (experimental-to-code ratio of 1.02 to 2.1, mean 1.53), a trend attributable mainly to the partial safety factors embedded in the design equation, together with the higher achieved concrete strength and the confining effect of closely spaced ties, which the equation does not include. These findings offer insight into the reserve strength margin present in code-based design methods for short RC columns.

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333. SMART IV BAG MONITORING AND FALL DETECTION ALERT SYSTEM

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Intravenous (IV) therapy still depends largely on nurses visually checking the fluid level, while elderly or weak patients walking with an IV stand are at constant risk of falling. An IV bag that runs dry can cause air entry, blood backflow and interrupted medication, and an unnoticed fall can delay emergency care. This paper presents an Internet of Things (IoT) based Smart IV Bag Monitoring and Fall Detection Alert System. A load cell with an HX711 amplifier measures the bag weight continuously, from which the remaining fluid percentage and infusion rate are computed. A wearable node with an MPU6050 accelerometer–gyroscope detects falls using a threshold-based algorithm with post-fall inactivity verification. Both nodes use the ESP32 microcontroller and send data over Wi-Fi through MQTT to a web server whose dashboard shows the live status of every bed. When the fluid level falls below a configurable threshold or a fall is confirmed, a local buzzer and LED are activated and the nursing staff are notified through the dashboard and SMS. The system is low-cost and scalable, reduces manual rounds, prevents empty- bag incidents and shortens the response time to patient falls.

334. INTELLIGENT WEATHER-AWARE DECISION SUPPORT SYSTEM FOR CROP MANAGEMENT USING MACHINE LEARNING

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Agriculture is the backbone of the global food supply, yet crop productivity is increasingly threatened by unpredictable weather, soil degradation, and poorly informed farm decisions. Recent advances in machine learning (ML) have shown promising results in crop recommendation, yield prediction, and agricultural advisory. However, many existing systems focus primarily on prediction accuracy while neglecting weather integration, reliability assessment, and explainability, which are critical for real-world adoption by farmers. To address these limitations, this paper presents an Intelligent Weather-Aware Decision Support System with Reliability-Aware Advisory for crop management. The proposed system uses a hybrid architecture that combines real-time weather data with soil and crop parameters. Weather attributes (temperature, humidity, rainfall, wind speed) and soil attributes (nitrogen, phosphorus, potassium, pH, moisture) are preprocessed and fed into ensemble machine learning models such as Random Forest and XGBoost to recommend suitable crops and estimate expected yield, along with a confidence score. This approach keeps computational cost low while maintaining high prediction performance, making the system suitable for real-time web-based deployment. To enhance transparency

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and trust, the system integrates SHAP-based feature importance to explain which weather and soil factors influenced each recommendation

335. SATELLITE IMAGE-BASED CROP YIELD PREDICTION USING AI

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Accurate estimation of agricultural production before harvest is challenging because crop development is affected by weather, soil properties, field management, and changing growing conditions. This paper proposes a satellite-supported artificial intelligence framework for estimating crop yield by combining repeated multi spectral observations with environmental and historical production information. Sentinel-2 and Landsat-8 imagery are considered as primary remote-sensing sources. Rainfall, temperature, humidity, soil moisture, and previous yield observations are incorporated as complementary variables. Spectral information is transformed into vegetation indicators such as NDVI, EVI, SAVI, and NDWI. A CNN-LSTM model is proposed to learn spatial characteristics from image data and temporal patterns from observations collected during the crop cycle. The resulting prediction can be presented through a dashboard or mobile interface for planning irrigation, fertilizer application, harvesting, and regional supply. The manuscript describes the proposed methodology rather than claiming experimental accuracy. Numerical results should be reported only after the framework is trained and tested using a documented datasets.

336 .BLOCKCHAIN-BASED VOTING SYSTEM

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Voting is the foundation of democracy, yet conventional paper-based and electronic voting machine systems suffer from long queues, booth capturing, impersonation, record tampering, delayed counting, and limited transparency. Remote electronic voting can widen participation, but centralised servers create a single point of failure and require complete trust in administrators. This paper presents a blockchain-based voting system in which every vote is recorded as a cryptographically signed transaction on a permissioned blockchain, with election rules enforced by smart contracts. Voters are verified through a three-factor process of voter ID, one-time password, and biometric match. Each ballot is encrypted with the election public key, stored immutably in hash-linked blocks, and acknowledged with a receipt that lets the voter confirm inclusion without revealing their choice. Validator nodes run by independent parties, such as the election commission, observers, and auditors, reach agreement through proof-of-authority consensus, so no single insider can alter the ledger. Decryption keys are released under a threshold arrangement, which prevents early counting, and tallying is automatic and publicly verifiable. A security analysis shows how the design counters impersonation, double voting, tampering, loss of secrecy, server failure, and insider manipulation. The proposed system aims to improve transparency, integrity, and accessibility while preserving ballot secrecy. A prototype and field evaluation are left as future work.

337.AN INTELLIGENT MACHINE LEARNING FRAMEWORK FOR PROACTIVE ROAD SAFETY ANALYTICS

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Road traffic crashes remain a leading cause of death and injury worldwide, yet most road-safety decisions are still taken reactively, after crashes have already occurred. This paper presents an Intelligent Machine Learning Framework for Proactive Road Safety Analytics that converts heterogeneous crash, road, weather and traffic data into early, explainable risk intelligence. The framework is organised into five layers: data sources, data processing, machine learning, explainability and validation, and decision support. It predicts crash severity with a pool of interpretable and ensemble classifiers (logistic regression, decision tree, random forest and gradient boosting), mines spatial crash hotspots with density-based clustering (DBSCAN), and explains model behaviour through permutation feature importance, with recall for severe crashes treated as the primary selection criterion. A prototype was evaluated on a synthetic dataset of 20,000 crash records described by 14 spatio-temporal, environmental and infrastructure features. Logistic regression gave the highest ROC-AUC (0.730) with a severe-crash recall of 65.7%, closely followed by gradientboosting (AUC 0.720, recall 64.1%); night-time driving, speed limit and visibility were the most influential factors, and DBSCAN recovered four of the eight planted hotspot centres. Because the data are synthetic, the results demonstrate the working of the pipeline rather than field performance; validation on real crash records is the immediate next step. The framework is modular and can support patrol allocation, black-spot treatment and targeted safety interventions.

338. PERFORMANCE EVOLUTION OF TRADITIONAL MUD CONSTRUCTION TOCSEB

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Traditional mud houses construction are known for their excellent thermal performance and low environmental impact. However, their use in contemporary architecture is often limited due to concerns regarding structural strength, durability, maintenance, and environmental sustainability Traditional mud houses construction are known for their excellent thermal performance and low environmental impact. However, their use in contemporary architecture is often limited due to concerns regarding structural strength, durability, maintenance, standardization, and construction practices. Compressed Stabilized Earth Blocks (CSEB) offer a modern and standardized form of earth construction. This study investigates whether CSEB can retain the environmental and thermal benefits of traditional mud construction while improving overall building performance.The study compares a traditional mud house construction with a CSEB block house under similar climatic and functional conditions. The comparison considers four major parameters: thermal comfort, structural strength, maintenance, and embodied carbon. Thermal performance is evaluated through wall thickness, thermal conductivity, thermal mass, and U-value. Structural performance is assessed through compressive strength and load-bearing capacity. Maintenance is studied based on durability, resistance to weathering, surface protection, and repair requirements. Embodied carbon is compared by considering material production, stabilizer content, transportation, and construction energy.The comparison helps identify how CSEB performs against traditional mud construction. CSEB provides greater standardization, improved structural reliability, and easier construction, while retaining significant thermal-mass benefits of earth

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construction. However, cement stabilization can increase embodied carbon compared with unstabilized mud.CSEB can act as a contemporary alternative to traditional mud construction when appropriately designed, providing a balance between thermal comfort, structural performance

339.SYNTHETIC DATA AND IMAGE ENHANCEMENT FOR BRAIN TUMOR DETECTION: A REVIEW OF GENERATIVE AI ADVANCES

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Accurate brain tumor detection using Magnetic Resonance Imaging (MRI) requires large, diverse, well-labelled datasets and good-quality scans. However, medical images can be limited because of privacy concerns, rare tumor types, high annotation costs, and differences between scanners. Generative Artificial Intelligence (Generative AI) can help address these challenges through synthetic data generation and image enhancement. Techniques such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models can create additional MRI-like images to support AI training. Image-enhancement methods can reduce noise, improve resolution, correct image variations, and estimate missing MRI sequences. These methods may help AI systems learn more effectively and support tumor classification and segmentation. However, generated images may contain false details or bias, so they must be carefully evaluated using real, unseen patient scans. Generative AI should be used as a decision-support tool for radiologists and not as a replacement for medical professionals.

340. A SURVEY ON GRAMMATICAL ERROR CORRECTION, REGIONAL DIALECT DIVERSITY, AND DATASET CREATION FOR LOW-RESOURCE TAMIL LANGUAGE UNDER DIGLOSSIA

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Tamil, one of the oldest classical Dravidian languages, is characterized by a profound socio-linguistic condition known as diglossia—a sharp division between a formal literary register (Centamil) and informal colloquial spoken registers (Koduntamil). While Centamil is strictly preserved in literature, formal journalism, and official institutions, daily communication and digital interactions take place in spoken colloquial forms. This diglossic divide is the primary root cause of low-resource data scarcity in modern Tamil natural language processing: although the language boasts tens of millions of native speakers, nearly all digitized and annotated text corpora are restricted to formal literary Tamil, leaving real-world conversational and spoken varieties virtually unannotated. As a consequence, state-of-the-art computational systems and grammatical error correction (GEC) pipelines are trained almost exclusively on standard written Tamil. When deployed on everyday text, these systems fail to recognize conversational grammar, misinterpret colloquial morphology, and incorrectly flag valid spoken expressions as errors. This limitation is compounded by rich regional dialectal diversity, spanning distinct linguistic zones—including Northern Tamil (Madras Bashai with voiced plosive shifts), Western Kongu Tamil (marked by lateral mergers, distinctive intonation, and auxiliary markers), Eastern Cauvery Delta Tamil (exhibiting intervocalic spirantization), and Southern varieties across Madurai, Nellai, and Kanyakumari. Because each dialect introduces unique phonological shifts, localized vocabulary, and non-standard inflectional suffixes, no existing benchmark dataset adequately

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covers this regional spectrum. To overcome this severe data scarcity without relying on expensive manual annotation, this survey highlights critical strategies for low-resource training: linguistically informed synthetic error generation, rule-based morphological and phonological perturbation, back-translation and noise injection, and cross-lingual transfer learning via pre-trained multilingual transformers (such as mT5 and mBART) fine- tuned on curated multi-dialectal samples. Grounded in the responsible hybrid BERT-RNN framework of Jabbar et al. [1], this survey establishes an end-to-end blueprint integrating dialect normalization, contextual transformer error localization, and sequential copying networks. This provides the foundational framework for building the first dialect-inclusive, robust Tamil NLP and error correction ecosystem.

341.THE FARMER'S TEXT ALERT (OFFLINE WARNING SYSTEM)

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Climate change and extreme weather events such as droughts, heatwaves, floods, and heavy rainfall can seriously affect farmers and their income. Small and remote farmers are often the most vulnerable because many of them do not have smartphones, reliable internet access, or access to timely climate information. Farmer's Text Alert is an offline-friendly early warning system designed to bridge this digital gap. The proposed system collects official weather and climate alerts and identifies risks affecting registered farming regions. Based on the location, crop, and preferred language of each farmer, the system generates simple and actionable agricultural advice. Instead of sending only a general weather warning, it informs farmers about the expected risk and suggests suitable preventive actions. These alerts are delivered through ordinary SMS, allowing farmers with basic mobile phones to receive important information without requiring an internet connection or mobile application. By combining climate information, farmer profiles, localized recommendations, and SMS communication, the system aims to support timely decision-making, reduce crop losses, and improve climate resilience among vulnerable farming communities. The project contributes towards Sustainable Development Goal 13 (Climate Action) and promotes inclusive access to climate- smart agriculture.

342.CIAIR: A COGNITIVE INTELLIGENCE AUTONOMOUS INSPECTION ROBOT FOR INDUSTRY 5.0

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The advancement of Industry 5.0 emphasizes human-robot collaboration, intelligent automation, safety, and adaptability in industrial environments. Conventional industrial inspection robots primarily perform

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predefined tasks and lack the ability to integrate contextual information, historical events, and risk-based decision-making. To address these limitations, this paper proposes CIAIR (Cognitive Intelligence Autonomous Inspection Robot), an intelligent mobile robotic system designed to support industrial inspection, workplace safety, and autonomous decision- making. The proposed system integrates computer vision, environmental sensing, autonomous navigation, and cognitive reasoning to monitor industrial environments. Using technologies such as ROS 2, OpenCV, YOLO-based object detection, LiDAR-based mapping, and sensor fusion, CIAIR aims to identify potential hazards, monitor machine conditions, detect safety violations, and navigate industrial workspaces autonomously. Its cognitive intelligence layer combines sensor observations with contextual information, historical records, risk assessment, and predefined industrial safety rules to prioritize inspection tasks and recommend appropriate actions. Explainable decision-making provides reasons for generated alerts, while adaptive inspection planning enables the system to focus on higher-risk areas. The architecture also supports future extensions such as human–robot interaction, multi-robot collaboration, predictive maintenance, and digital twin integration. CIAIR aims to move beyond conventional detection-based inspection by combining robotic perception, memory, reasoning, planning, and explainability within a unified robotic framework. The proposed system has the potential to improve industrial safety awareness, inspection efficiency, and decision support in human-centric manufacturing environments aligned with Industry 5.0 principles.

343.DESIGN AND IMPLEMENTATION OF A ROCKET LAUNCHPAD ENVIRONMENTAL MONITORING SYSTEM

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This project monitors environmental condition around a rocket launchpad and displays real-time data. Here, an ESP32 microcontroller receives data from DHT22 temperature and humidity sensor and HC-SR04 ultrasonic proximity sensor and displays it on an LCD. The system checks if the environmental conditions are suitable for the safe launch of a rocket. If the threshold conditions are crossed, LED indicators are triggered. The LEDs are connected through logic gates which ensure fail-safe mechanism. This system demonstrates the integration of hardware with software at the microcontroller level. Testing confirmed that the prototype could detect environmental changes and activate warnings within acceptable response times.

344.PRIVACY-PRESERVING FEDERATED LEARNING FOR COLLABORATIVE CYBER THREAT INTELLIGENCE SHARING

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Network security has become a big problem today, as the number of cyber-attacks have increased exponentially. Intrusion Detection Systems (IDS) are one of the most efficient ways to monitor network traffic for possible attacks. However, in current IDS architecture, data is collected from diverse resources and processed on a single server which compromises the privacy of the data and overloads the server due to the high data traffic. Our project aims at proposing a Federated Learning (FL) based intrusion detection system that will help preserve data privacy and reduce the load on the server. In FL, multiple clients (can be organizations) participate and train the model on their local data (CIC-IDS2017, CSE-CIC-IDS2018, UNSW-NB15, Bot-IoT) and only send the necessary model updates to the server (consensus). Thus, the sensitive data (network traffic) remains safe and is not shared between clients. We have applied the attention-based Bidirectional Gated Recurrent Unit (Attention-BiGRU) network to acquire the traffic features. The features are extracted using the algorithm of Feature Importance, which removes irrelevant information. The traffic is classified using Extreme Gradient Boosting (XGBoost). Furthermore, we have applied FedProx algorithm on the server-side to ensure faster convergence while training with non-IID data (different data on each client) and heterogenous models. We have also designed a web dashboard to keep track of the detected threats and the performance of the model during training after deploying it in the production environment. In addition, we have created a collaborative security model that helps clients to maintain the confidentiality of their network data and cooperatively train the detection system without sharing sensitive information. We believe our model can be helpful for organizations to increase their security level and detect threats in time.

345.CYBERTWIN AI 2.0 EXPLAINABLE MULTI-AGENT TWIN FOR AUTONOMUS CYBER THREAT DETECTION AND RESPONSE

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Industrial Control Systems (ICS) play an important role in critical infrastructure, manufacturing and industrial automation. The increasing connectivity of these systems has created new opportunities for cyber-physical attacks that can affect both digital operations and physical processes. Traditional cybersecurity approaches mainly depend on predefined rules and isolated monitoring techniques, which may be limited in detecting complex and evolving threats. Therefore, there is a need for an intelligent, proactive and explainable cybersecurity framework for continuous industrial system monitoring. In existing systems, Digital Twin technology and knowledge-based approaches are used to monitor industrial environments and identify abnormal behaviour. However, many conventional approaches focus mainly on threat detection and provide limited support for explaining the detected threats, evaluating their severity, predicting future attacks and coordinating response decisions. Furthermore, the decision-making process of intelligent cybersecurity systems may be difficult to understand without proper explanation mechanisms. To address these challenges, the proposed CyberTwin AI 2.0 – Explainable Multi-Agent Digital Twin for Autonomous Cyber Threat Detection and Response framework integrates Digital Twin technology with Artificial Intelligence, Machine Learning and Explainable AI. The system continuously monitors the virtual representation of an industrial environment to identify abnormal activities and potential cyber threats. Explainable AI is incorporated to provide understandable reasons behind threat predictions, while risk scoring is used to assess the severity and potential impact of detected attacks. Attack prediction supports proactive threat analysis, and multiple intelligent agents coordinate detection, analysis and response decision support. A centralized dashboard presents system status, detected threats, explanations, risk levels and alerts. The proposed framework aims to improve transparency, situational awareness and proactive cyber threat management in Industrial Control Systems.

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346. DESIGN AND DEVELOPMENT OF A LOW-COST VIBRATION-BASED BONE HEALTH SCREENING PROTOTYPE

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Bone health assessment is important for the early identification of changes in bone strength and the risk of fractures. Conventional bone mineral density (BMD) assessment methods, such as dual-energy X-ray absorptiometry (DXA), require specialized equipment and access to clinical facilities. This project presents Bone Sense AI, a low-cost prototype designed to investigate vibration-response measurements as a potential approach for preliminary wrist bone-stiffness screening. The prototype uses an ESP32 development board and a piezoelectric sensor to collect electrical responses associated with mechanically induced vibrations. A software interface guides the measurement process, activates the vibration motor, records repeated sensor readings, and calculates their average. Ten preliminary readings were collected and analyzed using descriptive statistics. The readings ranged from 274 to 339 arbitrary units, with a mean of 299.5 arbitrary units and a sample standard deviation of approximately 17.0 arbitrary units. These results demonstrate preliminary sensor data acquisition and basic statistical processing. However, the prototype has not been validated against clinical bone mineral density measurements. Further hardware development, calibration, and clinical validation are required before its suitability for bone health screening can be determined.

347.ENERGY-EFFICIENT CRYPTOGRAPHIC FRAMEWORK FOR RESOURCE-CONSTRAINED IOT ENVIRONMENTS

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The rapid expansion of digital systems has significantly increased global energy consumption, raising critical concerns regarding the sustainability of computation- intensive technologies. Although cryptographic operations are indispensable for ensuring confidentiality, integrity, and authentication, their computational complexity introduces substantial energy overheads. Consequently, embedding energy efficiency as a first-order design constraint in cryptographic systems has become imperative. Green cryptography addresses this challenge by minimizing power consumption while preserving rigorous security guarantees. In this paper, we propose a VLSI-inspired optimization framework for sustainable cryptography that systematically evaluates workloads across algorithmic, architectural, and hardware layers. The framework leverages lightweight primitives, instruction-set architecture (ISA) enhancements, and parallel processing platforms to mitigate computational overhead. Evaluated across key application domains including the Internet of Things (IoT), cryptocurrency ecosystems, and quantum computing environments the proposed model demonstrates that coordinated, cross-layer software–hardware optimization substantially improves energy efficiency while maintaining robust security, making it highly suitable for future sustainable digital infrastructures.

348.AI-ENABLED AUTONOMOUS UNDERWATER ROBOTIC TURTLES FOR RIVER PLASTIC DETECTION AND SELECTIVE COLLECTION: A STRUCTURED NARRATIVE REVIEW AND CONCEPTUAL FRAMEWORK

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Plastic pollution in rivers requires practical approaches to finding and removing waste without harming aquatic life. Most automated cleaning systems operate at or near the water surface, whereas submerged plastic is harder to handle because of turbidity, changing light, optical distortion, vegetation, sediment, currents, and irregular obstacles. This paper presents a structured narrative review of research on artificial intelligence, underwater computer vision, autonomous underwater robotics, navigation, and waste-collection mechanisms, with attention to their relevance for rivers, together with an evidence-informed conceptual framework. The literature shows that deep-learning detectors and segmentation models can support underwater litter perception, but their performance depends heavily on visibility, background complexity, occlusion, dataset quality, and domain shift. Underwater-robotics research offers sound foundations for sensing, localization, navigation, and control, and existing cleaning robots demonstrate several ways of retrieving waste. However, few studies treat perception, river-specific autonomy, selective collection, and ecological safety as one closed-loop problem. Building on this gap, we outline a conceptual bio-inspired robotic-turtle framework that combines camera, sonar, depth, and inertial sensing with image enhancement, AI-based detection, target localization, path planning, obstacle avoidance, selective collection, storage, communication, and emergency recovery. A gated decision rule for selective collection and a hypothesis-level comparison with propeller-driven vehicles are also presented. The proposed turtle is a conceptual research framework; it has not been built or experimentally validated. We close by identifying research priorities: river-specific datasets, robust perception in poor visibility, energy-aware autonomy, organism-safe collection, and staged validation from simulation to controlled-water and field trials in real rivers.

349.AI-BASED REAL-TIME FINANCIAL FRAUD DETECTION USING HYBRID MACHINE LEARNING

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Financial fraud has become a major challenge with the rapid growth of online banking, digital payments, and electronic transactions. Traditional fraud detection methods mainly depend on predefined rules, which may not effectively identify new and complex fraudulent patterns. This project, titled "AI-Based Real-Time Financial Fraud Detection Using Hybrid Machine Learning," proposes a machine-learning-based system to detect potentially fraudulent financial transactions in real time.

350.DEEP LEARNING-BASED ACCIDENT DETECTION IN LOW-QUALITY CCTV VIDEO FEEDS

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Roads in tunnels vary in many ways from certain sections of open roads. For many drivers, the subway is an unusual driving area in a road network that can create even pressure. In conjunction with the Advanced Learning Network known as the Object Detection and Monitoring System, a quick and standard tracking system used for detecting automatically and unexpected incidents monitors on CV tunnels, Writing weight acquisition, (2) standings, (3) people outside the tunnel (4) fire. Odts accepts the video frames as input to obtain the location binding (BBox) result for object binding and compares BBoxes of current and previous video images to give each moving object a unique identification number. With this system, you can track a moving object over time, which is unfamiliar with the normal object detection settings. An indepth learnin model was developed in the ODTS with a series of graphically designed (AP) data sets of 0.8479, 0.7161, and 0.9085 values for direct, automatic, human, and product items, respectively. Subsequently, based on the in-depth learning model developed, the ODTS video risk assessment system was evaluated using four risk video recordings for each risk. This allows the system to detect all crashes in 10 sec. The ODTS acquisition capabilities can be automatically upgraded without changes to program codes if the training database is improved.

351.SMART AIR QUALITY MONITORING NETWORK USING IOT

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Air quality is an important environmental and public-health concern because people may be exposed to harmful pollutants without being able to identify them through ordinary observation. This paper presents SkyShield, an Internet of Things (IoT)-based air-quality monitoring and alert platform that integrates an ESP32 device layer, wireless communication, a Java Spring Boot backend, relational data

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persistence, REST APIs, and a browser- based dashboard. The application supports readings associated with PM2.5, PM10, volatile organic compounds (VOC), oxygen concentration, carbon dioxide (CO_2), carbon monoxide (CO), nitrogen dioxide (NO_2), temperature, and humidity. Incoming readings are validated and evaluated against application-defined thresholds, then assigned one of five categories: GOOD, MODERATE, UNHEALTHY_SENSITIVE, UNHEALTHY, or HAZARDOUS. An application-level alert is recorded when a reading is classified as UNHEALTHY or HAZARDOUS. Sensor nodes can be registered and managed, historical readings can be retrieved, and alerts can be acknowledged through the application. A Wokwi-based ESP32 simulation is used to demonstrate the data-transfer and software workflow without requiring a complete physical sensor assembly. In the current demonstration, several environmental values are simulated; therefore, the work validates an integrated software and communication prototype rather than a calibrated air-quality instrument. The paper describes the system architecture, implementation, functional evaluation approach, limitations, and future development directions.

352.HEALTHSHIELD AN INTELLIGENT MEDICATION ADHERENCE AND DIGITALPRESCRIPTION MANAGEMENT SYSTEM USING JAVA FULL STACK

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Medication non-adherence is a persistent healthcare problem that contributes to treatment failure, hospital readmissions, escalating costs, and preventable deaths. Most existing reminder applications deliver only basic notifications and lack prescription management, adherence analytics, caregiver oversight, and clinician integration. This paper presents HealthShield, a centralized, web-based Java full stack platform that securely manages digital prescriptions, generates personalized dose reminders, records patient compliance, and gives doctors and caregivers real-time visibility into adherence. The system follows a layered architecture built on Spring Boot, Spring Security with JSON Web Tokens, Spring Data JPA, MySQL, and a React.js front end. A rule-based adherence scoring model and a missed-dose escalation workflow convert raw dose logs into actionable alerts for caregivers and physicians. Role-based access control, encrypted storage, and audit logging protect sensitive health information. The paper describes the design, implementation, and a functional evaluation plan, and discusses how the platform can improve continuity of care and reduce preventable non-adherence.

353. EXPLAINABLE SKIN DISEASE CLASSIFICATION USING CONVNEXT V2

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Skin diseases can exhibit similar visual patterns, making image-based classification challenging. Differences in skin tone, illumination, camera quality, and lesion appearance can further affect automated predictions. This paper presents a proposed explainable skin disease classification framework using ConvNeXt V2 and Gradient-weighted Class Activation Mapping (Grad-CAM). The framework targets five classes: acne, eczema, psoriasis, vitiligo, and tinea (ringworm). Input images are

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resized and preprocessed, while augmentation operations such as flipping, rotation, and brightness variation are used to improve training-data diversity. ConvNeXt V2 is used as the image- classification backbone, and Grad-CAM generates class- specific heatmaps to indicate image regions that influence predictions. The heatmaps are intended to help inspect whether the model attends to the lesion region or irrelevant background. The supplied project abstract describes the proposed workflow but does not provide dataset size, experimental settings, or quantitative results. Therefore, this paper defines the method and evaluation plan without claiming unreported performance. The framework is intended as a decision- support research prototype, not a replacement for professional diagnosis.

354.THE IMPACT OF TRADITIONAL COURTYARD PLANNING OF TAMIL NADU FOR SUSTAINABLE ARCHITECTURE SOLUTION

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Traditional courtyard houses of Tamil Nadu incorporate spatial planning strategies developed in response to local climate. However, rapid changes in contemporary housing have resulted in the reduced application of these traditional principles. This study investigates the potential of traditional courtyard planning as a sustainable and climate-responsive architectural solution. A performance-based approach is adopted to analyze selected traditional courtyard houses based on orientation, courtyard configuration, spatial organization, natural ventilation, and daylighting. The analysis indicates that courtyards function as both social and environmental spaces, contributing to natural ventilation, daylighting, solar protection, and moderation of indoor environmental conditions. The study concludes that traditional courtyard planning provides valuable principles for contemporary sustainable housing when appropriately adapted to present-day requirements. The findings demonstrate the potential of integrating vernacular spatial knowledge with modern sustainable design strategies to develop climate-responsive housing solutions in Tamil Nadu.

355.NEUROSPHERE: A HANDHELD MULTIMODAL SMART STRESS BALL FOR REAL-TIME STRESS MONITORING AND BIOFEEDBACK

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Most stress assessment still relies on self-report questionnaires or laboratory equipment, neither of which suits day-to-day use. NeuroSphere is a handheld stress ball that tracks stress while the user holds and squeezes it. Three sensors work together: galvanic skin response (GSR) sensing picks up sympathetic arousal, a MAX30102 photoplethysmography (PPG) sensor gives heart rate and heart rate variability, and a force- sensitive resistor (FSR) records how hard the ball is squeezed. An ESP32-C3 microcontroller reads the sensors, processes the signals, and sorts the user's state into low, medium, or high stress using fixed thresholds. The result is shown through a WS2812 LED strip and a vibration motor. The prototype is built on a perf board and runs on a rechargeable battery. GSR acquisition and I2C sensor communication have been tested. Finishing the firmware and adding Bluetooth Low Energy logging to a phone dashboard are the next steps.

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356.WIRELESS AUDIO TELEMETRY ACQUISITION SYSTEM FOR HUMAN-MACHINE INTERFACE AUTOMATION

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Traditional Human-Machine Interfaces (HMIs) for automated control rely on physical host tethering, local sound card drivers, and browser permissions, creating hardware constraints and disk I/O latency. This paper presents a wireless audio telemetry system featuring hardware-decoupled acoustic control and zero-disk-write processing. A remote mobile telemetry node acts as a wireless transducer, streaming uncompressed PCM/WAV audio over a local Wi-Fi IP socket (/audio.wav). A central host streams incoming network chunks directly into volatile RAM, eliminating local disk write latency and storage wear. To bypass transcription error propagation in multi-stage speech pipelines, the volatile audio buffer is evaluated in a single pass using multimodal generative AI to extract structured JSON intent parameters without intermediate speech-to-text conversion. User credentials are encrypted locally using symmetric Fernet cryptography within an SQLite database before background task execution via SMTP. Experimental evaluation verifies stream throughput, memory overhead, intent accuracy, and execution reliability.

357.PIXEL ARCHITECT: AN AI-ASSISTED FRAMEWORK FOR AUTOMATED 3D MODEL TRANSFORMATION AND COST-EFFECTIVE REAL-TIME INTERACTIVE ARCHITECTURAL VISUALIZATION

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Architectural visualization plays a significant role in communicating design concepts to clients through 2D drawings, photorealistic renderings, and walkthrough videos. However, conventional visualization methods often rely on fixed viewpoints, require time-consuming preparation, and provide limited user interaction. To address these challenges, this paper proposes Pixel Architect, an AI-assisted framework for automated 3D model transformation and cost- effective real-time interactive architectural visualization. The proposed system aims to convert existing architectural 3D models into optimized, browser-based interactive environments that enable users to explore architectural spaces through orbit inspection and first-person walkthrough modes. The framework integrates geometry validation, AI-assisted semantic scene analysis, mesh and material optimization, navigation and collision generation, and real-time rendering. Technologies such as Python, machine learning, Three.js, Babylon.js, WebGL, and GLB/glTF are considered for implementation. The proposed approach focuses on reducing repetitive manual preparation, improving accessibility, and enhancing spatial understanding through interactive exploration. System performance is planned to be evaluated using conversion time, manual intervention, loading speed, rendering performance, navigation accuracy, visual fidelity, and user

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satisfaction. The framework aims to provide an accessible and cost-effective solution for architects, designers, and visualization professionals while supporting future developments in immersive architectural experiences.

358. SYNTHETIC DATA AND IMAGE ENHANCEMENT FOR BRAIN TUMOR DETECTION: A REVIEW OF GENERATIVE AI ADVANCES

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Accurate brain tumor detection using Magnetic Resonance Imaging (MRI) requires large, diverse, well-labelled datasets and good-quality scans. However, medical images can be limited because of privacy concerns, rare tumor types, high annotation costs, and differences between scanners. Generative Artificial Intelligence (Generative AI) can help address these challenges through synthetic data generation and image enhancement. Techniques such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and diffusion models can create additional MRI-like images to support AI training. Image-enhancement methods can reduce noise, improve resolution, correct image variations, and estimate missing MRI sequences. These methods may help AI systems learn more effectively and support tumor classification and segmentation. However, generated images may contain false details or bias, so they must be carefully evaluated using real, unseen patient scans. Generative AI should be used as a decision-support tool for radiologists and not as a replacement for medical professionals.

359.PREDICTING CYBER-ATTACKSTHROUGH DATA SCIENCETECHNIQUES

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Anomaly detection is the process of identifying data points that do not follow regular patterns. This might indicate odd network activity, damaged sensors, or data quality issues. It may use it on both labelled and unlabeled datasets. When labels are not accessible, unsupervised techniques are often used. In this study, machine learning approaches are utilized to detect anomalies and assess model performance to improve forecast accuracy. The approach uses supervised machine learning to examine datasets and derive insights such as variable identification, univariate, bivariate, and multivariate analysis, and how to cope with missing data. The system distinguishes between four sorts of assaults: attacks include DoS, R2L, U2R, and probe. We examine different machine learning algorithms using metrics such as accuracy, entropy, recall, performance, and F1-score to determine which one is better for detecting and classifying cyber hazards. The evaluation results show that the Random Forest technique had the highest accuracy and F1-score, indicating that it performed better than the other models. This suggests that in this scenario, Random Forest is the best model for identifying and categorizing cyber hazards.

360. DIGITAL TWIN AND PREDICTIVE INTELLIGENCE FOR SMART CITIES

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Cities generate data at unprecedented scale, yet most urban systems still react to problems only after they occur. This paper presents a Digital Twin and Predictive Intelligence framework for smart cities that combines Internet of Things (IoT) sensing, a continuously synchronized virtual replica of the city and machine learning models for forecasting and decision support. Distributed sensors measure traffic flow, energy consumption, air quality and water usage and stream the data through edge gateways to a data platform. The digital twin maintains the live state of city assetsina georeferenced 3D model. Time-series and graph-based predictive models forecast congestion, energy demand and pollution levels ahead of time, and a what-if simulation engine evaluates candidate interventions such as signal retiming, load shifting and traffic diversion on the twin before they are applied in the real city. Predictions, alerts and recommended actions are delivered to a command dashboard used by city operators and to a citizen-facing application. The proposed architecture moves urban management from reactive to proactive operation reduces the risk of trial-and-error decisions in the physical city and provides a scalable foundation for sustainable smart city services.

361.SMARTBUS: CROWDSOURCED BUS TRACKING, CROWD ESTIMATION AND DESTINATION ALERTING FOR PUBLIC TRANSIT WITHOUT ON-BOARD GPS

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In many developing regions, a large share of public and institutional buses operate without on-board GPS hardware, so passengers cannot tell when a bus will arrive, how crowded it is, or where to alight in an unfamiliar area. This paper presents SmartBus, a passenger-centric framework that couples crowdsourced smartphone-based bus tracking with a decision-support layer. Anonymized location reports contributed by riders are map-matched to known routes to estimate bus position and predict arrival time. A lightweight estimator classifies on-board crowd level as Low, Medium or High, and a route-constrained geofencing module issues destination-approach and wrong-direction alerts. A configurable multi-factor score then recommends the most suitable bus instead of simply the earliest one. The framework will be evaluated on Salem city bus routes using ETA mean absolute error, alert accuracy and false-alarm rate, and crowd- classification accuracy, against schedule-based and earliest-bus baselines. SmartBus is designed to deliver these services for GPS-less fleets at near-zero infrastructure cost.

362. NEURO-SYMBOLIC AI: INTEGRATING NEURAL LEARNING AND LOGICAL REASONING FOR EXPLAINABLE CYBER THREAT DETECTION

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Cybersecurity threats are becoming increasingly sophisticated, creating a need for intelligent systems that can identify suspicious network activities and explain their decisions. Traditional machine learning techniques can recognize patterns in network traffic but may provide limited explanations for their predictions. Rule-based systems offer explicit reasoning but may struggle to identify complex patterns that have not been anticipated by predefined rules. Neuro-Symbolic Artificial Intelligence (Neuro-Symbolic AI) offers a promising approach by combining neural learning with symbolic reasoning. This paper explores a neuro-symbolic framework for cyber threat detection in which a neural model identifies potentially suspicious network activities and a symbolic reasoning component applies predefined security rules to interpret the results. The proposed framework aims to improve the interpretability of threat detection by connecting model predictions with understandable security rules. A prototype can demonstrate this approach using simulated network traffic and a rule-based reasoning engine. The paper discusses the proposed architecture, working methodology, potential applications, and challenges, including false positives, rule maintenance, data quality, and computational complexity. The framework is intended as an educational prototype rather than a validated production security system. Further experimental evaluation is required to determine whether combining neural learning and symbolic reasoning improves detection performance and explanation quality compared with individual approaches.

363.AN IOT-ENABLED GREEN COMPUTING FRAMEWORK FOR WATER QUALITY MANAGEMENT, LAKE RESTORATION, AND WASTE OPTIMIZATION

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Water pollution, declining freshwater availability, and the degradation of natural water bodies have become significant environmental challenges. Conventional water quality monitoring systems often rely on periodic sampling, manual inspection, and energy-intensive data processing, limiting their ability to provide continuous monitoring and timely intervention. This paper proposes an Internet of Things (IoT)-enabled green computing framework for integrated water quality management, lake restoration, and water resource optimization. The proposed framework combines low-power IoT sensor networks, edge computing, energy-efficient communication, cloud- based data management, and intelligent analytical techniques to monitor water quality parameters and support sustainable water management decisions. Sensors measure parameters such as pH, temperature, turbidity, electrical conductivity, and dissolved oxygen, while optional sensors monitor water levels and selected nutrients. Edge devices perform preliminary data processing, filtering, and threshold-based anomaly detection to reduce unnecessary data transmission and computational overhead. A centralized analytical platform integrates sensor observations, historical records, and environmental information to identify potential pollution events, assess lake conditions, and support water allocation and restoration planning. Green computing principles are incorporated through low-power hardware, adaptive sensing intervals, lightweight analytics, renewable-energy integration, and resource- efficient computing infrastructure. The framework is designed to improve monitoring continuity, reduce avoidable energy consumption, and support evidence- based environmental decisions. The proposed architecture is evaluated conceptually through defined performance indicators, including monitoring accuracy, communication energy, data-processing latency, system availability, and water-use efficiency. Experimental validation using real-world datasets and field deployments is identified as future work. The framework offers a scalable foundation for sustainable water monitoring and environmentally responsible lake management.

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364. A COMPREHENSIVE SURVEY OF DATASETS, FEATURE EXTRACTION , MACHINE LEARNING AND DEEP LEARNING APPROACHES FOR HANDWRITTEN SIGNATURE

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Handwritten signature verification is an important biometric authentication issue with diverse applications in banking, financial transactions, legal documentation, insurance, governmental administration, and identity authentication. Compared with online verification, which deals with a dynamic process involving pen pressure, speed, and order of strokes, offline verification works with still images obtained after a signature is written and scanned or photographed. Thus, the system has to distinguish between genuine and forged signatures based only on their spatial and visual features. The last decades have witnessed a shift from a manual selection of features and standard classifiers to convolutional neural networks, transfer learning, Siamese convolutional networks, capsule networks, and hybrid deep learning systems. This paper aims to present a comprehensive overview of offline handwritten signature verification, paying attention to publicly available datasets, preprocessing techniques, manually designed feature descriptors, traditional machine learning approaches, deep learning architectures, and evaluation methods. The analysis comprehensively reviews existing datasets, such as CEDAR, GPDS, UTSig, BHSig260, MCYT-75, SigComp2011, and PHBC, in terms of their respective populations of writers and genuine forgery samples. Additionally, the analysis covers existing and traditional features like Histogram of Oriented Gradients (HOG), Scale-Invariant Features Transform (SIFT), texture features, projection profiles, wavelet-based features, and hybrid representations. Then, CNN- based methods are analysed, including metric-learning techniques. The results presented indicate that deep learning techniques appear to be increasingly effective; however, results depend strongly on dataset characteristics, degree of variations between writers, forgery types, implementation of the training method, and data availability. Ultimately, some future research issues are raised, including writer-independent verification, generalization across different databases, learning-with-limited-resources problems, verification of data, synthetic data generation, lightweight models, and expert forgery detection.

365.A COMPREHENSIVE SURVEY OF DEEP LEARNING TECHNIQUES FOR MEDICAL IMAGE ANALYSIS WITH APPLICATIONS TO DIABETIC RETINOPATHY DETECTION AND GRADING

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Diabetic retinopathy (DR) is a major diabetes-related complication and an important cause of preventable visual impairment worldwide. The increasing availability of retinal fundus datasets and advances in artificial intelligence (AI) have led to substantial research in automated DR detection, classification, severity grading, and lesion analysis. This survey reviews recent developments in machine learning and deep learning-based DR analysis, with an emphasis on studies published primarily during 2024–2026. Thirty representative studies were examined in terms of the datasets used, methodological approaches, underlying technologies, positive findings, limitations, and research gaps. The reviewed studies include conventional convolutional neural networks (CNNs), transfer-learning approaches, ensemble models, attention mechanisms, Vision Transformers, CNN–Transformer hybrid architectures, multimodal learning, explainable AI, uncertainty-aware models, and privacy-preserving federated learning. The findings indicate that recent approaches have improved automated retinal feature representation, lesion-focused analysis, severity grading, class-imbalance handling,

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interpretability, and robustness across selected datasets. However, several limitations remain, including dataset imbalance, variations in image quality, annotation inconsistencies, limited geographical and demographic diversity, computational complexity, insufficient external and prospective validation, and difficulties in distinguishing adjacent DR severity stages. Emerging datasets and multimodal and ultra-wide-field imaging approaches address some of these limitations but introduce additional acquisition and deployment requirements. Overall, the survey identifies a transition from benchmark-oriented CNN classification toward more robust, interpretable, multimodal, and clinically deployable AI systems. Future research should prioritize standardized and diverse datasets, multicenter clinical validation, lightweight architectures, reliable uncertainty estimation, improved explainability, and models capable of maintaining performance across imaging devices and real-world healthcare environments.

366.SECUREDROID: A DETERMINISTIC MOBILE SECURITY FRAMEWORK FOR EXPECTED-VERSUS-OBSERVED SECURITY STATE EVALUATION, ATTRIBUTION, AND PRIVACY-PRESERVING RISK ASSESSMENT

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Modern mobile operating environments represent the primary locus of personal communications, financial transactions, and enterprise identity management. While the Android platform incorporates isolation primitives such as process sandboxing and a runtime permission architecture, overall endpoint posture remains contingent upon end-user configuration decisions, third-party software lifecycles, and permission assignments. Conventional mobile antivirus tools often rely on opaque cloud heuristics, invasive telemetry aggregation, or unsubstantiated threat claims that degrade user trust and violate basic data minimization principles. This paper presents SecureDroid, an on-device, privacy-preserving mobile security and event monitoring framework implemented natively in Kotlin for Android (minSdk 24, targetSdk 37, compileSdk 37; validated on API level 34). SecureDroid centers upon the Expected User Security State Architecture (EUSSTA), evaluating observed device attributes, target SDK levels, and dangerous permission grants against baseline operational policies through a four-valued deterministic logic. Discrepancies are transformed into structured, deduplicated findings, mapped to evidence records with explicit attribution confidence ratings, and coupled with user-controlled remediation routing via explicit Android Settings intents. Identity management is partitioned to Firebase Authentication, while cloud persistence is provided optionally through Supabase REST endpoints governed by Row-Level Security and explicit client- side token sanitization. Evaluated across 159 automated unit and integration tests (100% pass rate) and validated on physical Android hardware, SecureDroid demonstrates deterministic execution, zero credential leakage, and rigorous adherence to local-first privacy boundaries.

367.DESIGN AND IMPLEMENTATION OF A ZERO-KNOWLEDGE HYBRID POST-QUANTUM CRYPTOGRAPHIC CLOUD VAULT USING NIST ML-KEM-768 AND X25519

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Public-key schemes such as RSA, elliptic-curve cryptography and Diffie–Hellman are vulnerable to Shor's algorithm on a sufficiently large fault-tolerant quantum computer. Adversaries can therefore harvest encrypted traffic and stored ciphertext today and decrypt it later ("Harvest Now, Decrypt Later", HNDL). This paper presents the design and implementation of a zero-knowledge cloud vault that protects stored data against both classical and quantum adversaries. The system uses a hybrid Key Encapsulation Mechanism that combines classical X25519 with the NIST-standardized ML- KEM-768 (FIPS 203), whose security rests on the Module Learning-with-Errors problem. The two ephemeral shared secrets are concatenated and passed through HKDF-SHA256 with a 16-byte random salt to derive a 256-bit key, and payloads are encrypted with AES-256-GCM for authenticated encryption. Master private keys are generated and kept only on the client, protected by a SHA-256 integrity fingerprint that causes deterministic rejection of any mutated key before decapsulation. The backend, built on asynchronous FastAPI and Turso edge SQLite and deployed on Vercel, receives only opaque ciphertexts, public keys, salts and nonces. The resulting architecture remains confidential if either X25519 or ML-KEM-768 remains unbroken, and the server learns nothing about plaintext or private keys.

368.A COMPREHENSIVE REVIEW OF FINANCIAL TECHNOLOGY (FIN TECH) AND FINANCIAL INCLUSION: TRENDS, CHALLENGES, AND FUTURE RESEARCH DIRECTIONS

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Financial Technology (Fin Tech) has moved from a peripheral support function inside traditional banks to a driving forcere shaping how financial services are designed and delivered. Its core buildingblocks artificial intelligence, blockchain, big data analytics, cloud infrastructure, and mobile applications have reduced the costand friction of reaching customers, and in doing so have opened formalfinancialservicestopopulationsthatconventionalbankinglargelybypassed. Digital payments, mobile banking, and app-based lending sit at the centre of this shift. At the same time, the growth of FinTech has not been friction-free cybersecurity exposure,data-privacy concerns, uneven regulation, gaps in digital literacy, and fragile consumer trust all continue to slow its progress toward becoming a fully inclusive system. This paper draws together research published between 2020 and 2025 across five interlockingareas—artificial intelligence, blockchain, digitalpayments, financial literacy, and regulatory technology (RegTech)—to map where the field currently stands. Rather than simply cataloguing individual studies, it compares their methods and conclusions, flags where findings diverge, and surfaces gaps that remain largely untouched, including ethical AI governance, cyber security resilience, ruralfinancial inclusion, and cross-border digital finance. It closes by outlining directions future research could usefully take to build financial ecosystems that are more secure, more inclusive, and more sustain- able—offering a reference point for researchers, policymakers, financial institutions, and technology developers alike.

369.DEEP LEARNING -BASED CYCLONE DETECTION AND INTENSITY ESTIMATION USING SATELLITE IMAGERY AND MULTIMODAL WEATHER DATA

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Tropical cyclones are among the most destructive natural disasters, causing severe damage to human life, infrastructure, and coastal environments. Accurate cyclone detection and intensity estimation are essential for improving weather monitoring and disaster preparedness. Traditional approaches rely on satellite image interpretation and meteorological observations, which may not fully exploit the complementary information available from different data sources. This paper proposes a deep learning-based framework for cyclone detection and intensity estimation by integrating satellite imagery with multimodal meteorological data. A Convolutional Neural Network (CNN)

is employed to extract spatial features from satellite images, while a Multilayer Perceptron (MLP) processes numerical weather variables such as wind speedatmospheric pressure, temperature, and humidity. The extracted features are combined using a multimodal feature fusion technique to generate cyclone detection and intensity estimation predictions. The proposed framework is designed to support both classification and regression tasks. Its performance will be evaluated using precision, recall, F1-score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). Comparative experiments with image-only and weather-only models will be used to investigate the effectiveness of multimodal integration. The frame work aims to support automated cyclone monitoring and provide useful information for disaster management and early warning applications.

370.BEHAVIORAL DATA IN GAME-BASED COGNITIVE ASSESSMENT: A REVIEW OF RESPONSE TIME, INTERACTION PATTERNS, AND PERFORMANCE FEATURES

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Digital games can not only provide a quantitative measure of whether a task was completed successfully, but can also provide information about how it was completed. This paper synthesizes behavioral markers of game-based cognitive assessment, such as response time, sequence of actions, attempts, errors, persistence, and changes in performance across levels and sessions. Building on prior work in stealth assessment and evidence-centered game design, we understand gameplay behaviors as indicators of cognition, and we draw on recent work in machine learning and game learning analytics to discuss how these behavioral traces can be scaled up to identify patterns. At the same time, we draw upon meta-analyses and other evidence to specify the complexities of the relations between indicators and each of at least seven factors–actual ability, age, prior gaming experience, subject-matter knowledge, motivation, game interface, and extreme low and high scores. We review 20 tightly related studies, each of which holds one of at least seven distinct research gaps, including narrow, single construct experimental designs; classic-practice and relevant- won't-know-to-know samples of unrepresentative ages; conceptual weak theories; and field-wide small samples, inadequate telemetry standards, and nonstandardized normalizations. Based on this analysis, we describe a child-centered approach to cognitive assessment using game-related telemetry of behavior, simple adaptive levels, interpretable analysis, and longitudinal tracking, as well as a concrete proposal for the design of a

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feasibility study: the data schema, the evaluation plan, and the protective measures, in a pilot with about 30 children aged 6-16. Throughout, we emphasize the distinction between and the limitations of decision support and measurement, with measurable goals and scope.

371. ECOBATT: IOT-ENABLED BATTERY HEALTH MONITORING AND CARBON- AWARE CHARGING FOR SUSTAINABLE REUSE

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Lithium-ion packs in electric two-wheelers and small storage units are protected by a battery management system (BMS), yet the person using the pack rarely sees what it is actually doing. This paper presents EcoBatt, a low-cost add-on platform that makes the electrical and thermal behaviour of a pack visible and uses that information to support reuse decisions. An ESP32 reads voltage and current from an INA219 monitor and temperature from a DS18B20 probe on a 3S 18650 pack, applies alert rules, and streams the readings over Bluetooth to an Android dashboard. A laptop-hosted analytics layer consumes the logged data to estimate state of health (SOH) and remaining useful life (RUL) and to rank candidate charging windows by grid carbon intensity under health-aware constraints. The BMS stays solely responsible for cell protection; EcoBatt observes and advises. The design is positioned against fifteen recent SOH/RUL studies and a body of classical BMS work, and is compared feature by feature with conventional threshold monitoring and with data-driven estimators. The prototype demonstrates stable acquisition and live display of voltage, current and temperature. Quantitative benchmarking of the SOH/RUL and scheduling modules on larger cycling datasets is identified as the next step.

372. LAWLENS: SMART LEGAL DOCUMENT PROCESSING AND AUTOMATED RISK ASSESSMENT FOR DISTRICT COURTE-SERVICES

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Legal documents such as affidavits, lease agreements, commercial contracts, and judicial petitions often contain dense legal terminology, complex sentence structures, and conditional obligations that are difficult for non-expert users to understand. Citizens interacting with District Court e-Services may face difficulties identifying important responsibilities, penalties, payment conditions, and other significant clauses. This paper proposes LawLens, a citizen-oriented legal document processing and risk assessment framework designed for District Court e-Services. The proposed framework combines multi-format document ingestion, text extraction, optional optical character recognition for scanned documents, contextual text chunking, large language model-based clause analysis, rule augmented risk categorization, and plain-language multilingual output. Legal clauses are organized into Low, Medium, an High operational risk categories based on model confidence an predefined rule-based indicators. A lightweight FastAPI backend and SQLite persistence layer are proposed for managing document processing and generated reports. Since validated experimental measurements are not available for the present version, this paper defines an evaluation framework covering extraction quality, clause identification, risk classification, processing latency, groundedness, and usability. LawLens is intended as an assistive information system and not as a replacement for qualified legal advice or judicial interpretation.

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373. LUMIFARM: LOW-COST ESP32-BASED SUNLIGHT-FREE CULTIVATION WITH REAL-TIME SOIL MOISTURE, HUMIDITY AND TEMPERATURE MONITORING AND CONTROL

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Anyone growing plants indoors without sunlight ends up checking the soil, the air and the lamps by hand, or leaving the job to a timer that has no idea what the plants need at that moment. Commercial indoor-farm controllers fix this but cost far more than a small grower will spend. We built LumiFarm, a compact controller around a single ESP32. A capacitive soil-moisture sensor and a DHT22 report the conditions in the tray, and the ESP32 turns an LED grow light, a water pump and a ventilation fan on and off through relays, using threshold rules with hysteresis and a daily light cycle. Readings and device states are pushed over Wi-Fi to a dashboard. On the bench, the pump started 10 s after soil moisture dropped below its lower limit, the fan reacted in 50 s to high humidity and 60 s to high temperature, and the relays did not chatter around the thresholds. Everything in the build is a common, cheap part and every action follows a plain rule, so students and hobby growers can copy and retune it.

374. LEAFGUARD: A VISION SYSTEM FOR EARLY PLANT DISEASE DETECTION

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Timely detection of crop disease is one of the most decisive factors in preventing yield loss, yet the AI-based diagnostic tools reported in the literature are predominantly cloud-dependent, computationally heavy, or restricted to a single sensing modality. This paper presents CROPGURU, an edge-AI system that performs crop-leaf disease classification directly on an ESP32-S3 microcontroller using a lightweight TensorFlow Lite convolutional neural network (CNN), without requiring a persistent internet connection. An OV2640 camera module captures leaf images, which are preprocessed on-device (resizing, denoising, and RGB normalization) before inference. A capacitive soil-moisture sensor is sampled in parallel to augment the visual diagnosis with a root-zone moisture indicator, and an advisory engine maps the combined disease class and moisture reading to a treatment or preventive recommendation, shown on a local OLED display and, when Wi-Fi is available, mirrored to a mobile application and cloud database. We describe the system architecture, hardware and firmware design, the disease-detection pipeline, and a functional test protocol for validating the device under field-representative conditions. The design targets small-scale and rural farmers, for whom cost, portability, and offline operation are more critical constraints than raw model accuracy.

375. DESIGN OF FIRE SAFETY PROTECTION SYSTEM FOR ELECTRIC VEHICLE

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Due to the high rate of growth in electric vehicle technology, efficient and reliable battery management systems have become essential to ensure the safe and efficient operation of the vehicle. Batteries play a major role in the operation of electric vehicles. Inappropriate operation of the batteries may result in reduced efficiency, safety issues, and reduced life expectancy. Issues like deep discharge, overcharging, overcurrent, dry running, and high temperature changes have a major impact on the life expectancy of the vehicle's batteries. Therefore, intelligent operation and monitoring have become essential. In this project, a dual battery management system using the Raspberry Pi board will be implemented. In this system, one battery will be used to supply power to the load, and the other will be used to charge the battery using solar power. Users can also monitor the status of the battery using Wi-Fi and Blynk.

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13	AMRUTA INSTITUTE OF ENGINEERING & MANAGEMENT SCIENCES, BIDADI
14	ANNASAHEB DANGE COLLEGE OF ENGINEERING AND TECHNOLOGY, ASHTA
15	ANNASAHEB DANGE COLLEGE OF ENGINEERING AND TECHNOLOGY, ASHTA & TATYASAHEB KORE INSTITUTE OF ENGINEERING AND TECHNOLOGY, WARANANAGAR
16	ANU, GUNTUR
17	ASHOKRAO MANE GROUP OF INSTITUTION, KOLHAPUR
18	ATME COLLEGE OF ENGINEERING , MYSURU
19	AURORA'S PG COLLEGE - AUTONOMOUS, HYDERABAD
20	AVCOE, SANGAMNER
21	B. S. ABDUR RAHMAN CRESCENT INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI
22	BANGALORE INSTITUTE OF TECHNOLOGY, BENGALURU
23	BANNARI AMMAN INSTITUTE OF TECHNOLOGY, SATHYAMANGALAM
24	BMS COLLEGE OF ENGINEERING, BANGALORE
25	BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT, VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI
26	BRAINWARE UNIVERSITY, BARASAT
27	CEG CAMPUS, ANNA UNIVERSITY, CHENNAI
28	CGC UNIVERSITY, MOHALI
29	CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY, HYDERABAD

30	CHANDIGARH COLLEGE OF TECHNOLOGY, MOHALI
31	CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI
32	CMR UNIVERSITY, BENGALURU
33	COLLEGE OF ENGINEERING GUINDY, ANNA UNIVERSITY, CHENNAI
34	D. Y. PATIL COLLEGE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR
35	D.K.T.E. SOCIETY'S TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI
36	DATTAJIRAO KADAM TECHNICAL EDUCATION (DKTE) SOCIETY'S TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI
37	DAYALBAGH EDUCATIONAL INSTITUTE, AGRA
38	DAYANADA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT, BENGALURU
39	DAYANANDA SAGAR COLLEGE OF ENGINEERING, BANGALURU
40	DEOGIRI INSTITUTE OF ENGINEERING AND MANAGEMENT STUDIES, CHH. SAMBHAJINAGAR
41	DINABANDHU ANDREWS COLLEGE, KOLKATA
42	DR. AMBEDKAR INSTITUTE OF TECHNOLOGY, BENGALURU
43	DR. BABASAHEB AMBEDKAR MARATHWADA UNIVERSITY, CHHATRAPATI SAMBHAJINAGAR
44	DR. BAPUJI SALUNKHE INSTITUTE OF ENGINEERING AND TECHNOLOGY, KOLHAPUR
45	DR. D. Y. PATIL'S RAMRAO ADIK INSTITUTE OF TECHNOLOGY, NAVI MUMBAI
46	DY PATIL DEEMED TO BE UNIVERSITY, NAVI MUMBAI
47	E.G.S. PILLAY ENGINEERING COLLEGE (AUTONOMOUS), NAGAPATTINAM
48	EASWARI ENGINEERING COLLEGE, CHENNAI
49	FINOLEX ACADEMY OF MANAGEMENT AND TECHNOLOGY, RATNAGIRI
50	FUTURE INSTITUTE OF ENGINEERING AND MANAGEMENT, RAJPUR SONARPUR
51	FUTURE UNIVERSITY, BAREILLY
52	IICT MGMU
53	INDRA GANESAN COLLEGE OF ENGINEERING, TIRUCHIRAPPALLI
54	J.C. BOSE UNIVERSITY, YMCA, FARIDABAD
55	JADAVPUR UNIVERSITY, KOLKATA
56	JAGANNATH KISHORE COLLEGE, PURULIA, WEST BENGAL
57	JAYPEE UNIVERSITY, ANOOPSHAR
58	K. E. SOCIETY'S RAJARAMBAPU INSTITUTE OF TECHNOLOGY, SAKHARALE
59	K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY, TRICHY
60	K.K.WAGH INSTITUTE OF ENGINEERING EDUCATION AND RESEARCH, NASHIK
61	K.S.INSTITUTE OF TECHNOLOGY, BENGALURU
62	KAMARAJ COLLEGE OF ENGINEERING AND TECHNOLOGY, VELLAKULAM
63	KARPAGA VINAYAGA COLLEGE OF ENGINEERING AND TECHNOLOGY, CHENGALPATTU

64	KASEGAON EDUCATION SOCIETY'S RAJARAMBAPU INSTITUTE OF TECHNOLOGY, SAKHRALE
65	KINGS COLLEGE OF ENGINEERING (AUTONOMOUS), PUDUKKOTTAI
66	KINGSTON ENGINEERING COLLEGE, VELLORE
67	KJ'S TRINITY COLLEGE OF ENGINEERING AND RESEARCH, PUNE
68	KLE COLLEGE OF ENGINEERING & TECHNOLOGY, CHIKODI
69	KONERU LAKSHMAIAH EDUCATION FOUNDATION, GUNTUR
70	KRISHNASAMY COLLEGE OF ENGINEERING & TECHNOLOGY, SENJIKUMARAPURAM
71	LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING(AUTONOMOUS), MYLAPURAM
72	M S RAMAIAH INSTITUTE OF TECHNOLOGY, BANGALORE
73	MADRAS INSTITUTE OF TECHNOLOGY, ANNA UNIVERSITY, CHENNAI
74	MAHARAJA SURAJMAL INSTITUTE OF TECHNOLOGY, NEW DELHI
75	MAHARASHTRA INSTITUTE OF TECHNOLOGY, CHHATRAPATI SAMBHAJINAGAR
76	MAHENDRA COLLEGE OF ENGINEERING, SALEM
77	MANGALORE INSTITUTE OF TECHNOLOGY AND ENGINEERING, MANGALURU
78	MEENAKSHI SUNDARARAJAN ENGINEERING COLLEGE, CHENNAI
79	MGM UNIVERSITY, CHHATRAPATI SAMBHAJINAGAR
80	MIT MANIPAL, KARNATAKA
81	MIT, MANIPAL, KLE COLLEGE OF ENGINEERING & TECHNOLOGY, CHIKODI
82	MMMC & H, MMU SOLAN, HIMACHAL PRADESH
83	MUTHAYAMMAL ENGINEERING COLLEGE, RASIPURAM
84	NAVKIS COLLEGE OF ENGINEERING, HASSAN
85	NCERC, KERALA
86	NEW INSTITUTE OF TECHNOLOGY, KOLHAPUR
87	NIRWAN UNIVERSITY, JAIPUR
88	NOIDA INTERNATIONAL UNIVERSITY, GREATER NOIDA
89	P E S COLLEGE OF ENGINEERING, MANDYA
90	P. A. COLLEGE OF ENGINEERING AND TECHNOLOGY, COIMBATORE
91	P.S.R. ENGINEERING COLLEGE, SIVAKASI
92	PANIMALAR ENGINEERING COLLEGE, CHENNAI
93	PERUNTHALAIVAR KAMARAJAR INSTITUTE OF ENGINEERING AND TECHNOLOGY(PKIET), KARAIKAL
94	PES COLLEGE OF ENGINEERING, MANDYA
95	PES UNIVERSITY, BENGALURU
96	PREC, LONI, MAHARASHTRA
97	PRESIDENCY UNIVERSITY, BENGALURU
98	PSNA COLLEGE OF ENGINEERING AND TECHNOLOGY, DINDIGUL

99	QUEEN MARY'S COLLEGE, CHENNAI
100	R P SARATHY INSTITUTE OF TECHNOLOGY, SALEM
101	RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI
102	RAJARAMBAPU INSTITUTE OF TECHNOLOGY, RAJARAMNAGAR
103	RAJASTHAN TECHNICAL UNIVERSITY, KOTA
104	RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY, PUDUCHERRY
105	RAMANAND INSTITUTE OF PHARMACY AND MANAGEMENT, HARIDWAR
106	RAMRAO ADIK INSTITUTE OF TECHNOLOGY, D. Y. PATIL DEEMED TO BE UNIVERSITY, NAVI MUMBAI
107	RMD ENGINEERING COLLEGE, KAVARAIPETTAI
108	RNSIT, BENGALURU
109	RVR & JC COLLEGE OF ENGINEERING, GUNTUR
110	SANDIP UNIVERSITY, NASHIK,
111	SANJAY GHODAWAT INSTITUTE, KOLHAPUR
112	SANJEEVAN GROUP OF INSTITUTIONS, PANHALA
113	SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI
114	SAVEETHA SCHOOL OF ENGINEERING, SIMATS, CHENNAI
115	SCHOOL OF ARCHITECTURE, DSATM
116	SHARDA UNIVERSITY, GREATER NOIDA
117	SHIV NADAR UNIVERSITY, CHENNAI
118	SHIVAJI UNIVERSITY, KOLHAPUR
119	SHOBHIT UNIVERSITY, GANGOH
120	SHRI RAMSWAROOP MEMORIAL UNIVERSITY, LUCKNOW
121	SIDDHARTHA ACADEMY OF HIGHER EDUCATION, VIJAYAWADA
122	SJC INSTITUTE OF TECHNOLOGY, CHICKBALLAPUR
123	SKN SINHGAD COLLEGE OF ENGINEERING, PANDHARPUR
124	SNJB'S LATE SAU KANTABAI BHAVARLALJI JAIN COLLEGE OF ENGINEERING, NASHIK
125	SRI CHANDRASEKHANDRA SARASWATHI VISHWAMAHA VIDHYALAYA, KANCHIPURAM
126	SRI ESHWAR COLLEGE OF ENGINEERING, COIMBATORE
127	SRI KRISHNA COLLEGE OF ENGINEERING AND TECHNOLOGY, COIMBATORE
128	SRI KRISHNA COLLEGE OF TECHNOLOGY, COIMBATORE
129	SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE
130	SRI SAI RAM ENGINEERING COLLEGE, CHENNAI
131	SRI SHAKTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY, COIMBATORE
132	SRI SIVASUBRAMANIYA NADAR COLLEGE OF ENGINEERING, CHENNAI
133	SRI VENKATESWARA UNIVERSITY, TIRUPATI

134	SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, CHENNAI
135	SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, DELHI-NCR CAMPUS
136	SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, KATTANKULATHUR CAMPUS
137	SRM UNIVERSITY, AP
138	SSBT'S COLLEGE OF ENGINEERING AND TECHNOLOGY, JALGAON
139	ST FRANCIS INSTITUTE OF TECHNOLOGY
140	SWAMI VIVEKANANDA UNIVERSITY, BARRACKPORE, WEST BENGAL
141	SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT), PUNE CAMPUS
142	TATYASAHEB KORE INSTITUTE OF ENGINEERING AND TECHNOLOGY(DIPLOMA), WARANANAGAR
143	TECHNO INDIA UNIVERSITY, KOLKATA
144	TECHNO INTERNATIONAL NEW TOWN, KOLKATA
145	TERNA ENGINEERING COLLEGE, NAVI MUMBAI
146	UNIVERSITY OF DELHI, DELHI
147	VANITA VISHRAM WOMEN'S UNIVERSITY, SURAT
148	VELLORE INSTITUTE OF TECHNOLOGY, VELLORE
149	VIDYA PRATISHTHAN'S KAMALNAYAN BAJAJ INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNE
150	VIDYALANKAR INSTITUTE OF TECHNOLOGY, MUMBAI
151	VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE
152	VISTAS, CHENNAI
153	VISVESVARAYA TECHNOLOGICAL UNIVERSITY (VTU), BELAGAVI
154	VIT, CHENNAI
155	VIVEKANANDHA COLLEGE OF ENGINEERING FOR WOMEN, TIRUCHENGODE
156	WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
157	XAVIER INSTITUTE OF ENGINEERING, MUMBAI
158	YASHWANTRAO CHAVAN COLLEGE OF SCIENCE, KARAD